



Mathematics



Student's book

Version edited in 2026

Experimental version

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FOREWORD

Dear students,

The Rwanda Basic Education Board (REB) is honoured to present this Mathematics textbook for Senior Four (S4). This textbook is designed to support the effective implementation of the Competence-Based Curriculum (CBC). It promotes learner-centred teaching and learning practices that align with Rwanda's vision of empowering learners to reach their full potential, actively participate in society, and pursue meaningful careers.

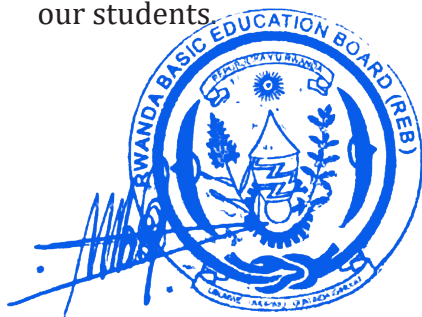
As Rwanda advances towards building a knowledge-based economy, education plays a pivotal role. Teachers are central to this transformation, guiding learners to connect classroom knowledge with real-life experiences, thereby fostering both individual growth and national development.

This student book serves as a comprehensive resource for the teaching and learning of Mathematics at Senior Four. Mathematics is an integral subject within the CBC, aimed at equipping learners with essential knowledge, skills, values, and attitudes necessary to become responsible and competent citizens. Through this subject, you are encouraged to develop problem-solving skills and contribute positively to your communities. In this regard, the only way to learn mathematics effectively is by doing mathematics through active problem-solving rather than passive reading. You are expected to work through problems, attempt proofs, and practise regularly, rather than merely reviewing theorems or looking up answers.

In its pursuit of quality education, the Government of Rwanda emphasizes the importance of using this kind of teaching and learning materials aligned with the curriculum.

I sincerely thank everyone who contributed to the development and refinement of this textbook. Special recognition goes to the REB team that coordinated the entire process. We also thank all partners, teachers, lecturers, and individuals who were involved from the beginning to the end.

Every feedback was highly valued and will contribute to improving the learning outcomes of our students.



Dr. MBARUSHIMANA Nelson

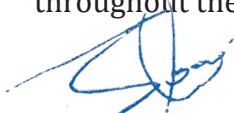
Director General

ACKNOWLEDGEMENT

I would like to extend my sincere appreciation to everyone who contributed to the refinement of the **Senior Four (S4) Mathematics textbook**. This achievement would not have been possible without the active involvement of various stakeholders.

Special thanks go to Rwanda Basic Education Board (REB) staff, subject experts, university lecturers and teachers for their valuable work in the editing of this textbook.

I am also grateful to all our education development partners whose support was instrumental throughout the process.



MURUNGI Joan

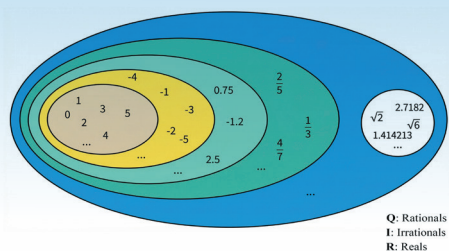
Head of CTRLR department

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UNIT 1

OPERATIONS ON THE SET OF REAL NUMBERS



Key Unit Competence:

Apply operations on real numbers, radicals, and logarithms in solving mathematical and real-life problems.

Learning objectives

By the end of this unit, you should be able to:

Knowledge and understanding

1

- Explain absolute value and describe its properties.
- Recall and explain laws of powers and radicals.
- State the properties of decimal logarithms.
- Identify relationships between exponential and logarithmic expressions.
- Explain how logarithmic models apply to real-life problems.

Skills

2

- Compute absolute values, powers, and radicals.
- Simplify and rationalize expressions with radicals.
- Apply logarithmic properties to solve problems.
- Model and solve real-life problems involving growth, decay, and compound interest.

Attitudes and values

3

- Appreciate the importance of real numbers in daily life
- Show accuracy and persistence in problem-solving.
- Demonstrate logical reasoning and teamwork during discussions
- Value the use of mathematics in explaining real world phenomena.
- Show respect, teamwork and unity when working with colleagues.

Introduction

This unit bridges foundational algebra with powerful mathematical tools used to model natural and financial phenomena.

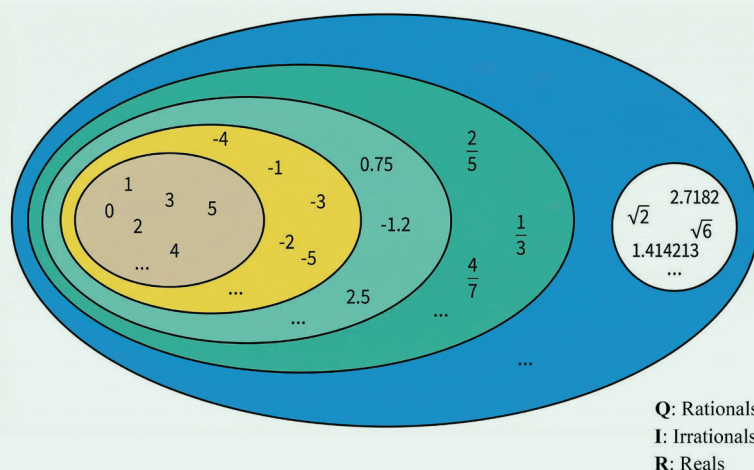
You will begin by exploring the absolute value as a measure of distance and its key properties, then move into the manipulation of powers and radicals including operations like addition, multiplication, and rationalization of denominators.

Building on this foundation, the unit introduces decimal logarithms (base 10), their properties, and how to solve equations involving them. These concepts are not purely abstract; you will apply them directly to real-world situations, learning to model growth (e.g., population, bacteria), decay (e.g., radioactive decay, drug concentration in the body), compound interest (e.g., savings accounts, loans) and earthquake magnitude (using the Richter scale, a logarithmic measure).

Introductory activity



From the following diagram, discuss and work out the given tasks:



- 1) How many sets of numbers do you know? List them down and give reasons for your answer.
- 2) Using a mathematical dictionary or the internet, define the sets of numbers you listed in (1).
- 3) Give an example of element for each set of numbers you listed.
- 4) Establish the relationship between the set of numbers that you listed.

1.1 Absolute value and its properties

Activity 1.1



Carry out a research on internet and Mathematics dictionary on the meaning of absolute value. Present your findings to the rest of the class for discussion. Use suitable examples.



Content Summary

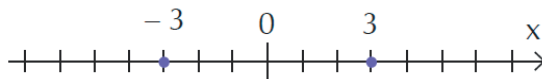
The **absolute value** of a number is the **distance of that number from zero on the number line**.

It is written using vertical bars: $|a|$ then $|6|=6$ and $|-6|=6$.

The absolute value of a real number is a positive number.

Definition (rigorous): $|x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$

The absolute value of x , denoted “ $|x|$ ” (and which is read as “the absolute value of x ”), is the distance of x from zero. This is why absolute value is never negative; absolute value only asks “how far?”, not “in which direction?” This means not only that $|3|=3$, because 3 is three units to the right of zero, but also that $|-3|=3$, because -3 is three units to the left of zero.



Geometric meaning: $|x-c|$ = distance between x and C .



Example: $|0-c| = |0-4| = 4$

Properties of Absolute Value

- 1) **Non-negativity:** $|a| \geq 0$: The absolute value of any number is always **positive or zero**.
- 2) Absolute value of zero: $|a| = 0$
- 3) Opposite numbers property $|a| = |-a|$

Example: $|7| = |-7| = 7$

- 4) Product property $|ab| = |a| \times |b|$

Example: $|(-3) \times (4)| = |-12| = 12$ or $|-3| \times |4| = 3 \times 4 = 12$

- 5) Quotient property: $\left| \frac{a}{b} \right| = \frac{|a|}{|b|}$ where $b \neq 0$

- 6) Triangle inequality: $|a+b| \leq |a|+|b|$ and $|a-b| \geq |a|-|b|$

- 7) $|X| = k$ ($k > 0$) $\Leftrightarrow X = k$ or $X = -k$.

Example, $|x| = 3$ means $x=3$ or $x=-3$.

- 8) $|X| < k$ ($k > 0$) $\Leftrightarrow -k < X < k$.

Example $|x| < 2 \Leftrightarrow -2 < x < 2$

9) $|x| > k (k > 0) \Leftrightarrow x < -k \text{ or } x > k$.

Example $|x| > 5 \Leftrightarrow x < -5 \text{ or } x > 5$

10) **Alternative:** $|A| = |B| \Leftrightarrow A^2 = B^2 \Leftrightarrow A = B \text{ or } A = -B$

Example: $|x| = 5 \Leftrightarrow x^2 = 25 \Leftrightarrow x = 5 \text{ or } x = -5$

More examples:

1) Solve $|2x-1| \leq 3x+2$ by considering domain issues.

Solution

Condition 1: $3x+2 \geq 0 \Rightarrow x \geq -\frac{2}{3}$

In this case, $|2x-1| \leq 3x+2$ means $-(3x+2) \leq 2x-1 \leq 3x+2$

Case 1 or condition 2:

$$-(3x+2) \leq 2x-1$$

$$-3x-2 \leq 2x-1$$

$$-3x-2x \leq -1+2$$

$$-5x \leq 1$$

$$x \geq -\frac{1}{5}$$

Case 2 or condition 3:

$$2x-1 \leq 3x+2$$

$$2x-3x \leq 2+1$$

$$-x \leq 3$$

$$x \geq -3$$

Considering the 3 conditions, we find that the intersection is the set of all real numbers $x \geq -\frac{1}{5}$.

Therefore, $S = \left\{ x \in \mathbb{R} : x \geq -\frac{1}{5} \right\} = \left[-\frac{1}{5}, +\infty \right)$.

2) Prove: $|a+b| \geq |a|-|b|$ for all real numbers a and b .

Solution:

By triangle inequality $|(a+b)-b| \leq |a+b|+|b| \Rightarrow |a|-|b| \leq |a+b|$

Similarly, $|b|-|a| \leq |a+b|$. Thus $|a|-|b| \leq |a+b|$.

Application activity 1.1



1) Simplify: a) $|-5|$ b) $|-4|+|-5|$ c) $|-7|+|4|$ d) $|-4 \times 6|$ d) $|-6+8|$

2) Solve the following equations: a) $|5-x| = 5$ b) $|5-x| = \frac{5}{2}$

3) Prove $|x+y| \leq |x|+|y|$ if and only if $xy \geq 0$. Interpret your findings verbally

1.2 Powers and radicals

Activity 1.2



a) In each of the following expression, state the number of factors and find the result:

1) $7 \times 7 \times 7$

2) $(-2) \times (-2)$

3) $4 \times 4 \times 4 \times 4$

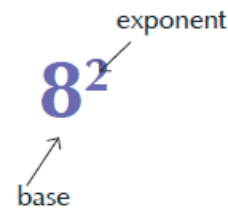
b) Evaluate the following powers 1) $(81)^{\frac{1}{2}}$ 2) $(256)^{\frac{1}{2}}$



Content Summary

A **power** is a mathematical expression that shows repeated multiplication of the same number or quantity.

General form: a^n where: a is the **base**, n is the **exponent**, a^n is the **power** read as “ a to the power of n ”. It can also be read as: “ a raised to the power n .”



Example: In the expression 8^2 we have:

Base: 8— the number being multiplied repeatedly.

Exponent (or index): 2— tells us how many times the base is used as a factor.

Power: 8^2 — the whole expression representing repeated multiplication.

Value of the power: $8 \times 8 = 64$.

Common powers: Square: a^2 : a squared and cube: a^3 : a cubed.

Properties of powers:

1) Product rule: $(a^n)(a^m) = a^{n+m}$

2) Quotient rule: $\frac{a^m}{a^n} = a^{m-n}, (a \neq 0)$

3) Power of power rule: $(a^m)^n = (a)^{m \times n}$

4) Power of product rule: $(ab)^m = a^m \times b^m$

5) Power of a quotient rule: $\left(\frac{a}{b}\right)^m = \frac{a^m}{b^m}, (b \neq 0)$

6) For the quotient rule: If $m=n$ we must agree that, $\frac{a^m}{a^n} = \frac{a^m}{a^m} = a^{m-m} = a^0 = 1$ where $a \neq 0$

7) $\frac{1}{a^n} = \frac{a^0}{a^n} = a^{0-n} = a^{-n}$ where $a \neq 0$ and gives the meaning to negative exponents.

Radicals (Roots): A radical represents the opposite operation of powers. The $\sqrt{a}$ means a number that multiplied by itself gives a .

$$\sqrt[n]{a} : \begin{cases} n \text{ is called the index} \\ a \text{ is called the base or radicand} \\ \sqrt{} \text{ is called the radical sign} \end{cases}$$

Examples: $\sqrt{9} = 3$ because $3 \times 3 = 9$.

Types of radicals: **Square root:** $\sqrt{a}$

Cube root: $\sqrt[3]{a}$.

The n^{th} root of real number is $\frac{1}{n}$ power of that real number. It is noted by $\sqrt[n]{a}$,

$$b \in R, n \in N \setminus \{1\}. \forall a, b \in R, \sqrt[n]{b} = a \Leftrightarrow b^{\frac{1}{n}} = a \Leftrightarrow b = a^n$$

If $n=2$, we say square root and $\sqrt[2]{b}$ is written as $\sqrt{b}$. Here b must be a positive real number or zero. If $n=3$, we say cube root noted $\sqrt[3]{b}$. Here b can be any real number. If $n=4$, we say 4th root noted $\sqrt[4]{b}$. Here b must be a positive real number or zero.

Generally, if $n=n$, we say n^{th} root noted $\sqrt[n]{b}$. Here if n is even, b must be a positive real number or zero and if n is odd b can be any real number (positive or negative)

Properties of radicals: $\forall n \in N \setminus \{1\}, m \in R$.

$$\text{a) } \sqrt[n]{a^m} = a^{\frac{m}{n}} \quad \text{b) } \sqrt[n]{ab} = \sqrt[n]{a} \times \sqrt[n]{b} \quad \text{c) } \sqrt[n]{\frac{a}{b}} = \frac{\sqrt[n]{a}}{\sqrt[n]{b}} \quad \text{d) } \sqrt[n]{\sqrt[m]{a}} = \sqrt[nm]{a} = a^{\frac{1}{nm}}.$$

Examples: Simplify

a) $\sqrt{46656}$

b) $\sqrt[3]{\sqrt{64}}$

c) $\sqrt[3]{ab} \times \sqrt[3]{a^2b^2}$

d) $\sqrt{\frac{36}{81}}$

Solution

a) $\sqrt{46656} = \sqrt{6^6} = 6^{\frac{6}{2}} = 6^3 = 216$

b) $\sqrt[3]{\sqrt{64}} = \sqrt[3 \times 2]{64} = \sqrt[6]{2^6} = 2$

c) $\sqrt[3]{ab} \times \sqrt[3]{a^2b^2} = \sqrt[3]{a^3b^3} = \sqrt[3]{(ab)^3} = ab$

d) $\sqrt{\frac{36}{81}} = \sqrt{\frac{6^2}{9^2}} = \frac{6}{9} = \frac{2}{3}$

Application activity 1.2



- 1) Simplify the following: a) $\sqrt[3]{27}$ b) $(\sqrt{8})^6$ c) $\sqrt[4]{64}$
- 2) Evaluate the following: a) $(2^3)(2^4)$ b) $\frac{7^8}{7^5}$ c) $(2^3)^4$
- 3) The area of a square garden is 36 cm^2 . Find the length of one side.
- 4) A cube has a volume of 64 cm^3 . Find the length of one edge.

1.3 Operation on radicals

Activity 1.3



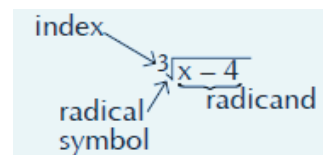
In group work out: 1) $\sqrt{4} + \sqrt{9}$ b) $2\sqrt{3} + 5\sqrt{3}$ c) $\sqrt{2} \times \sqrt{8}$



Content Summary

A radical is an expression that has a root, such as square root, cube root, etc. The symbol is $\sqrt{\quad}$

The square root of “ $x - 4$ ”.



The symbol $\sqrt{\quad}$ is used in connection with square roots, cube roots and n^{th} roots, for larger values of n . The notation $\sqrt{a}$ indicates square root $\sqrt[3]{a}$, indicates cube root and $\sqrt[n]{a}$ indicates n^{th} root of a .

Radicals can be added or subtracted only when they have the same radical part. The addition or subtraction of the radicals need to simplify if they have similar radicals. Similar radicals are the radicals with the same indices and same bases.

Fundamental Operations:

- 1) Multiplication of radicals (same index): $\sqrt[n]{a} \times \sqrt[n]{b} = \sqrt[n]{ab}$, $a, b \geq 0$ (for even n)
- 2) Division of radicals: $\frac{\sqrt[n]{a}}{\sqrt[n]{b}} = \sqrt[n]{\frac{a}{b}}$, $b \neq 0$
- 3) Addition/Subtraction: Only like radicals (same index and same radicand) can be combined: $2\sqrt{3} + 5\sqrt{3} = 7\sqrt{3}$
- 4) Expanding radicals with binomials (advanced):
 $(\sqrt{a} + \sqrt{b})(\sqrt{c} + \sqrt{d}) = \sqrt{ac} + \sqrt{ad} + \sqrt{bc} + \sqrt{bd}$

Example: Solve

1. $2\sqrt{5} + 3\sqrt{5}$
2. $7\sqrt{2} - 4\sqrt{2}$

Solution

1. $2\sqrt{5} + 3\sqrt{5} = 5\sqrt{5}$
2. $7\sqrt{2} - 4\sqrt{2} = 3\sqrt{2}$

Radicals with different bases **cannot be combined**. $\sqrt{3} + \sqrt{7}$ cannot be simplified.

When multiplying radicals, we may need to reduce to the same index.

Examples: Write as one radical $\sqrt[3]{3}\sqrt{7}$

Solution

$$\sqrt[3]{3} = \left(3^{\frac{1}{3}}\right)^{\frac{6}{6}} = \left(3^{\frac{6}{3}}\right)^{\frac{1}{6}} = \sqrt[6]{3^2} = \sqrt[6]{9}$$

$$\sqrt{7} = \left(7^{\frac{1}{2}}\right)^{\frac{6}{6}} = \left(7^{\frac{6}{2}}\right)^{\frac{1}{6}} = \sqrt[6]{7^3} \text{ Then } \sqrt[3]{3}\sqrt{7} = \sqrt[6]{9} \times \sqrt[6]{7^3} = \sqrt[6]{9 \times 7^3} = \sqrt[6]{3087}$$

Application activity 1.3



Simplify

1) $\sqrt{12} + \sqrt{3}$

2) $4\sqrt{3} - \sqrt{12}$

3) $5\sqrt{7} - \sqrt{28}$

4) A right triangle has legs $\sqrt{7} + 2$ and $\sqrt{7} - 2$. Find the area and the hypotenuse in simplest radical form.

1.4 Rationalization of the denominator

Activity 1.4



Discuss how these expressions could be rewritten without a radical in the denominator.

1) $\frac{1}{\sqrt{2}}$

2) $\frac{3}{\sqrt{5}}$



Content Summary

Rationalization is the process of removing radicals from the denominator of a fraction. This is done by multiplying the numerator and denominator by a suitable radical expression.

Note: If a, b, c and d are rational with b and d not both squares of rational numbers, then $a\sqrt{b} + c\sqrt{d}$ is a binomial surd.

The conjugate of the binomial surd $a\sqrt{b} + c\sqrt{d}$ is $a\sqrt{b} - c\sqrt{d}$ or $-a\sqrt{b} + c\sqrt{d}$.

Note that the product of a binomial surd and its conjugate is rational:

$(a\sqrt{b} + c\sqrt{d})(a\sqrt{b} - c\sqrt{d}) = a^2b - c^2d$. We often use this result to simplify surd expressions, particularly the quotient of binomial surds.

Types of rationalization:

Type 1: Monomial denominator with square root $\frac{a}{\sqrt{b}} = \frac{a}{\sqrt{b}} \times \frac{\sqrt{b}}{\sqrt{b}} = \frac{a\sqrt{b}}{\sqrt{b}}, b > 0$

Type 2: Monomial denominator with n^{th} root (extension) $\frac{a}{\sqrt[n]{b^m}} = \frac{a}{\sqrt[n]{b^m}} \times \frac{\sqrt[n]{b^{n-m}}}{\sqrt[n]{b^{n-m}}} = \frac{a\sqrt[n]{b^{n-m}}}{b}$

Type 3: Binomial denominator with square roots (conjugate method). For $\sqrt{a} \pm \sqrt{b}$,

$$\text{the conjugate } \sqrt{a} \mp \sqrt{b} \cdot \frac{c}{\sqrt{a} - \sqrt{b}} = \frac{c(\sqrt{a} + \sqrt{b})}{a - b} \text{ and } \frac{c}{\sqrt{a} + \sqrt{b}} = \frac{c(\sqrt{a} - \sqrt{b})}{a - b}$$

Type 4: Binomial with one radical and one rational term $\frac{c}{p + \sqrt{q}} = \frac{c(p - \sqrt{q})}{p^2 - q}$

Examples: 1. Express $\frac{\sqrt{2}}{\sqrt{5}}$ with a rational denominator.

Solution

$$1. \quad \frac{\sqrt{2}}{\sqrt{5}} = \frac{\sqrt{2}}{\sqrt{5}} \times \frac{\sqrt{5}}{\sqrt{5}} = \frac{\sqrt{10}}{5}$$

Express $\frac{3}{3\sqrt{2} - 2\sqrt{3}}$ with a rational denominator and in its simplest form.

Solution

$$\begin{aligned} \frac{3}{3\sqrt{2} - 2\sqrt{3}} &= \frac{3}{3\sqrt{2} - 2\sqrt{3}} \times \frac{3(3\sqrt{2} + 2\sqrt{3})}{3\sqrt{2} + 2\sqrt{3}} = \frac{3(3\sqrt{2} + 2\sqrt{3})}{18 - 12} \\ &= \frac{3(3\sqrt{2} + 2\sqrt{3})}{6} = \frac{3\sqrt{2} + 2\sqrt{3}}{2} \end{aligned}$$

Application activity 1.4



Simplify and give the answer in irrational form:

a) $\frac{3}{\sqrt{2}}$ b) $\frac{\sqrt{5}}{\sqrt{10}}$ c) $\frac{5}{2 - \sqrt{5}}$ d) $\frac{2\sqrt{3} - 1}{4 - \sqrt{3}}$

1.5 Decimal logarithms

Activity 1.5



1) By using internet and dictionary, discuss in groups and give two examples for each. a) What is a decimal? b) What is a logarithm?

2) Express these numbers as powers of 10:

a) $10 =$

c) $1000 =$

e) $0.1 =$

b) $100 =$

d) $1 =$

f) $0.01 =$



Content Summary

A decimal logarithm (or common logarithm) is a logarithm with base 10. For every positive real number N , $\log_{10}(N) = x$ when $10^x = N$.

Read: $\log_{10}(N) = x$: Log base 10 of N equals x . It represents the power x to which the base 10 must be raised to produce the number N .

Note: If no base is indicated we assume that its base is 10.

Key Values:

- a) $\log 1 = 0$ because $1 = 10^0$
- b) $\log 10 = 1$ because $10 = 10^1$
- c) $\log 10^n = n$ because $10^n = 10^n$ for the real number n .
- d) $\log 10^{-n} = -n$ for the real number n .

Examples: Without using a calculator, find:

- 1) $\log 1000$ 2) $\log \sqrt{10}$ 3) $\log \sqrt[3]{10}$

Solution

$$1) \log 1000 = \log 10^3 = 3 \quad 2) \log \sqrt{10} = \log 10^{\frac{1}{2}} = \frac{1}{2} \quad 3) \log \sqrt[3]{10} = \log 10^{\frac{1}{3}} = \frac{1}{3}$$

Application activity 1.5



Without using a calculator,

find: a) $\log 1,000,000$ b) $\log 10$ c) $\log 0.001$ d) $\log 1$

e) $\log \sqrt{10}$ f) $\log(\sqrt[4]{10})$ g) $\log\left(\frac{1}{\sqrt[4]{10}}\right)$

1.6 Properties of decimal logarithms

Activity 1.6



Discuss in groups and find:

- a) $\log (100 \times 10)$ b) $\log (1000 \div 10)$ c) $\log (10^3)$



Content Summary

Let us see properties of logarithms:

- 1) The logarithm of a product is equal to the sum of the logarithms:

$$\log(ab) = \log a + \log b$$

Example: $\log(100 \times 100) = \log 100 + \log 100 = 2 + 2 = 4$

This is true because $100 \times 100 = 10,000$ and $\log(100 \times 100) = \log 10,000 = \log 10^4 = 4$

Be careful:

$$\log 2x + 1 \neq \log(2x + 1) \text{ because}$$

$$\log 2x + 1 = (\log 2x) + 1$$

2) $\log \frac{1}{b} = -\log b$

- 3) The logarithm of a quotient equals the difference of the logarithms. $\log\left(\frac{a}{b}\right) = \log a - \log b$

Example: $\log\left(\frac{1000}{10}\right) = \log 1000 - \log 10 = 3 - 1 = 2$

- 4) The logarithm of a power equals the exponent multiplied by the logarithm. $\log(a^n) = n \log a$

Example: $\log(10^4) = 4 \log 10 = 4$

- 5) Logarithm of 1 $\log 1 = 0$ because $10^0 = 1$

- 6) Logarithm of 10: $\log 10 = 1$ because $10^1 = 10$

Example: Write as a single logarithm in the form $\log a$, $a \in R_0^+$.

a) $3 \log 5 - 2 \log 4$

b) $3 \log 4 + 6$

Solution

a) $3 \log 5 - 2 \log 4 = \log(5^3) - \log(4^2) = \log 125 - \log 16 = \log\left(\frac{125}{16}\right)$

b) $3 \log 4 + 6 = \log(4^3) + 6 \log 10 = \log 64 + \log(10^6)$

$$\log 64 + \log 1,000,000 = \log(64 \times 1,000,000) = \log 64,000,000$$

Application activity 1.6



1. Express each of the following as a single logarithm:

(a) $3 \log 2 + \log 4$ (b) $3 \log 2 + 5 \log 2$

(c) $2 \log 10 + \log 5$ (d) $2 \log 10 - 2 \log 2$

2. Simplify by writing each of the following as a single logarithm or write as an integer:

(a) $\log 6 + \log 5$ (b) $\log 16 - \log 2$

(c) $\log 24 - \log 6$ (d) $\log 7 + \log 5$

1.7 Solving problems about decimal logarithms

Activity 1.7



Discuss in group if it is true or false

1) $\log 1000 = 3$ 2) $\log 0.01 = -2$ 3) $\log(2+3) = \log 2 + \log 3$

4) $\log(20) = 1 + \log 2$ 5) $\log 1 = 0$



Content Summary

In general, $\log x = y \Leftrightarrow x = 10^y$, $\log_a(f(x))$ is defined iff $f(x) > 0, a > 0, a \neq 1$

$$\log(f(x)) + \log(g(x)) = k \Leftrightarrow \log[f(x) \cdot g(x)] = k$$

$$\text{and } \log f(x) = \log g(x) \Rightarrow f(x) = g(x) > 0$$

Example 1: Solve $\log(2x-3) + \log(x-1) = 1$.

Solution

Domain: $2x-3 > 0 \Rightarrow x > 1.5$ and $x-1 > 0 \Rightarrow x > 1$, Combined: $x > 1.5$.

$$\log[(2x-3)(x-1)] = 1 \Rightarrow (2x-3)(x-1) = 10 \quad 2x^2 - 5x - 7 = 0 \text{ or}$$

$$(2x-7)(x+1) = 0 \Rightarrow x = 3.5 \text{ or } x = -1$$

Domain: we reject $x = -1$ and finally, $x = 3.5$

Example 2: Solve $\log x + \log 5 = 3$

Solution

Using the product property: $\log x + \log 5 = \log(5x)$ $\log(5x) = 3$

$$5x = 10^3 \Rightarrow 5x = 1000 \Rightarrow x = 200$$

Example 3: Solve $\log(x-2) + \log(x+3) = \log 6$.

Solution

Using the quotient property: Domain: $x > 2$ and $x > -3 \rightarrow x > 2$.

$$\log[(x-2)(x+3)] = \log 6 \Rightarrow (x-2)(x+3) = 6 \quad (x+4)(x-3) = 0 \Rightarrow x = -4 \text{ (reject)}, x = 3 \text{ (accept)}$$

Application activity 1.7



1. Solve the following a) $\log x = 4$ b) $\log x = 1$ c) $\log(2x) = 3$
2. Use logarithm properties to solve a) $\log x + \log 4 = 2$ b) $\log x - \log 5 = 1$
3. a) If $\log N = 6$, find N b) If $\log(3x) = 4$, find x .

1.8 Model simple problems about growth

Activity 1.8



A plant grows by 2 cm every week.

Discuss in group how they could represent this growth using numbers, tables, or simple mathematical expressions.



Content Summary

Modelling growth means representing how a quantity increases over time using mathematical expressions, tables, or graphs. Two common types of growth models are:

1. Linear growth: The quantity increases by the same amount at each step.

Its general form: $y = a + b t$

Where:

a is the initial value

b is the constant increase

t is time.

2. Exponential growth

The quantity increases by the same factor or percentage over time. General form: $y = a \times b^t$

Where:

a is the initial amount

b is the growth factor

t is time

Growth models help us predict future values and understand patterns of change.

Example 1: Linear Growth

A plant is 10 cm tall and grows 3 cm each week. Find its height after 5 weeks.

Solution

Initial height = 10 cm

Growth per week = 3 cm

Height after 5 weeks = $10 + (3 \times 5) = 10 + 15 = 25$ cm

Example 2: Exponential Growth

A population of bacteria starts with 100 bacteria and doubles every hour. Find the number of bacteria after 3 hours.

Solution

Initial amount = 100

Growth factor = 2

$N = 100 \times 2^3 = 800$ So the population becomes 800 bacteria after 3 hours.

Application activity 1.8



1. A tree is 2 meters tall and grows 0.5 meters each year.
 - a) Find its height after 6 years.
 - b) Write a simple expression that represents its height after t years
2. A population starts with 50 animals and triples every year.
 - a) How many animals will there be after 2 years?
 - b) How many animals will there be after 3 years?
3. A student saves 1000 FRW each week starting with 500 FRW.
 - a) How much money will the student have after 5 weeks?
 - b) Write a formula representing the amount saved after t weeks.

1.9 Model simple problems about decay

Activity 1.9



Work out

"A candle loses 2 cm of its height every hour while burning."

Discuss in group how could represent this decrease using numbers, tables, or simple mathematical expressions.



Content Summary

Modelling decay means representing how a quantity **decreases over time** using mathematical expressions, tables, or graphs. Two common types of decay are:

1. Linear Decay

The quantity decreases by the same amount in each time period.

General form: $y = a - b t$

Where: a is the initial value; b is the amount of decrease per time unit; t is time.

Example 1: Linear Decay

A candle is 20 cm long and burns **2 cm** every hour. Find its length after **5** hours.

Solution

Initial length = 20 cm

Decrease per hour = 2 cm

Length after 5 hours = $20 - (2 \times 5) = 10 \text{ cm}$

2. Exponential Decay

The quantity decreases by the same factor or percentage over time.

General form: $y = a \times b^t$

Where a is the initial amount; b is the decay factor ($0 < b < 1$); t is time.

Decay models are used in many fields such as **radioactive decay, depreciation of assets, cooling processes, and medicine concentration in the body.**

Example 2: Exponential Decay

A machine worth **500,000** FRW loses **10%** of its value each year. Find its value after **2** years.

Solution

Initial value = 500,000 FRW

Decay factor = $1 - 0.10 = 0.9$

$V = 500000 \times (0.9)^2 = 500000 \times 0.81 = 405000 \text{ FRW}$.

Example 3

A medicine quantity in the body is **100 mg**, and each hour **half of it remains**. Find the amount after **3** hours.

Solution

Initial amount = 100 mg, decay factor = $\frac{1}{2}$, $A = 100 \times \left(\frac{1}{2}\right)^3 = 100 \times \left(\frac{1}{8}\right) = 12.5 \text{ mg}$

Application activity 1.9



1. A tank contains 80 liters of water, and 5 liters leak every hour.
 - a) How much water remains after 6 hours?
 - b) Write a formula representing the amount of water after t hours
2. A phone costs 300,000 FRW and loses 20% of its value each year.
 - a) Find its value after 1 year.
 - b) Find its value after 2 years.
3. A radioactive substance has 200 g, and half of it decays every hour.
 - a) How much remains after 2 hours?
 - b) How much remains after 4 hours?

1.10. Model simple problems about compound interest

Activity 1.10



Work out

"A student deposits 10,000 FRW in a bank that gives 10% interest per year."

- a) Discuss in group how the amount of money grows each year
- b) Do it for 5 years.



Content Summary

The formula for interest that is compounded is the compound interest formula when interest is compounded n times per year:

$$A = P \left(1 + \frac{R}{n \times 100} \right)^{n \times t}$$

Where: A = Total amount after t years; P = Principal amount; R = Annual interest rate (%);
 n = Number of compounding periods (times interest is compounded) per year; t = Time in years.

The above formula can be $A = P \left(1 + \frac{r}{n} \right)^{nt}$, where $r = \frac{R}{100}$ represents the interest rate and is always represented as a decimal number.

- if interest is compounded annually then $n = 1$
- if interest is compounded quarterly then $n = 4$
- if interest is compounded monthly then $n = 12$

Examples

1. Suppose your parents invest 10,000 FRW in a savings account for college at the time you are born. The average interest rate is 4% and is compounded quarterly. How much money will be in the college account when you are 18 years old?

Solution

We will use our formula $A = P\left(1 + \frac{r}{n}\right)^{nt}$ and let $P = 10,000$, $r = 0.04$, $n = 4$ and $t = 18$.

$$A = 10,000\left(1 + \frac{0.04}{4}\right)^{4 \times 18} = 20,471 \text{ FRW.}$$

2. Suppose your parents had invested that same 10,000 FRW in a money market account that averages 8% interest compounded monthly. How much would you have for college after 18 years?

Solution

$$P = 10,000, r = 0.08, n = 12 \text{ and } t = 18, A = P\left(1 + \frac{r}{n}\right)^{nt} = 10,000\left(1 + \frac{0.08}{12}\right)^{12 \times 18} = 42,005.7 \text{ FRW.}$$

Application activity 1.10



- 1) Mutesi invests 300,000 FRW at a bank that offers 5% compounded annually.
 - a) What is the growth factor for the investment?
 - b) Write an equation to model the growth of the investment.
 - c) How many years will it take for the initial investment to double
- 2) Murerwa bought a new car at a cost of 17, 850,000 FRW. The car depreciates by approximately 15% of its value each year
 - a) What is the depreciation factor for the value of this car?
 - b) Write an equation to model the depreciation value of this car.
 - c) What will the car be worth in 10 years?
- 3) Jose invests 500,000 FRW at a bank offering 10% compounded quarterly. Find the amount of the investment at the end of 5 years (if untouched).

1.11 Model simple problems about magnitude of an earthquake

Activity 1.11



Work out:

- 1) Use the internet and the Dictionary search the explanation of earthquake magnitudes
- 2) If "An earthquake has a magnitude of 5. What does this number represent?" discuss why scientists use a logarithmic scale instead of ordinary numbers.



Content Summary

The magnitude of an earthquake is measured using a logarithmic scale, often based on the Richter scale. This scale uses decimal logarithms to compare the intensity of earthquakes.

A simplified model for earthquake magnitude is:

$$M = \log\left(\frac{I}{I_0}\right) \text{ Where:}$$

M = magnitude of the earthquake

I = intensity of the earthquake

I_0 = reference intensity.

Key ideas:

An increase of **1 unit** in magnitude means the earthquake is **10 times more intense**.

An earthquake of magnitude **6** is **10 times stronger** than magnitude **5**.

This logarithmic model helps scientists compare very large differences in earthquake intensity.

Examples

- 1) An earthquake has an intensity **10 times greater** than the reference intensity I_0 . Find its magnitude.

Solution

$$M = \log\left(\frac{I}{I_0}\right) \text{ and } \frac{I}{I_0} = 10 \text{ then } M = \log 10 \text{ So, the magnitude is } \mathbf{1}.$$

- 2) An earthquake has intensity **1000 times** the reference intensity. Find the magnitude.

Solution

$$M = \log 1000 = 3 \text{ The earthquake magnitude is } \mathbf{3}.$$

- 3) If the magnitude of an earthquake is 5, find the intensity ratio $\frac{I}{I_0}$

Solution

$$M = \log\left(\frac{I}{I_0}\right) \text{ then } 5 = \log\left(\frac{I}{I_0}\right)$$

$$\frac{I}{I_0} = (10)^5 = 100,000 \text{ The earthquake intensity is } \mathbf{100,000 \text{ times the reference intensity.}}$$

Application activity 1.11



- 1) Find the magnitude of an earthquake if:
a) $\frac{I}{I_0}=10$ b) $\frac{I}{I_0}=100$ c) $\frac{I}{I_0}=10000$
- 2) Find the intensity ratio if the magnitude is:
a) $M = 2$ b) $M = 4$ c) $M = 6$
- 3) Two earthquakes have magnitudes 4 and 6.
a) Which earthquake is stronger?
b) How many times stronger is it?

End of unit assessment



Assessing knowledge and understanding:

1. Write **True (T)** or **False (F)** for each statement.
 - a) The absolute value of a number is always positive or zero.
 - b) $|-8|=-8$
 - c) $10^3=1000$
 - d) $\sqrt{25}=5$
 - e) Decimal logarithms are logarithms with base 10.
 - f) $\log(ab)=\log a + \log b$
 - g) Rationalization means removing a radical from the denominator of a fraction.
 - h) Earthquake magnitude is measured using a logarithmic scale.
2. Choose the correct answer.
 - a) The value of $|-7|$ is: A. -7 B. 7 C. 0 D. -1
 - b) Which of the following is equivalent to $\sqrt{a} \times \sqrt{b}$
A. $\sqrt{ab}$ B. $\sqrt{a+b}$ C. ab D. $\sqrt{a-b}$
 - c) If $\log(\mathbf{ab})$ is expanded, the result is:
A) $\log a - \log b$, B) $\log a + \log b$, C) $\log a \log b$ D) $\frac{\log a}{\log b}$
 - d) If a quantity increases by a constant percentage each year, the model is called:
A. Linear growth B. Exponential growth
C. Logarithmic growth
D. Constant growth

e) The formula for compound interest is:

- A. $A = P + rt$ B. $A = Prt$
C. $A = P(1+r)t$ D. $A = P(1-r)t$

f) Earthquake magnitude is measured using:

- A. Linear scale B. Logarithmic scale
C. Quadratic scale D. Cubic scale

Assessing skills

3. Evaluate: a) $\log 10$ b) $\log 100$ c) $\log 1000$
4. Simplify a) $\log 6 + \log 3$ b) $\log 20 - \log 4$
5. Write the general exponential model for decay

Assessing attitudes and values

6. A student deposits **200,000 FRW** in a bank at an interest rate of **5% per year compounded annually**. Workout and choose the correct answer:

a) What will be the amount after 3 years?

- A. 220,000 FRW
B. 231,525 FRW
C. 231,000 FRW
D. 215,250 FRW

b) Why is compound interest important when saving money?

- A. It reduces the amount of money saved each year
B. It helps money grow faster because interest is earned on previous interest
C. It only benefits banks, not savers
D. It keeps the savings amount constant

7. A student receives some money and decides to keep part of it in a savings account instead of spending it all. Which statement best shows a positive attitude toward financial responsibility?

- A. Spending all the money immediately to enjoy it now
B. Saving regularly to prepare for future needs
C. Ignoring the idea of keeping money in a bank
D. Borrowing money instead of saving.

UNIT 2

BINARY OPERATIONS

IDENTITY:
 $a * e = e * a = a$

ELEMENT:
 $a * e = a$

INVERSE:

INVERSE:
 $a * a^{-1} = e$



Key Unit Competence:

Apply the concepts of binary operations and algebraic structures such as groups, rings, and fields in analysing mathematical problems.

Learning objectives

By the end of this unit, you should be able to:

Knowledge and understanding

1

- Identify a binary operation and explain its meaning.
- State and describe the closure, commutative, identity, inverse, associative, and distributive properties.
- Explain how to construct and interpret a Cayley table for a given operation.
- Describe the basic algebraic structures and their properties.
- Identify examples of Groups, Rings, and Fields within number systems.

Skills

2

- Apply the properties of binary operations to simplify and verify algebraic expressions.
- Construct and use Cayley tables to test properties such as closure, commutativity, and associativity.
- Determine whether a given set and operation form a Group, Ring, or Field.
- Solve mathematical problems involving operations within algebraic structures.
- Analyse and compare different algebraic structures through guided activities and exploration.

Attitudes and values

3

- Appreciate the logical structure and consistency of algebraic systems.
- Develop accuracy, organization, and logical reasoning when performing mathematical operations.
- Show perseverance, curiosity, and teamwork in exploring abstract mathematical concepts.
- Value mathematical precision and the role of algebraic structures in understanding real-world systems.

Introduction

This unit introduces the foundational concepts of abstract algebra, moving from basic binary operations to the formal algebraic structures of groups, rings, and fields.

The purpose of this unit is to develop students' understanding of the definition and properties of binary operations, including closure and commutative properties, as well as identity and inverse properties, and associativity and distributive properties.

The unit also introduces the Cayley table of a binary operation and explores how these operations lead to important algebraic structures such as groups, rings, and fields. Through exploring activities on the structures of rings and fields, students learn to analyze and apply these concepts in mathematical reasoning.

By the end of this unit, you will be able to identify and verify the properties of binary operations, construct and interpret Cayley tables, and formally distinguish between the algebraic structures of groups, rings, and fields through guided exploration and problem-solving activities.

Introductory activity



1) Complete the following table

Operation	Example	Result
Addition	$4 + 3$	
Multiplication	5×25	
Subtraction	$7 - 4$	
Division	$8 \div 2$	

Work out the following questions.

- What do you notice about the results of the operations?
 - Does the result always belong to the same set of numbers used?
 - Try reversing the numbers
- 2) By using internet and Mathematics Dictionary explain these new words:
- Closure
 - Commutative property
 - Identity element
 - Inverse element
 - Associativity
 - Group properties.

2.1 Definition and properties.

2.1.1 Binary operation, Closure and commutative properties

Activity 2.1.1



Answer by true or false. If true, give an example and if false give a counter

- 1) The sum of two natural numbers is a natural number.
- 2) The product of two natural numbers is a natural number.
- 3) The difference of two natural numbers is not always a natural number.
- 4) The sum of two integers is an integer.
- 5) The difference of two integers is an integer.
- 6) The product of two integers is an integer.
- 7) The quotient of two integers is not always an integer.
- 8) The sum of two real numbers is a real number.
- 9) The product of two real numbers is a real number.
- 10) The quotient of two natural numbers is not always a natural number.



Content Summary

A **binary operation** in set F is any function that combines any two elements called **operands**, and produces another element of set F . A set equipped with a binary operation is called **magma**.

A **binary operation** on a set is a rule that combines two elements of the set to produce another element of the same set.

A set is **closed** under a binary operation if performing the operation on any two elements of the set always produces another element in the same set.

A set S is **closed** under an operation $*$ if: $a * b \in S$ for all $a, b \in S$. This means the result **remains in the same set**.

A binary operation is **commutative** if changing the order of the elements does not change the result. $a * b = b * a$

Example 1: Let S be a set and operation $*$ defined by: $a * b = a + b$. Check closure:

Solution:

Since $a + b \in S$ the **Closure holds**

Example 2: Check if the operation is commutative: $a * b = a + b$

Solution:

$a * b = a + b$ and $b * a = b + a$ then $a + b = b + a$ mean that $a * b = b * a$ hence it is **Commutative**

Example 3: Let: $a * b = a - b$ check commutativity:

Solution:

$a * b = a - b$, $b * a = b - a$ since: $a - b \neq b - a$ then it is not **commutative**

Example 4: Let: $a * b = ab$ check closure:

Solution:

Solution since product remains in the set, it is **closed**

Example 5: check commutativity: $a * b = ab$

Solution:

Since: $a * b = ab$ and $b * a = ba$ it is commutative.

Application activity 2.1.1



- 1) Fill in the blanks: a) $a * b = b * a \rightarrow$ _____ property
b) $a * b \in S \rightarrow$ _____ property
- 2) Given: $a * b = a + b$
 - a) Check if the operation is closed,
 - b) Check if it is commutative
- 3) Let $a * b = a - b$. Is it commutative?
- 4) Let $a * b = ab$. Is it closed?
- 5) Define a binary operation using letters and test closure
- 6) Can an operation be closed but not commutative?
- 7) Give an example using letters

2.1.2 Identity and inverse properties

Activity 2.1.2



- 1) In groups calculate these questions and discuss what you notice after the operations a) $a + \square = a$ b) $a \times \square = a$
- 2) Discuss the value of b be in each case a) $a + b = 0$ b) $a \times b = 1$.



Content Summary

Identity Property: An **identity element** of a binary operation is an element that leaves any element of the set unchanged when the operation is performed.

For a set S with operation $*$, an identity element e satisfies: $a * e = e * a = a$ for every $a \in S$

- For **addition**, the identity element is **0**: $a + 0 = 0 + a = a$
- For **multiplication**, the identity element is **1**: $a \times 1 = 1 \times a = a$

An **inverse element** of an element a is another element that combines with a to produce the identity element. For an operation $*$, the inverse of a is a^{-1} such that: $a * a^{-1} = e$ where e is the identity element.

Examples:

- For **addition**, the additive inverse of a is **$-a$** because $a + (-a) = 0$
- For **multiplication**, the inverse of a (when $a \neq 0$) is $\frac{1}{a}$ because $a \times \frac{1}{a} = 1$

1. Identify the identity element for addition on integers.

Solution:

For any integer a : $0 + a = a$ and $0 + a = a$. Therefore, **0 is the identity element for addition.**

2. Find the additive inverse of 7.

Solution:

The additive inverse of a number is the number that gives 0 when added. $7 + (-7) = 0$. So the additive inverse of **7 is -7 .**

3. Find the multiplicative inverse of 5.

Solution:

The multiplicative inverse of a number is the number that gives **1** when multiplied. $5 \times \frac{1}{5} = 1$, so the **multiplicative inverse of 5 is $\frac{1}{5}$**

4. Find the additive inverse of -4 .

Solution:

$-4 + 4 = 0$ So the **additive inverse of -4 is 4.**

Application activity 2.1.2



- 1) Explain why 0 is said to be the identity for addition.
- 2) Explain why 1 is the identity for multiplication
- 3) Find the additive inverse of x
- 4) Find the multiplicative inverse of y
- 5) Solve: $b + y = 0$
- 6) Solve: $c \times z = 1$
- 7) Why does 0 **not** have a multiplicative inverse?

2.1.3 Associativity and distributive properties

Activity 2.1.3



Work in pairs to **compare the following expressions** and discuss if they are equal for the real numbers a , b , and c :

- 1) $(a \times b) \times c$ and $a \times (b \times c)$
- 2) $a(b + c)$ and $ab + ac$



Content Summary

A binary operation $*$ is **associative** if the way numbers are grouped does not change the result. For an operation $*$, the associative property is: $(a * b) * c = a * (b * c)$ for all a, b, c .

Examples:

- Addition is associative: $(4 + 5) + 6 = 9 + 6 = 15$ and $4 + (5 + 6) = 4 + 11 = 15$
- Multiplication is also associative: $(3 \times 4) \times 5 = 12 \times 5 = 60$ and $3 \times (4 \times 5) = 3 \times 20 = 60$
- Not associative: Subtraction, division. $(a \div b) \div c \neq a \div (b \div c)$

The distributive property shows how multiplication distributes over addition or subtraction. $a(b + c) = ab + ac$ then $a(b - c) = ab - ac$. One element multiplies each term inside the bracket.

Why is this property useful?

- To expand brackets
- To factorize expressions (reverse of distribution)
- To simplify mental arithmetic (e.g., $7 \times 102 = 7 \times 100 + 7 \times 2$)

Note:

- Assuming distribution over division: $\frac{a}{b+c} \neq \left(\frac{a}{b} + \frac{a}{c}\right)$
- Assuming distribution of exponentiation: $(a+b)^2 \neq (a^2 + b^2)$

Example: $3(2+5) = 3 \times 7 = 21$ and $3(2+5) = (3 \times 2) + (3 \times 5) = 6 + 15 = 21$

Both results are the same.

Application activity 2.1.3



- 1) Let $a * b = a - b$. Test associativity
- 2) Expand: $x(y + z)$
- 3) Check: $(a + b)c = ac + bc$
- 4) Why is addition associative but subtraction is not?
- 5) Can an operation be distributive but not associative?
- 6) A rectangle has length $(2x+5)$ m and width 3 m. Write two expressions for area using distributive property.

2.2 Cayley table of binary operation

Activity 2.2



Work out in group

A set $S = \{1, 2, 3\}$ with addition as the operation combine each pair of elements and record the results in a table follow:

+	1	2	3
1			
2			
3			



Content Summary

A **Cayley table** is a table that shows the results of a **binary operation** performed on all possible pairs of elements in a set.

The elements of the set are written across the **top row** and down the **first column**. Each cell of the table shows the result of applying the operation to the corresponding pair of elements.

Example structure:

*	<i>a</i>	b	c
<i>a</i>	$a * a$	$a * b$	$a * c$
<i>b</i>	$b * a$	$b * b$	$b * c$
<i>c</i>	$c * a$	$c * b$	$c * c$

Examples:

- 1) The set $S = \{a, b, c\}$ with binary operation $*$ defines a group illustrated by the following Cayley table:

*	<i>a</i>	<i>b</i>	<i>c</i>
<i>a</i>	<i>a</i>	<i>b</i>	<i>c</i>
<i>b</i>	<i>b</i>	<i>c</i>	<i>a</i>
<i>c</i>	<i>c</i>	<i>a</i>	<i>b</i>

- a) Is $(S, *)$ closure?
 b) Find the identity element
 c) Find the inverse of each element of S .
 d) Is $(S, *)$ commutative?
 e) Is $a * (b * c) = (a * b) * c$?

Solution:

- a) $(S, *)$ is closure since every product inside the table is the element of S .
 b) The identity element is a since
 a) $a * a = a, a * b = b * a = b,$
 $a * c = c * a = c.$
 c) Since $a * a = a$ then $a^{-1} = a$.
 Since $b * c = a$, then $b^{-1} = c$.
 Since $c * b = a$ then $c^{-1} = b$
 (where a^{-1} means the inverse of a .
 d) Yes, $(S, *)$ is commutative since $a * b = b * a = b, a * c = c * a = c,$
 and $b * c = c * b = a$
 (the products inside the table are symmetric around the main diagonal.)
 e) Yes, $a * (b * c) = (a * b) * c$
 since $a * (b * c) = a * a = a$
 $= (a * b) * c = b * c = a.$

- 2) Table for $\bullet$ on $\{m, n, p\}$:

$\bullet$	<i>m</i>	<i>n</i>	<i>p</i>
<i>m</i>	<i>n</i>	<i>p</i>	<i>m</i>
<i>n</i>	<i>p</i>	<i>m</i>	<i>n</i>
<i>p</i>	<i>m</i>	<i>n</i>	<i>p</i>

Find identity element (if any).

Solution:

Check row m : $m \bullet m = n (\neq m)$, $m \bullet n = p (\neq n)$, $m \bullet p = m (\neq p) \rightarrow m$ not identity.

Row n : $n \bullet m = p (\neq m)$, $n \bullet n = m (\neq n)$, $n \bullet p = n (\neq p) \rightarrow$ no.

Row p : $p \bullet m = m (\neq p)$? Wait m is header column – check col

m : $p \bullet m = m$, $m \bullet m = n$, $n \bullet m = p \rightarrow$ no element gives m in all rows). Actually identity requires: $e \bullet x = x$ and $x \bullet e = x$ for all x . Check col m : $m \bullet m = n$, $n \bullet m = p$, $p \bullet m = m \rightarrow$ only $p \bullet m = m$ but fails for others. No identity.

Application activity 2.2



- 1) Construct the Cayley table for the set $S = \{0, 1, 2\}$ under **addition**.
- 2) Construct the Cayley table for the set $S = \{1, 2, 3\}$ under **multiplication**.
- 3) Given the set $S = \{0, 1\}$, under addition modulo.
 - a) Construct the Cayley table.
 - b) Check whether the operation is **commutative**.
 - c) Verify whether the set is **closed** under the operation.

2.3 Algebraic structure of a group

Activity 2.3



Work out:

The set $G = \{-2, -1, 0, 1, 2\}$ under the operation of addition. Perform the following tasks:

- 1) Add different pairs of numbers from the set and check whether the result still belongs to the set.
- 2) Choose three numbers from the set and compare $(a+b)+c$ with $a+(b+c)$
- 3) Find a number in the set that does not change other numbers when added to them.
- 4) For each number in the set, try to find another number in the set that gives when added together.



Content Summary

A **group** is an algebraic structure made of a **set** together with a **binary operation** that satisfies four important properties. Let G be a set with a binary operation $*$. The structure $(G, *)$ is called a **group** if the following properties hold:

- 1) **Closure:** For every $a, b \in G \Rightarrow a * b \in G$. The result of the operation remains in the set.
- 2) **Associativity:** For all $a, b, c \in G$, $(a * b) * c = a * (b * c)$
- 3) **Identity Element:** There exists an element $e \in G$ such that: $a * e = e * a = a$ for every $a \in G$
- 4) **Inverse Element:** For every $a \in G$, there exists an element a^{-1} in G such that: $a * a^{-1} = e$ where e is the identity element.

Note: If the operation also satisfies: $a * b = b * a$ then the group is called an **Abelian group** or **commutative group**.

Example 1: Show that the set of integers $\mathbb{Z}$ under addition forms a group.

Solution:

- 1) **Closure:** $a + b \in \mathbb{Z}$ for all integers a and b
- 2) **Associativity:** $(a + b) + c = a + (b + c)$
- 3) **Identity:** The identity element is **0** since: $a + 0 = a$
- 4) **Inverse:** For every integer a , the inverse is $-a$: $a + (-a) = 0$. Thus, integers under addition form a **group**.

Example 2: Determine whether natural numbers under addition form a group.

Solution:

- 1) Closure holds.
- 2) Associativity holds.
- 3) Identity element exists (0).
- 4) However, the inverse of a natural number is not always a natural number: Because: $3 + (-3) = 0$ But -3 is not a natural number.

Therefore, natural numbers under addition do not form a group.

Application activity 2.3



1. For the set $G = \{-1, 0, 1\}$ under addition:
 - a) Check whether closure holds.
 - b) Identify the identity element.
 - c) Find the inverse of each element.
2. Given: $a * b = a + b$. Verify all group properties
3. Let $a * b = a - b$. Is it a group?
4. Let $a * b = ab$. Identify identity element
5. Find inverse of x under addition
6. Find inverse of y under multiplication
7. Can a structure be a group without identity?
8. Why is associativity important in defining a group?

2.4 Algebraic structure of Ring

Activity 2.4



Complete this table

a	b	$a + b$	$a \times b$
2	3	?	?
-4	5	?	?
6	-2	?	?

Perform the following tasks:

1. What do you notice about the results of **addition**?
2. What do you notice about the results of **multiplication**?
3. Do the answers always remain **integers**?



Content Summary

The set of real number R with two operations Addition (+) and Multiplication ($\times$) written $(R, +, \times)$ is a **ring** when it satisfies the following properties:

1) Properties of Addition: For any $a, b, c \in R$

a) **Closure:** $a + b$ belongs to R

b) **Associative law:** $(a + b) + c = a + (b + c)$

c) **Additive identity:** There exists 0 such that $a + 0 = a$

d) **Additive inverse:** For every a , there exists $-a$ such that $a + (-a) = 0$

e) **Commutative law:** $a + b = b + a$

2) Properties of Multiplication

a) **Closure:** $a \times b$ belongs to R

b) **Associativity:** $(ab)c = a(bc)$

c) **Distributive Property:** Multiplication distributes over addition. $a(b + c) = ab + ac$

Example: Prove that $(\mathbb{Q}, +, \times)$ is a ring.

Solution:

Let $x = \frac{a}{b}, y = \frac{c}{d}, z = \frac{e}{f}$ be rational numbers where $a, b, c, d, e, f \in \mathbb{Z}$ and $b, d, f \neq 0$

1) Closure under Addition $x + y = \frac{a}{b} + \frac{c}{d} = \frac{ad + bc}{bd} \in \mathbb{Q}$ then it is closure.

2) Closure under Multiplication $x \times y = \frac{a}{b} \times \frac{c}{d} = \frac{ac}{bd}$

3) **Associativity of Addition:** For all $x, y, z \in \mathbb{Q}$, $(x + y) + z = x + (y + z)$ This property holds because addition of rational numbers inherits associativity from integers.

4) **Commutativity of Addition:** $x + y = y + x$

5) **Additive Identity:** The number **0** acts as the additive identity. $x + 0 = x$

6) **Additive inverse.** For any rational number $x = \frac{a}{b}$, $-x = -\frac{a}{b}$ which is also rational. **Every element has an additive inverse.**

7) **Distributive Property:** Multiplication distributes over addition:

8) $x(y + z) = xy + xz$ and $(x + y)z = xz + yz$

All ring axioms are satisfied:

- Closure under $+$ and $\times$
- Associativity
- Commutativity of addition
- Additive identity 0
- Additive inverse
- Distributive law

Therefore, $(\mathbb{Q}, +, \times)$ is a ring.

Application activity 2.4



- 1) Can multiplication lack identity in a ring?
- 2) Why must addition be commutative?
- 3) What is the difference between a group and a ring?
- 4) Why must addition be commutative in a ring?
- 5) Can multiplication fail to be commutative in a ring?

2.5 Exploring activities on the structure of ring

Activity 2.5



With the set $R = \{-2, -1, 0, 1, 2\}$ with the operations of **addition (+)** and **multiplication ($\times$)**. Perform the following tasks:

1. Add different pairs of numbers from the set and check whether the result belongs to the set.
2. Multiply different pairs of numbers and observe whether the result remains in the set.
3. Check whether addition satisfies the associative and commutative properties.
4. Identify the identity element for addition.
5. Check whether multiplication distributes over addition by testing examples such as $2(1 + 2)$ and $(2 \times 1) + (2 \times 2)$



Content Summary

A ring is an algebraic structure consisting of a set $\mathbf{R}$ equipped with two binary operations, usually addition (+) and multiplication ($\times$).

Example 1:

Verify if the set of integers with ordinary addition and multiplication $(\mathbb{Z}, +, \cdot)$ is a ring or not.

Solution:

$(\mathbb{Z}, +, \cdot)$ is a commutative group

$(\mathbb{Z}, \cdot)$ is closed since: For every $a, b \in \mathbb{Z}$, then $(a \cdot b) \in \mathbb{Z}$

$(\mathbb{Z}, \cdot)$ is associative since: For every $a, b, c \in \mathbb{Z}$, then $a \cdot (b \cdot c) = (a \cdot b) \cdot c$

The operation is distributive over +: For every $a, b, c \in \mathbb{Z}$, then $a \cdot (b + c) = a \cdot b + a \cdot c$ and $(a + b) \cdot c = a \cdot c + b \cdot c$. We conclude that $(\mathbb{Z}, +, \cdot)$ is a ring.

Example 2: Determine whether natural numbers form a ring.

Solution:

Natural numbers are closed under addition and multiplication. However, natural numbers **do not have additive inverses**. Because: $5 + (-5) = 0$ But **-5 is not a natural number**. Therefore, natural numbers **do not form a ring**.

Application activity 2.5



Let $S = \{0, 1, a, b\}$. Define addition (+) and multiplication ($\times$) as per the Cayley tables below. Is this a ring? If so, what kind?

+	0	1	a	b	$\times$	0	1	a	b
0	0	1	a	b	0	0	0	0	0
1	1	0	b	a	1	0	1	a	b
a	a	b	0	1	a	0	a	b	1
b	b	a	1	0	b	0	b	1	a

- 1) Verify additive group: Is it abelian?
- 2) Verify multiplicative structure: Associative?
- 3) Is there a multiplicative identity?
- 4) Are there zero divisors? Check all products:

2.6 Algebraic structure of a field

Activity 2.6



For the following table.

Set	Set and operations
$\mathbb{Z}$	Integers under + and $\times$
$\mathbb{Q}$	Rational numbers under + and $\times$
$\mathbb{R}$	Real numbers under + and $\times$

In groups discuss and answer the following questions:

- 1) Is this a commutative ring with unity? (Yes/No)
- 2) Does every **non-zero** element have a multiplicative inverse within the set? (Yes/No)
- 3) If you answered “No” to the second question, identify a specific non-zero element that lacks an inverse.



Content Summary

A field $(F, +, \cdot)$ is a field if it satisfies the following conditions:

1. $(F, +, \cdot)$ is a ring
2. **Commutativity** under ' $\cdot$ ': For every $a, b \in F$, $a \cdot b = b \cdot a$,
3. **Identity element e** under ' $\cdot$ ': For every $a \in F$, there exists $e \in F$ such that $a \cdot e = e \cdot a = a$.
4. **Inverse element** under ' $\cdot$ ': For every $a \in F$ and $a \neq 0$, there exists $a^{-1} \in F$ such that $a \cdot a^{-1} = a^{-1} \cdot a = e$

A field F is a non-empty set defined by the following properties. For all $a, b, c \in F$:

Under + (addition property)	Under $\times$ (multiplication property)
a) Closure: $a + b \in F$ b) Associativity: $(a + b) + c = a + (b + c)$ c) Additive inverse: $a + a^{-1} = a^{-1} + a = e$ for every $a \in F$, e is the identity element d) Additive identity: there exists an identity element $a \in F$ such that $a + e = e + a = a$ e) Commutative: $a + b = b + a$ f) Distributive property: $a \times (b + c) = (a \times b) + (a \times c)$ $(b + c) \times a = (b \times a) + (c \times a)$	a) Closure: $a \times b \in F$. b) Associativity: $(a \times b) \times c = a \times (b \times c)$. c) Multiplicative inverse for every $a \neq 0$, there exists an element an element $a^{-1} \in F$ such that $a^{-1} \cdot a = e$, e is the identity element. d) Multiplicative identity, there exists an identity element $e \in F$ such that $a \times e = e \times a = a$. (e) e) Commutative $a \times b = b \times a$. f) Cancellation law $a \neq e$ $a \times b = a \times c$ implies $b = c$.

Example: Show that **rational numbers** $(\mathbb{Q}, +, \times)$ form a field.

Solution:

For rational numbers $(\mathbb{Q}, +, \times)$:

1. Closure: Let $a, b \in \mathbb{Q}$ where $a = \frac{p}{q}, b = \frac{r}{s}, q \neq 0, s \neq 0$

$$\text{Addition: } a + b = \frac{ps + rq}{qs} \in \mathbb{Q} \text{ and Multiplication: } a \times b = \frac{pr}{qs} \in \mathbb{Q}$$

Those show Closed under both operations.

2. Associativity

For all $a, b, c \in \mathbb{Q}$: Addition: $(a + b) + c = a + (b + c)$, and Multiplication: $(a \times b) \times c = a \times (b \times c)$

3. Commutativity: For all $a, b \in \mathbb{Q}$, Addition: $a + b = b + a$ and Multiplication: $a \times b = b \times a$

4. Identity Elements: Additive identity: $a + 0 = a$ and Multiplicative identity: $a \times 1 = a$

Where $0, 1 \in \mathbb{Q}$ then Identities exist

5. Inverses: Additive inverse: For every $a \in \mathbb{Q}$, $a + (-a) = 0$. Multiplicative inverse: For

every $a \neq 0$ $a \times \frac{1}{a} = 1$. Since $a = \frac{p}{q}$, its inverse is $\frac{q}{p} \in \mathbb{Q}$ then Inverses exist

6. Distributive Property: For all $a, b, c \in \mathbb{Q}$: $a \times (b + c) = (a \times b) + (a \times c)$ then Distributive law holds

Conclusion: $(\mathbb{Q}, +, \times)$ is a **field**.

Application activity 2.6



1. Fill in the blanks.

- a) A field has two operations: ____ and ____
- b) The additive identity in a field is ____.
- c) The multiplicative identity in a field is ____.
- d) Every non-zero element in a field has a _

2. True or False

- a) Division by zero is allowed in a field.
- b) Every field is also a ring.
- c) Rational numbers form a field.

2.7 Exploring activities on the structure of Field

Activity 2.7



- Find x for the following questions in $\mathbb{Z}$ (integers), $\mathbb{Q}$ (rational) and $\mathbb{R}$ (reals)
 - Equation $A: 2x = 1$
 - Equation $B: 3x = 2$
 - Equation $C: x^2 = 2$ (if possible)
 - Equation $D: x^2 + 1 = 0$ (if possible)
- Record these equations, which have solutions within the system.
- Identify any patterns about them when solutions exist.



Content Summary

Example 1: Show that the set of real numbers $(\mathbb{R}, +, \times)$ satisfies all the field axioms.

Solution:

- Closure:** Addition: For any $a, b \in \mathbb{R}$, $a + b \in \mathbb{R}$ for example $2.5 + 3.1 = 5.6 \in \mathbb{R}$
Multiplication: For any $a, b \in \mathbb{R}$ for example $2 \times 4 = 8 \in \mathbb{R}$ Therefore, $\mathbb{R}$ is closed under addition and multiplication.
- Associative Property:** Addition: $(a + b) + c = a + (b + c)$ for example
 $(2 + 3) + 4 = 2 + (3 + 4) = 9$
Multiplication: $(ab)c = a(bc)$ for example $(2 \times 3) \times 4 = 2 \times (3 \times 4) = 24$ Thus, associativity holds.
- Commutative Property:** Addition: $a + b = b + a$ for example $3 + 5 = 5 + 3$
Multiplication: $ab = ba$ for example $4 \times 2 = 2 \times 4$ Thus, commutativity holds.
- Identity Elements:** Additive Identity: There exists **0** such that $a + 0 = a$ for example $7 + 0 = 7$
Multiplicative Identity: There exists **1** such that $a \times 1 = a$ for example $9 \times 1 = 9$
- Inverse Elements:** Additive Inverse: For every $a \in \mathbb{R}$, there exists $-a$ such that $a + (-a) = 0$
for example $5 + (-5) = 0$. Multiplicative Inverse: For every $a \neq 0$ then $a \times \frac{1}{a} = 1$
- Distributive Property:** Multiplication distributes over addition. $a(b + c) = ab + ac$ for example $2(3 + 4) = 2 \times 7 = 14$ or $(2 \times 3) + (2 \times 4) = 6 + 8 = 14$ Both sides are equal.

The set of real numbers $\mathbb{R}$ satisfies:

Closure

Commutativity

Inverse elements

Associativity

Identity elements

Distributive law

Therefore, $(\mathbb{R}, +, \times)$ is a field.

Example 2: Determine whether integers $(\mathbb{Z}, +, \times)$ form a field.

Solution:

Integers satisfy:

- Closure under addition and multiplication ✓
- Additive identity and inverse ✓

But check multiplicative inverse. $2^{-1} = \frac{1}{2}$ is **not an integer**. Therefore, integers **do not have multiplicative inverses for all non-zero elements**. $\mathbb{Z}$ **is not a field**.

Application activity 2.7



Determine whether $\mathbb{Q}(\sqrt{2}) = \{a + b\sqrt{2} : a, b \in \mathbb{Q}\}$ form a **field**.

End of unit assessment



Assessing knowledge and understanding:

Write **True (T)** or **False (F)** for each statement.

- 1) A binary operation on a set combines two elements of the set to produce another element of the same set.
- 2) Closure property means the result of an operation on elements of a set is always outside the set.
- 3) In a commutative operation, $a * b = b * a$
- 4) The identity element leaves any element unchanged after the operation.
- 5) Every element in a group has an inverse.
- 6) Associative property means $(a * b) * c = a * (b * c)$
- 7) In a ring, multiplication must always be commutative.
- 8) A field allows division by any non-zero element.
- 9) The Cayley table shows the results of a binary operation on a finite set

Assessing skills

- 10) Let the operation $*$ on integers be defined by $a * b = a + b + 1$
 - a) Find $2 * 3$
 - b) Check whether the operation is commutative and associative.

Assessing values and attitudes

- 11) When working in a group to explore algebraic structures, your team realizes that a specific mathematical operation only works properly if everyone strictly follows the defined rules (like closure or commutativity). Which of the following best describes how this collaborative experience reflects the nature of groups, rings, or fields?
- a) It shows that mathematical structures, like teamwork, rely on a random set of individual preferences rather than established rules.
 - b) It demonstrates that just as a mathematical group requires elements to follow strict properties (like identity and inverse) to function, a team requires every member to cooperate under shared rules to reach a correct solution.
 - c) It proves that group work is only useful for finding shortcuts, whereas algebraic structures are entirely about working completely alone.
 - d) It shows that individual elements in a mathematical set can change the operational rules whenever they want, similar to how group members work.
- 12) Imagine your group is creating a Cayley table for a finite set, but two members disagree on the result of a specific binary operation, causing an error in the table. What is the most constructive attitude to resolve this and ensure the accuracy of your algebraic structure?
- a) Ignore the disagreement and let each student write down their own preferred version of the table to save time.
 - b) Let the loudest group member decide the answer so the team can finish the assessment quickly.
 - c) Work together to test the specific elements against the formal algebraic definition of the operation, valuing logical proof and precision over personal opinion.
 - d) Delete that specific element from the set entirely so the group does not have to deal with the disagreement.

UNIT 3

INTRODUCTION TO LOGIC



Key Unit Competence:

Analyse statements and arguments using principles of logic in solving mathematical and real-life problems.

Learning objectives

By the end of this unit, you should be able to:

Knowledge and understanding

1

- Explain statements, propositions, arguments, and propositional functions (predicates and variables).
- Describe and use logical connectives and construct and interpret truth tables for simple and compound propositions.
- Explain conditional and biconditional statements and determine their truth conditions.
- State and apply the laws of logic, including the use and negation of quantifiers ($\forall$, $\exists$).
- Apply logical reasoning in problem-solving and in contexts such as set theory and electric circuits.

Skills

2

- Explain statements, propositions, arguments, and propositional functions (predicates and variables).
- Describe and use logical connectives and construct and interpret truth tables for simple and compound propositions.
- Explain conditional and biconditional statements and determine their truth conditions.
- State and apply the laws of logic, including the use and negation of quantifiers ($\forall$, $\exists$).
- Apply logical reasoning in problem-solving and in contexts such as set theory and electric circuits.

Attitudes and values

3

- Show accuracy, clarity, and consistency in reasoning and communication.
- Demonstrate honesty and fairness in logical argumentation.
- Develop curiosity and critical thinking when evaluating statements or arguments.
- Appreciate the importance of logical reasoning in mathematics, science, and daily decision-making.
- Work collaboratively and respect others' reasoning in discussions and group work.

Introduction

In this unit, you will learn the basic ideas of mathematical logic, which help you understand and analyse reasoning and arguments in mathematics. You will begin by studying statements and propositions and how they are used to form arguments.

You will then explore logical connectives such as negation, conjunction, and disjunction, and learn how to represent their truth values using truth tables. You will also study conditional and biconditional statements and understand different types of logical statements such as tautology, contradiction, and contingency.

In addition, you will learn about logical equivalence and important logical laws such as De Morgan's laws, distributive laws, and associative laws. The unit will also introduce quantifiers and their negation, as well as propositional functions involving predicates and variables. Finally, you will apply logical reasoning to real situations, including determining the validity and soundness of arguments, solving problems, and understanding applications of logic in set theory and electric circuits.

By the end of this unit, you will develop skills in clear reasoning and logical thinking that are important in mathematics and many other fields.

Introductory activity



In the following sentences, which ones are propositions, and which are not propositions?

- Don't eat the daisies!
- My class is senior four.
- Do you enjoy reading novels?
- The jokes are great.
- Masamba composed classical music.
- The camera is not a Kodak.
- This statement is false.
- Use the quadratic formula on that one.

3.1 Statements and propositions

Activity 3.1



From the following expressions, give your answer by using true or false

- 1) Every integer greater than 1 is positive.
- 2) Kampala is in Rwanda.
- 3) How old are you?
- 4) Every liquid is water.
- 5) Write down the names of the president of Uganda.
- 6) $1 - x^2 = 0$
- 7) Rwanda is an African country, or Rwanda is a member of Commonwealth.



Content Summary

A **statement (or proposition)** is a declarative sentence that has a truth value, meaning it is either true or false, but not both.

A statement that cannot be broken into two or more sentences is called **simple statement**. Combining two or more simple statements we form a **compound statement**.

A **compound proposition** is a proposition formed by connecting two or more propositions or by negating a single proposition. The words and phrases (or symbols) used to form compound propositions are called **connectives**.

Here are some sentences that are not propositions:

1. "Do you want to go to the market?" Since a question is not a declarative sentence, it fails to be a proposition.
2. "Clean up your room." Likewise, this is not a declarative sentence; hence, fails to be a proposition.
3. " $2x = 2 + x$." This is a declarative sentence, but unless x is assigned a value or is otherwise prescribed, the sentence is neither true nor false, hence, not a proposition.

Each proposition can be assigned one of two truth values. We use T or 1 for true and use F or 0 for false.

Example: Identify which of the following are **statements**.

a) $6+2=8$

b) How old are you?

c) $9>5$

d) Stop talking.

Solution:

Sentence	Is it a Statement?	Reason
a) $(6+2=8)$	✓ Yes	It is true, so it has a truth value.
b) How old are you?	✗ No	It is a question and has no truth value.
c) $(9>5)$	✓ Yes	It is true, so it has a truth value.
d) Stop talking.	✗ No	It is a command and has no truth value.

Application activity 3.1



- 1) Find out which of the following sentences are statements and which are not. Justify your answer.
 - a) Uganda is a member of East African Community.
 - b) The sun is shining.
 - c) Come to class!
 - d) The sum of two prime numbers is even.
 - e) It is not true that China is in Europe.
 - f) May God bless you!
- 2) Write down the truth value (T or F) of the following statements
 - a) Paris is in Italy.
 - b) 1,3 is a prime number.
 - c) Kigeri IV Rwabugiri was the King of the Kingdom of Rwanda
 - d) Lesotho is a state of South Africa.
 - e) Tanzania is in East of Rwanda and it is in SADC (Southern African Development Community).

3.2 Argument

Activity 3.2



For the following sets of statements:

Set A:

- a) All humans are mortal.
- b) I am human.
- c) Therefore, I am mortal.

Set B:

- a) All birds can fly.
- b) Penguins are birds.
- c) Therefore, penguins can fly.

Discuss in group: (1) Which set presents a **valid argument**?

(2) Which set has **faulty reasoning**?



Content Summary

Topic	Key Points
Argument Definition	– An argument is a set of statements where some statements (premises) support another statement (conclusion).
Premises and Conclusion	– Premises: Statements that give reasons. – Conclusion: Statement supported by premises.
Types of Arguments	– Deductive: Conclusion necessarily follows from premises. – Inductive: Conclusion likely follows from premises but not guaranteed.

Validity of an Argument	– Valid Argument: If premises are true, the conclusion must be true. – Invalid Argument: Premises may be true, but conclusion is false.
Symbolic Representation	– Arguments can be written using symbols like p, q, r. – Example: $p \rightarrow q$, $p \vdash q$ (p entails q).
Everyday Examples	– Arguments are everywhere: advertisements, debates, instructions, news, politician speeches, etc.

Example 1: Identify Premises and Conclusion

Statements:

- If a number is even, it is divisible by 2.
- 8 is even.
- Therefore, 8 is divisible by 2.

Solution:

- Premise 1: If a number is even, it is divisible by 2
- Premise 2: 8 is even
- Conclusion: 8 is divisible by 2

This is a valid argument.

Example 2: Check for Validity in the following argument:

“All birds can fly. Penguins are birds. Therefore, penguins can fly.”

Solution:

Premises are partly false (not all birds can fly). Conclusion is **false**, so this is an **invalid argument**.

Application activity 3.2



1. Identify premises and conclusion in these statements:
 - a) “If I study hard, I will pass. I study hard. Therefore, I will pass.”
 - b) “All fruits have seeds. Tomatoes have seeds. Therefore, tomatoes are fruits.”
2. Analyze whether these arguments are valid or invalid:
 - a) “If I am in Kigali, I am in Rwanda. I am in Rwanda. Therefore, I am in Kigali.”
 - b) “If a number is even, it is divisible by 2. 8 is divisible by 2. Therefore, 8 is even.”
3. write a short argument about any school rule, then swap with a partner to identify premises and conclusion.

3.3. Logical connectives

Activity 3.3



For these statements, form new statements using words: “and,” “or,” “not.”

- p: “I go to school.”
- q: “I do my homework.”



Content Summary

Connective	Symbol	Definition / Meaning	Example
Negation	$\neg p$ (or $\sim P$)	“ Not p ” – reverses the meaning of a statement	p: “I am happy.” $\Rightarrow \neg p$: “I am not happy.”
Conjunction	$p \wedge q$	“p AND q” – combines two statements; both must be true for the combined statement to be true	p: “I study math.” q: “I study science.” $\Rightarrow p \wedge q$: “I study math and science.”
Disjunction	$p \vee q$	“p OR q” – combines two statements; at least one must be true for the combined statement to be true	p: “I will play football.” q: “I will swim.” $\Rightarrow p \vee q$: “I will play football or swim.”

Key Points:

- **Negation** changes a statement to its opposite meaning.
- **Conjunction** links two statements to say **both are true**.
- **Disjunction** links two statements to say **at least one is true**.
- These connectives are used in **everyday language**, instructions, rules, and reasoning.

Example 1: Negation

Statement: “It is sunny today.” **Negation:** “It is **not** sunny today.”

Example 2: Conjunction

Statements

p: “I brush my teeth.” q: “I wash my hands.” Combined ($p \wedge q$): “I brush my teeth and wash my hands.”

Example 3: Disjunction

Statements: p: “I will eat rice.” q: “I will eat beans.” and Combined ($p \vee q$): “I will eat rice **or** beans.”

Example 4: Mixed use

Statements: p: “I do my homework.” q: “I go to school.”

Combined: “I do my homework **and** I go to school.”

Application activity 3.3



1. Form new statements using negation, conjunction, and disjunction:
 - a) "I like mangoes."
 - b) "I like oranges."
2. Discussion: Combine statements about school rules using and, or, not.
3. Write 2–3 sentences describing your morning routine using and, or, not.
4. Write a statement, partner converts it using negation or combines it with another using and/or.

3.4 Truth tables for basic operations

Activity 3.4



For the following statements:

- p: "It is raining."
- q: "I have an umbrella."

Discuss in group these questions:

- What if both are true?
- What if one is false?



Content Summary

Connective	Symbol	Meaning / Use
Negation	$\neg p$	"Not p": It reverses the truth value of p
Conjunction	$p \wedge q$	"p and q": It is true only if both p and q are true
Disjunction	$p \vee q$	"p or q": It is true if at least one of p or q is true

Example 1: Negation

Statement: p = "It is sunny."

p	$\neg p$	Explanation
T	F	Negation reverses the truth value of p
F	T	

Example 2: Conjunction

Statements: p: "I am in class." q: "I have my notebook."

p	q	$p \wedge q$	Explanation
T	T	T	"I am in class AND I have my notebook" is only true if both p and q are true.
T	F	F	
F	T	F	
F	F	F	

Example 3: Disjunction

Statements: p: "I study math." q: "I study science."

p	q	$p \vee q$	Explanation
T	T	T	"I study math OR I study science" is true if at least one of p or q is true
T	F	T	
F	T	T	
F	F	F	

Application activity 3.4



- Translate each of the following compound statements into symbolic form
 - Bwenge reads News Paper or Mathematics book.
 - Rwema is a student or not a book seller.
- Suppose that p is a false statement, and q is a true statement.
 - What is the truth-value of the compound statement $\neg p \vee q$?
 - What is the truth-value of the compound statement $p \vee \neg q$?
- Let p and q be two propositions. Construct the truth table of
 - $p \vee q$,
 - $p \vee \neg q$
 - $\neg(p \vee q) \wedge (\neg p \vee \neg q)$
 - $(\neg q \vee p) \wedge (\neg p \vee q)$

3.5 Conditional and biconditional statements

Activity 3.5



For the following statements

- If it rains, then the ground becomes wet.
- If you study hard, then you pass the exam.
- If a number is divisible by 2, then it is even.

Discuss in group.

- What word begins these sentences?
- What happens after if?
- What happens after then?



Content Summary

Definition:

A **conditional statement** is a logic statement used when the statement is in the form “if ...then ...” It can be **written** as “If p then q ” or $p \rightarrow q$, (or $p \Rightarrow q$)

Read in different ways: p implies q ; If p , then q ; q follows from p ; q if p ; p only if q ., p is sufficient for q , q is necessary for p .

In this case the proposition p is called **antecedent, hypothesis or premise** while the proposition q is the **conclusion or the consequent**.

In language, p can be called the first clause and q is the second clause. Where: p = hypothesis (condition) and q = conclusion (result).

Generally, $p \Rightarrow q$ is not the same as $q \Rightarrow p$. It may happen, however, that both $p \Rightarrow q$ and $q \Rightarrow p$ are true. The statement $q \Leftrightarrow p$ defined by $(q \Rightarrow p) \wedge (p \Rightarrow q)$ is a compound statement called the **bi-conditional connective**.

For this reason, the double headed arrow $q \Leftrightarrow p$ or $p \Leftrightarrow q$ is called the **bi-conditional**.

We can restate the definition of the bi-conditional connective by using its truth table:

- 1) $p \Rightarrow q$ is false only when the antecedent p is true and consequence q is false
- 2) $q \Leftrightarrow p$ is true only when p and q have the same truth value

p	q	$p \Rightarrow q$	$q \Rightarrow p$	$(p \Rightarrow q) \wedge (q \Rightarrow p)$	$p \Leftrightarrow q$
T	T	T	T	T	T
T	F	F	T	F	F
F	T	T	F	F	F
F	F	T	T	T	T

Example: If a number is divisible by 2, then it is even.

Hypothesis (p): The number is divisible by 2.

Conclusion (q): The number is even

Parts of a conditional Statement in the example If a student studies, then the student passes.

Part	Meaning	Example
Hypothesis (p)	First part after “if”	If a student studies,
Conclusion (q)	Second part after “then”	then the student passes

A **bi-conditional statement** combines two conditional statements. It is written as: **p if and only if q** . Symbolically: $p \leftrightarrow q$. This means **both statements imply each other**.

Examples:

- 1) A number is even **if and only if** it is divisible by 2.

Meaning: If a number is even $\Rightarrow$ it is divisible by 2. If a number is divisible by 2 $\Rightarrow$ it is even.

2) Form the conditional statement:

Let: **p**: "It is raining." And **q**: "The ground is wet."

Solution

Conditional statement: **If it is raining, then the ground is wet. Symbolically: $p \Rightarrow q$**

p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

3) Form this biconditional Statement

Let: **p**: "A number is even." And **q**: "The number is divisible by 2."

Solution:

Biconditional statement:

A number is even if and only if it is divisible by 2.

Symbolically: **$p \leftrightarrow q$ or $p \Leftrightarrow q$.**

p	q	$p \leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T

4) Construct the truth value of the following statement: "If either John takes Calculus or Betty takes Sociology then Peter will take English."

Solution:

Let denote the statements as: **p**: John takes Calculus, **q**: Betty takes Sociology and **r**: Peter takes English. The given statement can be symbolized as follow: **$(p \vee q) \Rightarrow r$** and the truth values are in the following truth table

p	q	r	$p \vee q$	$(p \vee q) \Rightarrow r$
T	T	T	T	T
T	T	F	T	F
T	F	T	T	T
T	F	F	T	F
F	T	T	T	T
F	T	F	T	F
F	F	T	F	T
F	F	F	F	T

Application activity 3.5



1) Suppose that **r** is a false statement, and **s** is a true statement.

a) What is the truth-value of the compound statement **$(\neg r) \Leftrightarrow s$** ?

b) What is the truth-value of the compound statement **$r \Leftrightarrow (\neg s)$** ?

c) What is the truth-value of the compound statement **$r \Leftrightarrow s$** ?

d) What is the truth-value of the compound statement **$\neg(r \Leftrightarrow (\neg s))$** ?

2) Construct the truth table for

- a) $p \leftrightarrow q$ and $(p \rightarrow q) \wedge (q \rightarrow p)$ b) $p \leftrightarrow q$ and $(\neg p \vee q) \wedge (\neg q \vee p)$
 c) $p \leftrightarrow q$ and $(p \vee q) \wedge \neg(q \wedge p)$ d) $\neg(p \leftrightarrow q)$ and $(p \wedge \neg q) \vee (\neg q \wedge p)$

3) Show that neither the converse nor the inverse of an implication are equivalent to the implication.

4) Formulate a conditional statement made by two compound statements p and q . Use that statement to express different forms of $\neg p \vee q$ and $p \Rightarrow q$ and show that $\neg p \vee q$ and $p \Rightarrow q$ are equivalent. Relate to the daily life situation.

3.6 Tautology, contradiction and contingency

Activity 3.6



For the following statements

1. "Either it rains today or it does not rain today."
2. "It is raining and it is not raining at the same time."
3. "I will study mathematics tomorrow."

Discuss in group the following questions:

- Which statement is always true?
- Which statement is always false?
- Which statement may be true or false depending on the situation?



Content Summary

- 1) **A tautology** is a compound statement that is always true for all possible truth values of its components. $p \vee \neg p$ **This statement means: "p or not p".** This is always true.
- 2) **A contradiction** is a compound statement that is always false for all possible truth values. $p \wedge \neg p$. **This means: "p and not p".** This can never be true.
- 3) **A contingency** is a statement that is sometimes true and sometimes false depending on the truth values. $p \wedge q$. This is true only when both p and q are true.

Examples

1: Determine whether the statement $p \vee \neg p$ is a tautology.

Solution:

p	$\neg p$	$p \vee \neg p$
T	F	T
F	T	T

The statement is **always true**, therefore:
 $p \vee \neg p$ is a **Tautology**.

2: Determine whether $p \wedge \neg p$ is a contradiction.

Solution:

p	$\neg p$	$p \wedge \neg p$
T	F	F
F	T	F

The statement is **always false**, therefore: $p \wedge \neg p$ is a **Contradiction**.

3: Determine whether $p \wedge q$ is a contingency.

Solution:

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

The statement is **sometimes true and sometimes false**. Therefore: $p \wedge q$ is a **Contingency**.

4: Determine whether $p \vee q$ is a contingency

Solution:

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

The statement is **not always true and not always false**. Therefore: $p \vee q$ is a **Contingency**.

5: Show that the following is a tautology: $(p \vee q) \vee [(\neg p) \wedge (\neg q)]$

Solution:

p	q	$\neg p$	$\neg q$	$p \vee q$	$\neg(p) \wedge (\neg q)$	$(p \vee q) \wedge [(\neg p) \wedge (\neg q)]$
T	T	F	F	T	F	T
T	F	F	T	T	F	T
F	T	T	F	T	F	T
F	F	T	T	F	T	T
						↑
						All T

The truth value is **always true**, $(p \vee q) \vee [(\neg p) \wedge (\neg q)]$ is a tautology.

Application activity 3.6



- 1) From the following compound statements, indicate which is tautology or contradiction
 - a) $p \wedge \neg(p \wedge q)$
 - b) $\neg q \wedge (q \wedge r)$
 - c) $(p \wedge q) \wedge \neg(p \vee q)$
 - d) $(p \vee r) \vee \neg r$
 - e) $p \wedge [(\neg q) \vee (\neg r)] \Rightarrow [p \Rightarrow (\neg q)]$
- 2) Show that the statement $(p \vee q) \wedge [(\neg p) \wedge (\neg q)]$ is a contradiction.
- 3) Show that the statement $[(p \vee q) \Rightarrow r] \vee [\neg(p \vee q) \Rightarrow r]$ is a tautology.
- 4) In your own words, formulate
 - a) 2 propositional statements expressing the tautology and show where these tautologies can be avoided in everyday practice of argumentation.
 - b) 2 propositional statements expressing the contradiction and show where these contradictions can be avoided in everyday practice of argumentation.

3.7 Logical equivalence, De Morgan's, distributive and associative laws

Activity 3.7



For the following sentences;

- "It is not true that I study math and science."
- "I do not study Mathematics or I do not study science."

Work out in group.

- Do the two statements express the same idea?
- Are they different sentences but with the same meaning?



Content Summary

1) Logical Equivalence: Two statements are logically equivalent if they have the same truth values in all cases. Symbolically written as: $p \equiv q$. This means p and q are logically equivalent.

2) Important Logical Laws

A. De Morgan's Laws: These laws explain how negation works with conjunction and disjunction.

The first law states that saying "it is not the case that both A and B are true" is exactly the same as saying "either A is not true, or B is not true."

- **In words:** The negation of an "AND" statement is the same as the "OR" of the negations.
- **The Logic:** If someone tells you, "It is not true that I am wearing a hat and a scarf," it means they are either not wearing a hat, not wearing a scarf, or not wearing either one.

The second law states that saying “it is not the case that either A or B is true” is the same as saying “A is not true AND B is not true.”

- **In words:** The negation of an “OR” statement is the same as the “AND” of the negations.
- **The Logic:** If someone says, “It is not true that it is raining or snowing,” they are confirming two specific things: it is not raining, and it is also not snowing.

Formal expression:

Law	Statement
First Law	$\neg(p \wedge q) \equiv (\neg p \vee \neg q)$
Second Law	$\neg(p \vee q) \equiv (\neg p \wedge \neg q)$

In summarized words:

Statement	Negation (De Morgan’s Equivalent)
Not (A AND B)	(Not A) OR (Not B)
Not (A OR B)	(Not A) AND (Not B)

B. Distributive Laws: These laws are similar to distributive laws in algebra.

The first law states that “AND” distributes over “OR.” It means that saying A is true, **AND** either B or C is true” is the same as saying “Either (A and B) are true, **OR** (A and C) are true.”

- **In words:** A and (B or C) is equivalent to (A and B) or (A and C).
- **The Logic:** Imagine you are told: “To enter the lab, you must have a Key **AND** either a Badge or a Fingerprint.” This is logically identical to saying: “You must have (a Key and a Badge) **OR** (a Key and a Fingerprint).”

The second law states that “OR” distributes over “AND.” This is often less intuitive than the first law because it doesn’t have a direct parallel in standard arithmetic, but it is equally valid in logic.

- **In words:** A or (B and C) is equivalent to (A or B) and (A or C).
- **The Logic:** Imagine a rule: “To pass the course, you must have an A on the final **OR** (perfect attendance and a completed project).” Logically, this means you must satisfy both of these conditions: You must have (an A or perfect attendance) **AND** you must have (an A or a completed project).

Formal Expression:

Law	Statement
First Law	$p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$
Second Law	$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$

In words:

Law	Statement	Distributed Form
First Law	A and (B or C)	(A and B) or (A and C)
Second Law	A or (B and C)	(A or B) and (A or C)

C. Associative Laws: These laws show that **grouping of statements does not change the result**.

First law (Conjunction): This law states that when connecting three or more statements with **AND**, the order in which you group them doesn't matter.

- **In words:** (A and B) and C is the same as A and (B and C).
- **The Logic:** If three things must all be true for a result to happen, it doesn't matter which two you verify first.
- **Example in Speech:** > "To get the promotion, you must (be on time **AND** work hard) **AND** meet your targets." is exactly the same as saying: "To get the promotion, you must be on time **AND** (work hard **AND** meet your targets)."

The Second (Disjunction): This law states that when connecting statements with **OR**, the grouping has no effect on the outcome.

- **In words:** (A or B) or C is the same as A or (B or C).
- **The logic:** If you only need one of several options to be true, it doesn't matter how you categorize those options.
- **Example in Speech:** > "For dessert, you can have (cake **OR** ice cream) **OR** fruit." is exactly the same as saying: "For dessert, you can have cake **OR** (ice cream **OR** fruit)."

Formal Expressions:

Law	Statement
First Law	$(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$
Second Law	$(p \vee q) \vee r \equiv p \vee (q \vee r)$

Example 1: De Morgan's Law. Verify that: $\neg(p \wedge q) \equiv (\neg p \vee \neg q)$

Solution:

p	q	$p \wedge q$	$\neg(p \wedge q)$	$\neg p$	$\neg q$	$\neg p \vee \neg q$
T	T	T	F	F	F	F
T	F	F	T	F	T	T
F	T	F	T	T	F	T
F	F	F	T	T	T	T

Columns $\neg(p \wedge q)$ and $\neg p \vee \neg q$ are identical. Therefore: $\neg(p \wedge q) \equiv (\neg p \vee \neg q)$

Example 2: Second De Morgan's Law. Verify: $\neg(p \vee q) \equiv (\neg p \wedge \neg q)$

Solution:

p	q	$p \vee q$	$\neg(p \vee q)$	$\neg p$	$\neg q$	$\neg p \wedge \neg q$
T	T	T	F	F	F	F
T	F	T	F	F	T	F
F	T	T	F	T	F	F
F	F	F	T	T	T	T

The last two columns are identical. Therefore: $\neg(p \vee q) \equiv (\neg p \wedge \neg q)$

Example 3: Distributive Law. Verify: $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$

Solution:

p	q	r	$q \vee r$	$p \wedge (q \vee r)$	$p \wedge q$	$p \wedge r$	$(p \wedge q) \vee (p \wedge r)$
T	T	T	T	T	T	T	T
T	T	F	T	T	T	F	T
T	F	T	T	T	F	T	T
T	F	F	F	F	F	F	F
F	T	T	T	F	F	F	F
F	T	F	T	F	F	F	F
F	F	T	T	F	F	F	F
F	F	F	F	F	F	F	F

Both expressions give the same truth values. Therefore, they are **logically equivalent**.

Example 4:

A security rule states: “It is not true that a visitor can enter if they lack an ID card AND lack a visitor pass.” Using De Morgan’s Laws, how would you state this rule more simply?

Solution:

According to De Morgan’s First Law, the negation of an **AND** statement is the **OR** of the negations.

- **Original:** Not (No ID **AND** No Pass)
- **Equivalent:** (Not No ID) **OR** (Not No Pass)
- **Simplified Speech:** “A visitor can enter if they have an ID card **OR** they have a visitor pass.”

Example 5: A restaurant offers a deal: “You get a free drink if you order a Burger OR a Pizza, AND it is a Monday.” How can you rewrite this using the Distributive Law?

Solution:

This follows the pattern $A \wedge (B \vee C) \Leftrightarrow (A \wedge B) \vee (A \wedge C)$, where A is “Monday,” B is “Burger,” and C is “Pizza.”

- **Original:** Monday **AND** (Burger **OR** Pizza)
- **Equivalent:** (Monday **AND** Burger) **OR** (Monday **AND** Pizza)
- **Simplified Speech:** “You get a free drink if you order a Burger on Monday, **OR** if you order a Pizza on Monday.”

Example 6: In a software program, a function only runs if: “(User is Logged In AND WiFi is Connected) AND Server is Active.” Does the logic change if we check the WiFi and Server status together first?

Solution:

No, the logic remains the same due to the Associative Law for Conjunction.

- **Original:** (Logged In **AND** WiFi) **AND** Server
- **Equivalent:** Logged In **AND** (WiFi **AND** Server)
- **Explanation:** Since all three conditions must be true for the function to run, the order in which the computer “groups” or checks them does not change the final result.

Example 7: Simplify the following logical expression: $\neg(A \vee (B \wedge C))$

Solution:

This requires using both De Morgan’s and Distributive properties.

Apply De Morgan’s: Change the $\neg(\dots \vee \dots)$ to $(\neg A) \wedge \neg(B \wedge C)$.

Apply De Morgan’s again: Change $\neg(B \wedge C)$ to $(\neg B \vee \neg C)$.

Result: $(\neg A) \wedge (\neg B \vee \neg C)$.

Distribute (optional): $\neg(A \vee (B \wedge C)) = (\neg A \wedge \neg B) \vee (\neg A \wedge \neg C)$.

Application activity 3.7



1) Verify the following logical equivalences:

a) $\neg(p \wedge q) \equiv \neg p \vee \neg q$

b) $\neg(p \vee q) \equiv \neg p \wedge \neg q$

2) State which logical law is used:

a) $\neg(p \wedge q) \equiv \neg p \vee \neg q$

b) $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$

c) $(p \vee q) \vee r \equiv p \vee (q \vee r)$

3) Use logical laws to simplify:

a) $\neg(p \wedge q)$

b) $p \wedge (q \vee r)$

c) $(p \vee q) \vee r$

3.8 Quantifiers and their negation

Activity 3.8



For the following statements

1. All students in this class study mathematics.
2. Some students in this class play football.

Discuss in group the following questions.

- What word describes the whole group?
- What word describes only part of the group?



Content Summary

1) A **quantifier** is a symbol or word used to show **how many elements of a set satisfy a statement**. There are two main types of quantifiers.

2) The **universal quantifier** means “for all” or “for every”.

Symbol: $\forall$

Meaning $\forall x, P(x)$ This means: **P(x) is true for every value of x**. “All students wear uniforms.” Symbolically: $\forall x, x \text{ is a student} \Rightarrow x \text{ wears a uniform}$

3) The **existential quantifier** means “there exists” or “there is at least one.”

Symbol: $\exists$

Meaning: $\exists x, P(x)$ this means: **There exists at least one x such that P(x) is true**. “Some students play football.” Symbolically: $\exists x, x \text{ is a student who plays football}$

4) Negation changes the meaning of quantified statements.

Statement	Negation
$\forall x P(x)$	$\exists x \neg P(x)$
$\exists x P(x)$	$\forall x \neg P(x)$

Meaning:

The negation of “All” is “There exists at least one not.”

The negation of “There exists” is “For all not.”

Original Statement	Negation
All A are B	Some A are not B
Some A are B	No A are B (or All A are not B)
No A are B	Some A are B
Some A are not B	All A are B

Example 1: Universal Quantifier

Let the set be: $A = \{1, 2, 3\}$. Statement: $P(x)$: “x is greater than 0”: $\forall x \in A, x > 0$

Solution:

x	$P(x): x > 0$	Since P(x) is true for all elements , the statement is true .
1	TRUE	
2	TRUE	
3	TRUE	

Example 2: Existential Quantifier

Let: $A = \{2, 4, 5\}$. Statement: $P(x)$: “x is odd”: $\exists x \in A, x \text{ is odd}$

Solution:

x	P(x): x is odd
2	FALSE
4	FALSE
5	TRUE

Since **at least one value (5)** satisfies the condition, the statement is **true**

Example 3: How do you negate the statement: “All students in the class passed the exam”?

Solution:

To prove an “All” statement is false, you only need to find **one** exception. Therefore, the negation of “All” is “Some are not.”

- **Original:** All S are P.
- **Negation:** Some S are **not** P. (Or: “There is at least one student who did not pass.”)
- **Common Mistake:** Saying “No students passed.” (This is a much stronger claim than necessary to disprove the original).

Example 4: How do you negate the statement: “Some mountains are easy to climb”?

Answer:

If it is false that “at least one” mountain is easy, it means that **every** single mountain must be difficult. The negation of “Some” is “None” (or “All are not”).

- **Original:** Some M are E.
- **Negation:** No M are E. (Or: “All mountains are **not** easy to climb.”)
- **Formal Rule:** $\neg(\exists x, P(x)) \Leftrightarrow \forall x, \neg P(x)$

Example 5: Consider the statement: “Every employee must attend the workshop AND some employees will receive a bonus.” How do you negate this entire logical expression?

Answer:

Negate the “AND”: Change it to “OR” and negate both sides.

Negate the first part: The negation of “Every employee must attend” is “Some employees do not have to attend.”

Negate the second part: The negation of “Some employees receive a bonus” is “No employees receive a bonus.”

Final Result: “Some employees do not have to attend the workshop **OR** no employees will receive a bonus.”

Application activity 3.8



- 1) Identify whether the following statements use **universal** or **existential quantifiers**.
 - a) All birds have wings.
 - b) Some students like mathematics.
 - c) Every triangle has three sides.
- 2) Write the symbolic form using quantifiers.
 - a) All numbers greater than 5 are positive.
 - b) Some students like science.
- 3) Find the negation of the following statements.
 - a) All students wear uniforms.
 - b) Some numbers are negative.
 - c) Every triangle has three sides.
- 4) Consider the predicates $r(x): x - 7$ and $s(x): x > 9$. If the universe of discourse is the real numbers, give the truth value of propositions: $(\exists x)s(x) \wedge \neg(\forall x)r(x)$.

3.9 Introduction to propositional functions (predicate and variables)

Activity 3.9



For the following sentences:

1. $x + 3 = 7$
2. $x > 5$
3. x is an even number

Discuss in group the following questions:

- Are these sentences **true or false**?
- What information is missing?



Content Summary

- 1) A **propositional function** is a statement that **contains a variable** and becomes **true or false when the variable is given a value**. It is often written as: $P(x)$
Where: **P** = predicate (the property or condition) and **x** = variable.
 $P(x): x + 2 = 5$, If $x = 3 \rightarrow$ statement is **true** and if $x = 4 \rightarrow$ statement is **false**
- 2) A **variable** is a symbol (usually a letter like x, y, z) that represents an **unknown element** of a set.
- 3) A **predicate** describes a **property or condition** about the variable.

$P(x): x$: is an even number

4) The **domain** is the set of values that the variable can take.

If the domain is: $A = \{1, 2, 3, 4\}$. Then we test each value to determine if the statement is **true or false**.

Example 1: Let: $P(x) : x + 2 = 5$ Domain: $A = \{1, 2, 3, 4\}$.

Solution:

x	$x + 2 = 5$	Truth Value of $P(x)$
1	$3 = 5$	FALSE
2	$4 = 5$	FALSE
3	$5 = 5$	TRUE
4	$6 = 5$	FALSE

The predicate $P(x)$ is **true only when $x = 3$** .

Example 2: Let: $P(x) : x$ is an even number and Domain: $A = \{1, 2, 3, 4\}$.

Solution:

x	x is even	Truth Value of $P(x)$
1	No	FALSE
2	Yes	TRUE
3	No	FALSE
4	Yes	TRUE

The statement is **true for 2 and 4**.

Example 3

Let: $P(x) : x > 4$ and Domain: $A = \{2, 4, 6, 8\}$

Solution:

x	$x > 4$	Truth Value of $P(x)$
2	No	FALSE
4	No	FALSE
6	Yes	TRUE
8	Yes	TRUE

The predicate is **true for values greater than 4**.

Application activity 3.9



1. Identify the **variable** in each propositional function.

a) $x + 5 = 9$

b) $y < 7$

c) n is a prime number

2. Let: $P(x): x + 1 = 6$ and Domain: $A = \{3, 4, 5, 6\}$. Complete the table:

x	$x + 1 = 6$	Truth Value of $P(x)$
3	?	?
4	?	?
5	?	?
6	?	?

3. Let: $P(x): x$ is divisible by 3 and Domain: $A = \{3, 5, 6, 9\}$. Determine the truth value for each element.

4. Students create a predicate such as: $P(x): x > 10$. Choose a domain and determine the truth values using a table.

3.10 Applications of logic in real life

In real life, logics is applied in different situations. Let us discuss the use of logic in reasoning (validity and soundness of an argument), problem-solving, set theory and electric circuits.

Activity 3.10



1) Discuss the following statement and say what the conclusion follows from the statement. Explain.

Statements A	Statements B
1. If it rains, the road becomes wet.	1. If I study hard, I pass the exam.
2. It is raining.	2. I passed the exam.
3. Therefore, the road is wet.	3. Therefore, I studied hard.

2) Conduct research in the books, use AI or other internet browsers and find the examples of application of logic in real life. Present your findings to the class.



Content Summary

1) Logic in Reasoning: Logic helps us determine whether an **argument is valid or sound**.

An argument is valid when the truth of the premises guarantees the truth of the conclusion.

Example:

1. If it rains, the ground gets wet.
2. It rains.
3. Therefore, the ground gets wet.

2) An argument is sound if:

- a) The argument is valid (the conclusion logically follows from the premises), and
- b) All the premises are actually true.

An argument is sound when it is valid *and* all its premises are true.

Example:

- a) All humans are mortal.
- b) Socrates is human.
- c) Therefore, Socrates is mortal.

This argument is valid and sound.

Clear comparison:

Term	What it means
Valid	If the premises are true, the conclusion must be true
Sound	Valid and all premises are true
Invalid	The conclusion does not logically follow

Example of a valid but not sound argument

Premise 1: All birds are mammals. (*false*)

Premise 2: All mammals have four legs. (*false*)

Conclusion: Therefore, all birds have four legs.

- The reasoning is **valid**
- But the argument is **not sound** (false premises).

3) Logic in Problem Solving: Logical reasoning helps us analyse problems step by step.

Example: If: A number is divisible by 4 $\rightarrow$ it is even. 16 is divisible by 4.

Therefore, 16 is even.

4) Logic in Set Theory:

The following table summarizes similarities of operations in set theory and logic

Let p, q and r be any propositions and A, B and C 3 sets

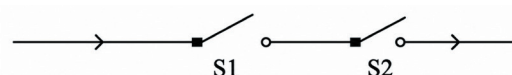
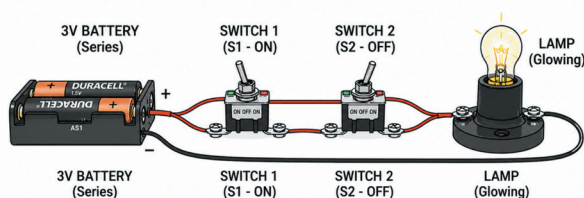
Logic	Set Theory	Law
$p \wedge q = q \wedge p$	$A \cap B = B \cap A$	Commutative Law
$p \wedge p = p$	$A \cap A = A$	Idempotent Law
$p \vee q = q \vee p$	$A \cup B = B \cup A$	Commutative Law
$p \vee p = p$	$A \cup A = A$	Idempotent Law
$(p \wedge q) \wedge r = p \wedge (q \wedge r)$	$(A \cap B) \cap C = A \cap (B \cap C)$	Associative Law
$(p \vee q) \vee r = p \vee (q \vee r)$	$(A \cup B) \cup C = A \cup (B \cup C)$	Associative Law
$p \wedge (q \vee r) = (p \wedge q) \vee (p \wedge r)$	$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$	Distributive Law
$p \vee (q \wedge r) = (p \vee q) \wedge (p \vee r)$	$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$	Distributive Law
$\neg(p \wedge q) = \neg p \vee \neg q$	$(A \cap B)' = A' \cup B'$	De Morgan's Law
$\neg(p \vee q) = \neg p \wedge \neg q$	$(A \cup B)' = A' \cap B'$	De Morgan's Law

5) Logic in Electric Circuits

Logical operations can represent **electric circuits using switches**.

If in an electric network two switches are used, then the two switches S_1 and S_2 can be in one of the following two cases;

a) Switches in series: then the current will flow through the circuit only when the two switches S_1 and S_2 are on (i.e. closed).



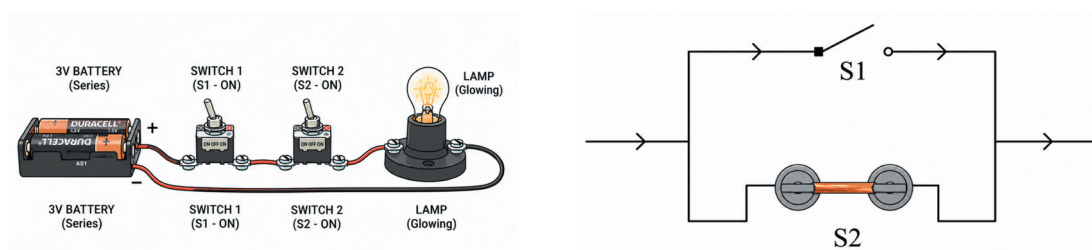
In this cases, the current passes when we have S_1 and S_2 . This is the same as the application of the connective and. $S_1 \wedge S_2$ is true only when S_1 is true and S_2 is true.

S1 (A)	S2 (B)	$A \wedge B$	Electricity
False (0)	False (0)	FALSE	✗ No
False (0)	True (1)	FALSE	✗ No
True (1)	False (0)	FALSE	✗ No
True (1)	True (1)	TRUE	✓ Yes

Conclusion: A series circuit represents the logical connective **AND ($\wedge$)** because the output is true **only when both inputs are true**.

b) Parallel Switches and the Logical Connective OR ($\vee$)

When switches **S1** and **S2** are connected **in parallel**, the circuit is closed and electricity flows **if at least one switch is ON**.



This exactly matches the truth conditions of the logical connective **OR**.

1 (A)	S2 (B)	$A \vee B$	Electricity
False (0)	False (0)	FALSE	✗ No
False (0)	True (1)	TRUE	✓ Yes
True (1)	False (0)	TRUE	✓ Yes
True (1)	True (1)	TRUE	✓ Yes

Conclusion: A parallel circuit represents the logical connective **OR ($\vee$)** because the output is true **whenever at least one input is true**.

Therefore, the following table shows logic, Truth Values, and Electricity

Electrical setup	Logical connective	Symbol	Truth condition
Series switches	AND	$\wedge$	True only if S1 and S2 are true
Parallel switches	OR	$\vee$	True if S1 or S2 (or both) is true
Switch ON	Truth value	1 / True	Current flows
Switch OFF	Truth value	0 / False	No current

Examples

1) Let: $A = \{1,2,3\}$ and $B = \{2,3,4\}$. Find intersection.

Solution:

Element	In A	In B	$A \cap B$
1	Yes	No	No
2	Yes	Yes	Yes
3	Yes	Yes	Yes
4	No	Yes	No

Intersection: $A \cap B = \{2,3\}$

2) Let: p = switch A closed and q = switch B closed for an **AND circuit (series)**

Solution:

p	q	Light
T	T	ON
T	F	OFF
F	T	OFF
F	F	OFF

The light turns **ON only when both switches are closed.**

Application activity 3.10



1) For switches p and q , complete the table for an OR circuit.

p	q	Light
T	T	?
T	F	?
F	T	?
F	F	?

2) Construct a circuit for each of the following statements:

1. $p \vee (q \wedge r)$

2. $(p \vee q) \vee r$

3) When two switches **S1** and **S2** are connected **in series**, which logical connective do they represent, and when does electricity flow?

4) When two switches **S1** and **S2** are connected **in parallel**, which logical connective do they represent, and when does electricity flow?

End of unit assessment



Assessing knowledge and understanding:

A. Write **True (T)** or **False (F)** for each statement.

1. A proposition is a statement that is either true or false.
2. The statement "Close the door!" is a proposition.
3. The negation of a statement always has the opposite truth value.
4. In a conjunction $p \wedge q$, the statement is true only when both p and q are true.
5. In a disjunction $p \vee q$, the statement is false only when both p and q are false.
6. A conditional statement has the form "if p , then q ".
7. A biconditional statement is true only when both statements have the same truth value.
8. A tautology is a statement that is always false.
9. A contradiction is a statement that is always false.
10. De Morgan's laws are used to simplify logical expressions.

B. Fill in the Missing Word

1. The logical connective **AND** is called _____.
2. The logical connective **OR** is called _____.
3. A statement with variables such as $P(x)$ is called a _____.
4. The quantifier $\forall$ means _____.
5. The quantifier $\exists$ means _____.

Assessing skills

C) Find the disjunction of the propositions p and q , where p is the proposition "Today is Sunday" and q : "The moon is made of cheese".

D) How many rows, not counting the top one, are needed to construct the truth table for a compound statement made from n statements?

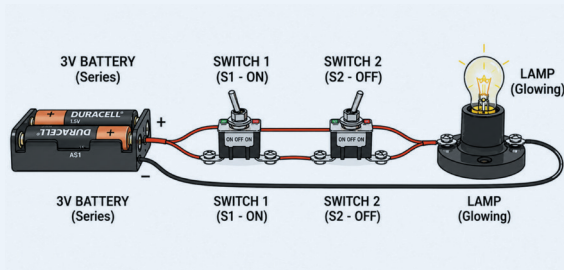
E) Show that $[p \wedge (p \rightarrow q)] \rightarrow q$ is a tautology. Is this argument **valid or invalid**? Explain briefly.

F) Two Switches **A** and **B** control a lamp. The lamp lights only when **both switches are ON**.

1. Write the logical expression.
2. Name the logical operation represented

Assessing values & attitudes

1. Why is **logical thinking** important in solving problems?
2. Give one example of how **logic helps in real life decision making**.
3. The diagram below shows two switches **S1** and **S2** connected in **series** to a lamp.



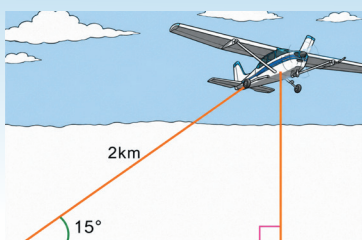
- a) Under what condition will the lamp light?
- b) Which logical connective does this circuit represent?
- c) State the corresponding truth value condition.

Answer in the TG

- a) The lamp lights **only when both S1 and S2 are ON**.
- b) This circuit represents the logical connective **AND ($\wedge$)**.
- c) The output (lamp ON) is **true only when both inputs are true**: $(S1 = \text{True}) \wedge (S2 = \text{True}) = \text{True}$.

UNIT 4

FUNDAMENTALS OF TRIGONOMETRY



Key Unit Competence:

Apply trigonometric circle and identities in determining trigonometric ratios and solving related problems in daily life.

Learning objectives

By the end of this unit, you should be able to:

Knowledge and understanding

1

- Explain the trigonometric functions sine, cosine, tangent, cotangent, secant, and cosecant.
- State and use trigonometric ratios for special angles (30° , 45° , 60° , and 90°).
- Explain and apply basic trigonometric identities to simplify expressions.
- Describe the reduction of trigonometric functions to positive acute angles using equivalent, complementary, supplementary, and negative angles.
- Apply the sine and cosine laws to solve triangles and solve real-life problems involving bearings, air navigation, angles of elevation and depression, and inclined planes.

Skills

2

- Compute trigonometric ratios of given angles using a calculator or trigonometric tables.
- Simplify and manipulate trigonometric expressions using standard identities.
- Solve triangles using sine and cosine laws.
- Apply trigonometric concepts to solve practical problems in navigation, bearings, and height-distance measurements.
- Draw and interpret diagrams involving angles of elevation, depression, and inclinations.

Attitudes and values

3

- Show accuracy, logical reasoning, and persistence when solving trigonometric problems.
- Appreciate the importance of trigonometry in solving real-life problems and in various fields such as engineering, geography, and physics.
- Develop curiosity and confidence in exploring geometric and spatial relationships.
- Work collaboratively and responsibly while performing practical trigonometric tasks.
- Demonstrate honesty and precision in recording and interpreting measurements.

Introduction

In this unit, you will learn important concepts of **trigonometry** that help describe relationships between angles and sides of triangles and solve real-life problems involving angles and distances.

You will begin by studying **angles and their measurements**, then explore the **unit circle**, which introduces the trigonometric functions **sine, cosine, tangent, secant, cosecant, and cotangent**. You will learn the **trigonometric ratios of special angles** and be introduced to **trigonometric identities**. You will also study the **graphs of trigonometric functions** to understand how these functions behave.

In addition, you will learn how to reduce trigonometric functions to positive acute angles using relationships between equivalent angles, negative or opposite angles, complementary angles, and supplementary angles. The unit will also teach you how to solve triangles using the cosine law and the sine law. Finally, you will apply trigonometry to practical situations such as bearing, air navigation, angles of elevation and depression, and inclined planes.

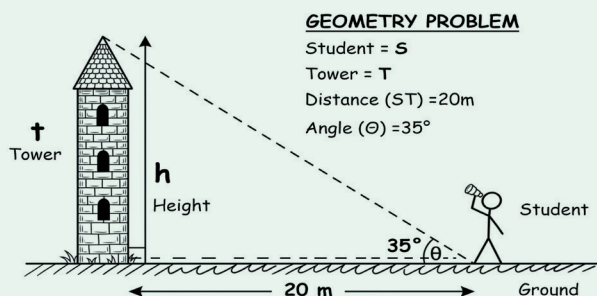
By the end of this unit, you will develop the skills to analyse and solve problems involving angles, triangles, and real-life applications of trigonometry.

Introductory activity



For the following image discuss in group “How can we find the height of the tower without climbing it?”

A student is standing 20 m away from a tall tower. The student measures the angle from the ground to the top of the tower and finds it is 35° .



4.1 Angle and its measurements

Activity 4.1



Discuss in pairs.

1. When the clock shows **3:00**, what angle is formed between the two hands?
2. When the clock shows **6:00**, what type of angle is formed?
3. If a student turns **half of a full turn**, how many degrees is that?
4. If a student rotates **one complete turn**, how many degrees does the student rotate?

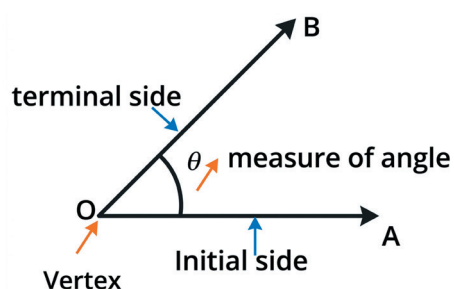


Content Summary

Definition of an Angle: An **angle** is formed when **two rays meet at a common point**.

- The common point is called the **vertex**.
- The two rays are called the **arms (or sides)** of the angle.

Angles are usually named using **three letters**, where the **vertex is the middle letter**.

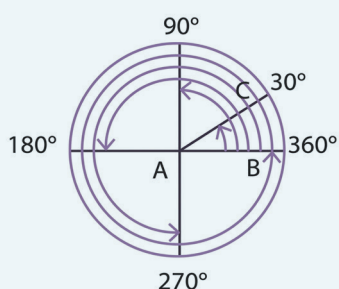


Degree measure:

To measure an angle in degrees, we imagine the circumference of a circle divided into 360 equal parts. We call each of those equal parts a “degree.” Its symbol is a small o: $1^\circ = \text{“1 degree.”}$

From Figure, angle BAC is 30° means that its sides enclose 30 of those equal divisions. Arc BC is $\frac{30}{360}$ of the entire circumference.

Types of Angles:



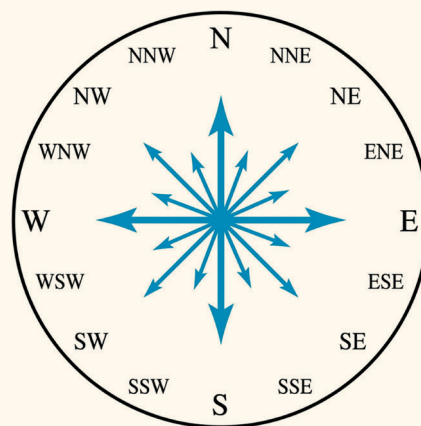
Angle Type	Measure
Acute angle	Less than 90°
Right angle	Exactly 90°
Obtuse angle	Between 90° and 180°
Straight angle	180°
Reflex angle	Between 180° and 360°
Complete angle	360°

Example 1:

Refer to the compass bearings shown aside North corresponds to an angle of 0° , and other angles are measured clockwise from north.

Find the angle in degrees that describes the compass bearing.

- a) NE (northeast)
- b) NNE (north-northeast)
- c) WSW (west-southwest)



Solution:

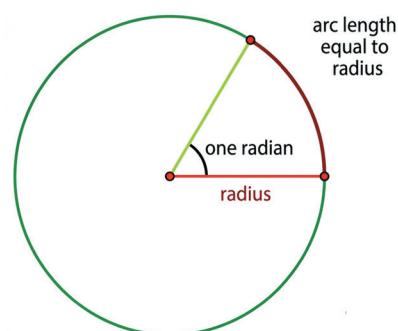
From the given compass bearing, we note that between two consecutive bearings, there are 22.5° (since $\frac{360^\circ}{16} = 22.5^\circ$). Then, measurements in degrees for NE (North-East) is $22.5^\circ \times 2 = 45^\circ$

- a) for NNE (north-northeast) is $22.5^\circ \times 1 = 22.5^\circ$
- b) for WSW (west-southwest) is $22.5^\circ \times 11 = 247.5^\circ$

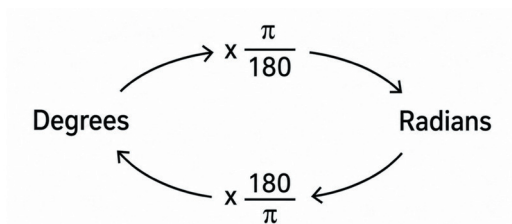
Radian measure:

When a central angle intercepts an arc that has the same length as a radius of the circle, the measure of the angle is defined to be one **radian**.

Like degrees, radian measures the amount of the rotation from the initial side to the terminal side of an angle. The relationship between **degrees and radians** is: $180^\circ = \pi$ radians. 1 radian is *about* 57.2958 **degrees**. The **radian** is a pure measure based on the **radius** of the circle.



Using this relationship: $1^\circ = \frac{\pi}{180^\circ}$ radians; then 1 radian = $\frac{180^\circ}{\pi}$



Example 2: Through how many radians does the minute hand of a clock rotate in 50 min? In 7 hours?

Solution:

It is known that, the minute hand of a clock rotates circumference of a circle in 60min, through 2π radians. Here, we are required to find the radian travelled by minute hand of a clock for a given time.

a) Minute hand of a clock rotates the circle in 50min, through $\frac{2\pi \times 50}{60}$ radians or $\frac{5\pi}{3}$ radians

b) Minute hand of a clock rotates the circle in 7 hours or 420 minutes, through $\frac{2\pi \times 420}{60}$ radians or 14π radians.

Example 3: Determine the type of angle:

a) 45°

b) 120°

c) 180°

Solution:

From the given compass bearing, we note that between two consecutive bearings,

a) $45^\circ < 90^\circ \rightarrow$ **Acute angle**

b) 120° is between 90° and $180^\circ \rightarrow$ **Obtuse angle**

c) $180^\circ \rightarrow$ **Straight angle.**

Application activity 4.1



1) Convert the following angles into radians:

1. 45°

2. 90°

3. 120°

4. 180°

5. 270°

2) State whether the following angles are **acute, right, obtuse, straight or reflex**.

1. 30°

2. 90°

3. 150°

4. 210°

5. 360°

3) A wheel makes **one complete turn**. How many degrees does it rotate?

4) A student rotates **half a turn**. How many degrees is that?

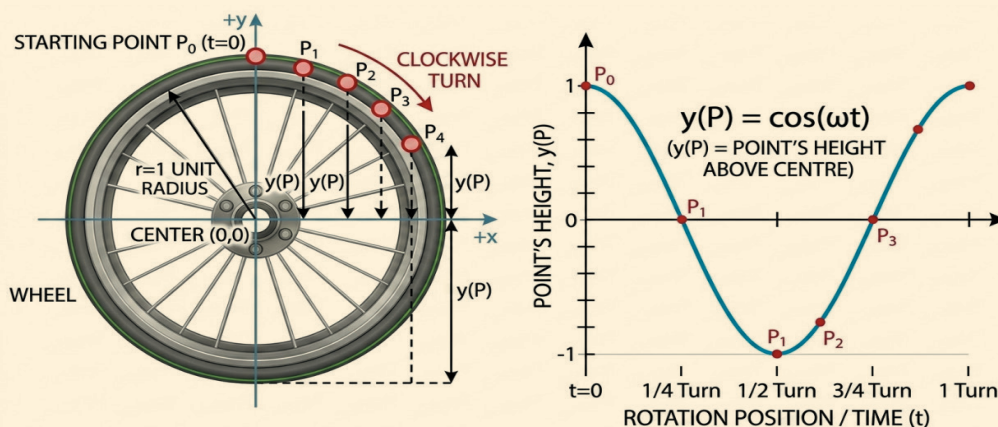
5) A radar antenna rotates $\frac{\pi}{2}$ radians. Convert it into degrees.

4.2 The unit circle, sine, cosine, secant, cosecant, cotangent and tangent.

Activity 4.2



Imagine a point on the edge of a wheel with radius $R=1$ unit of length.

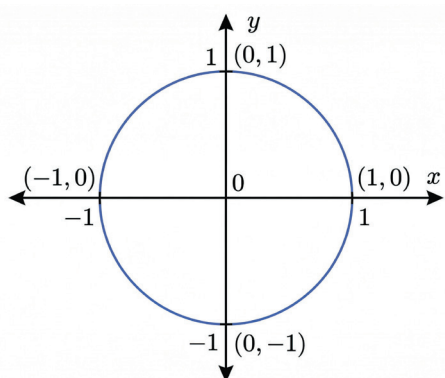


As the wheel turns the point P in clockwise direction from $(0,0)$, how is the point above the centre? Represent this using a drawing.



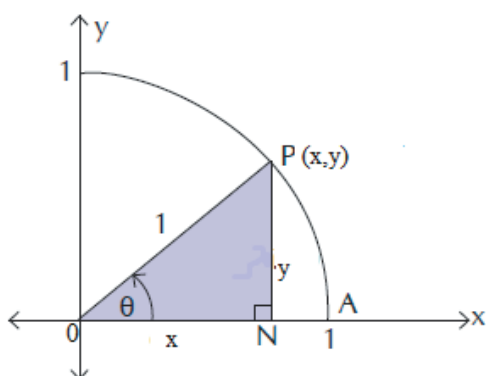
Content Summary

The **unit circle** is the circle with centre $(0, 0)$ and radius 1 unit.



Any point $P(x, y)$ on the unit circle corresponds to an **angle** θ measured from the positive x -axis. The equation of the unit circle is: $x^2 + y^2 = 1$ where x is the abscise and y is coordinate in Cartesian plan.

On the Cartesian plane the radius is 1 unit. Pick any point $P(x, y)$. What is the value of $\frac{x}{1}$ and $\frac{y}{1}$. Pick different points on the circumference and calculate $\frac{x}{1}$ and $\frac{y}{1}$.

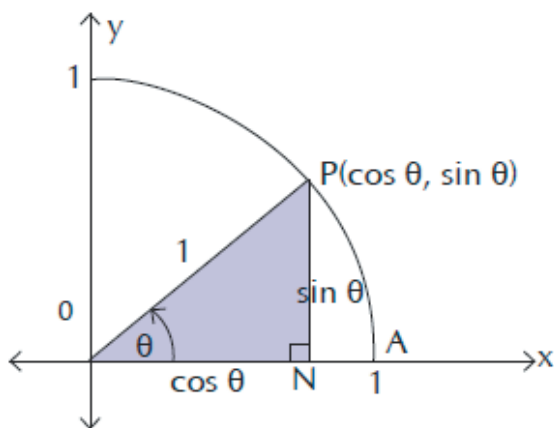


The x is the side adjacent to the angle θ and y is the side opposite the angle θ . The radius of 1 unit is the hypotenuse. Using right-angled triangle trigonometry:

$$\cos x = \frac{\text{Hypotenuse side}}{\text{Adjacent side}} = \frac{x}{1} = x, \text{ and}$$

$$\sin x = \frac{\text{Hypotenuse side}}{\text{Opposite side}} = \frac{y}{1} = y$$

$$\tan \theta = \frac{\text{Adjacent side}}{\text{Opposite side}} = \frac{y}{x} = \frac{\sin \theta}{\cos \theta}$$



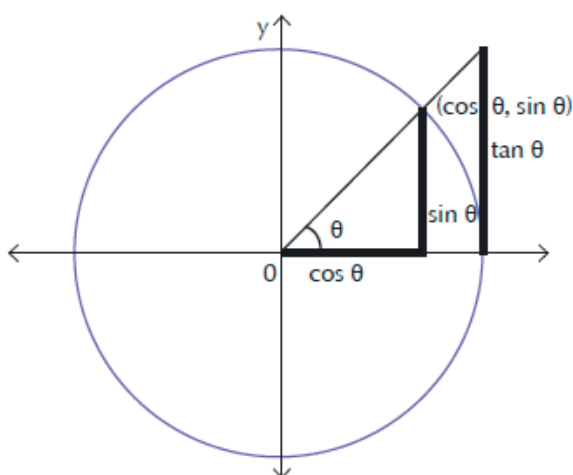
On the quarter circle, if $\angle AOP = \theta^\circ$, then the coordinates of P are $(\cos \theta, \sin \theta)$.

The x-and y-coordinates of P each have a special name.

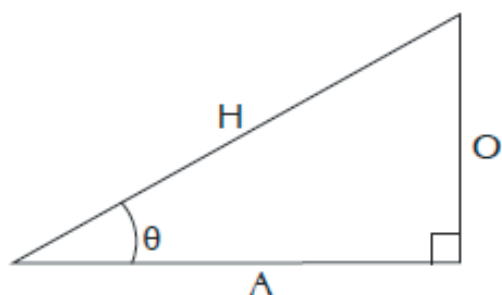
- The y-coordinate is called **the sine of angle θ** and noted $\sin \theta$.
- The x-coordinate is called **the cosine of angle θ** and noted $\cos \theta$. Notice also that in the triangle ONP, $x^2 + y^2 = 1$ (Pythagoras) and so

$$(\cos \theta)^2 + (\sin \theta)^2 = 1$$

$$\text{or } \cos^2 \theta + \sin^2 \theta = 1$$



We can also use the quarter unit circle to get another ratio. This is tangent which is written as 'tan' in short.



In this triangle the side H opposite the right angle is called the **hypotenuse**.

Relative to the angle θ , the side O opposite the angle θ is called the **opposite side**.

The remaining side A is called the **adjacent side**. Trigonometric ratios provide relationships between the sides and angles of a right angle triangle.

The three most commonly used ratios are sine, cosine and tangent.

Sine	$\sin \theta = \frac{O}{H}$
Cosine	$\cos \theta = \frac{A}{H}$
Tangent	$\tan \theta = \frac{O}{A}$

Note that $\tan \theta = \frac{\sin \theta}{\cos \theta}$

Other ratios are defined by using the above three:

Cosecant	$\csc \theta = \frac{1}{\sin \theta} = \frac{H}{O}$
Secant	$\sec \theta = \frac{1}{\cos \theta} = \frac{H}{A}$
Cotangent	$\cot \theta = \frac{1}{\tan \theta} = \frac{\cos \theta}{\sin \theta} = \frac{A}{O}$

These six ratios of the lengths of the sides of a right triangle are called **trigonometric functions**

Example: Identifying Sine and Cosine

A point on the unit circle is: P(0.8,0.6). Find: a) $\cos \theta$ and b) $\sin \theta$

Solution:

The coordinates are: (x, y) = (0.8,0.6). Therefore: $\cos \theta = 0.8$ and $\sin \theta = 0.6$

Application activity 4.2



- 1) Fill in the missing information from the unit circle:

Function	Meaning
$\sin \theta$	_____ coordinate
$\cos \theta$	_____ coordinate
$\tan \theta$	$\sin \theta \div$ _____
$\sec \theta$	$1 \div$ _____
$\csc \theta$	$1 \div$ _____
$\cot \theta$	$\cos \theta \div$ _____

- 2) True or False

- On the unit circle, sine is the **y-coordinate**.
- Cosine is the **x-coordinate**.
- Tangent compares sine and cosine.
- Secant is the reciprocal of cosine.
- Cotangent equals $\frac{\sin \theta}{\cos \theta}$

- 3) Think and discuss

- Why is the circle called a unit circle?
- What is the center of the unit circle?
- What do the coordinates of a point on the circle represent?

4.3 Trigonometric ratios of special angles (0° , 30° , 45° , 60° , 90°)

Activity 4.3

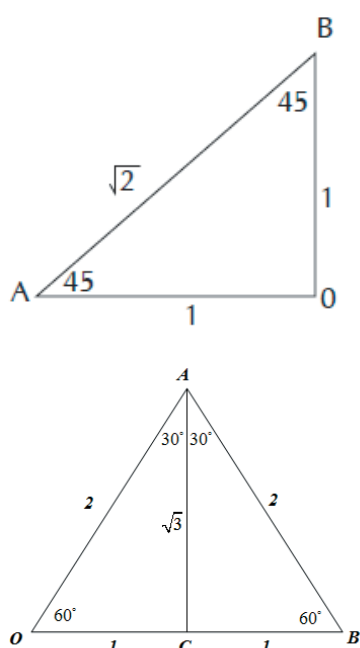


By using calculator discuss in group and Complete the table below.

Angle	0°	30°	45°	60°	90°
$\sin \theta$	?	?	?	?	?
$\cos \theta$	?	?	?	?	?



Content Summary



Consider the isosceles triangle AOB.

Hence the opposite side and adjacent sides are equal, say 1 unit. The hypotenuse is therefore of length $\sqrt{2}$ units (using Pythagoras' theorem).

$$\text{We have } \sin 45^\circ = \frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2} \text{ and } \cos 45^\circ = \frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2}$$

$$2 \tan 45^\circ = 1$$

Consider the equilateral triangle ABC, of sides 2 units in length, a line AD from A perpendicular to BC.

AD bisects BC giving $BD = CD = 1$ From this we can determine the following trigonometric ratios for the special angles 30° and 60°

$\sin 60^\circ = \frac{\sqrt{3}}{2}$	$\sin 30^\circ = \frac{1}{2}$
$\cos 60^\circ = \frac{1}{2}$	$\cos 30^\circ = \frac{\sqrt{3}}{2}$
$\tan 60^\circ = \sqrt{3}$	$\tan 30^\circ = \frac{1}{\sqrt{3}} = \frac{\sqrt{3}}{3}$

We can now complete the following table:

θ	0°	30°	45°	60°	90°
$\sin \theta$	0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1
$\cos \theta$	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0
$\tan \theta$	0	$\frac{\sqrt{3}}{3}$	1	$\sqrt{3}$	∞

Example: Compute the exact value of $\sin 30^\circ + \sin 60^\circ - \sin 90^\circ$

Solution:

$$\sin 30^\circ + \sin 60^\circ - \sin 90^\circ = \frac{1}{2} + \frac{\sqrt{3}}{2} - 1 = \frac{1+\sqrt{3}}{2} - 1 = \frac{1+\sqrt{3}}{2} - \frac{2}{2} = \frac{\sqrt{3}-1}{2}$$

Application activity 4.3



1. Fill in the following table from your memory

θ	0°	30°	45°	60°	90°
$\sin \theta$					
$\cos \theta$					
$\tan \theta$					

2. Compute the exact value of

a) $\sin 30^\circ + \sin 60^\circ - \sin 90^\circ$

b) $\sin^2 45^\circ + \cos^2 45^\circ - \tan 45^\circ$

3. Without using calculator, find out the following trigonometric ratios

a) $\cot 45^\circ$ b) $\cot 30^\circ$ c) $\cot 60^\circ$

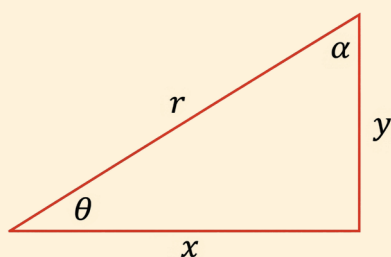
d) $\cot 0^\circ$ c) $\cot 90^\circ$ $\cot 90^\circ$

4.4 Introduction to trigonometric identities

Activity 4.4



Here is a right triangle



In this triangle, find out $\sin \theta$, $\cos \theta$, $\sin \alpha$ and $\cos \alpha$

Remember that in this triangle, Pythagoras' theorem states $x^2 + y^2 = r^2$. Dividing each term by r^2 yields

$$\frac{x^2}{r^2} + \frac{y^2}{r^2} = 1$$

Express $\frac{x^2}{r^2} + \frac{y^2}{r^2} = 1$ in terms of $\sin \theta$ and $\cos \theta$.

Find the value of $\cos^2 \alpha + \sin^2 \alpha$



Content Summary

In mathematics, an identity is an equation which is always true. There are many trigonometric identities, but the one you are most likely to see and use is, $\cos^2 \theta + \sin^2 \theta = 1$.

This relation is called the **fundamental formula of trigonometry** and is the most frequently used identity in trigonometry.

Dividing this identity by $\cos^2 \theta$ and $\sin^2 \theta$ gives two other identities.

$$1 + \tan^2 \theta = \sec^2 \theta$$

$$1 + \cot^2 \theta = \csc^2 \theta$$

Example 1: Simplify $\frac{\csc x}{\sec x}$.

Solution:

$$\frac{\csc x}{\sec x} = \frac{\frac{1}{\sin x}}{\frac{1}{\cos x}} = \frac{\cos x}{\sin x} = \cot x$$

Example 2: Simplify $\left(\frac{1}{\tan x} + \frac{1}{\cot x}\right) \sin x \cos x$

Solution:

$$\begin{aligned} \left(\frac{1}{\tan x} + \frac{1}{\cot x}\right) \sin x \cos x &= \left(\frac{1}{\frac{\sin x}{\cos x}} + \frac{1}{\frac{\cos x}{\sin x}}\right) \sin x \cos x = \left(\frac{\cos x \cos x + \sin x \sin x}{\sin x \cos x}\right) \sin x \cos x \\ &= \cos x \cos x + \sin x \sin x = \cos^2 x + \sin^2 x = 1 \end{aligned}$$

Application activity 4.4



1. Prove the following trigonometric identities

- $\tan \theta + \cot \theta = \sec \theta \csc \theta$
- $\sin^2 \theta - 4 \cos^2 \theta + 1 = 3 \sin^2 \theta - 2 \cos^2 \theta - 1$
- $\frac{\cos^2 A}{1 + \tan^2 A} - \frac{\sin^2 A}{1 + \tan^2 A} = 1 - 2 \sin^2 A$

2. Simplify:

- $\sec^4 \theta (1 - \sin^4 \theta) - 2 \tan^2 \theta$
- $\cot^2 A - \cot^2 B + \frac{\sin^2 A - \sin^2 B}{\sin^2 A \sin^2 B}$
- $\frac{\cos^3 \theta + \cos \theta \sin^2 \theta + \sin \theta}{3 \cos^3 \theta + 3 \cos \theta \sin^2 \theta - \sin \theta}$

4.5 Graphs of Trigonometric Functions

Activity 4.5



You are invited to draw a graph of the function $y = \sin x$. Start by preparing a table of values by choosing suitable values of x . Then, find the corresponding values of y . Now complete the following table

x	...	$-\pi$	$-\frac{3\pi}{4}$	$-\frac{\pi}{2}$	$-\frac{\pi}{4}$	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π	
$\sin x$											



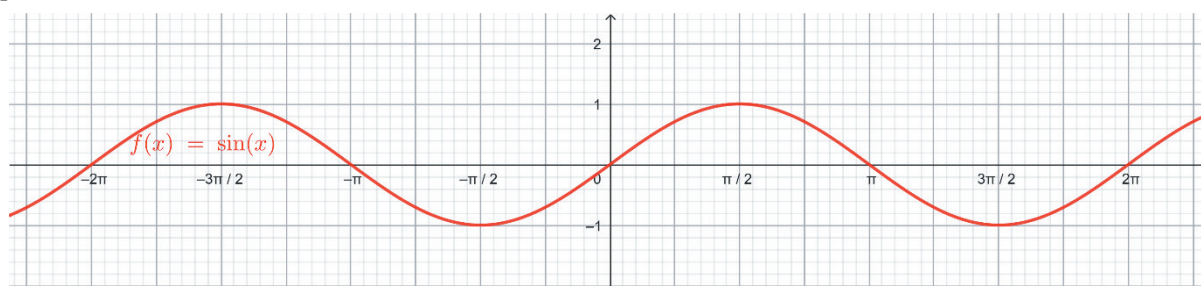
Content Summary

Sine function: $\sin x = y$

For the graph of $y = \sin x$ in $-\pi \leq x \leq \pi$ consider the following table. Note that the function is periodic with period 2π .

x	...	$-\pi$	$-\frac{3\pi}{4}$	$-\frac{\pi}{2}$	$-\frac{\pi}{4}$	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π	
$\sin x$		0	-0.71	-1	0.71	0	0.71	1	0.71	0	

Plot the ordered pairs $(x, \sin x)$. Join these points with a smooth curve to obtain the following graph:

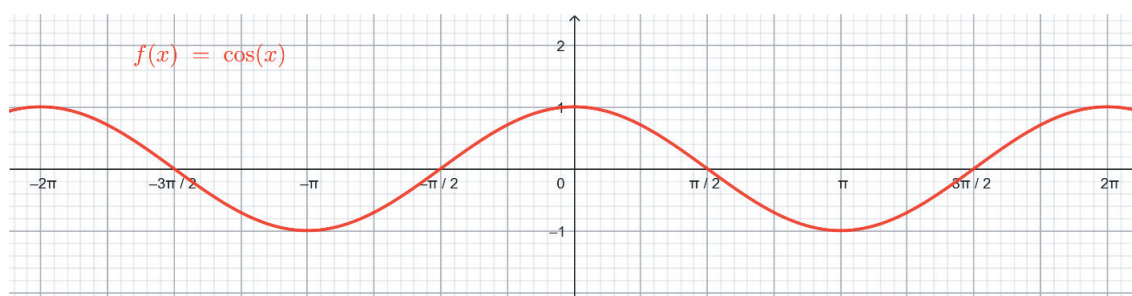


Cosine function: $\cos x = y$

To draw a graph of the function $y = \cos x$, start by completing a table of values for x and y . For the graph of $y = \cos x$ in $-\pi \leq x \leq \pi$, consider the following table. Note that the function is periodic with period 2π .

x	...	$-\pi$	$-\frac{3\pi}{4}$	$-\frac{\pi}{2}$	$-\frac{\pi}{4}$	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π	
$\cos x$		-1	-0.71	0	0.71	1	0.71	0	0.71	-1	

Plot the ordered pairs $(x, \cos x)$ and then join these points with a smooth curve. The following graph is obtained:



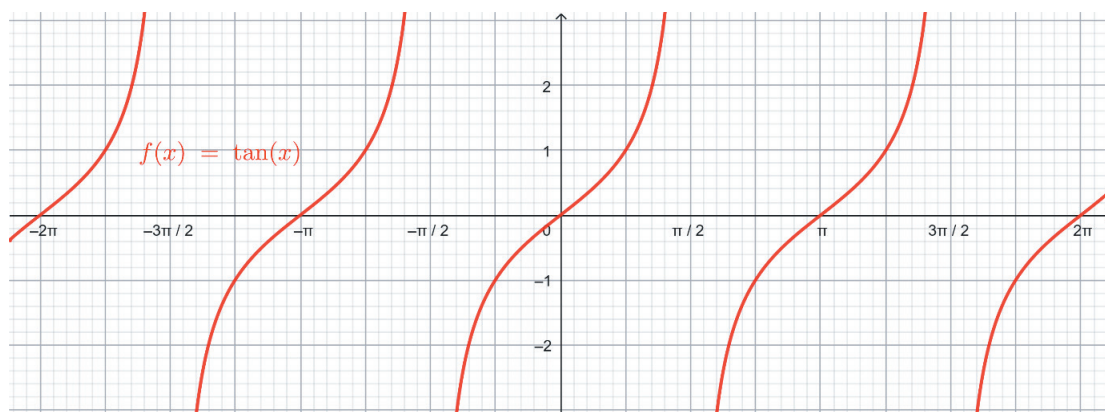
Tangent function: $\tan x = y$

Draw a graph of the function $y = \tan x$, prepare a table by choosing suitable values of x . Then find the corresponding values of y . For the graph of $y = \tan x$ in $-\pi \leq x \leq \pi$, note that the function is periodic with period π .

In addition, Since $\tan x = \frac{\sin x}{\cos x}$, $\tan x$, will be undefined whenever $\cos x = 0$. The zeros of the function $y = \cos x$ correspond to vertical asymptotes of the function $y = \tan x$.

x	...	$-\pi$	$\frac{3\pi}{4}$	$\frac{\pi}{2}$	$\frac{\pi}{4}$	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π	
$\tan x$		0	1	$+\infty$	1	0	1	$+\infty$	1	0	

Plot the ordered pairs $(x, \tan x)$ and then join these points with a smooth curve. The following graph is obtained:



Application activity 4.5



1: Complete the table for the sine function.

x	0°	90°	180°	270°	360°
$\sin x$					
$\cos x$					

- Complete the missing values.
- Plot the points on graph paper.
- Draw the sine curve.

2. Tangent Graph: Complete the table.

x	0°	30°	45°	60°
$\cot x$				

- Plot the points.
- Draw the cotangent curve.

Identify where the function is **undefined**

4.6 Reduction to functions of positive acute angles

4.6.1. Equivalent angles

Activity 4.6.1

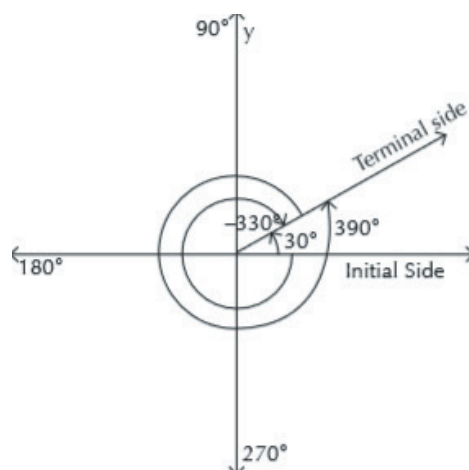


- Which of the following angles are coterminal with 120° ?
a) -270° b) 600° c) -270° d) 840°
- Write two angles coterminal with each angle
a) 180° b) -141° c) 129° d) 0°
- How are they, the trigonometric values of two coterminal angles?



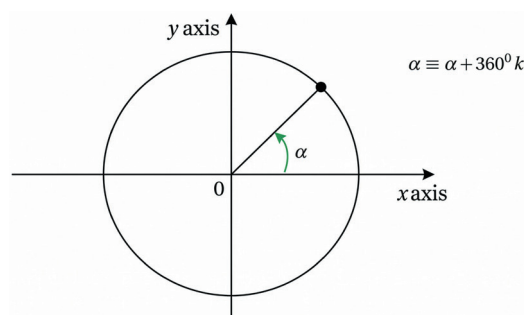
Content Summary

Coterminal angles are angles in standard position (angles with the initial side on the positive x-axis) that have a common terminal side. For example, 30° , -330° and 390° are all coterminal.



Two angles are equivalent if their difference is $2k\pi$, $k \in \mathbb{Z}$ (or $360^\circ k$, $k \in \mathbb{Z}$). This means that the angle α and $\alpha + 2k\pi$ are equivalent angles.

Note that equivalent angles are **coterminal**



The equivalent angles have the same trigonometric values. α and $\alpha + 360^\circ k$, $k \in \mathbb{Z}$ are equivalent since

$$\left. \begin{aligned} \sin(\alpha + 360^\circ k) &= \sin \alpha \\ \cos(\alpha + 360^\circ k) &= \cos \alpha \\ \tan(\alpha + 360^\circ k) &= \tan \alpha \\ \cot(\alpha + 360^\circ k) &= \cot \alpha \end{aligned} \right\} k \in \mathbb{Z}$$

Example

a) The angle 60° and 780° are equivalent since 780° can be written as $60^\circ + (2 \times 360^\circ)$, so that $780^\circ = 60^\circ + \underbrace{2 \times 360^\circ}_{2\pi}$

b) The angle 30° and -1050° are equivalent since

$$-1050^\circ = 30^\circ - \underbrace{3 \times 360^\circ}_{2\pi}$$

c) The angle $\frac{2\pi}{3}$ and $\frac{14\pi}{3}$ are equivalent since $\frac{14\pi}{3} = \frac{2\pi}{3} + 2(2\pi)$

If α and β are equivalent, we write $\alpha \equiv \beta$. Since 60° and 780° are equivalent, we may write $60^\circ \equiv 780^\circ$

Application activity 4.6.1



1. Find a positive and a negative angle coterminal with a 55° angle.
2. Find a positive and a negative angle coterminal with a $\frac{\pi}{3}$ angle.

4.6.2 Negative/Opposite angles

Activity 4.6.2



In the following diagrams, the line segment is the terminal side of a given angle in standard position.

- Using protractor, find the given angle and its principal trigonometric values.
- The line segment is symmetrical about the x-axis, copy each graph and draw the remaining part (reflection on the x-axis).
- Indicate the value of new angle in standard position and give its principal trigonometric values.
- Complete the given table and give your observations

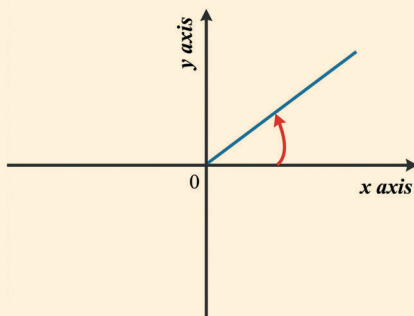


Figure 1

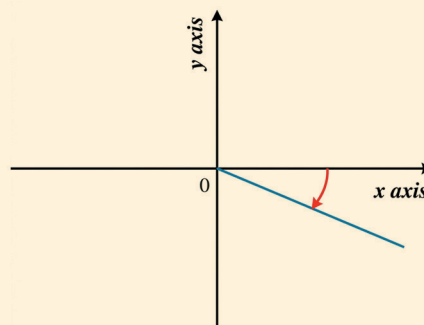


Figure 2

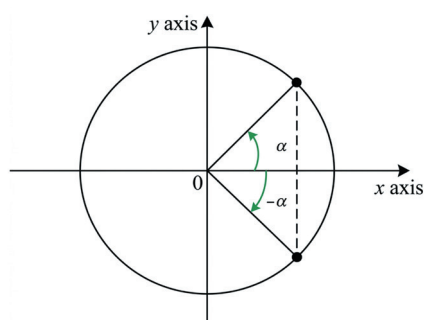
	The given angle (in degree)		New angle(in degree) after reflection		Observations from trigonometric values of both angles
	Figure 1	Figure 2	Figure 1	Figure 2	
sin					
cos					
tan					



Content Summary

From set theory, angle $-\alpha$ is opposite of angle α

Geometrically: The terminal sides of angle and its opposite angles are **symmetric to x-axis**.



For two angles, α and $-\alpha$, the following identities are true

$$\sin(-\alpha) = -\sin \alpha$$

$$\cos(-\alpha) = \cos \alpha$$

$$\tan(-\alpha) = -\tan \alpha$$

$$\cot(-\alpha) = -\cot \alpha$$

Examples

a) -10° is opposite of 10°

b) $-\frac{\pi}{4}$ is opposite of $\frac{\pi}{4}$

Application activity 4.6.2



Work out

1) If $\sin \theta = -0.1903$, find $\sin(-\theta)$

2) If $\cos(\theta) = 0.0133$, find $\cos(-\theta)$

3) If $\sec(\theta) = -1.753$, find $\cos(-\theta)$

4) If $\csc(-\theta) = \sqrt{3}$, find $\sin \theta$

5) If $\cos(-\theta) = \frac{1}{7}$, find $\sec \theta$

6) If $\cot(\theta) = -5.4219$, find $\tan(-\theta)$

4.6.3 Complementary angles

Activity 4.6.3



Work out

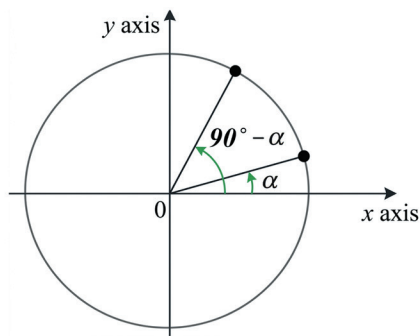
- 1) Give examples of 3 pairs of two angles whose sum is a right angle.
- 2) Present those paired angles on trigonometric circle and compare their trigonometric ratios cosine and sine. How can you conclude?



Content Summary

Two angles are said to be complementary if their sum is 90° (or $\frac{\pi}{2}$).

The angles α and $90^\circ - \alpha$ are complementary.



For two complementary angles α and $90^\circ - \alpha$, the following identities are true

$$\sin(90^\circ - \alpha) = \cos \alpha$$

$$\cos(90^\circ - \alpha) = \sin \alpha$$

$$\tan(90^\circ - \alpha) = \cot \alpha$$

$$\cot(90^\circ - \alpha) = \tan \alpha$$

Example 1: The angles 30° and 60° are complementary

Example 2: If $\cos \theta = 0.975$, find $\sin\left(\theta - \frac{\pi}{2}\right)$.

Solution:

As $\sin(-\theta) = -\sin \theta$, thus $\sin\left(\theta - \frac{\pi}{2}\right) = -\sin\left(\frac{\pi}{2} - \theta\right)$, from identity of negative angles
 $-\sin\left(\frac{\pi}{2} - \theta\right) = -\cos \theta$, from identity of complementary angles then $= -\cos \theta = -0.975$,
 from the given value of $\cos \theta$.

Application activity 4.6.3



Work out

- 1) If $\sin \theta = 0.00213$, find $\cos\left(\frac{\pi}{2} - \theta\right)$
- 2) If $\tan\left(\frac{\pi}{2} - \theta\right) = -0.11221$, find $\cot \theta$
- 3) If $\cot(-\theta) = 1.1482$, find $\tan\left(\theta - \frac{\pi}{2}\right)$
- 4) If $\cos(\theta) = 0.5$, find $\csc\left(\theta - \frac{\pi}{2}\right)$
- 5) If $\sin\left(\frac{\pi}{2} - \theta\right) = \frac{1}{2}$, find $\sec \theta$
- 6) If $\sec\left(\theta - \frac{\pi}{2}\right) = 7$, find $\csc \theta$

4.6.4 Supplementary angles

Activity 4.6.4



In the following diagrams, the line segment is the terminal side of a given angle in standard position.

- Using protractor, find the given angle and its principal trigonometric values.
- The line segment is symmetrical about the y-axis, copy each graph and draw the remaining part (reflection on the y-axis).
- Indicate the value of new angle in standard position and give its principal trigonometric values.
- Complete the given table and give your comments

	The given angle (in degree)		New angle (in degree) after reflection		Observations from trigonometric values of both angles
	Figure 1	Figure 2	Figure 1	Figure 2	
sin					
cos					
tan					

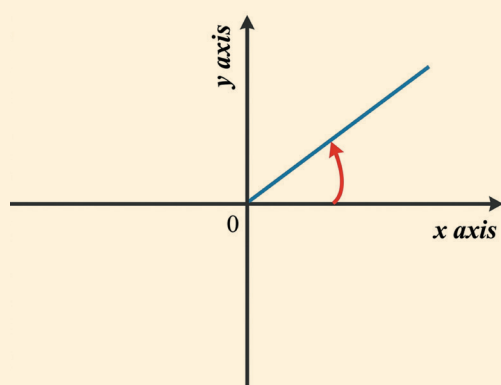


Figure 1

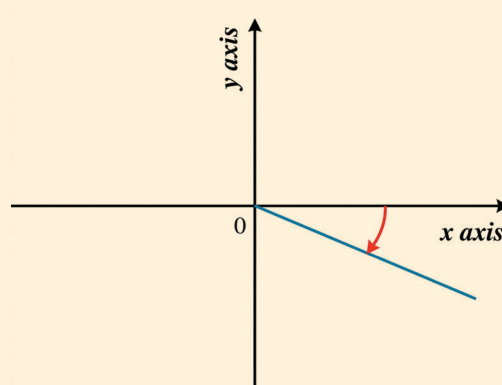


Figure 2



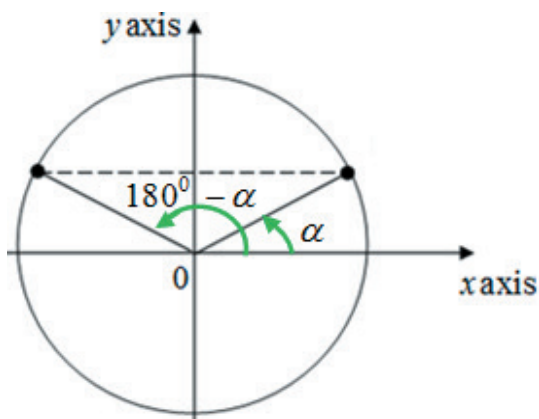
Content Summary

Two angles are said to be supplementary if their sum is 180° (or π).

From this definition, we note that α and $180^\circ - \alpha$ are supplementary.

Geometrically:

Two angles in standard position are supplementary, if their terminal sides are **symmetric to y-axis**.



For two supplementary angles α and $180^\circ - \alpha$, the following identities are true

$$\sin(180^\circ - \alpha) = \sin \alpha$$

$$\cos(180^\circ - \alpha) = -\cos \alpha$$

$$\tan(180^\circ - \alpha) = -\tan \alpha$$

$$\cot(180^\circ - \alpha) = -\cot \alpha$$

Example

- The angles 160° and 20° are supplementary
- The angles 110° and 70° are supplementary

Application activity 4.6.4



Work out

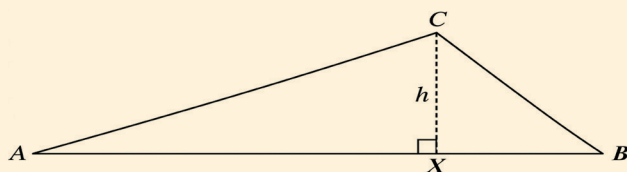
- If $\sin \theta = 0.0312$, find $\sin(\pi - \theta)$
- If $\tan(\pi - \theta) = -0.11221$, find $\tan \theta$
- If $\cot(\theta) = -2.5148$, find $\cot(\theta - \pi)$
- If $\sec(\theta) = 0.5$, find $\cos(\theta - \pi)$
- If $\csc(\pi - \theta) = \frac{1}{2}$, find $\csc \theta$
- If $\sec(\theta - \pi) = 3$, find $\sec \theta$

4.7. Solving a triangle by cosine law

Activity 4.7



Consider the following triangle



CX is perpendicular to side AB. Let $AB=c$, $AC=b$, $BC=a$, $CX=h$

- In triangle AXC find $\cos A$
- In triangle AXC use Pythagoras' theorem to find h^2
- In triangle BCX use Pythagoras' theorem to find h^2
- Combine results obtained in 1., 2., and 3. and give conclusion.



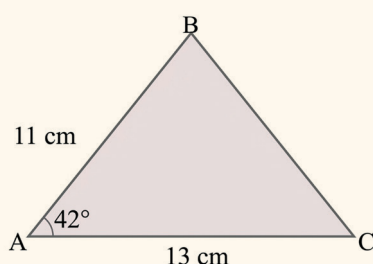
Content Summary

The Law of Cosines is an important extension of the Pythagorean theorem, with many applications. It is often called the “generalized Pythagorean theorem” because it contains that classic theorem. While the Law of Sines is the tool, we use to solve triangles in the AAS and ASA cases, the Law of Cosines is the required tool for SAS and SSS.

Let ABC be any triangle with sides and angles labelled in the usual way, then

$$a^2 = b^2 + c^2 - 2bc \cos A \quad b^2 = a^2 + c^2 - 2ac \cos B \quad c^2 = a^2 + b^2 - 2ab \cos C$$

Example 1: Find the length of BC from the following triangle.



Solution:

Let a be the length of side BC , from cosine law

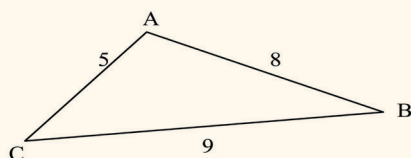
$$a^2 = b^2 + c^2 - bc \cos A, \text{ we have,}$$

$$a^2 = (13)^2 + (11)^2 - 2(13)(11)\cos 42^\circ$$

$$a^2 = 169 + 121 - 286 \cos 42^\circ \Leftrightarrow a = \sqrt{290 - 286 \cos 42^\circ} = 8.8$$

Finally, the length of side BC is 8.8 cm

Example 2: Find the angle C



Solution:

Let C be the required angle the given triangle;

From cosine law $c^2 = a^2 + b^2 - 2ab \cos C$, we have,

$$8^2 = (9)^2 + (5)^2 - 2(9)(5)\cos C$$

$$\Leftrightarrow 64 = 81 + 25 - 90 \cos C$$

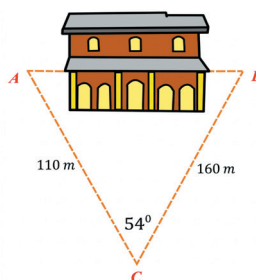
$$\text{Or } \cos C = \frac{42}{90}, \text{ thus } C = 62.2^\circ.$$

The required angle is $C = 62.2^\circ$.

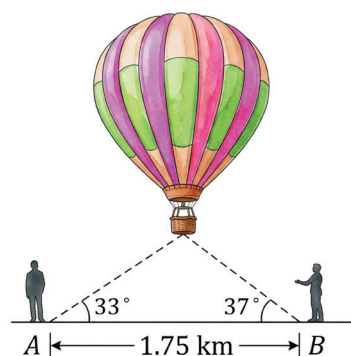
Application activity 4.7



- Find the length of the side BC of triangle ABC in which $AB=7\text{cm}$, $AC=9\text{cm}$ and $\angle BAC = 71^\circ$
- Find the length of the side AB of triangle ABC in which $BC=15.3\text{cm}$, $AC=9.4\text{cm}$ and $\angle ACB = 121^\circ$.
- Angelus wants to find the distance between two points A and B on opposite sides of a building. He locates a point C that is 110 m from A and 160 m from B , as illustrated in the figure. If the angle at C is 54° , find distance AB .



4. A hot-air balloon is seen over Tucson, Arizona, simultaneously by two observers at points A and B that are 1.75 km apart on level ground and in line with the balloon. The angles of elevation are as shown here. How high above ground is the balloon?

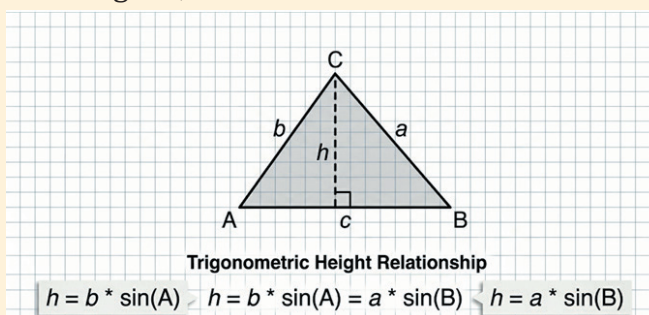
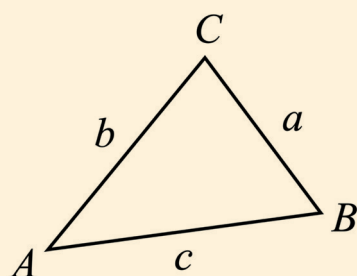


4.8 Solving a triangle by Sine law

Activity 4.8



- **Construct:** Draw a large acute triangle ABC on your paper. Ensure no two sides are the same length.
- **Measure:** Measure all three interior angles (A, B, C) and all three side lengths (a, b, c). Note that side *a* is opposite angle A, and so on.



- Draw an altitude (height) *h* from vertex C to the side *c*. Using the left side of the triangle, show that $h = b \sin A$. Using the right side of the triangle, show that $h = a \sin B$.

Comparison: Since both expressions equal *h*, set them equal: $b \sin(A) = a \sin(B)$. Now, answer to the following:

- 1) Divide both sides by $a \sin(A)$ and complete: $\frac{\sin A}{a} = \frac{?}{?}$
- 2) Complete what you found in the following table on the chalkboard

Group	$\sin(A)/a$	$\sin(B)/b$	$\sin(C)/c$
1	...	...	...
2	...	...	...
3			

- 3) Then, conclude regardless of the shape of their triangle: How are ratios?



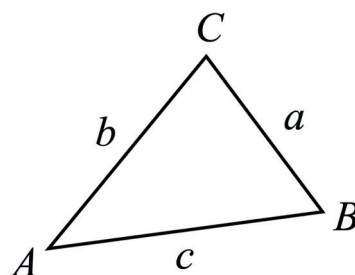
Content Summary

From Geometry, a triangle has six parts (three sides (S), three angles (A)), but its size and shape can be completely determined by fixing only three of those parts, provided they are the right three.

These three some that determine triangle congruence are known by their acronyms: AAS, ASA, SAS, and SSS. The other two acronyms represent match-ups that don't quite work: AAA determines similarity only, while SSA does not even determine similarity.

With trigonometry, we can find the other parts of the **triangle once congruence** is established. The tools we need are the Law of Sines and the Law of Cosines, the subjects of our last two trigonometric sections.

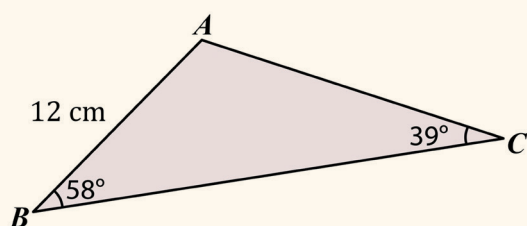
The sine law states that in any triangle ABC with angles A , B , and C , opposite sides a , b , and c , respectively, the following equation is true: $\frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c}$



A triangle with the usual labelling (angles A , B , C ; opposite sides a , b , c)

Example 1

Find the length of AC in the following triangle



Solution:

The acronym that we have for this case is AAS (two angles and one side).

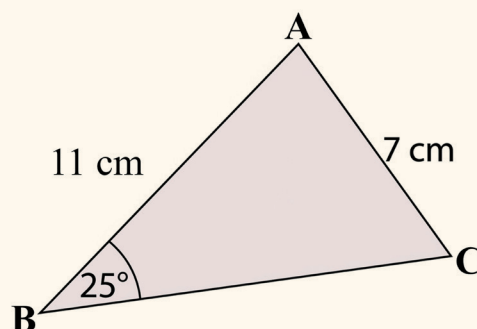
Let $AC = b$, using the sine rule, we get

$$\frac{\sin 58^\circ}{b} = \frac{\sin 39^\circ}{12} \text{ or } b = \frac{12 \sin 58^\circ}{\sin 39^\circ} = 16.17.$$

Therefore, $AC = 16.17 \text{ cm}$.

Example 2

Find the measure of angle C in triangle ABC if AC is 7 cm, AB is 11 cm and angle B measures 25° .



Solution:

For this case, the acronym SAS (two sides and one angle). From the sine rule, we have

$$\frac{\sin C}{c} = \frac{\sin B}{b} \Leftrightarrow \frac{\sin C}{11} = \frac{\sin 25^\circ}{7} \Leftrightarrow \sin C = \frac{11 \sin 25^\circ}{7}$$

or $C = 41.6^\circ$ or $180^\circ - 41.6^\circ = 138.4^\circ$. Then $A+B+C = A+25+138.4 > 180^\circ$ unless A is negative — impossible. So only $C \approx 41.6^\circ$ works. Thus, $C = 41.6^\circ$

Notice: While two angles and a side of a triangle are always sufficient to determine its size and shape, the same cannot be said for two sides and an angle. Perhaps unexpectedly, it depends on where that angle is. If the angle is included between the two sides (the SAS case), then the triangle is uniquely determined up to congruence. If the angle is opposite one of the sides (the SSA case), then there might be one, two, or zero triangles determined.

Application activity 4.8



1. A tower of 30 m of height stands on top of a hill. From a point some distance from the base of the hill, the angle of elevation to the top of the tower is 33° . From the same point the angle of elevation to the bottom of the tower is 32° . Find the height of the hill.
2. An aeroplane is 1200 m directly above one end of a field. The angle of depression of the other end of the field from the aeroplane is 64° . How long is the field?
3. Mutesi is standing on the bank of a river and observes that the angle subtended by a tree on the opposite bank is 60° . When she retreats 40 m from the bank, she finds the angle to be 30° . Find the height of the tree and the breadth of the river.

4.9. Applications of Trigonometry

4.9.1. Bearing

Activity 4.9.1



- 1) Refer to the content you learnt in ordinary level and explain the bearing of an object.
- 2) A boat leaves a harbour **A** and travels **8 km** on a **bearing of 060°** to reach point **B**. Another boat leaves the same harbour **A** and travels **10 km** on a **bearing of 120°** to reach point **C**.
 - a) Draw a simple diagram showing points **A, B, and C**.
 - b) What is the **angle $\angle BAC$** between the two directions?
 - c) What shape is formed by points **A, B, and C**?
 - d) Which trigonometric method can be used to find the **distance between B and C**?



Content Summary

A **bearing** is the **direction of one point from another measured clockwise from North**.

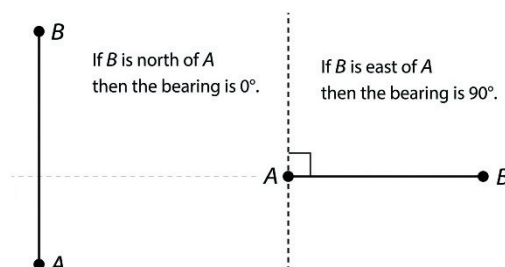
Important properties:

- Bearings are measured from **North (000°)**.
- They are measured **clockwise**.
- Bearings are written using **three digits**.

Examples:

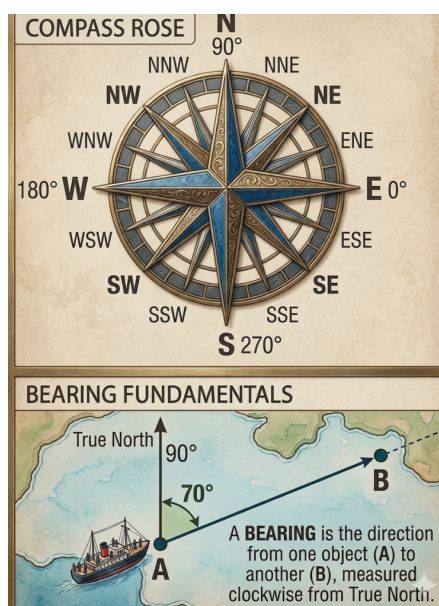
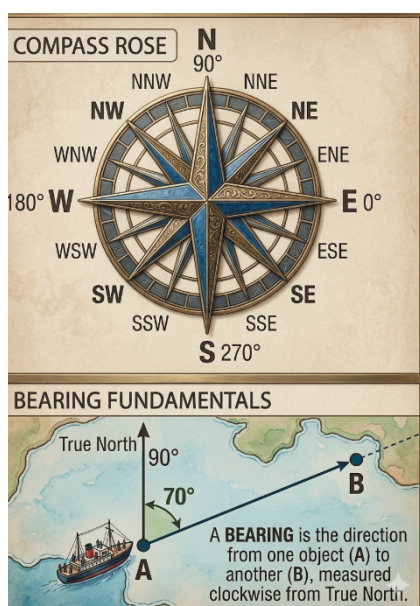
Direction	Bearing
North	000°
East	090°
South	180°
West	270°

We say that point B has a bearing of α degrees from point A if the line connecting A to B makes an angle of α with a vertical line drawn through A , the angle being measured clockwise.



Similarly, if B is south of A then the bearing is 180° , and if B is west of A then the bearing is 270° . The bearing can be any number between 0 and 360, because there are 360 degrees in a circle. We can also use right triangles to find distances using angles given as bearings.

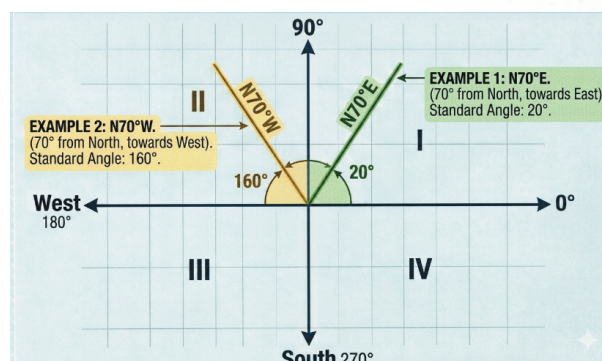
In navigation, a bearing is the direction from one object to another. Further, angles in navigation and surveying may also be given in terms of north, east, south, and west.



The Sailer knows the north and must orient the ship 70 degrees towards East.

Example 1:

$N70^{\circ}E$ refers to an angle from the north, towards the east, while $N70^{\circ}W$ refers to an angle from the north, towards the west. $N70^{\circ}W$ would result in an angle in the second quadrant.



The Rule for Quadrant Bearings: In navigation and surveying, quadrant bearings use the cardinal directions (North or South) as their starting point baseline.

- The first letter (**N**) tells you to start your look at the **North axis** (90° in standard mathematical position).
- The angle (70°) tells you how far to rotate.
- The second letter (**E**) tells you which direction to turn—in this case, towards the **East**.

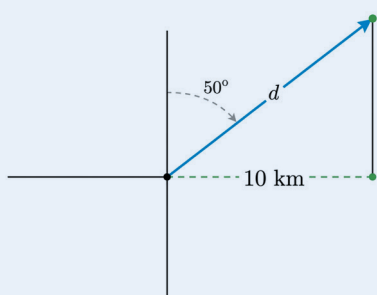
Therefore, you place your baseline on North and measure 70° **clockwise** towards East.

Bearing Notation	Starting Reference	Direction of Turn	Equivalent Standard Angle (from 0°)
$N70^{\circ}E$	North (90°)	Clockwise (towards East)	20° (Quadrant I)
$N70^{\circ}W$	North (90°)	Counter-Clockwise (towards West)	160° (Quadrant II)

Example 2:

A boat travels on a $N50^{\circ}E$ course. The ship travels until it is due north of a port which is 10 kilometres due east of the port from which the ship originated. How far did the boat travel?

Solution:



The angle between d and 10 km is the complement of 50° which is 40° .

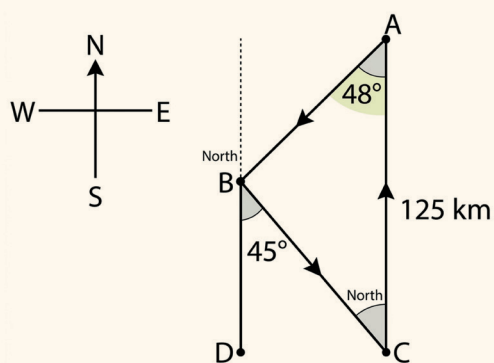
$$\cos 40^{\circ} = \frac{\text{Adjacent}}{\text{hypotenuse}} = \frac{10}{d} \Leftrightarrow \cos 40^{\circ} = \frac{10}{d}$$

$$d = \frac{10}{\cos 40^{\circ}} = 13.05 \text{ km}$$

Example 3:

The course for a boat race starts at point A and proceeds in the direction $S 48^{\circ} W$ to point B. Then in the direction $S 45^{\circ} E$ to point C, and finally back to A, as illustrated in the Figure. Point C lies 125 km directly south of point A.

Approximate the total distance of the racecourse.



Solution:

At B: Angle $ABC = 180 - (45 + 48) = 87$ degrees.

At C: Angle $BCA = 180 - (48 + 87) = 45$ degrees.

Therefore, as $AC = 125$ km, the rule of sine gives:

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

Distance BC (side a)

$$\frac{a}{\sin 48^\circ} = \frac{125}{\sin 87^\circ}, \text{ then}$$

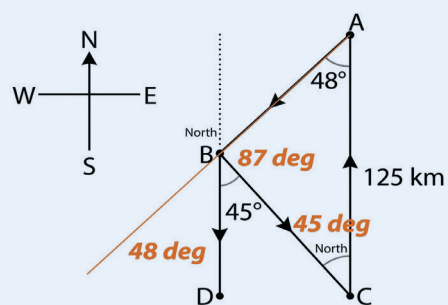
$$a = \frac{125 \cdot \sin 48^\circ}{\sin 87^\circ} \approx \frac{125 \cdot 0.7431}{0.9986} \approx 93.02 \text{ km}.$$

Find distance AB (side c): $\frac{c}{\sin 45^\circ} = \frac{125}{\sin 87^\circ}$ which

$$\text{gives } c = \frac{125 \cdot \sin 45^\circ}{\sin 87^\circ} \approx \frac{125 \cdot 0.7071}{0.9986} \approx 88.51 \text{ km}.$$

Total Distance = $AB + BC + CA =$

$$\text{Total Distance} \approx 88.51 + 93.02 + 125 = \mathbf{306.53 \text{ km}}$$



Application activity 4.9.1



- 1) A plane flies 20 km on a bearing of 030° from airport A to B. Another plane flies 25 km on a bearing of 110° from A to C.
 - a) Draw the diagram.
 - b) Find the angle at A.
 - c) Calculate the distance BC.
- 2) Two roads leave a town A: One road goes 18 km on a bearing of 050° to town B. Another goes 22 km on a bearing of 140° to town C.
 - a) What is $\angle BAC$?
 - b) Find the distance BC.

- 3) A ship sails 14 km on a bearing of 070° from port A to B. From B it sails 16 km on a bearing of 150° to C.
- Draw the triangle ABC.
 - Find angle ABC.
 - Calculate the distance AC.
- 4) Two airplanes leave the same airport: Plane 1 flies 30 km on bearing 040° and Plane 2 flies 40 km on bearing 120°
- Draw the triangle formed.
 - Find the distance between the airplanes.
 - Which trigonometric rule did you use?

4.9.2 Air Navigation

Activity 4.9.2



An airplane leaves an airport **A** and flies **80 km East**. At the same time, a second airplane leaves the same airport and flies **100 km** in a direction **60° North of East**.

- Draw a diagram showing the two flight paths.
- What geometric figure is formed by the paths of the airplanes?
- Which angle is formed between the two flight directions?
- How can we determine the **distance between the two airplanes**?



Content Summary

Air navigation is the process of determining the position and direction of an aircraft during flight.

Pilots use: Angles of direction, Distances and Trigonometric calculations to determine locations and flight paths

Directions are often expressed as: Degrees from North (bearings) and Angles relative to East or West.

Direction	Description
45° North of East	45° above the East direction
30° South of East	30° below East
60° West of North	60° toward West from North

These directions help pilots determine **flight paths**.

Trigonometric rules used

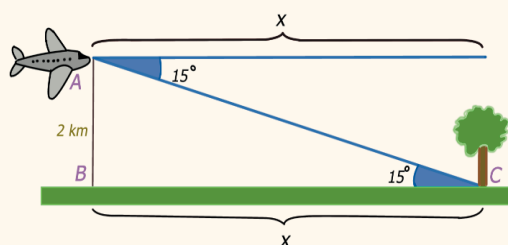
Cosine Rule $a^2 = b^2 + c^2 - 2bc \cos A$ Used when: Two sides and the **included angle** are known. This rule helps find the **distance between aircraft or airports**.

Sine Rule: $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$ Used when: Two angles and one side are known. It helps determine **unknown distances or directions**.

Steps for Solving Air Navigation Problems

1. Draw a diagram showing directions.
2. Represent the flight paths as sides of a triangle.
3. Identify the angles between directions.
4. Choose the appropriate rule: Cosine rule or Sine rule
5. Calculate the unknown distance or angle.

Examples 1: An airplane is flying at a height of 2 kilometres above the level ground. The angle of depression from the plane to the foot of a tree is 15° . Find the distance that the air plane must fly to be directly above the tree.



Solution:

Let x be the distance the airplane must fly to be directly above the tree. The level ground and the horizontal are parallel, so the alternate interior angles are equal in measure.

$\tan 15^\circ = \frac{2}{x} \Leftrightarrow x = \frac{2}{\tan 15^\circ} \approx 7.46$: So, the airplane must fly about 7.46 kilometres to be directly above the tree.

Example 2: Distance Between two aircraft

Two aircraft leave the same airport **A**. Aircraft **B** flies **120 km East** and Aircraft **C** flies **150 km at 50° North of East**. Find the **distance between the aircraft**.

Solution:

Step 1: Identify the angle between directions: 50°

Step 2: Apply cosine rule $BC^2 = 120^2 + 150^2 - 2(120)(150)\cos 50^\circ$

Step 3: Calculate $BC^2 = 120^2 + 150^2 - 2(120)(150)\cos 50^\circ$
 $= 14400 + 22500 - 36000(0.6428) = 117.3 \text{ km.}$

Distance between aircraft $\approx$ **117.3 km.**

Application activity 4.9.2



- 1) Aircraft **A** and **B** are **200 km apart**. A third aircraft **C** observes them such that: $\angle A = 45^\circ$ and $\angle B = 65^\circ$. Find the distance **AC**.
- 2) An aircraft flies 150 km on a bearing of 080° from airport A to airport B. Another aircraft flies 200 km on a bearing of 140° from airport A to airport C. Find the distance between B and C.
- 3) A pilot observes an airport at an angle of depression of 12° while flying at an altitude of 3 km. How far horizontally is the aircraft from the airport?
- 4) Two aircraft leave the same airport. One flies 180 km at 30° North of East. The other flies 220 km at 20° South of East. Find the distance between the two aircraft.

4.9.3. Angle of elevation

Activity 4.9.3



A student stands **20 m** away from a tall building. When the student looks up to the top of the building, the **angle formed between the ground and the line of sight is 40°** .

- a) Draw a diagram showing the student, the building, and the ground.
- b) What type of triangle is formed?
- c) Which side represents the **height of the building**?
- d) Which trigonometric ratio can help us find the height?

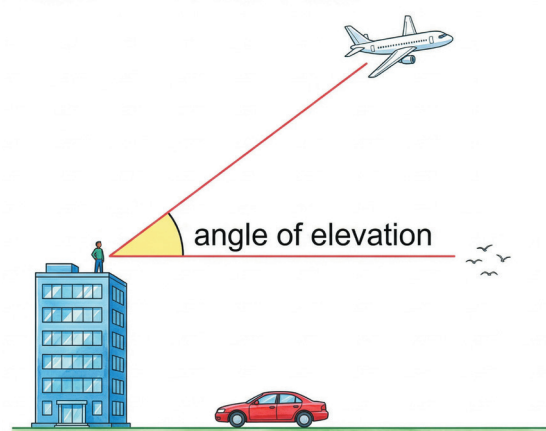


Content Summary

An angle of elevation is the angle between the horizontal line and the line of sight when looking upward at an object.

Suppose that an observer is standing at the top of a building and looking straight ahead at the birds (horizontal line).

The observer must raise his/her eyes to see the airplane (slanting line). The angle formed between the two lines is called the angle of elevation.



Examples: Looking at the top of a building, looking at the top of a tree, observing an airplane in the sky, Looking at the top of a tower. In these situations, a right triangle is formed.

Trigonometric ratios used

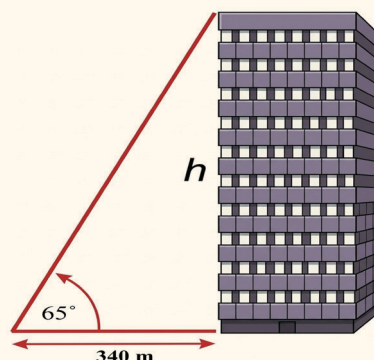
- 1) **Tangent Ratio** $\tan \theta = \frac{\text{opposite}}{\text{adjacent}}$, Used when we know: The **horizontal distance** or The **height**
- 2) **Sine Ratio** $\sin \theta = \frac{\text{opposite}}{\text{hypotenuse}}$. Used when we know: The **height** or The **line of sight distance**
- 3) **Cosine Ratio** $\cos \theta = \frac{\text{adjacent}}{\text{hypotenuse}}$, Used when we know: The **ground distance** or The **line of sight distance**

Steps for Solving Angle of Elevation Problems

1. Draw a right triangle representing the situation.
2. Identify: Opposite side (height); Adjacent side (horizontal distance); Hypotenuse (line of sight).
3. Identify the angle of elevation.
4. Choose the correct trigonometric ratio.
5. Substitute known values and solve.

Example: Finding the Height of a Building

From a point 340 m away from the base of the Peachtree Center Plaza in Atlanta, Georgia, the angle of elevation to the top of the building is 65° (See Figure) Find the height h of the building.



Solution:

We need a ratio that will relate an angle to its hypotenuse and adjacent sides. The cosine function is the appropriate choice.

$$\tan 65^\circ = \frac{h}{340} \text{ or } h = 340 \tan 65^\circ \approx 729 \text{ m}.$$

Application activity 4.9.2



- 1) A person stands 30 m away from a tower. The angle of elevation to the top is 40° .
a) Draw the diagram. b) Calculate the height of the tower.
- 2) A flagpole is 20 m high. The angle of elevation from a point on the ground is 35° . Find the distance from the observer to the base of the pole.
- 3) A building is observed from a point on the ground. The angle of elevation is 60° , and the distance to the base is 12 m. Calculate the height of the building.

4.9.4. Angle of depression

Activity 4.9.3



A student stands on the **top of a tower 20 m high**. The student looks down at a car on the ground and measures the **angle between the horizontal line and the line of sight to the car as 30°** .

Discuss in group the following questions

1. Draw a diagram showing the tower, the ground, and the car.
2. What type of triangle is formed?
3. Which side represents the **height of the tower**?
4. What is the angle formed when looking downward called?

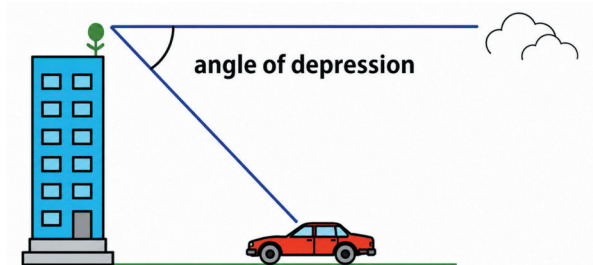


Content Summary

The angle of depression is the angle between the horizontal line and the line of sight when looking downward at an object.

You can use right triangles to find distances, if you know **an angle of elevation or an angle of depression**. The figure below shows each of these kinds of angles.

Suppose that an observer is standing at the top of a building and looking straight ahead at the birds (horizontal line). The observer must lower his/her eyes to see the car parked (slanting line). The angle formed between the two lines is called the angle of depression.



Looking down from a **building to a person on the ground**

In these situations, a **right triangle** is formed.

Examples: Looking down from a **tower to a car** and Looking down from a **cliff to a river**

Relationship Between Elevation and Depression.

Important property: The **angle of depression equals the angle of elevation** because they are **alternate interior angles** formed by parallel horizontal lines.

Trigonometric Ratios Used

1) **Tangent Ratio:** $\tan \theta = \frac{\text{opposite of } \theta}{\text{adjacent of } \theta}$ Used when we know: Height of the object and Horizontal distance

2) **Sine Ratio:** $\sin \theta = \frac{\text{opposite}}{\text{hypotenuse}}$

Used when we know: Height and Line of sight distance

Cosine Ratio: $\cos \theta = \frac{\text{adjacent}}{\text{hypotenuse}}$

Used when we know: Horizontal distance and Line of sight distance

Steps for Solving Angle of Depression Problems

1. Draw the **diagram** showing horizontal lines.
2. Identify the **angle of depression**.
3. Form a **right triangle**.
4. Identify the sides: Opposite (height); Adjacent (distance); Hypotenuse (line of sight)

Apply the appropriate **trigonometric ratio**.

Example: A lighthouse is **30 m high**. The **angle of depression to a boat is 40°** . Find the **distance from the base of the lighthouse to the boat**.

Solution:

Step 1: Identify the sides: Opposite side = 30 m,
Adjacent side = distance and Angle = 40° .

Step 2: Apply tangent ratio $\tan 40^\circ = \frac{30}{\text{distance}}$.

Step 3: Solve $\text{distance} = \frac{30}{\tan 40^\circ} = 35.8 \text{ m}$:

The boat is approximately **35.8 m from the lighthouse**.

Application activity 4.9.2



- 1) A tower is **40 m high**. The **angle of depression to a car on the ground is 30°** .
 - a) Draw the diagram.
 - b) Calculate the **distance from the base of the tower to the car**.
- 2) From the **top of a building**, the **angle of depression to a person on the ground is 45°** . The person is **20 m from the base of the building**. Find the **height of the building**.
- 3) A lighthouse **35 m high** observes a ship with an **angle of depression of 50°** . Find the **distance from the base of the lighthouse to the ship**.

4. 9.5. Inclined Plane

Activity 4.9.5



A ladder **5 m long** is leaning against a vertical wall making an **angle of 60°** with the ground.

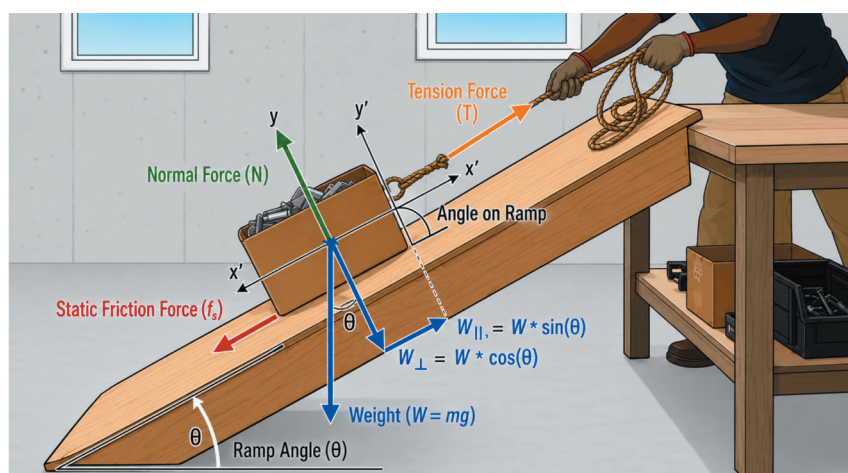
Discuss in groups:

1. Draw a diagram showing the ladder, the wall, and the ground.
2. What geometric figure is formed?
3. Which side represents the **height of the wall reached by the ladder**?
4. Which trigonometric ratio can be used to find the height reached?



Content Summary

An **inclined plane**, also known as a **ramp**, is a flat supporting surface tilted at an angle, with one end higher than the other, used as an aid for raising or lowering a load.



It can be also a sloping surface that forms an angle with the horizontal surface.

Examples include: A ladder leaning against a wall, A ramp, A sloping road, A roof

Trigonometric Ratios Used

- 1) **Sine Ratio:** $\sin \theta = \frac{\text{opposite}}{\text{hypotenuse}}$ Used to determine the height of the inclined object.
- 2) **Cosine Ratio:** $\cos \theta = \frac{\text{adjacent}}{\text{hypotenuse}}$, Used to determine the horizontal distance.
- 3) **Tangent Ratio:** $\tan \theta = \frac{\text{opposite}}{\text{adjacent}}$, Used when both height and horizontal distance are involved.

Steps for Solving Inclined Plane Problems

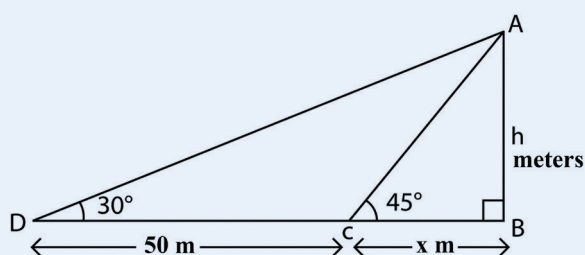
1. Draw a right triangle representing the situation.
2. Identify:
 - Opposite side
 - Adjacent side
 - Hypotenuse
3. Choose the appropriate trigonometric ratio.
4. Substitute known values.
5. Solve for the unknown quantity

Example

A person standing on the bank of a river observes that the angle subtended by a tree on the opposite bank is 45° . When he retreats 50 metres from the bank, he finds the angle to be 30° . Find the breadth of the river and the height of the tree.

Solution:

Let $AB = h$ metres be the height of the tree and $CB = x$ metres be the breadth of the river $\angle BCA = 45^\circ$. Consider C as the first position of the person and D as the second position of the person after he retreats.



From the above figure,

In triangle ABC , $\angle ABC = \angle ABD = 90^\circ$, $\angle BCA = 45^\circ$

$$\tan 45^\circ = \frac{AB}{BC} = \frac{h}{x} \Leftrightarrow 1 = \frac{h}{x} \text{ or } x = h.$$

In triangle ABD , $\angle BDA = 30^\circ$,

$$\tan 30^\circ = \frac{AB}{BD} = \frac{h}{x + 50}.$$

$$\text{As } x = h, \tan 30^\circ = \frac{h}{x + 50} \Leftrightarrow \frac{1}{\sqrt{3}} = \frac{h}{h + 50}$$

$$\Leftrightarrow h + 50 = h\sqrt{3} \Leftrightarrow h = \frac{50}{\sqrt{3} - 1} = 40.98$$

Therefore, the height of the tree is the same as the breadth of the river that is 40.98 m

Application activity 4.9.2



Work out

- 1) A 10 m ladder leans against a wall making an angle of 70° with the ground. Find the height reached and the distance from the wall.
- 2) A ramp rises 1.5 m over a horizontal distance of 6 m. Find the angle of inclination and the length of the ramp.
- 3) A roof has a pitch of 30° (angle with horizontal). If the horizontal span is 8 m, what is the vertical rise?

End of unit assessment



Assessing knowledge and understanding:

1) Write **True (T)** or **False (F)**.

- An angle can be measured in degrees or radians.
- In the unit circle, the radius is 1 unit.
- The value of $\sin 90^\circ$ is equal to 0.
- The cosine rule can be used when two sides and the included angle are known.
- Bearings are measured anticlockwise from North.
- The graph of the sine function is periodic.
- The sine rule is used when two angles and one side are known.
- The angle of depression is measured downward from a horizontal line.
- The cosine function has a maximum value of 1.
- A ladder leaning against a wall forms a right triangle.

2) Fill in the Missing Values

Angle	sin	cos	tan	ctg
30°	—	—	—	—
45°	—	—	—	—
60°	—	—	—	—
90°	—	—	—	—

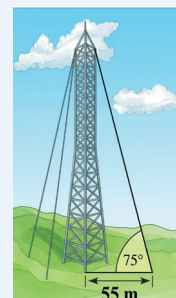
Assessing skills

3) Verify the following identities

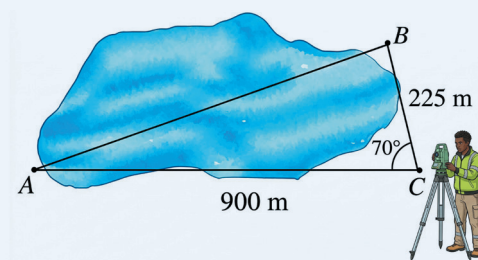
a) $\frac{\sin a}{1 - \cos a} = \frac{1 + \cos a}{\sin a}$ b) $\sec^2 a + \csc^2 a = \sec^2 a \csc^2 a$ c) $\sec^4 a - \tan^4 a = \sec^2 a + \tan^2 a$

d) $\sqrt{\frac{1 - \cos a}{1 + \cos a}} = \cos a - \cot a$

- 4) A guy wire from the top of the transmission tower at KARONGI forms a 75° angle with the ground at a 55-meters distance from the base of the tower. How tall is the tower?



- 5) In order to determine the distance between two points A and B on opposite sides of a lake, a surveyor chooses a point C that is 900 m from A and 225 m from B, as shown in the figure. If the measure of the angle at C is 70° , find the distance between A and B

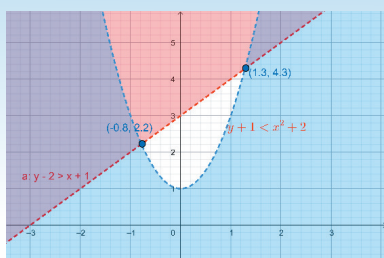


Assessing attitudes and values

- 6) A flight navigator uses trigonometry to continuously calculate a plane's precise position and drift caused by crosswinds. If a student claims, *"With modern GPS, pilots no longer need to understand the geometric relationships of trigonometry,"* which of the following responses best demonstrates a mature appreciation of the discipline?
- The student is correct; manual trigonometry is completely obsolete in modern aviation fields.
 - Trigonometry is only useful as a backup option in case the airplane experiences a total electrical failure.
 - Understanding trigonometry allows pilots to comprehend how navigation systems model physical space, fostering the critical thinking needed to verify instruments and handle unexpected tracking anomalies.
 - Trigonometry should only be studied by software engineers who program the GPS, not by the pilots who operate the aircraft.
- 7) During a school project to design a wheelchair ramp with a safe angle of inclination, two students disagree on the steepness. Jean wants to guess the length to save time, while Aline insists on using the tangent ratio to calculate the exact dimensions. Which approach best reflects the values of a responsible and mathematically minded team?
- Jean's approach, because saving time is more important than precision when building minor public structures.
 - Aline's approach, because exploring the geometric relationships confidently ensures the design meets safety standards and protects future users.
 - Leaving the project unfinished until a teacher provides the exact measurements to avoid any student arguments.
 - Choosing the steepness based on the visual appeal of the ramp rather than calculating the trigonometric ratios.
- 5) While conducting a practical outdoor activity to find the height of the school flagpole using a clinometer, a student notices that their calculated height varies slightly every time they re-measure the angle of elevation. What is the most honest and scientifically precise action for the student to take?
- Change the recorded angle measurements in the notebook so that every single trial yields identical, perfect results.
 - Copy the final answer from another group to ensure their group gets full marks for the exercise.
 - Stop measuring after the first attempt to avoid dealing with conflicting data points.
 - Persistently record all true variations, analyze potential sources of instrumental error, and use logical reasoning to find an accurate average.

UNIT 5

LINEAR AND QUADRATIC EQUATIONS AND INEQUALITIES



Key Unit Competence:

Solve daily life problems algebraically or graphically using linear and quadratic equations or inequalities.

Learning objectives

By the end of this unit, you should be able to:

Knowledge and understanding

1

- Explain and interpret inequalities involving absolute values.
- Explain the concept of parametric equations and parametric inequalities in one unknown.
- Recognize and classify equations reducible to quadratic form such as bi-quadratic, nested, irrational, and reciprocal equations.
- Apply methods for solving quadratic inequalities and simultaneous quadratic equations in two unknowns.
- Explain the relationship between equations, inequalities, and their graphs, and describe how they model real-life problems.

Skills

2

- Solve inequalities involving absolute values using both graphical and algebraic methods.
- Express and manipulate parametric equations to eliminate parameters and find relationships between variables.
- Transform complex expressions into quadratic form and solve bi-quadratic, nested, irrational, and reciprocal equations accurately.
- Solve quadratic inequalities and simultaneous quadratic equations using graphs, number lines, and algebraic methods.
- Apply equations and inequalities to solve real-life problems and use ICT tools or graphing software to visualize and verify results.

Attitudes and values

3

- Demonstrate accuracy, persistence, and logical reasoning in solving mathematical problems.
- Develop confidence, independence, and creativity in exploring different solution methods.
- Appreciate the practical importance of equations and inequalities in real-life decision-making.
- Show collaboration and effective communication when working in groups.
- Uphold honesty, precision, and intellectual curiosity in presenting mathematical results.

Introduction

In this unit, you will learn how to solve different types of **equations and inequalities** and apply them to real-life situations.

You will begin by studying **inequalities involving absolute values**, then review how to solve **equations and inequalities in one unknown**. After that, you will explore **parametric equations and parametric inequalities in one unknown**, where parameters are used to represent families of equations.

You will also learn how to solve **quadratic equations**, including **simultaneous quadratic equations**, and different types of equations that can be **reduced to quadratic form**, such as **bi-quadratic equations, nested equations, irrational equations, and reciprocal equations**. In addition, you will study **quadratic inequalities** and learn how to solve **simultaneous quadratic equations in two unknowns**.

Finally, you will apply these concepts to **solve real-life problems involving equations and inequalities**, helping you understand how algebra can be used to model and solve practical situations.

By the end of this unit, you will develop strong skills in analysing, solving, and interpreting different algebraic equations and inequalities.

Introductory activity



If x is the number of pens for a Student, the teacher decides to give everyone two more pens. What is the number of pens that a Student will have?

- a) Complete the following table called table of value to indicate the number $y = f(x) = x + 2$ of pens for a Student who had x pens for $x \geq 0$

x	-2	-1	0	1	2	3
$y = f(x) = x + 2$			2			
(x, y)			(0, 2)			

- b) Use the coordinates of points obtained in the table and complete them in the Cartesian plan.
- c) Join all points obtained. What is the form of the graph obtained?
- d) Suppose that instead of writing $f(x) = x + 2$ you write the equation $y = x + 2$. Is this equation a linear equation or a quadratic equation? What is the type of the inequality $x + 2 \geq 0$?
- e) Find out an example of problem from the real life situation that can be solved by the use of linear equation in one unknown.

5.1 Inequalities involving absolute values

Activity 4.1



Consider the inequality: $|x| < 4$. List all integers whose distance from 0 is less than 4. Discuss in group the following question:

- What interval represents these numbers?
- Can we write this as a compound inequality?



Content Summary

The absolute value of a number is its distance from zero on the number line.

$$|x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$$

Absolute Value Inequalities: Less Than Inequality $|x| < a$ If $a > 0$, $-a < x < a$

Meaning: The distance of x from 0 is **less than a** . Similarly, $|x| < a \Rightarrow -a < x < a$

Greater Than Inequality $|x| > a$, If $a > 0$, $x < -a$ or $x > a$

Meaning: The distance of x from 0 is **greater than a** . Similarly, $|x| > a \Rightarrow x < -a$ or $x > a$

Example: Solve the inequality $|3x - 2| < 1$.

Solution:

$$\begin{aligned} |3x - 2| < 1 & \quad 1 < 3x - 2 < 3 \\ -1 < 3x - 2 < 1 & \quad \frac{1}{3} < x < 1. \end{aligned}$$

Application activity 5.1



Solve the following:

a) $|5 - x| = \frac{5}{2}$ b) $|3 - 2x| = 5$ c) $|9 - 3x| \geq 5$ d) $|x - 2| > 1$ e) $|2x + 3| \geq x - 4$

5.2 Equations in one unknown

Activity 5.2



Research on internet or Mathematics dictionary “What is a linear equation?”

Present your findings, with clear examples, to the rest of the class.



Content Summary

1. A linear equation is an equation of a straight line, A linear equation is a polynomial of degree For example: 1) $y = 2x + 1$ 2) $5x = 6 + 3y$ 3) $\frac{y}{2} = 3$

The general form of the linear equation with one variable is $ax + b = 0$ Where $a, b \in \mathbb{R}$: and $a \neq 0$ the value of x in which the equality is verified is called the **root**. (solution of the equation).

Example: Find x if $2x + 4 = 10$

Solution:

a) Isolate “ x ” to one side of the equation by subtracting 4 from both sides:

$$2x + 4 - 4 = 10 - 4 \text{ the answer is } 2x = 6$$

b) Divide both sides by 2: $\frac{2x}{2} = \frac{6}{2}, x=3$

c) Check your work with the original equation: $2 \times 3 + 4 = 6 + 4 = 10$

2. Product equation: This is in the form $(ax + b)(cx + d) = 0$. Since the product of factors is null (zero) either one of them is zero. To solve this equation, we proceed as follows:

$$(ax + b)(cx + d) = 0 \text{ then } ax + b = 0 \text{ or } cx + d = 0, x = -\frac{b}{a} \text{ or } x = -\frac{d}{c}$$

Example: Solve $(2x + 4)(x - 1) = 0$.

Solution:

$$2x + 4 = 0 \text{ or } x - 1 = 0$$

$$x = -2 \text{ or } x = 1.$$

3. Fractional equation of the first degree: The general form is $\frac{ax + b}{cx + d} = 0. \Rightarrow ax + b = 0$

4. We solve $ax + b$ and we take the value(s) which verify the existence conditions.

Example: Solve $\frac{2x - 6}{x + 1} = 0$.

Solution:

Existence condition: $x \neq -1$

$2x - 6 = 0$ and $x = 3$.

5. Graph

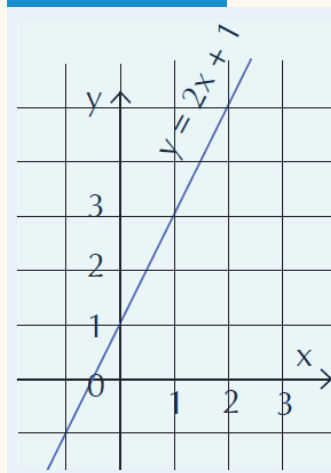
Let us look more closely at one example $y = 2x + 1$ is a linear equation.

The graph of $y = 2x + 1$ is a straight line.

When x increases, y increases **twice as much**, hence $2x$

When x is 0, y is already 1. Hence $+1$ is also needed. So, $y = 2x + 1$

Solution:



Application activity 5.2



Solve the following:

(a) $(x + 2)(2x - 1) = 0$

(b) $(5x - 15)(3x - 9) = 0$

(c) $x^2 - 5x = 0$

(d) $x + 1 \quad 3x - 6 = 0$

(e) $\frac{x - 2}{5x + 3} = 0$

(f) $3(x + 7) = 0$

(g) $\frac{3 - x}{2x - 7} = 0$

5.3 Inequalities in one unknown

Activity 5.3



Discuss the difference between: (a) linear equations and linear inequalities (b) solving linear equations and solving linear inequalities.



Content Summary

The inequalities of one unknown are in the form $ax + b \geq 0$ or $ax + b \leq 0$ with $a \in \mathbb{R}$, $b \in \mathbb{R}$

Solving inequalities: They are solved as linear equations except that:

- 1) when we multiply an inequality by a negative real number the sign will be reversed
- 2) when we interchange the right side and the left side, the sign will be reversed.

Example

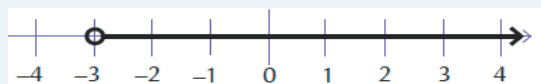
1. Solve: $-2(x + 3) < 10$

Solution:

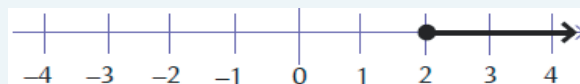
$$\begin{array}{lll} -2(x + 3) < 10 & -2x < 16 & x > -8 \\ -2x - 6 + 6 < 10 + 6 & \frac{-2x}{-2} > \frac{16}{-2} & \end{array}$$

Graphs: The graph of a linear inequality in one variable is a number line. We use an unshaded circle for $<$ and $>$ and a shaded circle for $\leq$ and $\geq$.

The graph for $x > -3$:

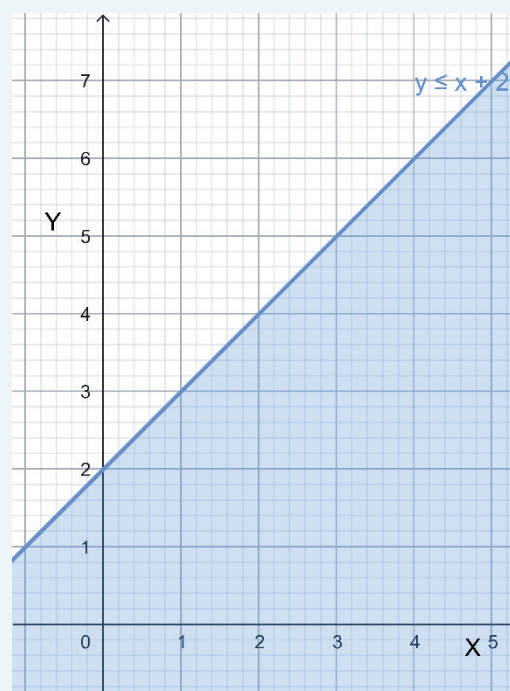


The graph for $x \geq 2$:



The inequality is $y \leq x + 2$. You can see the line, $y = x + 2$ and the shaded area is where y is less than or equal to $x + 2$

Solve



Examples: Algebraically solve the following inequalities in the set of real numbers

Example 1: $-2x + 5 \leq 0$

Solution:

$$-2x + 5 \leq 0$$

$$\Leftrightarrow -2x \leq -5$$

$$\Leftrightarrow x \geq \frac{5}{2} \quad S = \left[\frac{5}{2}, +\infty \right[$$

Example 2: $x - 4 > 0$

Solution:

$$x - 4 > 0$$

$$\Leftrightarrow x > 4$$

$$S =]4, +\infty[$$

Example 3: $x > 2x - 4$

Solution:

$$x > 2x - 4$$

$$\Leftrightarrow -x > -4$$

$$\Leftrightarrow x < 4$$

$$S =]-\infty, 4[$$

Application activity 5.3



1) $y \leq 3$ 2) $y = \frac{1}{2}x - 3$ 3) $y = -3 + 2$ 4) $y \geq -2$ 5) $2y - x \leq 6$ 6) $\frac{y}{2} + 2 > 0$

5.4 Parametric equations in one unknown

Activity 5.4



Carry out research. Find the meaning of parametric equations and inequalities. Discuss your findings using suitable examples.



Content Summary

A **parametric equation in one unknown** is an equation in which the variable is expressed in terms of another variable called a **parameter**.

General form: $x = f(t)$ where:

- x is the unknown variable
- t is the **parameter**

To determine values of x :

1. Choose values of t
2. Substitute them into the equation

Examples

1. Given $x = 2t + 5$ Find x when $t = -2, -1, 0, 1$

Solution:

(t)	$(x = 2t + 5)$
-2	1
-1	3
0	5
1	7

2. Solve and discuss the equation $(2 - 3m)x + 1 = m^2(1 - x)$.

Solution:

$$(2 - 3m)x + 1 = m^2(1 - x) \text{ then } 2x - 3mx + 1 = m^2 - m^2x$$

$$2x - 3mx + m^2x - m^2 + 1 = 0 \text{ then } x(2 - 3m + m^2) = m^2 - 1$$

$$x = \frac{m^2 - 1}{2 - 3m + m^2} = \frac{(m - 1)(m + 1)}{(m - 1)(m - 2)} = \frac{m + 1}{m - 2}$$

Solve and discuss, we have to find the set of possible values m can take (i.e. the domain)

and the corresponding solutions: $x = \frac{m + 1}{m - 2}$. The domain of $f(m) = \frac{m + 1}{m - 2}$ is $\mathbb{R} \setminus \{2\}$

and we can say that: If $m \neq 2$, then the solution is $x = \frac{m + 1}{m - 2}$. If $m = 2$, then there is no solution.

3. Find the values of k for which the equation $x^2 + (k + 1)x + 1 = 0$ has:

- two distinct real roots
- no real roots.

Solution:

$$\Delta = (k + 1)^2 - 4(1)(1) = (k + 1)^2 - 4$$

$$k^2 + 2k + 1 - 4 = k^2 + 2k - 3 = (k + 3)(k - 1) = 0$$

then $k = -3$ or $k = 1$

Table of sign of $\Delta = k^2 + 2k - 3 = (k + 3)(k - 1)$

k	$-\infty$	-3	1	$+\infty$
$k+3$	-	0	+	+
$k-1$	-	-	0	+
$(k+3)(k-1)$	+	0	-	+

a) For two distinct real roots; $\Delta > 0$ and so $k < -3$ or $k > 1$.

b) For no real roots $\Delta < 0$ and so $-3 < k < 1$.

Application activity 5.4



1. Find the value of t

- $x = 4t + 3$, when $x = 11$
- $x = 2t - 5$, when $x = 7$
- $x = t^2 + 4$, when $x = 13$

Find the range of values of m for which the equation $(m - 3)x^2 - 8x + 4 = 0$ has

- two real roots
- no real root
- one double root.

d) Find the set of values of m for which $x^2 + 3mx + m$ is positive for all real values of x .

2. The position of an object is given by $x=5t+2$ where t represents time. Find the position when: $t = 0$, $t = 3$ and $t = 5$

3. A quantity is given by $x = 2t^2 + 1$ Find the value of x when $t = 1, 2, 3$.

5.5 Parametric inequalities in one unknown

Activity 5.5



(k)	Inequality obtained
0	$(x > 1)$
1	$(x > 3)$
2	$(x > 5)$
-1	$(x > -1)$

Discuss in group:

1. What happens to the inequality when k changes?
2. Does the solution of x depend on the value of k ?



Content Summary

A **parametric inequality in one unknown** is an inequality that contains:

- an **unknown variable** (usually x)
 - a **parameter** (such as k, a, m)
- General form: $ax + b > k$

where:

- x is the unknown
- k (or another letter) is the **parameter**

Example: Solve $(m+3)x \geq 2$

Solution:

$$(m+3)x \geq 2$$

There are 3 cases:

Case 1

If $m = -3$ then the equation becomes $(-3+3)x \geq 2$

$0x \geq 2$ which is impossible

The solution set is $S = \{ \}$

Case 2

If $m > -3 \Rightarrow x \geq \frac{2}{m+3}$

, then solution set is

$$S = \left[\frac{2}{m+3}, +\infty \right[$$

Case 3

If $m < -3 \Rightarrow x \geq \frac{2}{m+3}$

, then the solution set is

$$S = \left] -\infty, \frac{2}{m+3} \right]$$

Application activity 5.5



1. Solve the inequalities
1) $x + a > 7$ 2) $x - k \leq 9$
3) $2x + m > 5$ 4) $3x - k \geq 6$
2. Solve in terms of the parameter
1) $x + 2k < 10$ 2) $x - 3a > 4$ 3) $4x + k \leq 12$
3. Solve the following depending on the value of the parameter.
1) $ax \geq 8$ 2) $mx < 10$
4. The temperature x must satisfy $x > 20 - k$ where k is a parameter.
 - Find the inequality when $k = 5$.
 - Find the inequality when $k = 10$.

5.6 Solving quadratic equations

Activity 5.6



1. The following mathematical statements are always true for only some values of x . Find out the real values of x in the following statement $x^2 + 2x - 24$
2. Suppose that you own a plot of land and want to build a fence there. Let the area of the plot be 900 m^2 . You want your fence to have a length equal to twice its width. Let X be the width, then the length is $2X$. Knowing that the Length times width is equal to the area, how will you optimise the area of land you have? Do you think the statement $2x^2 = 900$ can help you to find out the length and width of your fence? Explain your answer.



Content Summary

Equations of the type $ax^2 + bx + c = 0$ ($a \neq 0$) are called quadratic equations.

There are three main ways of solving such equations:

- a) By factorizing or finding square roots.
- b) By using the formula $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$
- c) By completing the square

1. Quadratic equations by factorizing or finding square roots

The method of solving quadratic equations by factorization should only be used if it is readily factorized by inspection.

To factorize a quadratic equation, one can use the sum and product of its roots.

Let x and y be two real numbers such that $x + y = s$ and $xy = p$.

Here $y = s - x$ and $x(s - x) = p \Rightarrow x^2 - sx + p = 0$. This equation is said to be quadratic equation and s, p are the sum and product of the two roots respectively.

Quadratic equation or equation of second degree has the form $ax^2 + bx + c = 0$, where the sum of two roots is $s = -\frac{b}{a}$ and their product is $p = \frac{c}{a}$

Example 1: Solve in $\mathbb{R} : x^2 + 2x - 24 = 0$:

Solution:

$$x^2 + 2x - 24 = 0 \Leftrightarrow (x + 6)(x - 4) = 0$$

So, either $x + 6 = 0$ or $x - 4 = 0$ giving $x = -6$ or $x = 4$.

Example 2: Solve in $\mathbb{R} : 5x^2 + 7x - 6 = 0$

Solution:

$$5x^2 + 7x - 6 = 0 \Leftrightarrow x(5x - 3) + 2(5x - 3) = 0$$

$(5x - 3)(x + 2) = 0$ So, either $5x - 3 = 0$ or $x + 2 = 0$ giving $x = \frac{3}{5}$ or $x = -2$.

Example 3: Solve in $\mathbb{R} : x^2 - 7x + 5 = -5$

Solution:

$$x^2 - 7x + 5 = -5 \Leftrightarrow x^2 - 7x + 10 = 0$$

As we saw it, in this equation the sum of two roots is -7 and the product is 10. To find those roots we can think about two numbers such that their sum is -7 and their product is 10. Those numbers are -2 and -5. Thus $S = \{2, 5\}$

2. Quadratic equations by the formula

To solve this equation, first we find the discriminant (delta): $\Delta = b^2 - 4ac$

In fact, $ax^2 + bx + c = 0 \Leftrightarrow ax^2 + bx = -c \Leftrightarrow a\left(x^2 + \frac{b}{a}x\right) = -c$ as $a \neq 0$ (making the

coefficient of x^2 one) $\Leftrightarrow \left(x + \frac{b}{2a}\right)^2 = -\frac{c}{a} + \frac{b^2}{4a^2} \leftarrow \frac{b^2}{4a^2}$ Is the square of half the

coefficient of x $\left(\frac{b}{a}\right)$ in $x^2 + \frac{b}{a}x \Leftrightarrow \left(x^2 + \frac{b}{a}x\right)^2 = \frac{b^2 - 4ac}{4a^2} \Rightarrow x + \frac{b}{a} = \pm \sqrt{\frac{b^2 - 4ac}{4a^2}}$

$\Leftrightarrow x + \frac{b}{2a} = \pm \frac{\sqrt{b^2 - 4ac}}{2|a|} \Leftrightarrow x + \frac{b}{2a} = \pm \frac{\sqrt{b^2 - 4ac}}{2a}, \text{ if } a > 0 \text{ or } x + \frac{b}{2a} = \mp \frac{\sqrt{b^2 - 4ac}}{2a} \text{ if } a < 0$

Simply $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ let $\Delta = b^2 - 4ac$ there are three case:

If $\Delta > 0$ there are two distinct real roots $x_1 = \frac{-b + \sqrt{\Delta}}{2a}$ and $x_2 = \frac{-b - \sqrt{\Delta}}{2a}$

If $\Delta = 0$ there is one repeated real root (one double root): $x_1 = x_2 = \frac{-b}{2a}$

If $\Delta < 0$ there is no real root.

Example 1: Solve in $\mathbb{R}$: $x^2 + 2x + 1 = 0$

Solution:

$$x^2 + 2x + 1 = 0 \Rightarrow \Delta = 2^2 - 4(1)(1) = 0$$

$$x_1 = x_2 = \frac{-2}{2} = -1 \quad S = \{-1, -1\}$$

Example 3: Solve in $\mathbb{R}$: $2x^2 + 3x + 4 = 0$

Solution:

$$2x^2 + 3x + 4 = 0 \Rightarrow \Delta = 3^2 - 4(2)(4) = -24$$

Since $\Delta < 0$, there is no real root.

Then, $S = \emptyset$

Example 2: Solve in $\mathbb{R}$: $x^2 - 7x + 5 = -5$

Solution:

$$x^2 - 7x + 5 = -5 \Leftrightarrow x^2 - 7x + 10$$

$$\Delta = (-7)^2 - 4(1)(10) = 9$$

$$x_1 = \frac{-(-7) + \sqrt{9}}{2} = 5, x_2 = \frac{-(-7) - \sqrt{9}}{2} = 2$$

$$S = \{2, 5\}$$

Example 4: For what value of k will the equation $2x^2 + 2x + k = 0$ have one double roots? Find that root.

Solution:

For one double root $\Delta = 0$.

$$\Delta = 4 - 4k = 0 \Rightarrow k = 1$$

Thus, the value of k is 1.

$$\text{That root is. } x = -\frac{2}{2} = -1$$

Quadratic equations by completing the square

Before solving quadratic equations by completing the square, let's look at some examples of expanding a binomial by squaring it. $(3+3)^2 = x^2 + 6x + 9$ and $(x-5)^2 = x^2 - 10x + 25$

Notice that the constant term (k^2) of the trinomial is the square of half of the coefficient of trinomial's x -term. Thus, to make the expression $(x^2 + kx)$ a perfect square, you must add $\left(\frac{1}{2}k\right)^2$ to the expression.

When completing the square to solve quadratic equation, remember that you must preserve the equality. When you add a constant to one side of the equation, be sure to add the same constant to the other side of equation.

Example 1: Solve in the set $\mathbb{R} : x^2 - 4x + 1 = 0$ by completing the square

Solution:

$x^2 - 4x + 1 = 0$ Rewrite original equation

$x^2 - 4x = -1$ Subtract 1 from both sides.

$x^2 - 4x + (-2)^2 = 1 + (-2)^2$ Add $(-2)^2 = 4$ to both sides.

$(x - 2)^2 = 5$ Binomial squared.

$(x - 2) = \pm\sqrt{5}$ Take square roots.

$x = 2 \pm \sqrt{5}$ Solve for x .

The equation has two solutions:

$x_1 = 2 - \sqrt{5}$ and $x_2 = 2 + \sqrt{5}$

Example 2: Solve in the set $\mathbb{R} : 4x^2 + 2x - 5 = 0$ by completing the square

Solution:

$4x^2 + 2x - 5 = 0$ Rewrite original equation

$4x^2 + 2x = 5$ Add 5 to both sides.

$4x^2 + \frac{1}{2}x = \frac{5}{4}$ Divide both sides by 4.

$4x^2 + \frac{1}{2}x + \left(\frac{1}{4}\right)^2 = \frac{5}{4} + \left(\frac{1}{4}\right)^2$ Add $\left(\frac{1}{4}\right)^2 = \frac{1}{16}$ to both sides.

$\left(x + \frac{1}{4}\right)^2 = \frac{21}{16}$ Binomial squared.

$\left(x + \frac{1}{4}\right) = \pm\frac{\sqrt{21}}{4}$ Take square roots.

$x = -\frac{1}{4} \pm \frac{\sqrt{21}}{4}$ Solve for x .

The equation has two solutions: $x_1 = \frac{1 + \sqrt{21}}{4}$

and $x_2 = \frac{1 - \sqrt{21}}{4}$

Application activity 5.6



1) Solve in set of real numbers the following equations by factorization

1) $x^2 + 6x + 8 = 0$

2) $x^2 - 2x = 3$

2) Solve in set of real numbers the following equations using the quadratic formula

1) $x^2 - 12x + 11 = 0$

2) $x^2 + 2x = 35$

3) Solve in set of real numbers the following equations by completing the square

1) $x^2 + 5x - 24 = 0$

2) $x^2 - 13x + 36 = 0$

5.7 Simultaneous quadratic equations in one unknown

Activity 5.7



Solve these **without a calculator**

- $x^2 - 7x + 12 = 0$
- $2x^2 - 8 = 0$
- If $A = x^2 + 2x$ and $B = 3x - 4$, find x such that $A = B$



Content Summary

A **simultaneous quadratic equation in one unknown** is a system where two quadratic expressions in x are **equal to each other** or both equal to zero: $ax^2 + bx + c = dx^2 + ex + f$

This reduces to a **single quadratic equation**: $(a - d)x^2 + (b - e)x + (c - f) = 0$

Once reduced to standard form $Ax^2 + Bx + C = 0$, you can use:

Method	Best when...	Example form
1. Factoring	A, B, C are integers with simple factors	$x^2 - 5x + 6 = 0$
2. Quadratic Formula	Factoring is difficult or impossible	$2x^2 - 4x - 7 = 0$
3. Completing the Square	Required for vertex form or when $A = 1, B$ is even	$x^2 - 6x + 5 = 0$

Example 1: Factoring Method

Problem: Solve simultaneously:
$$\begin{cases} x^2 + 5x + 6 = 0 \\ 2x^2 + 3x - 2 = 0 \end{cases}$$

Solution:

Step 1: Bring all terms to LHS: $x^2 + 5x + 6 - 2x^2 - 3x + 2 = 0$

Step 2: Multiply by -1 : $-x^2 + 2x + 8 = 0 \Leftrightarrow x^2 - 2x - 8 = 0$

Step 3: Factor: $(x - 4)(x + 2) = 0$ then $x = 4$ or $x = -2$

Step 4: Check (optional but good practice):

$$x = 4: LHS = 16 + 20 + 6 = 42, RHS = 32 + 12 - 2 = 42$$

$$x = -2: LHS = 4 - 10 + 6 = 0, RHS = 8 - 6 - 2 = 0$$

Answer: $\{-2, 4\}$

Example 2: Quadratic Formula Method

Problem: Solve:
$$\begin{cases} x^2 + 4x + 1 = 0 \\ 2x^2 - 3x + 6 = 0 \end{cases}$$

Solution:

Step 1: Rearrange: $x^2 + 4x + 1 = 2x^2 - 3x + 6 \Rightarrow x^2 - 2x^2 + 4x + 3x + 1 - 6 = 0$
 $-x^2 + 7x - 5 = 0$ Multiply by -1 : $x^2 - 7x + 5 = 0$

Step 2: Use quadratic formula: $a=1, b=-7, c=5$

$$x = \frac{7 \pm \sqrt{(-7)^2 - 4(1)(5)}}{2(1)} = \frac{7 \pm \sqrt{29}}{2}$$

Answer: $\left\{ \frac{7 - \sqrt{29}}{2}, \frac{7 + \sqrt{29}}{2} \right\}$

Example 3: Completing the Square Method

Problem: Solve:
$$\begin{cases} 2x^2 - 8x + 3 = 0 \\ x^2 - 2x + 7 = 0 \end{cases}$$

Solution:

Step 1: Rearrange: $2x^2 - 8x + 3 = x^2 - 2x + 7 \Rightarrow x^2 - 6x - 4 = 0$

Step 2: Complete the square: $x^2 - 6x - 4 = 0 \Rightarrow x^2 - 6x = 4$

$$x^2 - 6x = 4 \Rightarrow x^2 - 6x + 9 = 4 + 9 \Rightarrow (x - 3)^2 = 13 \Rightarrow x - 3 = \pm\sqrt{13}$$

Answer: $x = 3 - \sqrt{13}$ or $x = 3 + \sqrt{13}$

Application activity 5.7



1. Solve each using the **specified method**:

- a) (Factoring) $x^2 + 2x - 8 = 2x^2 - 3x + 4$
- b) (Quadratic Formula) $4x^2 - 2x + 1 = x^2 + 5x - 3$
- c) (Completing Square) $x^2 + 10x - 2 = 2x^2 + 4x + 1$

2. Two rockets' heights (in meters) are given by: $h_1 = -5t^2 + 40t + 10$ and $h_2 = -4t^2 + 35t + 20$ At what time(s) $t \geq 0$ are they at the same height?

5.8 Solving equations reducible to quadratic form: bi-quadratic equations

Activity 5.7



Let $u = x^2$ in each of the following equations of degree 4 and rewrite the given equation by making it an equation of degree 2.

1. $x^4 - 2x^2 + 4 = 0$

2. $6x^4 + 5x^2 + 1 = 0$



Content Summary

A **bi-quadratic equation** is an equation of the form: $ax^4 + bx^2 + c = 0$ where: $a \neq 0$ and a, b, c are constants. It is called **bi-quadratic** because the highest power is **4**, which is the square of a quadratic term.

The steps used to solve a bi-quadratic equation:

Step 1: Substitute $y = x^2$

Step 2: Transform the equation into a quadratic equation in y .

Step 3: Solve the quadratic equation.

Step 4: Replace y with x^2 .

Step 5: Solve for x .

General Procedure, For $ax^4 + bx^2 + c = 0$ let $y = x^2$ then $ay^2 + by + c = 0$

Then Solve for y , then find x .

Example

Solve $x^4 - 5x^2 + 4 = 0$

Solution:

Step 1: Let $y = x^2$

The equation is $y^2 - 5y + 4 = 0$

Step 2 Factorize $(y-1)(y-4) = 0$ and $y = 1$ or $y = 4$

Step 3: Substitute back $x^2 = 1 \Rightarrow x = \pm 1$ and $x^2 = 4 \Rightarrow x = \pm 2$

Solution $x = \pm 1, \pm 2$

Application activity 5.8



Solve the following biquadratic equations in the set of real numbers

1. $x^4 - 13x^2 + 36 = 0$
2. $x^4 - 10x^2 + 9 = 0$
3. $x^4 - 61x^2 + 900 = 0$

5.9 Solving equations reducible to quadratic form: Nested equations

Activity 5.9



For this equation: $(x^2 + 3)^2 - 5(x^2 + 3) + 6$, Discuss the following questions

1. What expression appears **repeatedly** in the equation?
2. Can we replace this repeated expression with another variable?



Content Summary

A **nested equation** is an equation where one algebraic expression is contained inside another expression repeatedly.

Example: $(x^2 + 1)^2 - 7(x^2 + 1) + 10 = 0$ The expression $x^2 + 1$ appears repeatedly.

The step used to solve nested equations reducible to quadratic form:

Step 1: Identify the repeated expression.

Step 2: Substitute a new variable.

Example: $y = x^2 + 1$

Step 3: Rewrite the equation in terms of y .

Step 4: Solve the resulting quadratic equation.

Step 5: Substitute back to obtain the values of x .

Nested equations often take a form similar to: $a(g(x))^2 + b(g(x)) + c = 0$ where $g(x)$ is an algebraic expression.

Let: $y = g(x)$ Then the equation becomes: $ay^2 + by + c = 0$

Example 1

Solve: $(x + 2)^2 - 5(x + 2) + 6 = 0$

Solution:

Step 1: Substitute

Let $y = x+2$

Then the equation becomes: $y^2 - 5y + 6 = 0$

Step 2: Factorize $y^2 - 5y + 6 = 0 \Leftrightarrow (y-2)(y-3) = 0$

Step 3: Substitute back

If $y = 2$ the $x+2 = 2$ then $x = 0$

If $y = 3$ the $x+2 = 3$ then $x = 1$

Solution $x = \{0, 1\}$

Application activity 5.9



Solve the following equations.

- 1) $(x+1)^2 - 6(x+1) + 8 = 0$
- 2) $(x^2 + 2)^2 - 7(x^2 + 2) + 10 = 0$
- 3) $(x^2 - 1)^2 - 5(x^2 - 1) + 6 = 0$
- 4) $(3x+1)^2 - 8(3x+1) + 15 = 0$
- 5) $(2x^2 + 1)^2 - 5(2x^2 + 1) + 6 = 0$

5.10 Solving equations reducible to quadratic form: Irrational equations

Activity 5.9



The equation: $\sqrt{x+5} + \sqrt{x+1} = 6$, In group discuss the following questions:

1. What makes this equation different from the equations previously studied?
2. What operation could remove the square roots?



Content Summary

An **irrational equation** is an equation in which the unknown variable appears **inside a radical (root)**.

The steps used to solve irrational equations reducible to quadratic form:

Step 1: Isolate the radical term if possible.

Step 2: Square both sides of the equation to eliminate the radical.

Step 3: Simplify the equation.

Step 4: Repeat squaring if another radical remains.

Step 5: Solve the resulting quadratic equation.

Step 6: Substitute solutions into the original equation to **check for extraneous solutions**.

NB: Squaring both sides may introduce **extraneous solutions** (solutions that do not satisfy the original equation).

Therefore, **all solutions must be verified** in the original equation.

Example

Solve $\sqrt{x+5} = x-1$

Solution:

Square both sides: $x+5 = (x-1)^2$

Expand: $x+5 = x^2 - 2x + 1$

Rearrange: $x^2 - 3x - 4 = 0$

Factorize: $(x-4)(x+1) = 0$ $x = 4$ or $x = -1$

Check: For $x=4$ $\sqrt{9} = 3$ Correct.

For $x=-1$ $\sqrt{4} = -2$ correct

Solution $x = -2$ and 4

Application activity 5.10



Solve these irrational questions

1) $\sqrt{2x+1} = x$

2) $\sqrt{x+6} = x-2$

3) $\sqrt{x+3} + \sqrt{x+1} = 5$

5.11 Solving equations reducible to quadratic form: Reciprocal equations

Activity 5.9



The equation: $x + \frac{1}{x} = 5$. Discuss in group the following questions.

1) Why x cannot be equal to 0?

2) Multiply both sides by x to eliminate the denominator.



Content Summary

A **reciprocal equation** is an equation that contains a variable and its reciprocal

The step used to solve reciprocal equations:

Step 1: Identify the reciprocal expressions.

Step 2: Multiply the equation by x (or another suitable power of x) to eliminate denominators.

Step 3: Transform the equation into a quadratic equation.

Step 4: Solve the quadratic equation.

Examples

1. Solve $x - \frac{1}{x} = 3$

Solution:

Multiply by x then $x^2 - 1 = 3x$

Rearrange: $x^2 - 3x - 1 = 0$

Quadratic formula: $x = \frac{3 \pm \sqrt{9+4}}{2}$

2. Solve $x^2 + \frac{1}{x^2} = 7$

Solution:

Use the identity: $\left(x + \frac{1}{x}\right)^2 = x^2 + \frac{1}{x^2} + 2$

Substitute: $\left(x + \frac{1}{x}\right)^2 = 7 + 2 \Leftrightarrow \left(x + \frac{1}{x}\right)^2 = 9$ then

$x + \frac{1}{x} = 3$ or -3 Solve each case.

Case 1: if $x + \frac{1}{x} = 3$ then $x^2 - 3x + 1 = 0$

Case 2: if $x + \frac{1}{x} = -3$ then $x^2 + 3x + 1 = 0$

Solve both quadratic equations.

Application activity 5.11



Solve

1) $x + \frac{1}{x} = 4$

2) $x^2 + \frac{1}{x^2} = 6$

3) $x^2 + \frac{1}{x^2} = 14$

5.12 Quadratic inequalities

Activity 5.12



The table below of equation $(x^2 - 5x + 6)$. the x is replaced by a number and the solution give the sign.

(x)	$(x^2 - 5x + 6)$	Sign
0	6	+
1	2	+
2	0	0
2.5	-0.25	-
3	0	0
4	2	+

Observe the table and discuss in group.

- At which values of x does the expression become **zero**?
- Where is the expression **positive**?
- Where is it **negative**?



Content Summary

A **quadratic inequality** is an inequality containing a quadratic expression.

Steps for Solving Quadratic Inequalities

- Write the inequality in the form $ax^2 + bx + c \leq$ or ≥ 0
- Solve the quadratic equation $ax^2 + bx + c \leq$ or ≥ 0
- Factorize the expression if possible.
- Determine the roots.
- Construct a **sign table**.
- Select the intervals that satisfy the inequality.

Example

Solve $x^2 - 5x + 6 > 0$

Solution:

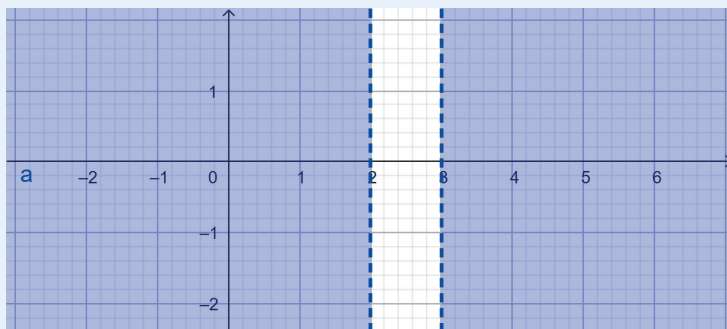
Step 1: Factorize $(x - 2)(x - 3) > 0$

Step 2: Determine the roots $x = 2, x = 3$

Step 3: Sign Table

x	$-\infty$	2	3	$+\infty$					
Factors									
$x-2$	-	-	0	+	+	+	+		
$x-3$	-	-	-	-	0	+	+	+	
$(x-2)(x-3)$	+	+	0	-	0	+	+	+	+

For any interval, the sign of the product of two linear functions is the product of the signs of its factors. The answer is $x < 2$ or $x > 3 \Leftrightarrow x \in]-\infty, 2[\cup]3, +\infty[$



Application activity 5.12



Solve the following inequalities

- 1) $(1 - x)(2x + 1) > 0$
- 2) $2x^2 - x - 3 < 0$
- 3) $x(x + 4) > x - 4$
- 4) $x(x + 4) > x - 1$
- 5) $x^2 + x + 1 < 0$

5.13 Simultaneous quadratic equations in two unknowns

Activity 5.13



Two numbers x and y satisfy the following conditions: $y = x^2$ and $x + y = 6$

Discuss in group

- a) Which value(s) of x make $x + y = 6$?
- b) How many pairs (x, y) satisfy both equations?



Content Summary

Simultaneous quadratic equations are a system of two equations in two unknowns (typically x and y) where at least one equation is of degree 2 (quadratic). The solution set consists of all ordered pairs (x, y) that satisfy **both** equations simultaneously.

General Form

Type	General Form	Geometric Interpretation	Max Number of Real Solutions
Linear-Quadratic	$\begin{cases} y = mx + c \\ y = ax^2 + bx + c \end{cases}$	A line intersecting a parabola (or circle).	2

Methods for Solving

1) Substitution Method (Best when one equation is linear or can be easily solved for one variable)

- Solve one equation for one variable (e.g., $y = \dots$).
- Substitute this expression into the second equation.
- Solve the resulting quadratic equation in one variable.
- Back-substitute to find the corresponding values of the other variable.

2) Elimination Method (Best when both equations are quadratic and have symmetric coefficients)

- Add or subtract the equations to eliminate the quadratic terms (e.g., x^2 or y^2).
- This often yields a linear equation.
- Solve the linear equation for one variable and substitute back into one of the original quadratics.

Example 1: Linear-Quadratic System (Substitution)

Solve the system:
$$\begin{cases} y = 2x - 1 \\ y = x^2 - 2x + 3 \end{cases}$$

Solution:

Substitute:

- 1) Since $y = 2x - 1$, substitute into the second equation: $2x - 1 = x^2 - 2x + 3$
- 2) **Rearrange to standard quadratic form:** $0 = x^2 - 2x + 3 - 2x + 1 \Leftrightarrow x^2 - 4x + 4 = 0$
- 3) **Solve for x:** $x = 2$ (double root) $\Leftrightarrow (x - 2)^2 = 0$
- 4) **Solve for y:** $y = 2(2) - 1 = 3$
- 5) **Solution Set:** $\{2, 3\}$

Example 2: Quadratic-Quadratic System (Elimination)

Solve the system:
$$\begin{cases} x^2 + y^2 = 25 \text{ (Circle)} \\ x^2 + 4y = 13 \text{ (Parabola)} \end{cases}$$

Solution

1. Eliminate x^2 : Subtract the second equation from the first:
2. $(x^2 + y^2) - (x^2 + 4y) = 25 - 13 \Leftrightarrow y^2 - 4y = 12$
3. Rearrange to standard quadratic: $y^2 - 4y - 12 = 0 \Leftrightarrow (y - 6)(y + 2) = 0 \Rightarrow y = 6 \text{ or } y = -2$
4. Substitute back to find x: if $y = 6$: $x^2 + 36 = 25 \Rightarrow x^2 = -11$ No real solution.
5. If $y = -2$: $x^2 + 4 = 25 \Rightarrow x^2 = 21 \Rightarrow x = \pm\sqrt{21}$
6. **Solution Set:** $S = (\sqrt{21}, -2)$ and $(-\sqrt{21}, -2)$

Geometric Interpretation: The circle and parabola intersect at two symmetric points in the lower half-plane.

Example 3: Quadratic-Quadratic System (Substitution after factoring)

Solve the system:
$$\begin{cases} x + y = 5 \\ xy = 6 \end{cases}$$

Note: While the first equation is linear, the second is quadratic. This is a classic problem.

Solution:

1. Substitution: From $x + y = 5$, get $y = 5 - x$.

2. Substitute into $xy = 6$:

$$x(5 - x) = 6 \Leftrightarrow 5x - x^2 = 6 \Leftrightarrow 0 = x^2 - 5x + 6 \Leftrightarrow 0 = (x - 2)(x - 3) \Rightarrow x = 2 \quad \text{or} \quad x = 3$$

3. Find y : If $x = 2$, $y = 5 - 2 = 3$. If $x = 3$, $y = 5 - 3 = 2$.

4. Solution Set: $(2, 3)$ and $(3, 2)$

Thus the solutions are $x = 3, y = 2$ or $x = 2, y = 3$.

Application activity 5.13



1. The perimeter of a rectangle is 20 cm , and its area is 24 cm^2 . Find the dimensions of the rectangle.
2. Two numbers are such that one number is the square of the other number and is also 8 more than twice that number. What are the two numbers?
3. Solve the following system
$$\begin{cases} y = x^2 - 4 \\ x^2 + y^2 = 10 \end{cases}$$

5.14 Solving problems from real life situations involving equations and inequalities

Activity 5.13



In groups, research on the importance and necessity of quadratic equations and inequalities.



Content Summary

Quadratic equations lend themselves to modelling situations that happen in real life. These include:

- a) Projectile motions
- b) The rise and fall of profits from selling goods
- c) The decrease and increase in the amount of time it takes to run a kilometre based on your age, and so on.

The wonderful part of having something that can be modelled by a quadratic is that you can easily solve the equation when set equal to zero and predict. If you throw a ball (or shoot an

arrow, fire a missile or throw a stone) it will go up into the air, slowing down as it goes, then come down again. A quadratic equation can tell you where it will be at any given time.

Example: A ball is thrown straight up, from 3 m above the ground, with a velocity of 14 m/s. After how long does it hit the ground?

Solution:

Ignoring air resistance, we can work out its height by adding up these three things:

The height starts at 3 m :	3
It travels upwards at 14 metres per second (14 m / s) :	14t
Gravity pulls it down, changing its speed by about 5 m / s per second (5 m / s ²)	-t ²
(Note: the -5t ² is simplified from $-\left(\frac{1}{2}\right)at^2$ with $a = 9.81 \frac{m}{s^2}$)	

Add them up and the height **h** at any time **t** is: $h = 3 + 14t - 5t^2$

And the ball will hit the ground when the height is zero: $0 = 3 + 14t - 5t^2$

Let us solve it; we can use factoring method. Multiply all terms by -1 to make it easier:

$5t^2 - 14t - 3 = 0$. Now we can factor it. $a \times c = -15$, and $b = -14$.

The positive factors of -15 are 1, 3, 5, 15, and one of the factors has to be negative. By trying a few combinations, we find that -15 and 1 work ($-15 \times 1 = -15$, and $-15 + 1 = -14$)

Rewrite middle with -15 and 1: $5t^2 - 15t + t - 3 = 0$

Factor first two and last two: $5t(t - 3) + 1(t - 3) = 0$, the common factor is $(t - 3)$: $(5t + 1)(t - 3) = 0$. And the two solutions are: $5t + 1 = 0$ or $t - 3 = 0$

The " $t = -0.2$ " is a negative time, impossible in our case. The " $t = 3$ " is the answer we want: The ball hits the ground after 3 seconds.

Application activity 5.14



1. The height of a ball t seconds after it is thrown into the air from the top of a building can be modelled by $h(t) = -16t^2 + 48t + 64$, where $h(t)$ is height in metres. How high is the building? How high does the ball rise before starting to drop downward, and after how many seconds does the ball hit the ground?
2. The profit function telling George how much money he will net for producing and selling x specialty umbrellas is given by $P(x) = -0.00405x^2 + 8.15x - 100$. What is George's loss if he doesn't sell any of the umbrellas he produces? How many umbrellas does he have to sell to break even, and how many does he have to sell to earn the greatest possible profit?
3. A ball is thrown upwards with a velocity of 40 m/s from a height of 6 m. At what times is the height of the ball greater than 12 m? Use the height formula $h = -16t^2 + v_0t + h_0$ and round your answers to the nearest hundredths of a second.

**Assessing knowledge and understanding:****1. Write True (T) or False (F).**

- i. Every quadratic equation has at most two real solutions.
- ii. An inequality solution is usually expressed as an interval.
- iii. Parametric equations always have two unknowns.
- iv. Absolute value inequalities often produce two solution cases.
- v. Simultaneous quadratic equations can have more than one solution pair.

2. Which of the following is a quadratic equation?

A) $x^3 + 2x + 1 = 0$ B) $2x + 5 = 0$ C) $x^2 - x^4 + 2 = 0$ D) $x^4 + 2 = 0$

3. A parametric equation is an equation that:

- A. Contains only one variable
- B. Contains a parameter (constant whose value may change)
- C. Has no solution
- D. Has only one solution

Assessing skills**4. Solve the following questions**

- a) $x^2 - 7x + 12 = 0$ b) $x^4 - 5x^2 + 4 = 0$ c) $\sqrt{x+5} = x-1$ d) $x + \frac{1}{x} = 5$
- e) $|x-3| < 4$ f) $x^2 - 9 > 0$

5. A rectangle has length 3 m more than its width and an area of 40 m^2 .

- i. Form a quadratic equation for the width.
- ii. Find the dimensions of the rectangle.

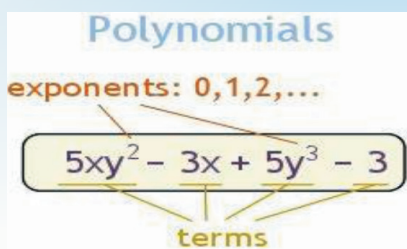
Assessing values & attitudes**6. When solving an irrational equation (such as $\sqrt{x+3} = x-3$), a student algebraically finds two potential solutions: $x = 1$ and $x = 6$. Upon substituting them back into the original equation, the student discovers that $x = 1$ creates an invalid statement ($2 = -2$), making it an extraneous root. Which of the following mindsets best explains why checking your answers is a vital mathematical value?**

- a) Checking solutions is a formal routine only done to satisfy grading rubrics, as the algebraic steps themselves never make mistakes.
- b) Verifying answers demonstrates logical reasoning and a commitment to precision, ensuring that the algebraic steps taken match the actual domain restrictions of the original problem.
- c) If an algebraic method yields an incorrect solution, the student should discard the entire method and try guessing random numbers instead.
- d) Checking solutions is only necessary when a student lacks confidence in their basic arithmetic or factoring skills.

- 7) A local cooperative wants to optimize its monthly budget. They set up a system of linear inequalities to find the acceptable range of expenditures on raw materials versus transport while keeping total costs below a specific threshold. Why is expressing this situation as an *inequality* more valuable to the cooperative than using a strict *equation*?
- a) Inequalities are better because they are easier to solve and require less precise arithmetic than equations.
 - b) Equations are completely useless in business contexts, so inequalities are the only math tools used by managers.
 - c) Real-life constraints often involve limits, maximum boundaries, and acceptable ranges rather than a single fixed outcome, making inequalities essential for flexible, realistic planning.
 - d) Using inequalities allows the cooperative to completely ignore their budget limits whenever they want to spend extra money.
- 8) While working in a study group to find the solution region for the quadratic inequality $x^2 - 4x + 3 \geq 0$, one team member gets stuck on testing intervals and wants to quit the assignment. Another team member suggests drawing a quick sketch of the parabola to visually verify where the curve is above the x -axis. Which action best demonstrates excellent collaborative values and persistence?
- a) The team should stop working immediately and wait for the teacher to give them the final solution regions.
 - b) The team should exclude the student who got stuck and turn in the project with only the remaining members' names.
 - c) The team should persistently combine their ideas, using the visual graph to confidently build their understanding of the test intervals and complete the task together.
 - d) The team should argue about which method is better until the time runs out, without writing down any final mathematical conclusions.

UNIT 6

POLYNOMIAL, RATIONAL AND IRRATIONAL FUNCTIONS



Key Unit Competence:

Analyse polynomial, rational, and irrational functions by determining their domains, representing them graphically, and solving related problems.

Learning objectives

By the end of this unit, you should be able to:

Knowledge and understanding

1

- Explain the concept of a polynomial and factorization of polynomials.
- State and interpret the domain and range of a function.
- Identify the domain of definition for polynomial, rational, and irrational functions.
- Describe operations on functions
- Explain composite and inverse functions and determine the parity (evenness/oddness) of a function.
- Explain the process of decomposing rational functions into partial fractions for different types of denominators.
- Recognize the importance and applications of rational and irrational functions in real-life contexts.

Skills

2

- Factorize polynomials accurately using different methods (grouping, formulas, etc).
- Determine domain and range of given functions both algebraically and graphically.
- Perform operations on functions and construct composite and inverse functions.
- Test and classify functions as even, odd, or neither.
- Decompose rational expressions into partial fractions involving linear and quadratic factors.
- Apply polynomial, rational, and irrational functions to solve real-life problems in various fields.

Attitudes and values

3

- Show logical reasoning, accuracy, and persistence when solving algebraic problems
- Develop appreciation for the use of functions and polynomials in modeling real situations
- Demonstrate curiosity and creativity in manipulating algebraic expressions
- Work collaboratively and share strategies respectfully during group tasks
- Exhibit confidence, honesty, and responsibility when handling mathematical operations and solutions

Introduction

This unit develops learners' understanding of polynomial, rational, and irrational functions. Learners explore algebraic manipulation, graphical representation, and real-life applications of different types of functions. The unit promotes problem-solving, logical reasoning, and mathematical modelling skills.

Introductory activity



Exploring Real-Life Relationships

A car travels at a constant speed. The distance covered is given by: $d = 60t$

- What type of relationship is this?
- What happens to the distance as time increases?

6.1. Factorization of polynomials

Activity 6.1



Given the following expressions

(a) $2a + 2b$

(b) $3r + 6r^2$

(c) $xy + axy$

(d) $9x^2y + 15xy^2$

In groups, answer the following two questions

1. What factor is **common** in each expression?
2. Can the expression be written as a **product of two expressions**?



Content Summary

A. Polynomials

A **polynomial** is an algebraic expression consisting of terms in the form $a_n x^n$, where a_n is a constant (coefficient) and n is a non-negative integer.

The general form of a polynomial is given as

$$f(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0.$$

Example 1: These are polynomials: (1) $7x^3 + 2$, (2) $\frac{x^2}{2}$

Example 2: These are not polynomials because the exponents are not positive integers:

$$(1) 3xy^{-2} \quad (2) \sqrt{x}.$$

A **polynomial function** is a function defined by a polynomial.

Example 3:

- $f(x) = 2x + 5$ (linear polynomial)
- $g(x) = x^2 + 5x + 6$ (quadratic polynomial)
- $h(x) = x^3 - 4x + 2$ (cubic polynomial)

Degree of a polynomial: The highest power of x .

Leading coefficient: The coefficient of the term with the highest power.

Constant term: The term without a variable.

B. Factorization

Factorization of a polynomial means **expressing the polynomial as a product of two or more simpler expressions called factors**. For example, $x^2 + 2x$ can be factorized into x and $x + 2$ i.e. $x^2 + 2x = x(x + 2)$. Hence, x and $x + 2$ are two factors of $x^2 + 2x$.

Types of factorizations

1. Factorization into monomials

Recall the distribution property $a(b + c) = ab + ac$.

Example: Factorize $ax + bx$.

Solution:

In ax and bx , x is common and also a common factor. Put then the common factor x outside as $x(a + b)$. Now, the polynomial $ax + bx$ can be written in terms of its factors, x and $(a + b)$, as $ax + bx = x(a + b)$.

2. By grouping of terms

Example: Factorize $a^2 + bc + ab + ac$

Solution:

Let us group terms in $a^2 + bc + ab + ac$. But, which terms to group? Those terms which yield a common factor on grouping. Group the terms like this: $a^2 + ab + bc + ac$, a is the common factor in $a^2 + ab$ and c is the common factor in $bc + ac$. $a^2 + ab = a(a + b)$ and $bc + ac = c(b + a) = c(a + b)$. Now, the common factor is the binomial $(a + b)$. So, factorization yields $a^2 + ab + bc + ac = (a + b)(a + c)$. Therefore $(a + b)$ and $(a + c)$ are two factors of $a^2 + ab + bc + ac$.

3. Factorization of trinomial perfect squares

Example:

- $a^2 + 2ab + b^2 = (a + b)^2$
- Factorize $9p^2 + 24pq + 16q^2$

Solution:

$9p^2 = (3p)^2$, just like a^2 . $16q^2 = (4q)^2$, just like b^2 . $24pq = 2(3p)(4q)$, just like $2ab$. Applying, $a^2 + 2ab + b^2 = (a + b)^2$ we have $9p^2 + 24pq + 16q^2 = (3p + 4q)^2$.

4. Factorization of trinomials of the form $x^2 + bx + c$

Example: Factorize $x^2 + 13x + 36$

Solution:

Leading coefficient is 1. Set $x^2 + 13x + 36 = (x + a)(x + b)$, where a and b are two numbers such that $a + b = 13$, and $a \times b = 36$. To find a and b , express 36 as product of pairs of its factors.

$$36 = 36 \times 1,$$

$$36 = 18 \times 2,$$

$$36 = 12 \times 3,$$

$$36 = 9 \times 4,$$

from factorization of 36 into the above four forms, in the pair 9 and 4, the sum is 13. Then, $x^2 + 13x + 36 = (x^2 + 4x) + (9x + 36) = x(x + 4) + 9(x + 4) = (x + 9)(x + 4)$

5. Factorization of difference of two perfect squares

Example: Factorize $9x^2 - 16y^2$

Solution:

$9x^2 = (3x)^2$ and $16y^2 = (4y)^2$. Using the algebraic formula: $(a^2 - b^2) = (a - b)(a + b)$, factorization of $9x^2 - 16y^2 = (3x + 4y)(3x - 4y)$.

Application activity 6.1



1. Identify the degree, leading coefficient, and constant term of:

$$f(x) = 5x^4 - 3x^2 + 7$$

2. Classify each of the following as monomial, binomial, or trinomial:

a) $7x^2$

b) $x^2 + 3x$

c) $x^3 + 2x + 1$

3. Factorize:

a) $ax + bx + ay + by$

b) $12x^2 + 18x^3$

c) $4x^2 + 12x + 9$

d) Area $A = x(30 - x)$.

i. Express as polynomial.

ii. Find A when $x = 10$.

6.2. Definition of domain and range of a function

Activity 6.2



Given the function $f(x) = 2x + 1$. In pairs

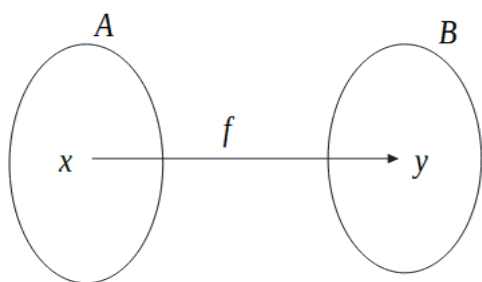
- a) compute $f(x)$ when $x = 0, -2, 1, 5$ and present the results using a Ven diagram,
b) identify the set of all inputs and the set of all outputs.



Content Summary

A. Definition of a function

A function is a rule that assigns each input value exactly one output value.



We write the function from A to B as $f : A \rightarrow B$. Most of time we do not indicate the two sets and we simply write $y = f(x)$. The sets A and B are called the **domain** and **range** of f , respectively.

B. Domain

The domain of a function is the set of all possible input values (x-values) for which the function is defined.

C. Range

The range of a function is the set of all output values (y-values) obtained from the function.

Example 1

State the domain and range of the following relations. Is the relation a function?

$$\{(2, -3), (4, 6), (3, -1), (6, 6), (2, 3)\}$$

Solution:

The above list of points, being a relationship between certain x 's and certain y 's, is a relation. The domain is all the x -values, and the range is all the y -values. To give the domain and the range, just list the values without duplication:

$$\text{domain: } \{2, 3, 4, 6\}$$

$$\text{range: } \{-3, -1, 3, 6\}$$

Since $x = 2$ gives me two possible destinations, then **this relation is not a function.**

Note that if you find a duplicate x -value, then the different y -values mean that you do not have a function.

Example 2

Find the domain and range of: $f(x) = x^2$

Solution:

Domain: All real numbers, $\mathbb{R}$.

Range: Since squares are never negative, then $\text{Range} = y \geq 0$

Application activity 6.2



1. Given the function: $f(x) = 3x - 2$

List the domain and range if: $x = -2, -1, 0, 1, 2$.

2. Find the domain of $f(x) = \frac{5}{x-3}$

6.3. Determination of domain of definition and range for polynomial and rational functions

Activity 6.3



For the following functions

1. $f(x) = x^2 + 3x + 1$

2. $g(x) = \frac{2}{x}$

Work in groups and test the images of the following values: $x = -2, x = 0, x = 3$.



Content Summary

A. Domain of a definition of a function

Definition: Domain of definition of a function is the set of all real numbers for which the expression of the function is defined as a real number.

The domain of f is denoted by $Dom(f)$.

- Polynomial functions of the form $f(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0$ are defined for **all real number**, i.e. $Dom f = \mathbb{R}$.
- A **rational function** is a function of the form $f(x) = \frac{P(x)}{Q(x)}$ where $P(x)$ and $Q(x)$ are polynomials and $Q(x) \neq 0$. When determining domain of definition of a rational function, the denominator must be different from zero.

Example

Find the domain of the function of each of the following:

1. $f(x) = 3x^3 - 2x^2 + 4x - 6$

2. $g(x) = \frac{2x+3}{x-3}$

Solution:

- The function $f(x)$ is defined for all real numbers x . Therefore, the domain of f is $\mathbb{R}$.
- The function $g(x)$ defined for all real numbers except the number that makes the denominator to be zero. We set $x - 3 \neq 0 \Rightarrow x \neq 3$. Therefore, the domain of $g(x)$ is $\{x \in \mathbb{R} : x \neq 3\} = \mathbb{R} \setminus \{3\}$.

B. Range

Let $f: A \rightarrow B$ be a function. The range of f , denoted by $Im(f)$ is the image of A under f , that is, $Im(f) = f[A]$. The range consists of all possible values the function f can have.

Steps to finding range of function

- Put $y = f(x)$
- Solve x in terms of y
- The range of f is the set of all real numbers y such that x can be solved.

Example

For each of the following functions, find the range.

1. $f(x) = x + 5$

2. $g(x) = \frac{8}{2x-4}$

Solution:

1) Set $y = f(x) = x + 5$. Solve for x ; $x = y - 5$. Note that x can be solved for any value of y . Therefore, the range is $\mathbb{R}$.

2) Put $y = g(x) = \frac{8}{2x-4}$. Solve for x ; $y(2x-4) = 8, 2xy - 4y = 8, 2xy = 8 + 4y \Rightarrow x = \frac{8+4y}{2y}$.

Note that x can be solved if and only if $2y \neq 0$. i.e. $y \neq 0$. Therefore, the range of the function g is $\{y \in \mathbb{R} : y \neq 0\} = \mathbb{R} \setminus \{0\}$.

Application activity 6.3



1. Find the domain of f in each of the following.

a) $f(x) = x^5 - 3x^4 + x^2 - 5$

b) $f(x) = \frac{x^2 + 1}{x^2 - x - 2}$

2. Find the range of each of the following functions.

a) $y = 2x - 3$ for $x \geq 0$

b) $f(x) = 1 - x$ for $x \leq 1$

c) $f(x) = \frac{1}{x}$ for $x \geq 2$

6.4. Determination of domain of definition of irrational functions

Activity 6.4



In small groups, substitute different values in the function $f(x) = \sqrt{x-4}$ and identify what happens if the expression inside the square root becomes negative.



Content Summary

- An **irrational function** is a function that contains a **variable inside a radical sign**, such as a square root. *Example:* $f(x) = \sqrt{x-2}$
- In general, for irrational functions, the $Dom f$ is obtained by avoiding a negative number under an even root. Particularly, for functions involving **square roots**, the expression inside the radical must be **greater than or equal to zero**.

Example: Find the domain of

a) $f(x) = \sqrt{x-2}$

b) $h(x) = \sqrt{6-3x}$

Solution:

a) Existence condition: $x - 2 \geq 0 \Rightarrow x \geq 2$, $Dom f = [2, \infty)$

b) The function $h(x)$ is defined if and only if

$6 - 3x \geq 0, -3x \geq -6 \Rightarrow 3x \leq 6 \Rightarrow x \leq 2$. Therefore, the domain of the function $h(x)$ is $\{x \in \mathbb{R} : x \leq 2\} = (-\infty, 2]$.

Application activity 6.4

Find the domain of f in each of the following.

a) $f(x) = \sqrt{2x-3}$

b) $f(x) = \frac{\sqrt{x}}{4x-7}$

c) $f(x) = \sqrt{\frac{x+1}{x-1}}$

d) $f(x) = \sqrt{x^2 - 5x + 6}$ e) $f(x) = \sqrt{x+7} - \sqrt{x^2 + 2x - 15}$ f) $f(x) = \frac{3x^2 + 5x - 2}{\sqrt[3]{7-3x}}$.

6.5. Operations on functions**Activity 6.5**

Consider the following functions:

$$f(x) = x^2, \quad g(x) = 2x + 1, \text{ and } h(x) = x^2 + 2x + 1.$$

Complete the following table

x	0	1	2	3
f(x)				
g(x)				
h(x)				

What do you observe about the relationship between the functions

**Content Summary**

If f and g are two functions, we can perform the following operations:

1. Addition: $(f + g)(x) = f(x) + g(x)$
2. Subtraction $(f - g)(x) = f(x) - g(x)$
3. Multiplication $(f \cdot g)(x) = f(x) \cdot g(x)$
4. Division $\left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)}, g(x) \neq 0$

Example

Given $f(x) = x^2 - 1$, $g(x) = x + 1$. Find

a) $f(x) + g(x)$

c) $f(x) \times g(x)$

b) $f(x) - g(x)$

d) $\frac{f(x)}{g(x)}$

Solution:

a) $f(x) + g(x) = x^2 - 1 + x + 1 = x^2 + x$

b) $f(x) - g(x) = x^2 - 1 - (x + 1) = x^2 - x - 2$

c) $f(x) \cdot g(x) = (x^2 - 1)(x + 1) = x^3 + x^2 - x - 1$

d) $\frac{f(x)}{g(x)} = \frac{x^2 - 1}{x + 1} = \frac{(x + 1)(x - 1)}{x + 1} = x - 1$

Application activity 6.5



1) Given: $f(x) = x^2 + 3x + 1$, $g(x) = x^2 - 2x + 4$.

Find:

a) $(f + g)(x)$

b) $(f - g)(x)$

2) Given: $f(x) = 3x + 2$, $g(x) = x^2$

Find:

a) $f(x) + g(x)$

c) $f(x) \times g(x)$

b) $f(x) - g(x)$

d) $\frac{f(x)}{g(x)}$

6.6. Composite and inverse functions

Activity 6.6



Given two functions $f(x) = 2x + 1$ and $g(x) = x^2$.

In small groups, discuss what happens when the **output of one function becomes the input of another function**? That is, when you first find $g(x)$ when $x = 3$ and then substitute the result into $f(x)$.

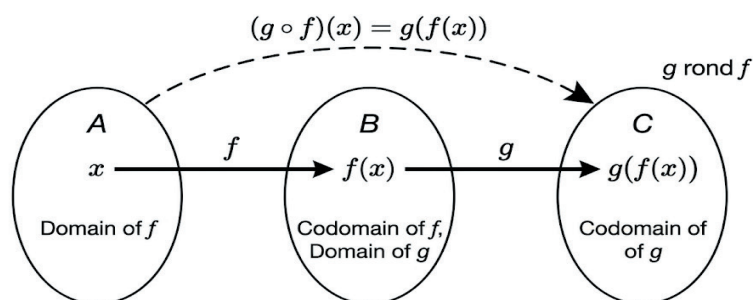


Content Summary

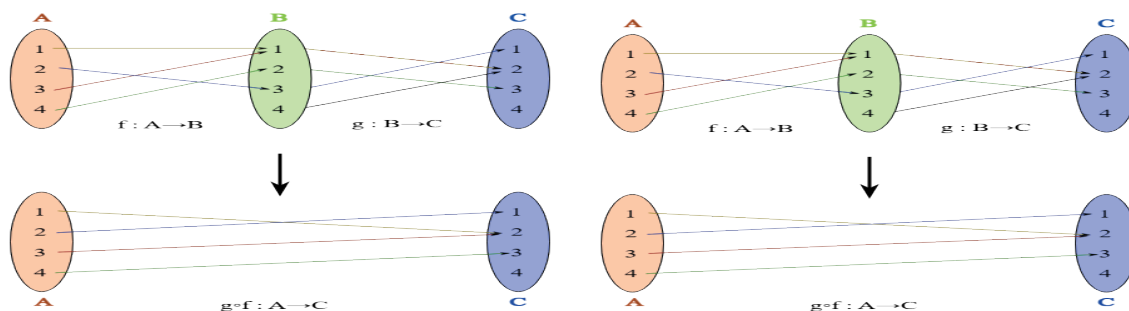
A. Composite functions

The composition of f and g , denoted $g \circ f$ is defined by the rule $g \circ f = g(f(x))$ provided that f is in the domain of g .

Illustration of Composite Function ($g \circ f$)



Composite function: the output of f becomes the input of g .



Note: $f \circ g$ is read as “ f compose g ,” and $g \circ f$ is also read as “ g compose f .” The functions $g \circ f$ and $f \circ g$ are called the **composite** or **compound** functions and in general, $g \circ f \neq f \circ g$.

Example

Given $f(x) = x^2 - 1$, $g(x) = x + 1$. Find

a) $f(x) + g(x)$

c) $f(x) \times g(x)$

b) $f(x) - g(x)$

d) $\frac{f(x)}{g(x)}$

Solution:

a) $f(x) + g(x) = x^2 - 1 + x + 1 = x^2 + x$

b) $f(x) - g(x) = x^2 - 1 - (x + 1) = x^2 - x - 2$

c) $f(x) \cdot g(x) = (x^2 - 1)(x + 1) = x^3 + x^2 - x - 1$

d) $\frac{f(x)}{g(x)} = \frac{x^2 - 1}{x + 1} = \frac{(x + 1)(x - 1)}{x + 1} = x - 1$

Given the functions defined by $f(x) = 4x + 3$ and $g(x) = 7x$. Find

a) $(f \circ g)(x)$

b) $(g \circ f)(x)$

c) $(f \circ g)(2)$

Solution:

a) $(f \circ g)(x) = f(g(x)) = f(7x) = 4(7x) + 3 = 28x + 3$

b) $(g \circ f)(x) = g(f(x)) = g(4x + 3) = 7(4x + 3) = 28x + 21$

c) $(f \circ g)(2) = 28(2) + 3 = 56 + 3 = 59$

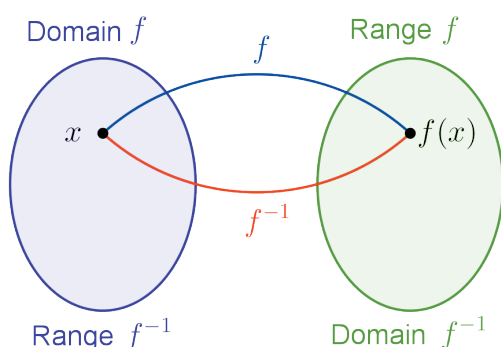
B. The inverse function

The **inverse of a function**, denoted as f^{-1} , is essentially the “undo” button for the original function f .

If the function f takes an input x and gives you a result y , the inverse function f^{-1} takes that y and brings you back to the original x .

Think of it as a reverse mapping:

Original function: $f(x) = y$ and Inverse function: $f^{-1}(y) = x$



For a one-to-one function defined by $y = f(x)$, the equation of the inverse can be found as follows:

- Replace $f(x)$ by y .
- Interchange x and y .
- Solve for y .
- Replace y by $f^{-1}(x)$.

An inverse function f^{-1} exists if and only if the function f is **bijective**. This means the function must satisfy two specific conditions: it must be **one-to-one** (injective) and **onto** (surjective).

Example

Find the inverse of the one-to-one functions defined by:

a) $f(x) = 3 - x^3$

b) $g(x) = \frac{2x+3}{x-1}$ for x different from 1.

Solution:

a) $y = 3 - x^3$

$x = 3 - y^3$

$x - 3 = -y^3$

$y^3 = 3 - x \Rightarrow y = \sqrt[3]{3 - x}$

Therefore $f^{-1}(x) = \sqrt[3]{3 - x}$.

b) $y = \frac{2x+3}{x-1} \Rightarrow y(x-1) = 2x+3$

$x = \frac{2y+3}{y-1} \Rightarrow x(y-1) = 2y+3$

$xy - x = 2y + 3 \Rightarrow xy - 2y = x + 3 \Rightarrow y(x-2) = x+3$

$y = \frac{x+3}{x-2}$

Therefore $f^{-1}(x) = \frac{x+3}{x-2}$

Note: $(f \circ f^{-1})(x) = (f^{-1} \circ f)(x) = x$ **Example**Determine whether $f(x) = \frac{1}{x-3}$ and $g(x) = \frac{3x+1}{x}$ are inverse functions by computing their compositions.**Solution:**

Since $(f \circ g)(x) = f(g(x)) = f\left(\frac{3x+1}{x}\right) = \frac{1}{\frac{3x+1}{x} - 3} = \frac{1}{\frac{3x+1-3x}{x}} = \frac{1}{\frac{1}{x}} = \frac{1}{1/x} = x$ and

$$(g \circ f)(x) = g(f(x)) = g\left(\frac{1}{x-3}\right) = \frac{3\left(\frac{1}{x-3}\right) + 1}{\frac{1}{x-3}} = \frac{\frac{3+x-3}{x-3}}{\frac{1}{x-3}} = \frac{\frac{x}{x-3}}{\frac{1}{x-3}} = \frac{x}{1} = x.$$

Thus, the functions f and g are a pair of inverse functions.**Application activity 6.6**1. Given: $f(x) = x^2 + 1$, $g(x) = x - 4$

Find:

a) $(f \circ g)(x)$

b) $(g \circ f)(x)$

2. Find the inverse of: $f(x) = x + 3$ 3. Find the inverse of: $f(x) = \frac{x+2}{3}$

6.7. Parity of a function

Application activity 6.7



Given the following functions

1. $f(x) = x^2$

2. $g(x) = x^3$

find $f(2)$ and $f(-2)$, $g(2)$ and $g(-2)$. Compare the results using $=$ or $\neq$.



Content Summary

The parity of a function describes the relationship between $f(x)$ and $f(-x)$

A function may be:

- Even
- Odd
- Neither even nor odd

A. Even function

A function $f(x)$ is **even** if: $f(-x) = f(x)$, for all x in the domain.

Example

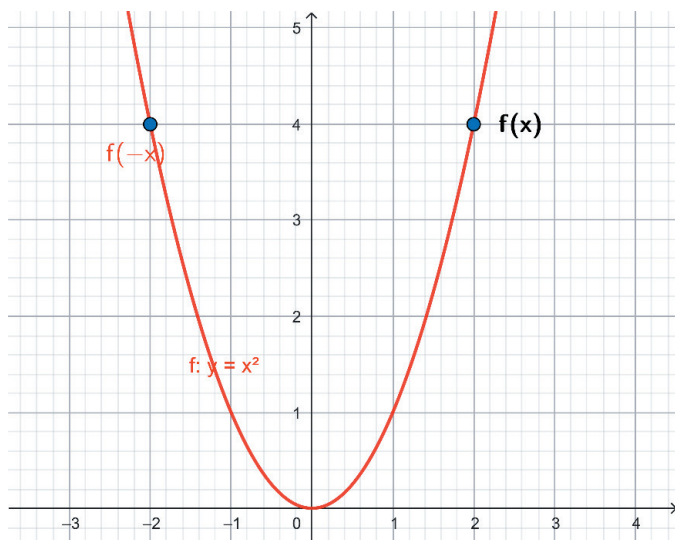
Study the parity of the function $f(x) = x^2$

Solution:

Test: $f(-x) = (-x)^2 = x^2 = f(x)$. Since $f(-x) = f(x)$, for all x in the domain, then $f(x)$ is an even function.

The graph of an even function is symmetric with respect to the y-axis.

Example: $f(x) = x^2$



B. Odd function

A function $f(x)$ is **odd** if: $f(-x) = -f(x)$, for all x in the domain.

Example

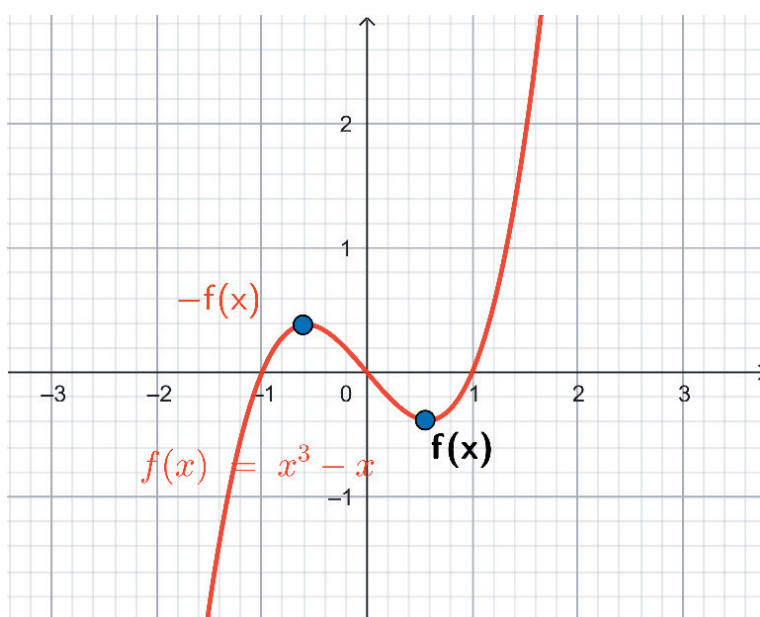
Study the parity of the function $f(x) = x^3$

Solution:

Test: $f(-x) = (-x)^3 = -x^3 = -f(x)$. Since $f(-x) = -f(x)$, for all x in the domain, then $f(x)$ is an odd function.

The graph of an odd function is symmetric with respect to the origin.

Example: $f(x) = x^3 - x$



C. Neither even nor odd function

Some functions do not satisfy either condition.

Example:

Study the parity of the function $f(x) = x + 1$

Test: $f(-x) = -x + 1$. This is not equal to $f(x)$ and not equal to $-f(x)$

Therefore the function is **neither even nor odd**.

Application activity 6.7



Given the following functions

1. $f(x) = x^2$

2. $g(x) = x^3$

find $f(2)$ and $f(-2)$, $g(2)$ and $g(-2)$. Compare the results using $=$ or $\neq$.

Study the parity of the following functions

$$1) f(x) = x^2 - 1 \quad 2) g(x) = \frac{x^3 + x}{5} \quad 3) h(x) = \frac{x^3 + x + 2}{x^2}$$

6.8 Partial fractions decomposition for denominator containing different linear factors

Activity 6.8



Given that $\frac{5}{(x+1)(x-2)} = \frac{A}{(x+1)} + \frac{B}{(x-2)}$; Work in small groups to find the constants A and B . Remember that in such conditions, $5 = A(x-2) + B(x+1)$.



Content Summary

Partial fraction decomposition is the process of expressing a rational function as a sum of simpler fractions.

If $\frac{P(x)}{(x-a)(x-b)}$ where $x-a$ and $x-b$ are different linear factors, then

$$\frac{P(x)}{(x-a)(x-b)} = \frac{A}{x-a} + \frac{B}{x-b} \text{ where } A \text{ and } B \text{ are constants to be determined.}$$

Steps for decomposition

- Express the rational function as a sum of partial fractions.
- Multiply both sides by the common denominator.
- Expand the expression.
- Compare coefficients or substitute convenient values of x .
- Determine constants A and B .

Example

Decompose the following into partial fraction $\frac{3}{(x+1)(x+2)}$

Solution:

$$\text{Assume } \frac{3}{(x+1)(x+2)} = \frac{A}{x+1} + \frac{B}{x+2}$$

Multiply both sides by $(x+1)(x+2)$.

$$3 = A(x+2) + B(x+1).$$

$$\text{Expand: } 3 = Ax + 2A + Bx + B \Rightarrow 3 = (A+B)x + (2A+B)$$

$$\text{Compare coefficients: } A+B=0 \text{ and } 2A+B=3$$

$$\text{Solving: } A=3, \quad B=-3$$

$$\text{Therefore, } \frac{3}{(x+1)(x+2)} = \frac{3}{x+1} - \frac{3}{x+2}$$

Application activity 6.8



1. Decompose the following functions into partial fraction

$$\frac{4}{(x+3)(x-1)}$$

$$\frac{3x+5}{(x-2)(x+4)}$$

2. Express the following in partial fractions

$$\frac{2x+9}{(x-4)(x+2)}$$

6.9 Partial fractions decomposition for denominator containing repeated linear factors

Activity 6.9



In your groups, find the values of the constants A and B from the following equation

$$\frac{3}{(x+1)^2} = \frac{A}{x+1} + \frac{B}{(x+1)^2}$$



Content Summary

When the denominator contains a repeated linear factor, the rational function can be decomposed as follows:

$$\frac{P(x)}{(x-a)^2} = \frac{A}{x-a} + \frac{B}{(x-a)^2}, \text{ where } A \text{ and } B \text{ are constants to be determined.}$$

Steps for decomposition

- Write the rational expression as a sum of partial fractions.
- Multiply both sides by the common denominator.
- Expand the expressions.
- Compare coefficients.
- Solve to find the constants.

Example

Decompose $\frac{5x+1}{(x-1)^2}$

Solution:

Assume $\frac{5x+1}{(x-1)^2} = \frac{A}{x-1} + \frac{B}{(x-1)^2}$

Multiply both sides by $(x-1)^2$ to obtain $5x+1 = A(x-1) + B$.

Expand: $5x+1 = Ax - A + B \Rightarrow 5x+1 = Ax + (-A+B)$

Compare coefficients: $A = 5$ and $-A+B = 1$

Substitute $A = 5$ into $-A+B = 1$ to get $B = 6$

Thus, $\frac{5x+1}{(x-1)^2} = \frac{5}{x-1} + \frac{6}{(x-1)^2}$.

Application activity 6.9

1. Decompose into partial fractions the following function

a) $\frac{4}{(x+2)^2}$

b) $\frac{3x+2}{(x+1)^2}$

2. Write the following in partial fractions

$$\frac{2x+5}{(x-3)^2}$$

6.10 Partial fractions decomposition for denominator containing linear factor and repeated linear factors

Activity 6.10

In your groups, find the values of the constants A , B and C from the following equation

$5 = A(x+2)^2 + B(x+1)(x+2) + C(x+1)$ and then replace them into the following fraction to form $\frac{A}{x+1} + \frac{B}{x+2} + \frac{C}{(x+2)^2}$.

**Content Summary**

When the denominator contains a linear factor and a repeated linear factor, the rational function can be decomposed as:

$$\frac{P(x)}{(x-a)(x-b)^2} = \frac{A}{x-a} + \frac{B}{x-b} + \frac{C}{(x-b)^2}, \text{ where } A, B \text{ and } C \text{ are constants to be determined.}$$

Steps for decomposition

- Write the rational function as a sum of partial fractions.
- Multiply both sides by the common denominator.

- Expand the expressions.
- Compare coefficients of like powers of x .
- Solve to find the constants.

Example

Decompose the following function into simpler fractions $\frac{3x+4}{(x+1)(x-1)^2}$

Solution:

Assume $\frac{3x+4}{(x+1)(x-1)^2} = \frac{A}{x+1} + \frac{B}{x-1} + \frac{C}{(x-1)^2}$

Multiply both sides by $(x+1)(x-1)^2$ to obtain $3x+4 = A(x-1)^2 + B(x+1)(x-1) + C(x+1)$.

Expand: $3x+4 = A(x^2-2x+1) + B(x^2-1) + Cx+C$

$$\Rightarrow 3x+4 = (A+B)x^2 + (-2A+C)x + (A-B+C)$$

Compare coefficients: $A+B=0$ (1), $-2A+C=3$ (2), and $A-B+C=4$ (3)

From (1) $A=-B$ and substitute into (2) and (3) to get $A=\frac{1}{2}$, $B=-\frac{1}{2}$, $C=4$

Thus, $\frac{3x+4}{(x+1)(x-1)^2} = \frac{1/2}{x+1} - \frac{1/2}{x-1} + \frac{4}{(x-1)^2}$

Application activity 6.10



1. Write the following into partial fractions $\frac{6}{(x+1)(x-2)^2}$
2. Decompose or expand into partial fractions the following function

a) $\frac{4}{(x+1)(x+3)^2}$ b) $\frac{2x+5}{(x+2)(x-1)^2}$

6.11 Partial fraction decomposition for denominator containing linear and quadratic (which cannot be factored) factors

Activity 6.10



In pairs, find the values of the constants A , B and C from the following equation

$3x+5 = A(x^2+1) + (Bx+C)(x+1)$. Substitute them into the following fraction

$$\frac{A}{x+1} + \frac{Bx+C}{x^2+1}$$



Content Summary

When the denominator contains a linear factor and a quadratic factor that cannot be factorized, the rational function is decomposed as

$$\frac{P(x)}{(x-a)(x^2+bx+c)} = \frac{A}{x-a} + \frac{Bx+C}{x^2+bx+c} \text{ where } A, B \text{ and } C \text{ are constants to be determined.}$$

Example 1

Decompose the following fraction into partial fraction

$$\frac{2x+3}{(x-1)(x^2+1)}$$

Solution:

$$\text{Assume } \frac{2x+3}{(x-1)(x^2+1)} = \frac{A}{x-1} + \frac{Bx+C}{x^2+1}$$

Multiply both sides by $(x-1)(x^2+1)$ to obtain $2x+3 = A(x^2+1) + (Bx+C)(x-1)$

Expand: $2x+3 = Ax^2 + A + Bx^2 - Bx + Cx - C$

$$\Rightarrow 2x+3 = (A+B)x^2 + (-B+C)x + (A-C).$$

Compare coefficients: $A+B=0$ (1), $-B+C=2$ (2), and $A-C=3$ (3)

Use elimination method in (1) and (2) to get $A+C=2$ (4). Again, eliminate C in equation (3) and (4) to get $A=\frac{5}{2}$. From equation (4), we have $C=-\frac{1}{2}$; and from

equation (1) we get $B=-\frac{5}{2}$.

$$\text{Thus, } \frac{2x+3}{(x-1)(x^2+1)} = \frac{\frac{5}{2}}{x-1} + \frac{-\frac{5}{2}x - \frac{1}{2}}{x^2+1}$$

Example 2

$$\text{Decompose } \frac{2x+1}{(x+1)(x^2+x+1)}$$

Solution:

$$\text{Assume } \frac{2x+1}{(x+1)(x^2+x+1)} = \frac{A}{x+1} + \frac{Bx+C}{x^2+x+1}$$

Multiply both sides by $(x+1)(x^2+x+1)$ to get $2x+1 = A(x^2+x+1) + (Bx+C)(x+1)$

Expand term terms: $2x+1 = Ax^2 + Ax + A + Bx^2 + Bx + Cx + C$

On the right-hand side, group terms with similar powers to get

$$2x+1 = (A+B)x^2 + (A+B+C)x + (A+C)$$

Compare coefficients of like powers of x to get the system
$$\begin{cases} A + B = 0 & (1) \\ A + B + C = 2 & (2) \\ A + C = 1 & (3) \end{cases}$$

To solve the system, one can use elimination method. From equation (1) we have $A = -B$, replace A by $-B$ in equation (2), we get $-B + B + C = 2 \Rightarrow C = 2$. Equation (3) yields $A = -1$. Hence, from (1) $B = 1$.

Therefore,
$$\frac{2x+1}{(x+1)(x^2+x+1)} = \frac{-1}{x+1} + \frac{x+2}{x^2+x+1}$$

Application activity 6.11



1. Write the following in partial fractions

a) $\frac{3x+1}{(x-2)(x^2+4)}$ b) $\frac{4x+5}{(x+1)(x^2+x+1)}$

6.12. Applications of rational and irrational functions in diary life

Activity 6.12



In groups of five, research on the different ways of applying functions of rational and irrational number. Discuss your findings in class.



Content Summary

A. Free falling objects

This diagram (not drawn to scale) shows the results of several distance calculations for a free-falling object dropped from a position of rest.

A free-falling object is accelerated toward the earth with a constant acceleration equal to $g = 9.8\text{m/s}^2$. The formulas used to describe the motion of a free-falling object are:

1. $h = h_0 + v_0t + \frac{1}{2}gt^2$

2. $v = v_0 + gt$



Sometimes time is not known. In this case the following formula is used $v^2 - v_0^2 = 2g(h - h_0)$. Where g represents the acceleration due to gravity, v_0 stands for the initial velocity, v is the final velocity, $y_0 = h_0$ is the initial height, and $y = h$ is the final height.

Example 1

A stone is dropped from a tower of height 50m. Find its height above the ground after 2 seconds.

Solution:

Since the stone is dropped, $v_0 = 0$.

Using $y = y_0 + v_0 t + \frac{1}{2}gt^2$, we have $y = 50 + 0(2) - \frac{1}{2}(9.8)(2)^2 = 50 - 19.6 = 30.4$

Thus, the stone is 30.4m above the ground after 2 seconds.

Example 2

A ball is thrown upward from the ground with an initial velocity of 15m/s. Find the velocity after 3 seconds.

Solution:

$$v = v_0 + gt, \quad v = 15 - 9.8(3) = -14.4 \text{ m/s}.$$

The negative sign shows that the object is moving downward.

Example 3

A ball is thrown upward with an initial velocity of 20m/s. Find the maximum height reached.

Solution:

At maximum height, $v = 0$. Using $v^2 - v_0^2 = 2g(y - y_0)$, we have

$$0 - (20)^2 = 2(-9.8)(y - 0) \Leftrightarrow -400 = -19.6y \text{ or } y = 20.4 \text{ m}$$

Thus, the maximum height reached is approximately 20.4m.

B. Determination of the equilibrium cost (price) of a commodity in an isolated market

In an isolated market, the price of a commodity is determined by the interaction between demand and supply.

Demand Function

The demand function shows the relationship between price and quantity demanded.

$$Q_d = -ap + b \quad (a, b > 0)$$

Where Q_d represents the quantity demand of the commodity, p is the (cost) price of the commodity, and a, b are positive constants. Demand usually decreases when quantity increases.

Supply Function

The supply function represents the relationship between price and quantity supplied.

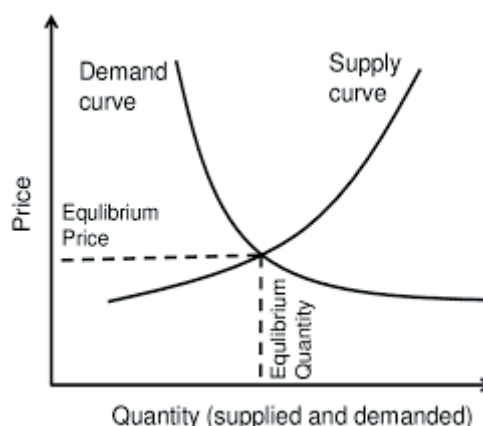
$$Q_s = cp - d \quad (c, d > 0)$$

Where Q_s stands for the quantity supplied of the commodity, and c, d are positive constants. Supply generally increases as quantity increases.

Equilibrium Price

The equilibrium occurs when

$$Q_d = Q_s$$



By solving this equation, we obtain the equilibrium quantity.

$$-ap + b = cp - d \Leftrightarrow (c + a)p = b + d \text{ and } p = \frac{b + d}{c + a}$$

Thus, the equilibrium cost (price) of the commodity is $\bar{p} = \frac{b + d}{c + a}$

After finding the equilibrium quantity, we substitute it into either the demand or supply equation to obtain the equilibrium price.

$$\bar{Q} = -a\left(\frac{b + d}{c + a}\right) + b = \frac{-a(b + d) + b(c + a)}{c + a} = \frac{-ab - ad + bc + ab}{c + a} = \frac{bc - ad}{c + a}$$

Since the denominator $(c + a)$ is positive, the positivity of Q requires that the numerator satisfies $bc - ad > 0$. Thus, to be economically meaningful, the model should contain the additional restriction $bc > ad$.

Example

Below is the model of an isolated market for a commodity x .

$$Q_d = -x + 6$$

$$Q_s = 4x - 4$$

Find:

- the equilibrium cost of the commodity,
- the equilibrium quantity of the commodity.

Solution:

a) Equilibrium cost

At equilibrium, $Q_d = Q_s$

Therefore, $-x + 6 = 4x - 4 \Leftrightarrow 6 + 4 = 4x + x$, so $10 = 5x \Leftrightarrow x = 2$

Thus, the equilibrium cost of the commodity is 2.

b) Equilibrium quantity :Substitute $x = 2$ into either the demand or supply equation.

Using the demand equation: $Q_d = -x + 6 = -2 + 6 = 4$

Thus, the equilibrium quantity of the commodity is 4

Application activity 6.12

1. A stone is dropped from a tower 80m high.
 - a) Find its height after 3 seconds.
 - b) Find the velocity after 3 seconds.
2. Mukansanga throws a ball vertically upward and it reaches the height of 4.9m . Find
 - a) the time taken by the ball to reach the maximum height
 - b) the initial speed by which the ball was thrown.
3. A ball is thrown vertically upward from the ground with an initial velocity of 24m/s .
 - a) Find the maximum height reached.
 - b) Determine the time taken to reach the maximum height.
4. Find the equilibrium cost of the commodity for the following model

$$Q_d = 15 - x^2$$

$$Q_s = 4x - 6$$

**A. Assessing Knowledge and Understanding**

1. Choose the correct answer

a) Which of the following is a factor of $x^2 - 9$?

- A. $(x+9)$ B. $(x-3)$ C. $(x+1)$ D. $(x-1)$

b) The domain of the polynomial function $f(x) = 3x^2 - 2x + 5$ is

- A. $\mathbb{R}$ B. $x > 0$ C. $x \neq 0$ D. $x \geq 0$

c) The domain of the rational function $f(x) = \frac{1}{x-4}$ is

- A. $x > 4$ B. $x \neq 4$ C. $x \geq 4$ D. $\mathbb{R}$

d) If $f(x) = 2x + 3$ and $g(x) = x^2$, then $(f \circ g)(x)$ is

- A. $2x^2 + 3$ B. $4x + 3$ C. $x^2 + 3$ D. $2x^2$

e) The function $f(x) = x^3 - x$ is

- A. even B. odd C. constant D. periodic

2. Factorize $x^2 + 4x + 3$

3. Explain the domain of a function.

B. Skills

1. Determine the domain of the function $f(x) = \frac{3}{x-2}$

2. Determine the domain and range of $f(x) = x^2$

3. Given $f(x) = x + 1$, $g(x) = x^2$. Find $(f \circ g)(x)$

4. Determine whether the function $f(x) = x^4$ is even or odd.

5. Decompose into partial fractions $\frac{5}{(x-1)(x-2)}$

6. In a small market the demand and supply functions are
 $Q_d = 15 - x^2$, and $Q_s = 4x - 6$

Find

- the equilibrium price
- the equilibrium quantity

C. Attitudes and Values

- Explain why understanding equilibrium price is important in real markets.
- Discuss how mathematics can help in solving real-life economic problems.

UNIT 7

LIMITS OF POLYNOMIAL, RATIONAL AND IRRATIONAL FUNCTIONS

$$\lim_{x \rightarrow a} f(x)$$



Key Unit Competence:

Apply correctly limits of polynomial, rational and irrational functions in solving related problems.

Learning objectives

By the end of this unit, you should be able to:

Knowledge and understanding

1

- Explain the neighbourhood of a real number and the concept of a limit of a variable
- State and interpret the limit of a function both algebraically and graphically
- Distinguish between one-sided limits, infinite limits, and limits at infinity
- Explain the Squeeze theorem and its use in finding limits.
- Identify and describe indeterminate cases and how they occur in limit evaluation
- Explain the applications of limits in determining continuity, discontinuity, and asymptotes of functions.

Skills

2

- Evaluate limits of functions algebraically, graphically, and numerically
- Determine left-hand and right-hand (one-sided) limits of given functions
- Compute infinite limits and limits at infinity using appropriate techniques
- Apply the Squeeze theorem to find limits of complex functions.
- Identify and resolve indeterminate cases using algebraic manipulation
- Use limits to analyze the continuity and discontinuity of functions
- Apply limits to determine vertical, horizontal, and oblique asymptotes of rational and irrational functions.

Attitudes and values

3

- Show persistence, accuracy, and logical reasoning when solving limit problems
- Demonstrate curiosity in exploring graphical behavior of functions near critical points
- Appreciate the role of limits as a foundation for calculus and real-world modeling
- Work collaboratively and share ideas respectfully.
- Exhibit confidence and responsibility in interpreting mathematical results.

Introduction

In mathematics, the concept of a **limit** describes the value that a function approaches as the variable gets closer to a certain number. Polynomial, rational, and irrational functions often appear in many real-life situations such as motion, economics, and population growth. Studying the limits of these functions helps us understand how they behave when the input values approach particular numbers.

Therefore, the study of limits provides an important tool for analyzing the behavior of polynomial, rational, and irrational functions and forms a foundation for more advanced topics such as continuity and calculus.

Introductory activity



Consider the function: $f(x) = \frac{x^2 - 1}{x - 1}$

1. Complete the table below:

x	0.9	0.99	0.999	1	1.001	1.01	1.1
$f(x)$							

2. Is $f(1)$ defined? What value does $f(1)$ seem to approach?

7.1. Neighbourhood of a real number and limit of a variable

Activity 7.1



Consider a number 5

- Write five numbers that are, on both sides, very close to 5 but not equal to 5.
- Write some numbers slightly less than 5 and some slightly greater than 5.
- What do you observe about these values compared to 5.

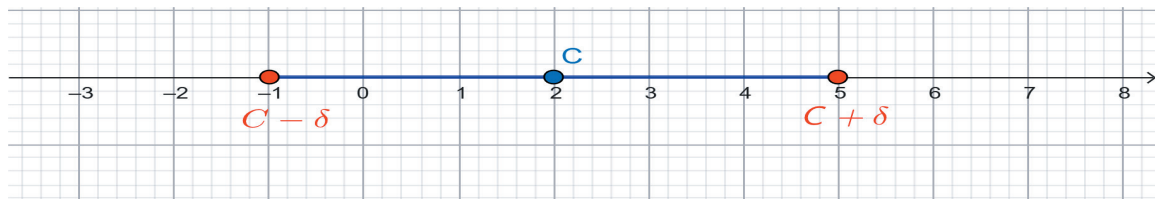


Content Summary

A. Neighbourhood of a real number

Let C be a real number and δ be a positive number.

On the real line, a neighbourhood of a real number is an open interval $(C - \delta, C + \delta)$ with its center at C . One can say that a neighbourhood of the real number is any interval that contains a real number C and some point below and above it.



Neighbourhood of C

Example 1: If $c = 5$ and $\delta = 1$, then the neighbourhood of 5 is $(5 - 1, 5 + 1) = (4, 6)$

This means that all numbers between 4 and 5 belong to the neighbourhood of 5.

Example 2

Find the 2-neighbourhood of 5.

Solution:

$(5 - 2, 5 + 2) = (3, 7)$. Thus the 2-neighbourhood of 5 is $(3, 7)$.

B. Limit of a variable

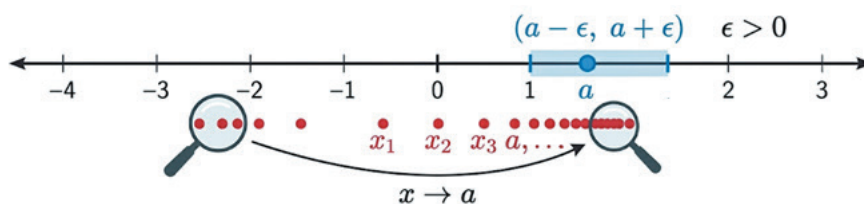
In mathematics, the **limit of a variable** describes the behavior of a variable as it gets closer and closer to a specific constant value without necessarily ever reaching it.

If a variable x takes a succession of values such that the difference between x and a constant a (expressed as $|x - a|$) becomes and remains less than any positive number ϵ (epsilon), however small, we say that **a is the limit of x** .

We express this using the following notation: $x \rightarrow a$

This is read as **" x approaches a "** or **" x tends to a "**.

In other words, if the values of x become arbitrarily close to a number a , we say that a is the **limit** of x , and we write $x \rightarrow a$, which means that x approaches the value a .



If the values of x approach a from numbers smaller than a , we say that x approaches a from the left. We write $x \rightarrow a^-$ and we call it **left-hand limit**.

If the values of x approach a from numbers greater than a , we say that x approaches a from the right. We write $x \rightarrow a^+$ and we call it **right-hand limit**.

Example 1: If $c = 5$ and $\delta = 1$, then the neighbourhood of 5 is $(5 - 1, 5 + 1) = (4, 6)$

This means that all numbers between 4 and 5 belong to the neighbourhood of 5.

Example 2

Find the 2 – neighbourhood of 5.

Solution:

$(5 - 2, 5 + 2) = (3, 7)$. Thus the 2 – neighbourhood of 5 is $(3, 7)$.

Example

A variable x takes the values 4.8, 4.9, 4.99, 4.999. Determine the limit of x .

Solution:

The values get closer and closer to 5. Thus $x \rightarrow 5$ and the limit of x is 5.

Application activity 7.1



1. Find the 1 – neighbourhood of the number 8.
2. Determine whether 4.8 belongs to the 1 – neighbourhood of 5.
3. A variable x takes the values 1.9, 1.99, 1.999. Determine the limit of x .
4. A variable x takes the values 6.2, 6.02, 6.002. Determine whether x approaches 6 from the left or from the right.

7.2 Definition and graphical interpretation of limit of a function

Activity 7.2



Consider a function $f(x) = x^2$

x	1.9	1.99	1.999	2	2.001	2.01	2.1
$f(x)$							

1. Calculate the corresponding values of $f(x)$ and complete the above table.
2. What value does $f(x)$ approach as x gets closer to 2?
3. Does the value of $f(x)$ approach a particular number?



Content Summary

A. Definition of the limit of a function

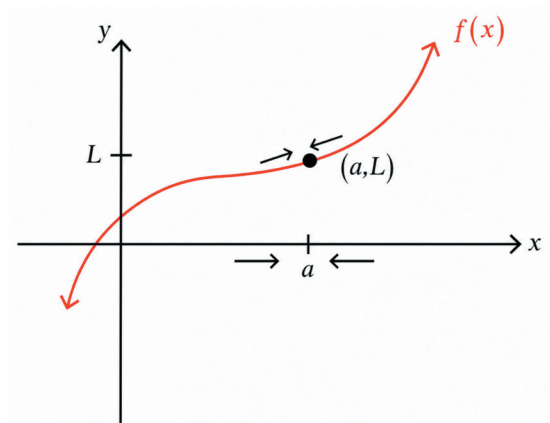
Let $f(x)$ be a function defined near a real number a .

We say that the function $f(x)$ has a limit L as x approaches a if the values of $f(x)$ become arbitrarily close to L whenever x is sufficiently close to a .

This is written as $\lim_{x \rightarrow a} f(x) = L$.

Where

x is the independent variable,
 a is the value that x approaches,
 L is the limit of the function.



Note that in the above case the limit is found by the direct substitution i.e.

$$\lim_{x \rightarrow 3} f(x) = \lim_{x \rightarrow 3} 2x + 1 = 2(3) + 1 = 7 = f(3).$$

Sometimes the direct substitution may appear to give an indeterminate case like

$\frac{0}{0}$ or $\frac{\infty}{\infty}, \dots$ and we have to first transform or simplify the function.

Example 1

Evaluate the following limit

$$\lim_{x \rightarrow 2} x^3 - 3x^2 + 4x + 5$$

Solution:

$$\begin{aligned}\lim_{x \rightarrow 2} x^3 - 3x^2 + 4x + 5 &= (2)^3 - 3(2)^2 + 4(2) + 5 \\ &= 8 - 12 + 8 + 5 = 9\end{aligned}$$

Example 2

Find $\lim_{x \rightarrow 1} \frac{x^2 + x - 2}{x - 1}$

Solution:

Direct substitution cannot be applied and we simplify first.

$$\frac{x^2 + x - 2}{x - 1} = \frac{(x - 1)(x + 2)}{x - 1} = \lim_{x \rightarrow 1} (x + 2) = 1 + 2 = 3$$

Example 3

Evaluate the limit $\lim_{x \rightarrow 0} \frac{x^2 + 3x}{x}$

Solution:

$$\lim_{x \rightarrow 0} \frac{x^2 + 3x}{x} = \lim_{x \rightarrow 0} \frac{x(x + 3)}{x} = \lim_{x \rightarrow 0} (x + 3) = 3$$

Example 4

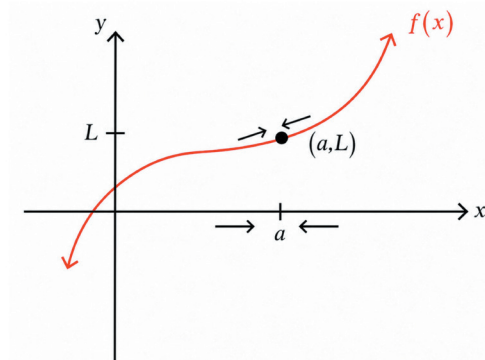
Compute $\lim_{x \rightarrow 3} \frac{x^2 - 9}{x - 3}$

Solution:

$$\lim_{x \rightarrow 3} \frac{x^2 - 9}{x - 3} = \lim_{x \rightarrow 3} \frac{(x + 3)(x - 3)}{x - 3} = \lim_{x \rightarrow 3} (x + 3) = 6$$

B. Graphical interpretation of the limit

Graphically, the limit of a function represents the value that the function approaches when the variable x gets closer to a . The equation of a limit is written as $\lim_{x \rightarrow a} f(x) = L$



Example

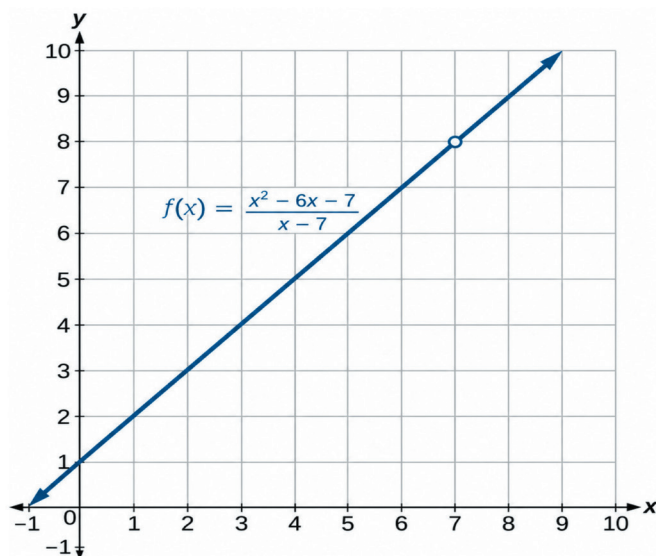
Graphically, find $\lim_{x \rightarrow 7} \frac{x^2 - 6x - 7}{x - 7}$

Solution:

The given function can be factorized as

$$f(x) = \frac{x^2 - 6x - 7}{x - 7} = \frac{(x - 7)(x + 1)}{x - 7} = x + 1 \text{ for } x \neq 7.$$

$$\text{Thus } \lim_{x \rightarrow 7} \frac{x^2 - 6x - 7}{x - 7} = \lim_{x \rightarrow 7} (x + 1) = 8$$



When x approaches 7 the values of $f(x)$ approach 8.

Graphically, the curve approaches the point $(7, 8)$.

Note: Even if the function is not defined at a , the limit may still exist if the function values approach a particular number.

Note: If we have to find $\lim_{x \rightarrow a} f(x)$, the function does not have to be defined at the point at which the limit is taken since we are only concerned with what happens around the point. The function must be defined in the neighbourhood of a .

C. Rules for limits of functions at a point

1. $\lim_{x \rightarrow a} k = k$ (where $a \in \mathbb{R}$ and k is a constant)
2. $\lim_{x \rightarrow a} x^n = a^n$ (where $a \in \mathbb{R}$ and n is a positive integer)
3. $\lim_{x \rightarrow a} \sqrt[n]{x} = \sqrt[n]{a}$ (where $a \in \mathbb{R}$ and n is an odd positive integer)

Application activity 7.2



1. Evaluate the limit

a) $\lim_{x \rightarrow 1^+} f(x) = 4$

b) $\lim_{x \rightarrow -3} \frac{x^2 - 9}{x + 3}$

2. Consider the function $f(x) = x^2$. Graphically find $\lim_{x \rightarrow 2} f(x)$

7.3 Operations on limits and Squeeze theorem

Activity 7.3



Consider three functions defined near $x = 0$:

$$-x^2 \leq x^2 \sin\left(\frac{1}{x}\right) \leq x^2$$

- Evaluate the limits

$$\lim_{x \rightarrow 0}(-x^2) \quad \text{and} \quad \lim_{x \rightarrow 0} x^2$$

- What do you notice about the function $x^2 \sin\left(\frac{1}{x}\right)$ compared with the other two functions?
- What value do you think the function approaches as x approaches 0 ?



Content Summary

A. Operations on Limits

If $\lim_{x \rightarrow a} f(x) = L$ and $\lim_{x \rightarrow a} g(x) = M$, then the following rules apply.

Sum Rule: $\lim_{x \rightarrow a} [f(x) + g(x)] = L + M$

Difference Rule: $\lim_{x \rightarrow a} [f(x) - g(x)] = L - M$

Constant Multiple Rule: $\lim_{x \rightarrow a} [cf(x)] = cL$, where c is a constant.

Product Rule: $\lim_{x \rightarrow a} [f(x)g(x)] = LM$

Quotient Rule: $\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{L}{M}$, provided that $M \neq 0$.

Example 1

Evaluate: $\lim_{x \rightarrow 2} (x^2 + 3x)$

Solution:

$$\lim_{x \rightarrow 2} x^2 = 4 \quad \text{and} \quad \lim_{x \rightarrow 2} 3x = 6$$

$$\text{Thus } \lim_{x \rightarrow 2} (x^2 + 3x) = 4 + 6 = 10$$

Example 2: Evaluate $\lim_{x \rightarrow 1} \frac{x^2 + 1}{x + 1}$

Solution:

$$\frac{1^2 + 1}{1 + 1} = \frac{2}{2} = 1$$

Example 3

Evaluate the following limit and justify each step $\lim_{x \rightarrow -2} \frac{x^3 + 2x^2 - 1}{5 - 3x}$

Solution:

$$\begin{aligned} \lim_{x \rightarrow -2} \frac{x^3 + 2x^2 - 1}{5 - 3x} &= \frac{\lim_{x \rightarrow -2} (x^3 + 2x^2 - 1)}{\lim_{x \rightarrow -2} (5 - 3x)} = \frac{\lim_{x \rightarrow -2} x^3 + 2 \lim_{x \rightarrow -2} x^2 - \lim_{x \rightarrow -2} 1}{\lim_{x \rightarrow -2} 5 - 3 \lim_{x \rightarrow -2} x} \\ &= \frac{(-2)^3 + 2(-2)^2 - 1}{5 - 3(-2)} = \frac{-8 + 8 - 1}{5 + 6} = -\frac{1}{11}. \end{aligned}$$

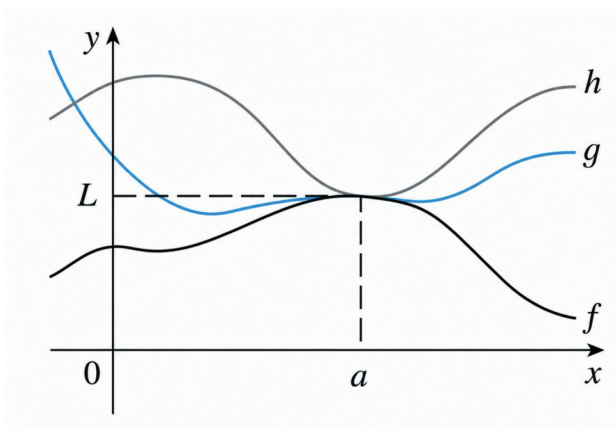
B. The Squeeze Theorem

Let $f(x)$, $g(x)$ and $h(x)$ be functions defined near a such that

$$f(x) \leq g(x) \leq h(x) \text{ for all } x \text{ near } a.$$

If $\lim_{x \rightarrow a} f(x) = L$ and $\lim_{x \rightarrow a} h(x) = L$ then

$$\lim_{x \rightarrow a} g(x) = L$$



Thus the function $g(x)$ is **squeeze** between two functions that approach the same limit.

Example: Evaluate: $\lim_{x \rightarrow 0} x^2 \sin\left(\frac{1}{x}\right)$

Solution:

Since $-1 \leq \sin\left(\frac{1}{x}\right) \leq 1$, multiplying by x^2 gives $-x^2 \leq x^2 \sin\left(\frac{1}{x}\right) \leq x^2$.

But $\lim_{x \rightarrow 0} (-x^2) = 0$ and $\lim_{x \rightarrow 0} x^2 = 0$. Therefore, $\lim_{x \rightarrow 0} x^2 \sin\left(\frac{1}{x}\right) = 0$.

Application activity 7.3



1. Evaluate

a) $\lim_{x \rightarrow 3} (x^2 + 2x)$

b) $\lim_{x \rightarrow 2} (4x - 5)$

c) $\lim_{x \rightarrow 1} (x^2 + x + 1)$

d) $\lim_{x \rightarrow 2} \frac{x^2 - 1}{x + 1}$

2. Find the limit of the following functions by squeeze theorem

a) $\lim_{x \rightarrow 0} x \cos 2x$

b) $\lim_{x \rightarrow +\infty} \frac{\sin x}{x}, \quad \forall x \in [0, +\infty[$

7.4. One-sided limits and existence of a limit of a function

Activity 7.4



Consider the function $\lim_{x \rightarrow 3} \frac{|x-3|}{x-3}$

- What value does $f(x)$ approach when x approaches 3 from the left?
- What value does $f(x)$ approach when x approaches 3 from the right?



Content Summary

A. Left-hand limit

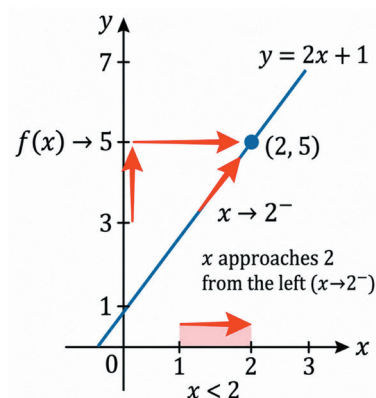
The left-hand limit of a function is the value that the function approaches when x approaches a number a from values less than a .

It is written as $\lim_{x \rightarrow a^-} f(x)$

This means that x approaches a from the left side ($x < a$).

Example

If $f(x) = 2x + 1$ then $\lim_{x \rightarrow 2^-} f(x) = 5$ because when x approaches 2 from values smaller than 2, the function values approach 5.



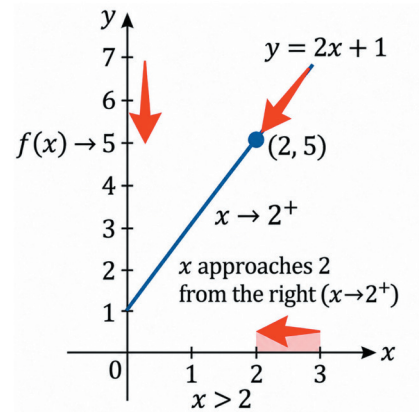
B. Right-hand limit

The right-hand limit of a function is the value that the function approaches when x approaches a number a from values greater than a . It is written as $\lim_{x \rightarrow a^+} f(x)$

This means that x approaches a from the right side ($x > a$).

Example

If $f(x) = 2x + 1$ then $\lim_{x \rightarrow 2^+} f(x) = 5$ because when x approaches 2 from values greater than 2, the function values approach 5.



C. Existence of limit of a function

The limit of a function at a point exists if and only if the left-hand limit and the right-hand limit are equal. That is, $\lim_{x \rightarrow a} f(x) = L$ if $\lim_{x \rightarrow a^-} f(x) = L$ and $\lim_{x \rightarrow a^+} f(x) = L$

If the two one-sided limits are different, then the limit of the function at that point **does not exist**.

Example 1: Find $\lim_{x \rightarrow 0} \frac{|x|}{x}$

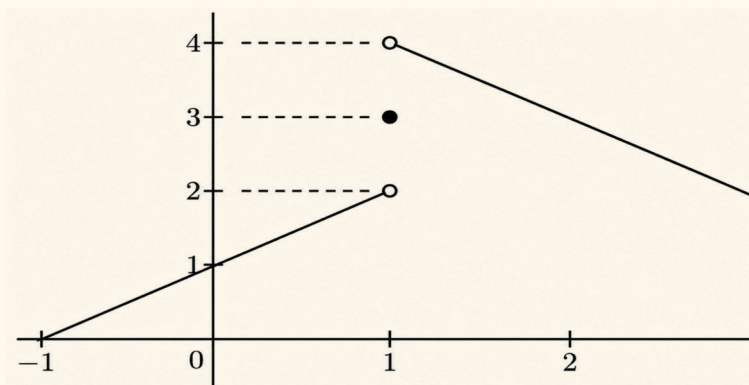
Solution:

$$\lim_{x \rightarrow 0^+} \frac{|x|}{x} = \lim_{x \rightarrow 0^+} \frac{x}{x} = \lim_{x \rightarrow 0^+} 1 = 1 \quad \text{and} \quad \lim_{x \rightarrow 0^-} \frac{|x|}{x} = \lim_{x \rightarrow 0^-} \frac{-x}{x} = \lim_{x \rightarrow 0^-} (-1) = -1$$

Therefore, $\lim_{x \rightarrow 0} \frac{|x|}{x}$ does not exist.

Example 2:

Using the following graph, study the limit of $f(x)$ at $x = 2$.



Solution:

The following graph shows that as x approaches 1 from the left, $y = f(x)$ approaches 2 and this can be written as $\lim_{x \rightarrow 1^-} f(x) = 2$.

As x approaches 1 from the right, $y = f(x)$ approaches 4 and this can be written as $\lim_{x \rightarrow 1^+} f(x) = 4$.

Observe that $\lim_{x \rightarrow 1^-} f(x) \neq \lim_{x \rightarrow 1^+} f(x)$. Therefore $\lim_{x \rightarrow 1} f(x)$ does not exist.

Application activity 7.4



1. Determine the left-hand and right-hand limits of $f(x) = 2x + 1$ at $x = 3$.
2. Find $\lim_{x \rightarrow 2^-} (x^2)$ and $\lim_{x \rightarrow 2^+} (x^2)$
3. Find $\lim_{x \rightarrow 0} \left(x + \frac{|x|}{x} \right)$
4. Consider the function $f(x) = \begin{cases} x+1 & x < 2 \\ x^2 & x \geq 2 \end{cases}$
5. Find the left-hand and right-hand limits at $x = 2$.
6. The function f is defined by $f(x) = \begin{cases} x^2 + 6x + 8, & \text{for } x \leq -1, \\ -x + 4, & \text{for } x > -1. \end{cases}$
7. Using a graphical method, determine $\lim_{x \rightarrow 1} f(x)$.

7.5 Infinite limits

Activity 7.5



Consider the function: $f(x) = \frac{1}{x-2}$

Complete the table below.

x	1.9	1.99	1.999	2	2.001	2.01	2.1
$f(x)$							

- What happens to the values of $f(x)$ as x approaches 2
- Do the values increase without bound or decrease without bound?



Content Summary

Definition of Infinite Limit

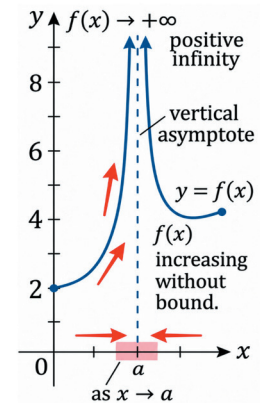
Let $f(x)$ be a function defined near a real number a .

If the values of $f(x)$ increase without bound as x approaches a , we say that the limit of $f(x)$ is **positive infinity**.

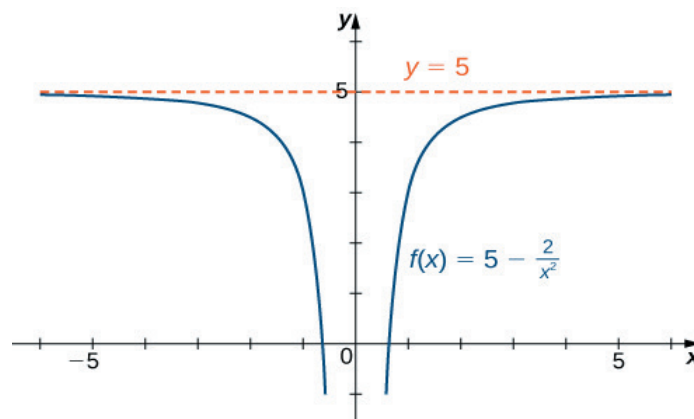
We write: $\lim_{x \rightarrow a} f(x) = +\infty$

If the values of $f(x)$ decrease without bound as x approaches a , we say that the limit of $f(x)$ is **negative infinity**.

$\lim_{x \rightarrow a} f(x) = -\infty$



For example, $\lim_{x \rightarrow 0} \left(5 - \frac{2}{x^2}\right) = -\infty$



Graphical interpretation

Graphically, infinite limits occur when the graph of a function rises or falls indefinitely near a particular value of x . This usually corresponds to a **vertical asymptote** of the function.

Example 1

Find: $\lim_{x \rightarrow 0} \frac{1}{x}$

Solution:

As x approaches 0 from positive values, $\frac{1}{x}$ becomes very large.

Therefore $\lim_{x \rightarrow 0^+} \frac{1}{x} = +\infty$

As x approaches 0 from negative values, $\frac{1}{x}$ becomes very large negative. Hence $\lim_{x \rightarrow 0^-} \frac{1}{x} = -\infty$

Thus, the limit at $x = 0$ does not exist because the two one-sided limits are different.

Example 2

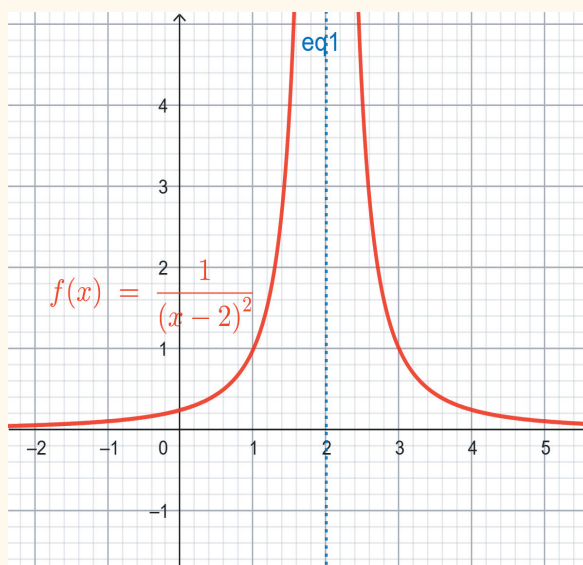
Evaluate:

$$\lim_{x \rightarrow 2} \frac{1}{(x-2)^2}$$

Solution:

As x approaches 0 from positive values,
As x approaches 2, the denominator
becomes very small but always positive.
Thus the value of the function becomes

very large. Then, $\lim_{x \rightarrow 2} \frac{1}{(x-2)^2} = +\infty$



Application activity 7.5



Work individually or in groups.

1. Evaluate $\lim_{x \rightarrow 0} \frac{1}{x^2}$

2. Determine $\lim_{x \rightarrow 4} \frac{1}{x-4}$

3. Evaluate:

a) $\lim_{x \rightarrow 1} \frac{1}{(x-1)^2}$

b) $\lim_{x \rightarrow 3} \frac{1}{x-3}$

4. Explain the meaning of $\lim_{x \rightarrow a} f(x) = +\infty$

7.6. Limits at infinity

Activity 7.6



Consider the function $f(x) = \frac{1}{x}$ and complete the table below.

x	10	100	1000	-10	-100	-1000
$f(x)$						

- What happens to the values of $f(x)$ as x becomes very large?

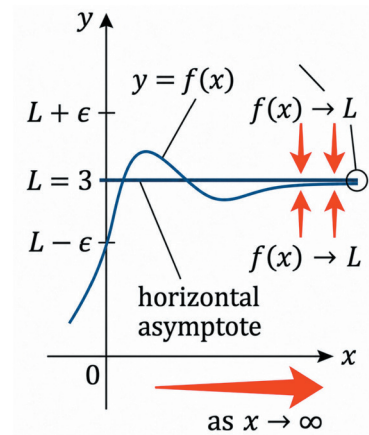


Content Summary

Definition of Limits at Infinity

If the values of $f(x)$ approach a number L as x increases without bound, we say that the limit of $f(x)$ as x approaches infinity is L .

This is written as $\lim_{x \rightarrow \infty} f(x) = L$



Similarly, if the values of $f(x)$ approach a number L as x decreases without bound, we write $\lim_{x \rightarrow -\infty} f(x) = L$

Graphical Interpretation

Graphically, limits at infinity describe the behaviour of the graph of a function as x moves far to the right or far to the left.

If $\lim_{x \rightarrow \infty} f(x) = L$ then the line $y = L$ is called a **horizontal asymptote** of the function.

Example 1 Evaluate: $\lim_{x \rightarrow \infty} \frac{1}{x}$

Solution:

As x becomes very large, the value of $\frac{1}{x}$ becomes very small, so $\lim_{x \rightarrow \infty} \frac{1}{x} = 0$

Example 2: Evaluate: $\lim_{x \rightarrow \infty} \left(2 + \frac{1}{x}\right)$

Solution:

Since $\frac{1}{x} \rightarrow 0$ we obtain $\lim_{x \rightarrow \infty} \left(2 + \frac{1}{x}\right) = 2$

Example 3

$$\begin{aligned} \lim_{x \rightarrow \infty} \frac{3x^2 - x - 2}{5x^2 + 4x + 1} &= \lim_{x \rightarrow \infty} \frac{x^2 \left(3 - \frac{1}{x} - \frac{2}{x^2}\right)}{x^2 \left(5 + \frac{4}{x} + \frac{1}{x^2}\right)} = \frac{\lim_{x \rightarrow \infty} \left(3 - \frac{1}{x} - \frac{2}{x^2}\right)}{\lim_{x \rightarrow \infty} \left(5 + \frac{4}{x} + \frac{1}{x^2}\right)} \\ &= \frac{\lim_{x \rightarrow \infty} 3 - \lim_{x \rightarrow \infty} \frac{1}{x} - \lim_{x \rightarrow \infty} \frac{2}{x^2}}{\lim_{x \rightarrow \infty} 5 + \lim_{x \rightarrow \infty} \frac{4}{x} + \lim_{x \rightarrow \infty} \frac{1}{x^2}} = \frac{3 - 0 - 0}{5 + 0 + 0} = \frac{3}{5} \end{aligned}$$

Note:

1. $\lim_{x \rightarrow \pm\infty} (ax + b) = \pm\infty$ if $a > 0$
2. $\lim_{x \rightarrow \pm\infty} (ax + b) = \mp\infty$ if $a < 0$
3. $\lim_{x \rightarrow \pm\infty} (ax^2 + bx + c) = +\infty$ if $a > 0$
4. $\lim_{x \rightarrow \pm\infty} (ax^2 + bx + c) = -\infty$ if $a < 0$

To compute the limit of a function as $x \rightarrow \pm\infty$, we use the following principles: $\forall a \in \mathbb{R}$

1. $a(\pm\infty) = \begin{cases} \pm\infty, & \text{if } a > 0, \\ \mp\infty, & \text{if } a < 0, \end{cases}$
2. $-\infty \pm a = -\infty$, $+\infty \pm a = +\infty$
3. $\frac{a}{\pm\infty} = 0$, $\frac{a}{0} = +\infty$ or $-\infty$.

Application activity 7.6



Work individually or in groups.

1. Evaluate: $\lim_{x \rightarrow \infty} \frac{5}{x}$
2. Determine: $\lim_{x \rightarrow \infty} \left(4 + \frac{2}{x}\right)$
3. Evaluate: $\lim_{x \rightarrow -\infty} \frac{7}{x}$
4. Determine the following limit $\lim_{x \rightarrow -\infty} \frac{4x^2 + 3x - 2}{x^2 + x - 1}$

7.7. Indeterminate forms

Activity 7.7



Evaluate the following limits by direct substitution.

$$\lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 2}$$

What value do you obtain when you substitute $x = 2$ into the expression?

- What difficulty do you observe?



Content Summary

A. Definition of indeterminate form

An **indeterminate form** occurs when direct substitution in a limit gives an expression that does not determine the value of the limit. To evaluate such limits, we simplify the expression before computing the limit.

Common indeterminate forms include: $\frac{0}{0}$, $\frac{\infty}{\infty}$, $0 \cdot \infty$, $\infty - \infty$.

The table below provides summary on these different forms of indeterminate, their description and the most common methods to resolve these indeterminate forms.

Indeterminate Form	Description	Common Methods to Resolve	Example
$\frac{0}{0}$	Both numerator and denominator approach 0. The ratio cannot be determined directly.	Factorization, simplification, rationalization.	$\lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 2}$
$\frac{\infty}{\infty}$	Both numerator and denominator grow without bound. Their ratio may approach a finite number, 0, or ∞	Divide by the highest power of x, simplification.	$\lim_{x \rightarrow \infty} \frac{2x^2 + 1}{x^2 + 3}$
$0 \cdot \infty$	One factor approaches 0 while the other approaches ∞ , making the product uncertain.	Rewrite the expression as a fraction and simplify.	$\lim_{x \rightarrow 0^+} x(\log x)$
$\infty - \infty$	The difference of two quantities that both approach infinity; the result is uncertain.	Combine fractions, rationalize, or factorize.	$\lim_{x \rightarrow \infty} (\sqrt{x^2 + x} - x)$

B. Indeterminant form of the type $\left(\frac{0}{0}\right)$

Factorization

By factorizing the expression and cancelling the common factor, the limit can be evaluated.

Example: Evaluate: $\lim_{x \rightarrow 3} \frac{x^2 - 9}{x - 3}$

Solution:

Factorize the numerator.

$$x^2 - 9 = (x - 3)(x + 3)$$

$$\text{Thus, } \frac{(x - 3)(x + 3)}{x - 3}$$

$$\text{For } x \neq 3, \frac{(x - 3)(x + 3)}{x - 3} = x + 3$$

Now compute the limit.

$$\lim_{x \rightarrow 3} (x + 3) = 6$$

Simplification

Sometimes algebraic simplification of the expression removes the indeterminate form.

Example: Evaluate $\lim_{x \rightarrow 2} \frac{x^2 + x - 6}{x^2 - 3x + 2}$

Solution:

$$\lim_{x \rightarrow 2} \frac{x^2 + x - 6}{x^2 - 3x + 2} = \frac{0}{0} \quad (\text{indeterminate form (I.F)})$$

$$\lim_{x \rightarrow 2} \frac{x^2 + x - 6}{x^2 - 3x + 2} = \lim_{x \rightarrow 2} \frac{(x-2)(x+3)}{(x-2)(x-1)} = \lim_{x \rightarrow 2} \frac{x+3}{x-1} = \frac{2+3}{2-1} = \frac{5}{1} = 5$$

Rationalization

Rationalization is useful when the expression contains radicals.

We multiply the numerator and denominator by the conjugate to simplify the expression.

Example Evaluate: $\lim_{x \rightarrow 0} \frac{\sqrt{1+x} - 1}{x}$

Solution:

$$\lim_{x \rightarrow 0} \frac{\sqrt{1+x} - 1}{x} = \frac{0}{0} \quad (\text{indeterminate form})$$

Multiply numerator and denominator by the conjugate.

$$\lim_{x \rightarrow 0} \frac{\sqrt{1+x} - 1}{x} = \lim_{x \rightarrow 0} \frac{(\sqrt{1+x} - 1)(\sqrt{1+x} + 1)}{x(\sqrt{1+x} + 1)} = \lim_{x \rightarrow 0} \frac{1+x-1}{x(\sqrt{1+x} + 1)} = \lim_{x \rightarrow 0} \frac{x}{x(\sqrt{1+x} + 1)} = \lim_{x \rightarrow 0} \frac{1}{\sqrt{1+x} + 1}.$$

Now compute the limit.

$$\lim_{x \rightarrow 0} \frac{1}{\sqrt{1+x} + 1} = \frac{1}{\sqrt{1+0} + 1} = \frac{1}{2}$$

C. Indeterminant form of the type $\frac{\infty}{\infty}$

Three cases may happen when resolving the indeterminate form of the type $\frac{\infty}{\infty}$.

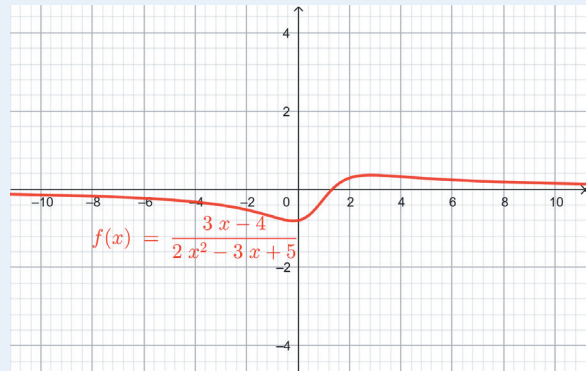
1. If the degree on the numerator is higher than the degree of the denominator, the limit is infinity
2. If both the degree of the numerator and the denominator are equal, the limit will be the ratio of the coefficients of the highest powers.
3. If the degree on the numerator is less than the degree of the denominator, the limit becomes zero

Example 1

Evaluate $\lim_{x \rightarrow \infty} \frac{3x-4}{2x^2-3x+5}$

Solution:

$$\begin{aligned}\lim_{x \rightarrow \infty} \frac{3x-4}{2x^2-3x+5} &= \lim_{x \rightarrow \infty} \frac{x^2 \left(\frac{3}{x} - \frac{4}{x^2} \right)}{x^2 \left(2 - \frac{3}{x} + \frac{5}{x^2} \right)} \\ &= \lim_{x \rightarrow \infty} \frac{\frac{3}{x} - \frac{4}{x^2}}{\left(2 - \frac{3}{x} + \frac{5}{x^2} \right)} = \frac{0-0}{2-0+0} = \frac{0}{2} = 0\end{aligned}$$



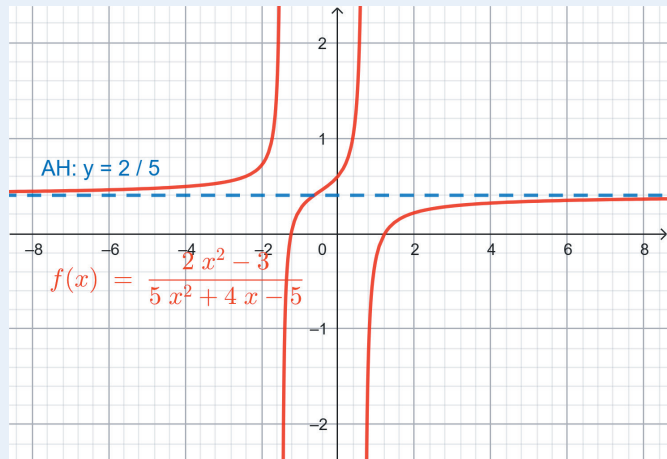
As x approaches $+\infty$, the value of $f(x)$ is approximately 0.

Example 2

Find $\lim_{x \rightarrow \infty} \frac{2x^2-3}{5x^2+4x-5}$

Solution:

$$\begin{aligned}\lim_{x \rightarrow \infty} \frac{2x^2-3}{5x^2+4x-5} &= \lim_{x \rightarrow \infty} \frac{x^2 \left(2 - \frac{3}{x^2} \right)}{x^2 \left(5 + \frac{4}{x} - \frac{5}{x^2} \right)} \\ &= \lim_{x \rightarrow \infty} \frac{2 - \frac{3}{x^2}}{5 + \frac{4}{x} - \frac{5}{x^2}} = \frac{2-0}{5+0-0} = \frac{2}{5}\end{aligned}$$



As x approaches $\pm\infty$, the value of $f(x)$ is approximately $\frac{2}{5}$.

Example 3

Determine $\lim_{x \rightarrow \infty} \frac{3x^3-2x+1}{-x^2+3x-5}$

Solution:

$$\lim_{x \rightarrow \infty} \frac{3x^3-2x+1}{-x^2+3x-5} = \lim_{x \rightarrow \infty} \frac{x^3 \left(3 - \frac{2}{x^2} + \frac{1}{x^3} \right)}{x^2 \left(-1 + \frac{3}{x} - \frac{5}{x^2} \right)} = \lim_{x \rightarrow \infty} \frac{x \left(3 - \frac{2}{x^2} + \frac{1}{x^3} \right)}{-1 + \frac{3}{x} - \frac{5}{x^2}} = \frac{3(\infty)}{-1} = -\infty$$

D. Indeterminant form of the type $0 \cdot \infty$

Example

Evaluate the limit

$$\lim_{x \rightarrow 0^+} x \ln x.$$

Solution:

$$\lim_{x \rightarrow 0^+} x \ln x = 0 \cdot \infty \quad (\text{I.F})$$

$$\lim_{x \rightarrow 0^+} x \ln x = \lim_{x \rightarrow 0^+} \frac{\ln x}{\frac{1}{x}}$$

As $x \rightarrow 0^+$, $\ln x \rightarrow -\infty$ and $\frac{1}{x} \rightarrow \infty$.

Since the denominator grows faster than the numerator, the fraction approaches 0.

Therefore,

$$\lim_{x \rightarrow 0^+} x \ln x = 0$$

E. Indeterminant form of the type $\infty - \infty$

Example

Evaluate the limit

$$\lim_{x \rightarrow \infty} (\sqrt{x^2 + x} - x).$$

Solution:

$$\lim_{x \rightarrow \infty} (\sqrt{x^2 + x} - x) = \lim_{x \rightarrow \infty} \sqrt{x^2 + x} - \lim_{x \rightarrow \infty} x = \infty - \infty \quad (\text{I.F})$$

Multiply and divide by the conjugate

$$\lim_{x \rightarrow \infty} (\sqrt{x^2 + x} - x) \frac{\sqrt{x^2 + x} + x}{\sqrt{x^2 + x} + x} = \lim_{x \rightarrow \infty} \frac{(\sqrt{x^2 + x})^2 - x^2}{\sqrt{x^2 + x} + x} = \lim_{x \rightarrow \infty} \frac{x^2 + x - x^2}{\sqrt{x^2 + x} + x} = \lim_{x \rightarrow \infty} \frac{x}{\sqrt{x^2 + x} + x}.$$

Divide numerator and denominator by x :

$$\lim_{x \rightarrow \infty} \frac{x}{\sqrt{x^2 + x} + x} = \lim_{x \rightarrow \infty} \frac{1}{\sqrt{1 + \frac{1}{x}} + 1} = \frac{1}{1 + 1} = \frac{1}{2}.$$

Application activity 7.7



Compute:

a) $\lim_{x \rightarrow 4} \frac{x^2 - 16}{x - 4}$

b) $\lim_{x \rightarrow 3} \frac{\sqrt{x+1} - 1}{x}$

c) $\lim_{x \rightarrow 2} \frac{x^2 + x - 6}{x^2 - x - 2}$

d) $\lim_{x \rightarrow 3} \frac{\sqrt{x+1} - 2}{x - 3}$

e) $\lim_{x \rightarrow \infty} \frac{3x^4 - 2x^2 + 1}{7x^4 + 6x^3 + x}$

7.8 Applications of limits on continuity and discontinuity of a function

Activity 7.8



Consider the function $f(x) = \frac{x^2 - 1}{x - 1}$

- Determine the value of $f(x)$ for $x = 0.9, 0.99, 1.01, 1.1$.
- What value do the function values approach as x gets closer to 1?
- Is the function defined at $x = 1$?



Content Summary

Continuity of a function

A function $f(x)$ is said to be continuous at a point $x = a$ if the following three conditions are satisfied:

1. $f(a)$ is defined.
2. $\lim_{x \rightarrow a} f(x)$ exists.
3. $\lim_{x \rightarrow a^-} f(x) = \lim_{x \rightarrow a^+} f(x) = f(a)$.

Graphical Interpretation

Graphically, a function is continuous at a point if the graph can be drawn without lifting the pencil at that point.



Continuous Function

Example

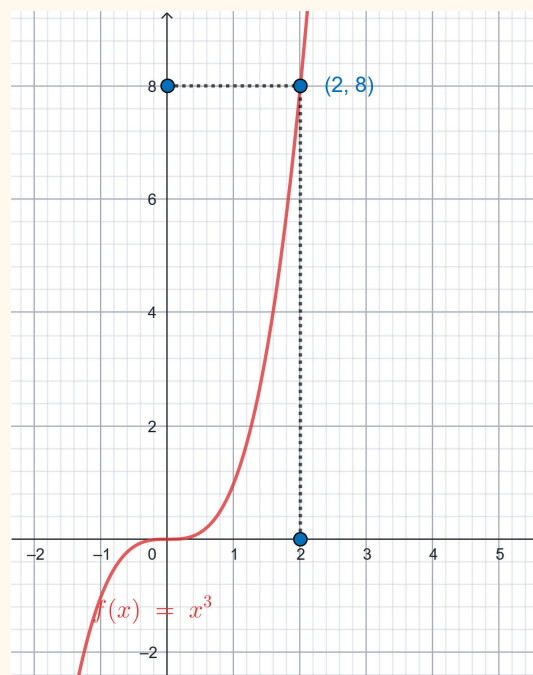
Determine whether the function $f(x) = x^3$ is continuous at $x = 2$.

Solution:

$$f(2) = 8$$

$$\lim_{x \rightarrow 2^+} x^3 = \lim_{x \rightarrow 2^-} x^3 = 8$$

Since $\lim_{x \rightarrow 2} f(x) = f(2)$, the function is continuous at $x = 2$.

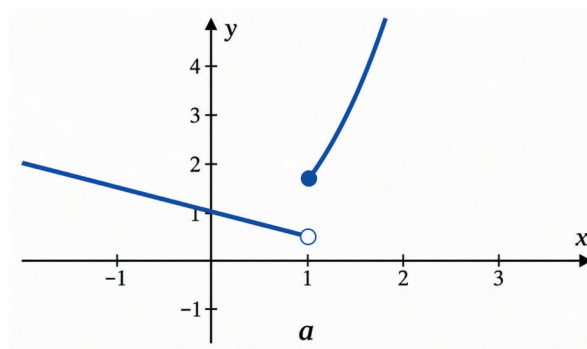


Discontinuity of a function

A function is said to be **discontinuous** at a point if any of the conditions for continuity is not satisfied.

A function $f(x)$ is said to be **discontinuous at a point** $x = a$ if any of the three criteria fail:

1. $f(a)$ is defined.
2. $\lim_{x \rightarrow a} f(x)$ exists.
3. $\lim_{x \rightarrow a^+} f(x) = \lim_{x \rightarrow a^-} f(x) = f(a)$.

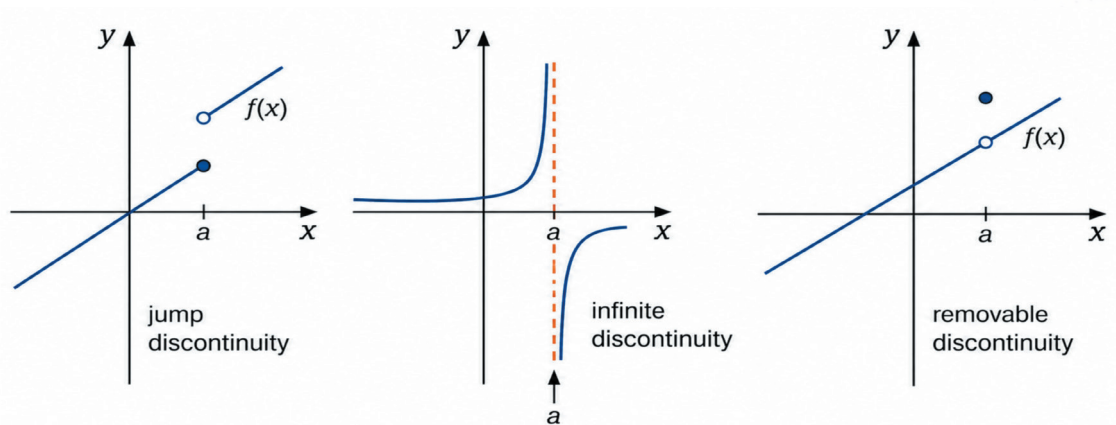


Example:

a) $\lim_{x \rightarrow a^-} f(x) = L_1$

b) $\lim_{x \rightarrow a^+} f(x) = L_2$

c) $f(a) = L_3$ where $L_1 \neq L_2$



- **Jump discontinuities** occur when a function has two ends that don't meet even if the hole is filled in. In order to satisfy the vertical line test and make sure the graph is truly that of a function, only one of the end points may be filled.
- **Infinite discontinuities** occur when a function has a vertical asymptote on one or both sides.
- **Removable discontinuities** occur when a rational function has a factor with an x that exists in both the numerator and the denominator.
- Removable discontinuities are shown in a graph by a hollow circle that is also known as a hole. For example, $f(x) = \frac{(x+2)(x-1)}{(x-1)}$ at $x=1$.

Example 1

State whether or not the following function is continuous at $x = 2$. $f(x) = \begin{cases} x^2, & x \neq 2, \\ 0, & x = 2. \end{cases}$

Solution:

$f(2) = 0$, $\lim_{x \rightarrow 2} f(x) = \lim_{x \rightarrow 2} x^2 = 4$. We observe that $\lim_{x \rightarrow 2} f(x) \neq f(2)$. Since $4 \neq 0$.

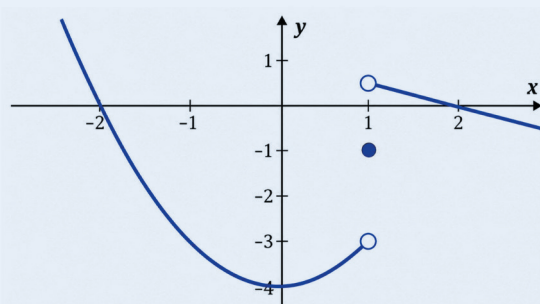
Therefore, the function $f(x)$ is not continuous at $x = 2$.

Example 2: Identify the discontinuity of the piecewise function graphically.

$$f(x) = \begin{cases} x^2 - 4 & x < 1 \\ -1 & x = 1 \\ -\frac{1}{2}x + 1 & x > 1 \end{cases}$$

Solution:

There is a jump discontinuity at $x=1$. The piecewise function describes a function in three parts; a parabola on the left, a single point in the middle and a line on the right.



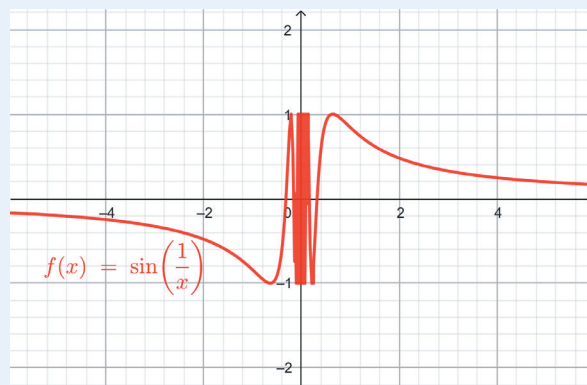
Example 3:

Describe the continuity or discontinuity of the

function $f(x) = \sin\left(\frac{1}{x}\right)$.

Solution:

The function seems to oscillate infinitely as x approaches zero. One thing that the graph fails to show is that 0 is clearly not in the domain. The graph does not shoot to infinity, nor does it have a simple hole or jump discontinuity.



Continuity over an interval

The function $f(x)$ is said to be continuous over interval $]a, b[$ if and only if $f(x)$ is continuous at any point of the interval $]a, b[$.

Example

Determine whether or not the function $f(x) = \frac{1}{x^2 - 1}$ is continuous over the interval $] -2, 2[$.

Solution:

$$\text{Dom} f = \mathbb{R} \setminus \{-1, 1\}.$$

The points $x = -1$ and $x = 1$ belong to the interval $] -2, 2[$ but they are not in the domain of $f(x)$. At $x = -1$ and $x = 1$, the denominator becomes zero and the function is not defined. Therefore, the function has discontinuities at these points.

Properties

1. The sum of two continuous functions is a continuous function.
2. The quotient of two continuous functions is a continuous function where the denominator is not zero.

Application activity 7.8



1. Determine whether the function $f(x) = 3x + 2$ is continuous at $x = 1$.
2. Determine whether the function $f(x) = \frac{x^2 - 4}{x - 2}$ is continuous at $x = 2$.
3. Consider the function $f(x) = \begin{cases} 2x & x < 1 \\ x + 2 & x \geq 1 \end{cases}$

Determine whether the function is continuous at $x = 1$.

7.9. Applications of limits on asymptotes

Activity 7.9



Consider the function $f(x) = \frac{1}{x-2}$.

Compute the following values:

x	1.5	1.9	1.99	2	2.01	2.1	2.5
$f(x)$							

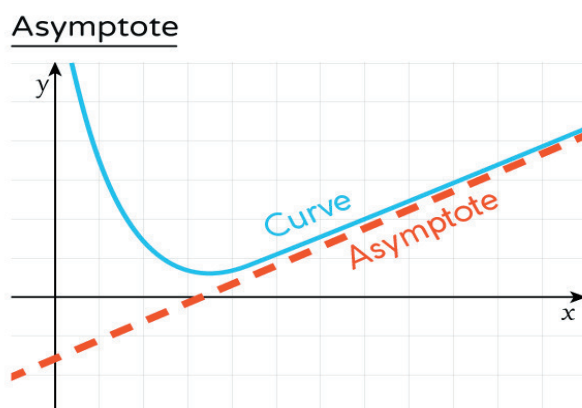
- What happens to $f(x)$ as x approaches 2 from the left?
- What happens to $f(x)$ as x approaches 2 from the right?
- Draw the graph of the function.
- What do you observe about the line $x = 2$.



Content Summary

Definition of an asymptote

An asymptote is a line that a curve approaches but never reaches as the variable approaches a certain value or infinity.



The main types of asymptotes are:

- Vertical asymptotes
- Horizontal asymptotes
- Oblique (slant) asymptotes

Limits are used to determine these asymptotes.

A. Vertical Asymptote (V.A)

A vertical line $x = a$ is a vertical asymptote of $f(x)$ if $\lim_{x \rightarrow a^-} f(x) = \pm\infty$ or $\lim_{x \rightarrow a^+} f(x) = \pm\infty$.

Example

Determine the vertical asymptote of

$$f(x) = \frac{1}{x-2}.$$

Solution:

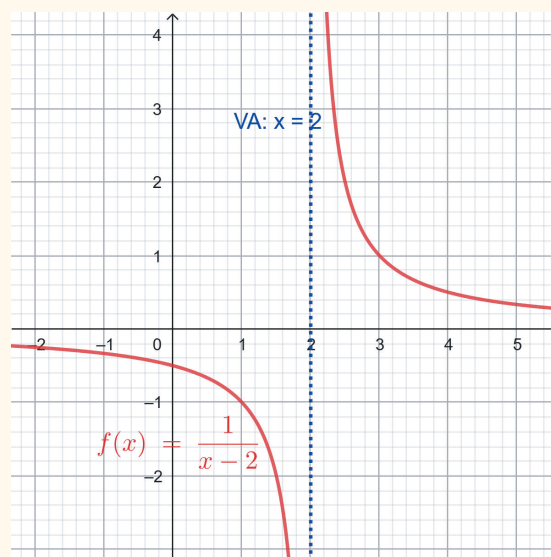
The denominator becomes zero when $x - 2 = 0 \Rightarrow x = 2$.

Evaluate the limits:

$$\lim_{x \rightarrow 2^-} \frac{1}{x-2} = -\infty, \quad \lim_{x \rightarrow 2^+} \frac{1}{x-2} = +\infty.$$

Thus, the vertical asymptote is

$$\text{V.A} \equiv x = 2.$$



B. Horizontal Asymptote (H.A)

A horizontal line $y = L$ is a horizontal asymptote if $\lim_{x \rightarrow \infty} f(x) = L$ or $\lim_{x \rightarrow -\infty} f(x) = L$.

Example

Determine the horizontal asymptote of $f(x) = \frac{x^2 - 1}{x^2 + 2}$.

Solution:

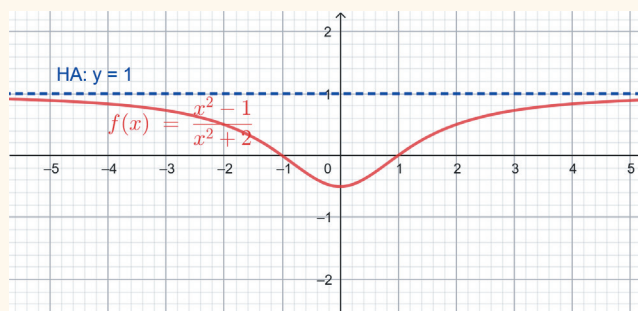
Divide numerator and denominator

$$\text{by } x^2: f(x) = \frac{1 - \frac{1}{x^2}}{1 + \frac{2}{x^2}}.$$

$$\text{Taking the limit: } \lim_{x \rightarrow \infty} f(x) = \frac{1-0}{1+0} = 1.$$

Thus, the horizontal asymptote is

$$\text{H.A} \equiv y = 1.$$



C. Oblique (Slant) Asymptote (O.A)

The line with equation $\text{O.A} \equiv y = ax + b$ is called the **oblique asymptote** of the curve representing f if $\lim_{x \rightarrow \pm\infty} [f(x) - (ax + b)] = 0$.

The real numbers a and b are obtained using the following limits: $a = \lim_{x \rightarrow \pm\infty} \frac{f(x)}{x}$,

$$b = \lim_{x \rightarrow \pm\infty} [f(x) - ax].$$

Example

Find the equation of the oblique asymptote to the curve representing

$$f(x) = \frac{x^2 + 3x - 1}{2x + 1}.$$

Solution:

The oblique asymptote has the form

$$y = ax + b.$$

First determine a :

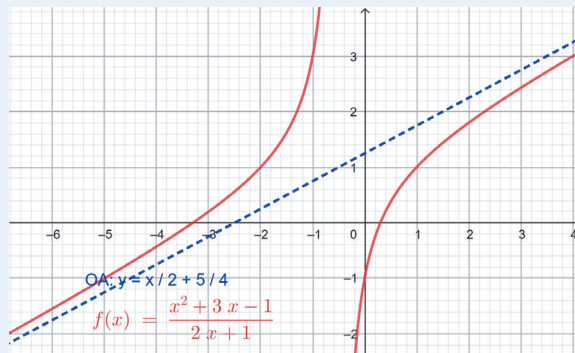
$$\begin{aligned} a &= \lim_{x \rightarrow \infty} \frac{f(x)}{x} \Rightarrow a = \lim_{x \rightarrow \infty} \frac{x^2 + 3x - 1}{x(2x + 1)} \\ &= \lim_{x \rightarrow \infty} \frac{x^2 + 3x - 1}{2x^2 + x} = \frac{1}{2} \end{aligned}$$

Now determine b :

$$\begin{aligned} b &= \lim_{x \rightarrow \infty} [f(x) - ax] = \lim_{x \rightarrow \infty} \left[\frac{x^2 + 3x - 1}{2x + 1} - \frac{1}{2}x \right] = \\ \lim_{x \rightarrow \infty} \left[\frac{2x^2 + 6x - 2 - 2x^2 - x}{4x + 2} \right] &= \lim_{x \rightarrow \infty} \frac{5x - 2}{4x + 2} = \frac{5}{4}. \end{aligned}$$

Hence the oblique asymptote is

$$\text{O.A} \equiv y = \frac{1}{2}x + \frac{5}{4}.$$



Note: For rational functions:

- There exists an oblique asymptote if the degree of the numerator is one greater than the degree of the denominator.
- An alternative way to determine the oblique asymptote is by using **polynomial long division**. In that case, the equation of the asymptote is simply the quotient.

Example

Find the oblique asymptote for $y = \frac{x^2 + 1}{x}$

Solution:

Using long division we have $f(x) = x + \frac{1}{x}$. The oblique asymptote is $\text{O.A} \equiv y = x$.

Notes:

- **Vertical asymptotes** are obtained by finding the values that make the denominator equal to zero.
- For the **horizontal asymptote**:
 - If the degree of the numerator is less than the degree of the denominator, then $\text{H.A} \equiv y = 0$.

- If the degree of the numerator is equal to the degree of the denominator, then

$$\text{H.A} \equiv y = \frac{\text{leading coefficient of numerator}}{\text{leading coefficient of denominator}}.$$
- The horizontal asymptote can be considered as a special case of the oblique asymptote.

Application activity 7.9



1. Determine the vertical asymptote of $f(x) = \frac{5}{x-3}$
2. Determine the horizontal asymptote of $f(x) = \frac{4x+1}{2x+3}$
3. Determine the asymptotes of $f(x) = \frac{x^2+3}{x}$.

End of unit assessment



A. Knowledge and Understanding

1. Choose the correct answer
 - a) A neighbourhood of a real number a is:
 - A. A set containing only the number a
 - B. An open interval containing a
 - C. Any finite set
 - D. A closed interval containing a
 - b) The limit of a function $f(x)$ at $x = a$ exists if:
 - A. $f(x)$ exists
 - B. $f(x)$ is positive
 - C. The left-hand and right-hand limits are equal
 - D. $f(x)$ is continuous.
 - c) Evaluate: $\lim_{x \rightarrow 2} (3x + 1)$
 - A. 5
 - B. 7
 - C. 6
 - D. 4
 - d) Which of the following represents an indeterminate form?
 - A. $\frac{5}{0}$
 - B. ∞
 - C. $\frac{0}{0}$
 - D. 5
 - e) If $\lim_{x \rightarrow a^-} f(x) \neq \lim_{x \rightarrow a^+} f(x)$, then
 - A. f is continuous
 - B. The limit does not exist
 - C. The function is constant
 - D. The function is differentiable

B. Skills

1. Determine whether the function

$$f(x) = \begin{cases} x^2 & x \neq 1 \\ 2 & x = 1 \end{cases} \text{ is continuous at } x = 1.$$

2. Find all the equations of the asymptotes of the following functions

$$y = \frac{x^3}{x^2 - 1}$$

3. Find the following limits, if they exist

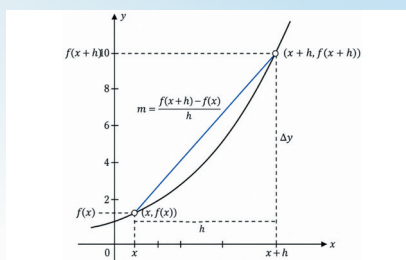
$$\begin{array}{llll} \text{a) } \lim_{x \rightarrow 2} \frac{x^2 - x - 2}{x^2 - 3x + 2} & \text{b) } \lim_{x \rightarrow 3} \frac{\sqrt{25 - x^2} - 4}{x - 3} & \text{c) } \lim_{x \rightarrow -\infty} \frac{4x + 1}{x - 6} & \text{d) } \lim_{x \rightarrow 3} \frac{\sqrt{x} - \sqrt{3}}{x - 3} \\ \text{e) } \lim_{x \rightarrow 1} \left(\frac{1}{1 - x} - \frac{3}{1 - x^3} \right) & \text{f) } \lim_{z \rightarrow 0} \frac{\sqrt{2z + 4} - 2}{z} & \text{g) } \lim_{x \rightarrow +\infty} (\sqrt{x^2 + x - 2} - x) & \text{h) } \lim_{x \rightarrow 0} \frac{x}{|x|} \end{array}$$

C. Attitudes and Values

- 1) During a group work session on finding the limit of a rational function, your group encounters an indeterminate form $\frac{0}{0}$. Two group members immediately say, “*The limit does not exist because we cannot divide by zero,*” and suggest skipping the problem. Which of the following actions best demonstrates the right attitude of a collaborative and resilient mathematics student?
- a) Agree with them to save time so that the group can move on to easier questions on the worksheet.
 - b) Work on the problem completely by yourself in silence, find the correct answer using factorization, and turn it in without explaining it to them.
 - c) Gently challenge the assumption by reminding the team that $\frac{0}{0}$ is just an indeterminate form, and suggest they work together to try algebraic simplification or factorization to find the actual behavior of the function.
 - d) Report the two group members to the teacher for wanting to give up on the assignment.
- 2) You are using a digital graphing tool to estimate $\lim_{x \rightarrow 2} f(x)$. As you zoom into the graph very close to $x = 2$, the line looks like it is hitting exactly $y = 3$. However, when your classmate does the algebraic calculation, they notice a tiny hole in the graph making it undefined at that exact point, though the values infinitely close to it approach 3. How should you value and report this finding?
- a) Ignore the algebraic proof because the digital tool is advanced and visual evidence is always perfectly accurate.
 - b) Appreciate the algebraic precision because limits describe the behavior of a function *near* a point, not necessarily *at* the point, and record the limit as 3 while acknowledging the discontinuity.
 - c) Change your algebraic steps artificially on your paper so that the math appears to show the function is perfectly solid at $x = 2$.
 - d) Conclude that because the function is undefined at $x = 2$, calculating the limit is a waste of time and has no practical value.

UNIT 8

DIFFERENTIATION OF POLYNOMIALS, RATIONAL AND IRRATIONAL FUNCTIONS AND THEIR APPLICATIONS



Key Unit Competence:

Apply derivatives of polynomial, rational, and irrational functions in solving related mathematical and real-life problems.

Learning objectives

By the end of this unit, you should be able to:

Knowledge and understanding

1

- Describe the derivative of a function and explain the concept of differentiation from first principles
- State and explain the rules of differentiation
- Describe higher-order derivatives and their meanings
- Explain geometric interpretations of derivatives (tangent, normal)
- State and interpret Lagrange's and Rolle's theorems
- Explain L'Hôpital's rule and its application in evaluating limits
- Describe the variation of a function in terms of increasing/decreasing intervals, stationary points, concavity, and inflection points
- Explain the relationship between derivatives and optimization problems in real-life contexts.

Skills

2

- Differentiate functions using first principles and standard rules
- Compute higher-order derivatives of given functions
- Determine equations of tangents and normal to curves
- Apply Lagrange's and Rolle's theorems to solve mathematical problems
- Use L'Hôpital's rule to evaluate indeterminate limits.
- Analyse functions to find stationary points, concavity, and inflection points
- Construct and interpret a table of variation.
- Sketch polynomial, rational, and irrational functions using derivative properties
- Solve rate of change and optimization problems using differentiation techniques.

Attitudes and values

3

- Show curiosity and critical thinking when exploring how functions change
- Demonstrate accuracy, patience, and logical reasoning when performing differentiation.
- Appreciate the importance of derivatives in modelling and solving real-life problems (e.g., motion, economics, and engineering)
- Develop perseverance and confidence in tackling challenging mathematical problems
- Value the use of mathematical reasoning and graphical representation in problem-solving.

Introduction

Differentiation is one of the fundamental concepts in calculus. It is used to determine the rate at which a quantity changes with respect to another quantity. In mathematics, the derivative of a function represents the rate of change of the function with respect to its variable and is closely related to the slope of the tangent line to the graph of the function at a given point.

In this unit, we study how to differentiate different types of functions such as polynomial, rational and irrational functions.

Understanding how to differentiate these functions allows us to solve many practical problems. Differentiation is widely used to determine the slope of curves, find maximum and minimum values of functions, analyze increasing and decreasing behavior of functions, and solve problems in fields such as physics, economics and engineering.

Through this unit, learners will develop the skills needed to compute derivatives of various functions and apply differentiation techniques to solve mathematical and real-life problems.

Introductory activity



A car moves along a straight road. The table below shows the distance covered over time:

Time (seconds)	Distance (meters)
0	0
1	5
2	15
3	30
4	50

1. Calculate the change in distance between each time interval.
2. Is the distance increasing at a constant rate? Explain.
3. Estimate how fast the car is moving between:

- $t = 1$ and $t = 2$
- $t = 2$ and $t = 3$

4. What do you notice about how the speed changes?
5. How can we describe the speed of the car at exactly $t = 2$?

8.1 Concepts of derivative of a function: Definition and differentiation from first principle

Activity 8.1



Consider the function $f(x) = x^2$

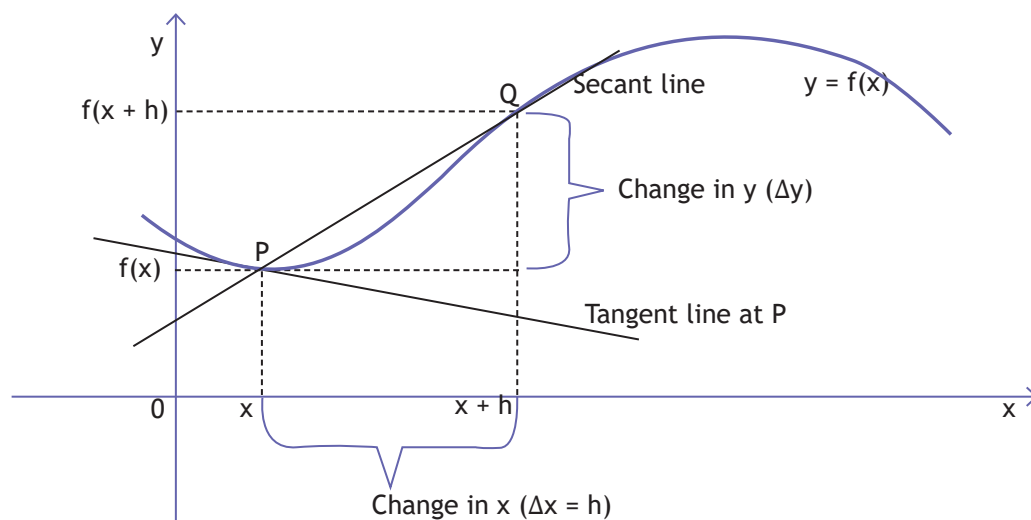
- Calculate the values of the function at $x = 1$, $x = 1.5$, $x = 2$.
- Find the slope of the line joining the points $(1, f(1))$ and $(2, f(2))$.
- Find the slope of the line joining the points $(1, f(1))$ and $(1.5, f(1.5))$.
- What do you observe about the slope as the second point gets closer to the first point $x = 1$?
- What do you think the slope represents on the graph of the function?



Content Summary

Slope of a function at a point

To define the slope of the curve C at P , take a point Q on the curve different from P . The line PQ is called a secant line at P . Its slope, denoted by m_{PQ} , can be found using the coordinates of P and Q . If we let Q move along the curve, the slope m_{PQ} changes.



Suppose that as Q approaches P , the number m_{PQ} approaches a fixed value and the increment $h = x_Q - x_P$ of x approaches 0. This value, denoted simply by m_P , is called the slope of C at P ; and the line with slope m_P and passing through P is called the tangent line to the curve C at P .

In view of the concept “*limit of a function at a point*”, the slope m_{PQ} of the secant line PQ is

$$m_{PQ} = \frac{f(x+h) - f(x)}{(x+h) - x} = \frac{f(x+h) - f(x)}{h}$$

Note that as Q approaches P , the number h approaches 0. From these, we see that the slope of C at P (denoted by m_P) is

$$m_P = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

If this limit exists, the number m_P is called **slope of tangent line** to the graph of f at P or at $x = x_0$.

Definition (Derivative from first Principles)

The derivative of a function $f(x)$ at a point x is defined by $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$, provided the limit exists.

This definition is called the **derivative from first principles**.

Alternative notations

The derivative: $f'(x) = y' = \frac{dy}{dx} = \frac{df}{dx} = \frac{d}{dx} f(x)$ where $\frac{dy}{dx}$ and $\frac{d}{dx}$ are called differentiation operators but $\frac{dy}{dx}$ should not be regarded as a ratio.

Example 1

Using first principles, find the derivative of $f(x) = 3x + 1$.

Solution:

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}, \text{ then } f(x+h) = 3(x+h) + 1 = 3x + 3h + 1$$

$$\text{Substitute: } f'(x) = \lim_{h \rightarrow 0} \frac{3x + 3h + 1 - (3x + 1)}{h} = \lim_{h \rightarrow 0} \frac{3h}{h} = \lim_{h \rightarrow 0} \frac{3h}{h} = \lim_{h \rightarrow 0} 3 = 3.$$

$$\text{Thus } f'(x) = 3$$

Example 2

Let $f(x) = x^2 + 1$. Find $f'(x)$.

Solution:

$$\text{The derivative of } f(x) \text{ is } f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{(x+h)^2 + 1 - x^2 - 1}{h} =$$

$$\lim_{h \rightarrow 0} \frac{x^2 + 2hx + h^2 + 1 - x^2 - 1}{h} = \lim_{h \rightarrow 0} \frac{2hx + h^2}{h} = \lim_{h \rightarrow 0} (2x + h) = 2x$$

Thus, $f'(x) = 2x$. From the definition, one can observe the following simple rules.

A. Derivative of a constant function

If f is a constant function, $f(x) = c$, for all x then $\frac{df}{dx} = \frac{d}{dx}(c) = 0$

Example: calculate the derivative of $f(x) = 8$

Solution:

$$\frac{df(x)}{dx} = \frac{d}{dx}(8) = 0$$

B. Derivative of identity function

The derivative of the identity function $f(x) = x$ is the constant function 1, that is $\frac{d(x)}{dx} = 1$

C. Multiplication by a scalar

If f is a differentiable function of x , and c is a constant, then $\frac{d}{dx}(cf(x)) = c \frac{d}{dx}f(x)$

Example: Find the derivative of $f(x) = 3x$

Solution:

$$f'(x) = 3 \frac{d}{dx}(x) = 3 \cdot 1 = 3$$

D. Derivative of a power

If n is any real number, then $\frac{d}{dx}x^n = nx^{n-1}$ for all x where the powers x^n and x^{n-1} are defined.

This holds for any function with power.

Example 1

Find the derivative of $f(x) = x^4$

Solution:

$$f'(x) = 4x^3$$

Example 2

Differentiate $f(x) = x^{-2}$ respecting the value of x .

Solution:

$$f'(x) = (x^{-2})' = -2(x)'(x)^{-2-1}$$

$$f'(x) = -2x^{-3}$$

Application activity 8.1



1. Determine from first principles the derivative of the following functions:

a) 2

b) $x^3 + 2x + 3$

c) $x^4 - \frac{1}{3}x$

d) $\frac{1}{x^2}$

e) $\sqrt{x+2}$

2. If $f(x) = 4x + 2x^2$, find:

a) $f'(x)$ using the definition of derivative

b) $f'(-2)$

3. Without using definition, find the derivative function, $f'(x)$, given that $f(x)$ is:

a) $2x^2$

b) x^{-1}

4. Given $f(x) = 5x^{-4}$, find $f'(-3)$.

8.2. Rules of differentiation

8.2.1 Sum and difference rule

Activity 8.2.1



In the previous lesson, you learned how to differentiate simple functions.

Consider the functions:

$$f(x) = x^2 \quad \text{and} \quad g(x) = x$$

- Find $f'(x)$ and $g'(x)$ using known rules.
- Form a new function: $h(x) = f(x) + g(x)$
- Try to find $h'(x)$
- Compare your result with: $f'(x) + g'(x)$
- What do you observe?



Content Summary

If a function is expressed as the sum of two functions: $h(x) = f(x) \pm g(x)$, then

$h'(x) = f'(x) \pm g'(x)$. That is $\frac{d}{dx}(u \pm v) = \frac{du}{dx} \pm \frac{dv}{dx}$ for two functions $u(x)$ and $v(x)$.

Example1: calculate the derivative of $y = x^4 + 12x$ respecting x

Solution:

$$y' = 4x^{4-1} + 12x^{1-1}$$

$$y' = 4x^3 + 12$$

Example 2: $y = x^4 - 2x^2$

Solution:

$$\frac{dy}{dx} = 4x^3 - 4x$$

Let $D(I, \mathbb{R})$ be the set of functions differentiable on interval I . Thus, if $f \in D(I, \mathbb{R})$ for positive and negative, and fractional value of n , then $[f^n(x)]' = nf^{n-1}(x)f'(x)$.

Example 3:

Differentiate $f(x) = (2x+1)^4$ respecting the value of x .

Solution:

$$f'(x) = ((2x+1)^4)' = 4(2x+1)'(2x+1)^{4-1}$$

$$f'(x) = 4(2)(2x+1)^3$$

$$f'(x) = 8(8x^3 + 12x^2 + 6x + 1)$$

$$f'(x) = 64x^3 + 96x^2 + 48x + 8$$

Derivative of polynomial

Let $y = a_n x^n + a_{n-1} x^{n-1} + \dots + a_2 x^2 + a_1 x^1 + a_0$ be a polynomial. Then, we have

$$\frac{dy}{dx} = na_n x^{n-1} + (n-1)a_{n-1} x^{n-2} + \dots + 2a_2 x^1 + a_1$$

Differentiation of radical functions

If we take any function in the square root function, then

$$y = \sqrt{f(x)} \Rightarrow \frac{dy}{dx} = \frac{1}{2\sqrt{f(x)}} \frac{d}{dx} f(x) = \frac{1}{2\sqrt{f(x)}} f'(x)$$

Example 4: Find the derivative of $y = \sqrt{x}$

Solution:

$$y = \sqrt{x} \Rightarrow \frac{dy}{dx} = \frac{1}{2\sqrt{x}}$$

Example 5: Differentiate the following function $y = \sqrt{4x}$

Solution:

$$y = \sqrt{4x} \Rightarrow \frac{dy}{dx} = \frac{4}{2\sqrt{x}} = \frac{2}{\sqrt{x}}$$

Example 6: Find the derivative of $y = \sqrt{2x^2 + 5}$

Solution:

We have the given function as $y = \sqrt{2x^2 + 5}$ and differentiating with respect to variable x , we get $\frac{dy}{dx} = \frac{d}{dx} \sqrt{2x^2 + 5}$.

Now using the formula derivative of a square root, we have $\frac{dy}{dx} = \frac{1}{2\sqrt{2x^2 + 5}} \frac{d}{dx} (2x^2 + 5)$
 $\frac{dy}{dx} = \frac{4x}{2\sqrt{2x^2 + 5}} = \frac{2x}{\sqrt{2x^2 + 5}}$

Remark: $\frac{d}{dx} (\sqrt[n]{f(x)}) = n \frac{\frac{d}{dx} (f(x))}{\sqrt[n]{f^{n-1}(x)}}$

Example 7: Differentiate: $y = \sqrt[7]{x^5 - 2x}$

Solution:

$$y = \sqrt[7]{x^5 - 2x} \Rightarrow y' = 7 \frac{5x^4 - 2}{\sqrt[7]{(x^5 - 2x)^6}}$$

Application activity 8.2.1



1. Find the derivative function, $f'(x)$, given that $f(x)$ is:

- a) $2x^2 - 9$
- b) $x^2 + 3x - 2$
- c) $x^3 + 3x^2 + 4x - 1$
- d) $5x^4 - 6x^2$

2. A function is given by $f(x) = x^3 + 2x^2 - 3x$

- a) Find $f'(x)$,
- b) Evaluate $f'(2)$.

8.2.2 The product rule

Activity 8.2.2



Consider the functions: $f(x) = x^2$ and $g(x) = x$

- Find $f'(x)$ and $g'(x)$.
- Form a new function: $h(x) = f(x) \cdot g(x)$
- Find $h'(x)$
- Compare your result with: $f'(x) \cdot g(x)$ and $f(x) \cdot g'(x)$
- Does $h'(x) = f'(x) \cdot g(x)$ or $h'(x) = f(x) \cdot g'(x)$?
- Try adding the two expressions. What do you observe?



Content Summary

If a function is expressed as a product of two functions:

$$y = f(x) \cdot g(x), \quad \frac{dy}{dx} = f'(x)g(x) + f(x)g'(x).$$

For two functions $u(x)$ and $v(x)$ we have $\frac{d}{dx}(uv) = u \frac{dv}{dx} + v \frac{du}{dx}$

Notice that this is not just the product of two derivatives

Example 1

Find the derivative of $y = (2x + 1)(x^2)$.

Solution:

$$\text{Let } f(x) = 2x + 1, \quad g(x) = x^2$$

$$\text{Then } f'(x) = 2, \quad g'(x) = 2x$$

Apply the product rule:

$$y' = 2(x^2) + (2x + 1)(2x) = 2x^2 + 4x^2 + 2x = 6x^2 + 2x$$

$$y' = 6x^2 + 2x$$

Example 2: Given $y = (x^2 + 3)(2x^3 + 5x)$, find $\frac{dy}{dx}$.

Solution:

$$\frac{dy}{dx} = \frac{d}{dx}[(x^2 + 3)(2x^3 + 5x)] = (x^2 + 3)'(2x^3 + 5x) + (x^2 + 3)(2x^3 + 5x)'$$

$$= 2x(2x^3 + 5x) + (x^2 + 3)(6x^2 + 5) = 4x^4 + 10x^2 + 6x^4 + 5x^2 + 18x^2 + 15$$

$$= 10x^4 + 33x^2 + 15$$

Application activity 8.2.2



Work in groups.

1. Find the derivatives of the following functions:

a) $y = x(x + 2)$

b) $y = (x^2)(x^3)$

c) $y = (3x + 1)(x^2 + 2)$

d) $y = x^2(2x + 5)$

2. A function is given by $y = (x + 1)(x^2 + 1)$

a) Find $\frac{dy}{dx}$

b) Evaluate the derivative at $x = 1$.

8.2.3 The quotient rule

Activity 8.2.3



Consider the functions: $f(x) = x^2$ and $g(x) = x + 1$

- Form the function: $h(x) = \frac{f(x)}{g(x)} = \frac{x^2}{x + 1}$
- Find $f'(x)$ and $g'(x)$
- Try to differentiate $h(x)$ by simplifying it first (if possible).
- Can you apply the sum or product rule directly to $\frac{x^2}{x + 1}$?
- What difficulty do you encounter?



Content Summary

If a function is expressed as a quotient of two functions:

$y = \frac{f(x)}{g(x)}$, $g(x) \neq 0$, $\frac{dy}{dx} = \frac{f'(x)g(x) - f(x)g'(x)}{[g(x)]^2}$. For two functions $u(x)$ and $v(x)$, we have

$$\frac{d}{dx}\left(\frac{u}{v}\right) = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2} \text{ or can be written as } d\left(\frac{u}{v}\right) = \frac{v du - u dv}{v^2}$$

Example 1

Find out the derivative for $f(x) = \frac{2x^3 + 5x}{x^2 + 3}$

Solution:

$$f'(x) = \frac{(2x^3 + 5x)'(x^2 + 3) - (2x^3 + 5x)(x^2 + 3)'}{(x^2 + 3)^2} \Leftrightarrow f'(x) = \frac{(6x^2 + 5)(x^2 + 3) - (2x^3 + 5x)(2x)}{(x^2 + 3)^2}$$

Example 2

Find the derivative of $f(x) = \frac{2x^2 + 3x}{4x^3 + x + 1}$

Solution:

$$\begin{aligned} f'(x) &= \left(\frac{2x^2 + 3x}{4x^3 + x + 1} \right)' = \frac{(2x^2 + 3x)'(4x^3 + x + 1) - (2x^2 + 3x)(4x^3 + x + 1)'}{(4x^3 + x + 1)^2} \\ &= \frac{(4x + 3)(4x^3 + x + 1) - (2x^2 + 3x)(12x^2 + 1)}{(4x^3 + x + 1)^2} = \frac{16x^4 + 4x^2 + 4x + 12x^3 + 3x + 3 - 24x^4 - 2x^2 - 36x^3 - 3x}{(4x^3 + x + 1)^2} \\ &= \frac{-8x^4 - 24x^3 + 2x^2 + 4x + 3}{(4x^3 + x + 1)^2} \end{aligned}$$

Application activity 8.2.3



Work in groups.

Find the derivatives of the following functions:

a) $y = \frac{x^2}{x+2}$ b) $y = \frac{3x+1}{x^2}$ c) $y = \frac{x^2+2}{x+3}$ d) $y = \frac{2x^2+x}{x-1}$.

8.2.4 The chain rule

Activity 8.2.4



Consider the functions:

$$f(x) = x^2 \quad \text{and} \quad g(x) = 2x + 1$$

Form a new function: $y = f(g(x)) = (2x + 1)^2$

- Expand $(2x + 1)^2$ and differentiate the result.
- Now, instead of expanding, try to think of y as a function of another variable u where:
 - $u = 2x + 1$, $y = u^2$
- Find $\frac{dy}{du}$ and $\frac{du}{dx}$
- Multiply the two derivatives. What do you observe?



Content Summary

If a function is a composition of two functions: $y = f(g(x))$, then its derivative is given by:

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx} \quad \text{or equivalently,} \quad \frac{d}{dx}[f(g(x))] = f'(g(x)) \cdot g'(x).$$

Remark: The chain rule is used when a function is inside another function (a composite function).

Example 1

Find the derivative of $y = (2x + 1)^2$.

Solution:

Let $u = 2x + 1$, then $y = u^2$ and $\frac{dy}{du} = 2u$, $\frac{du}{dx} = 2$

Apply the chain rule: $\frac{dy}{dx} = 2u \cdot 2 = 4(2x + 1) \Rightarrow \frac{dy}{dx} = 4(2x + 1)$

Example 2

Given $y = (3x^2 + 4x - 5)^3$, find $\frac{dy}{dx}$

Solution:

Let $y = u^3$ where $u = 3x^2 + 4x - 5$. Thus, $\frac{dy}{du} = 3u^2$ and $\frac{du}{dx} = 6x + 4$.

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx} = 3u^2(6x + 4) = 3(3x^2 + 4x - 5)^2(6x + 4)$$

Example 3

Find the derivative of $y = \frac{4}{1 - 2x}$

Solution:

Let $u = 1 - 2x$, then $y = \frac{4}{u} = 4u^{-1}$. Then $\frac{dy}{du} = -4u^{-2}$ and $\frac{du}{dx} = -2$.

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx} = (-4u^{-2})(-2) = 8u^{-2} = \frac{8}{(1 - 2x)^2}$$

Application activity 8.2.4



Work in groups.

Find the derivative function for each of the following:

a) $y = (4x - 5)^2$ b) $y = \sqrt{3x - x^2}$ c) $y = 6(5 - x)^3$

d) $y = \sqrt[3]{2x^3 - x^2}$ e) $y = \frac{6}{(5x - 4)^2}$ Top of Form

8.3 Higher order derivatives

Activity 8.3



Recall that the derivative of a function represents the rate of change.

Consider the function: $f(x) = x^3$

- Find the first derivative $f'(x)$
- Now differentiate $f'(x)$ to obtain $f''(x)$
- Continue the process to find $f'''(x)$.
- What do you observe as you continue differentiating?



Content Summary

The **first derivative** of a function $f(x)$ is denoted by: $f'(x)$ or $\frac{df}{dx}$

The **second derivative** is the derivative of the first derivative: $f''(x) = \frac{d^2 f}{dx^2}$

Similarly, the **third derivative** is: $f'''(x) = \frac{d^3 f}{dx^3}$

In general, the n^{th} **derivative** is written as: $f^{(n)}(x) = \frac{d^n f}{dx^n}$

Interpretation

- First derivative: rate of change (slope of tangent).
- Second derivative: rate of change of the first derivative (acceleration).
- Higher derivatives: describe further changes in motion or variation.

Example 1

Find the first, second and third derivatives of $f(x) = x^3$.

Solution:

$$f'(x) = 3x^2 \Rightarrow f'' = 6x \Rightarrow f''' = 6 \text{ and } f^{(4)} = 0$$

Example 2

Find the second derivative of $f(x) = 2x^4 - 3x^2 + x$.

Solution:

First derivative: $f'(x) = 8x^3 - 6x + 1$ and second derivative: $f''(x) = 24x^2 - 6$

Example 3

Find the second derivative of $y = (x^2 + 1)^2$.

Solution:

First derivative (using chain rule): $y' = 2(x^2 + 1)(2x) = 4x(x^2 + 1)$

Second derivative (using product rule): $y'' = 4(x^2 + 1) + 4x(2x) = 4x^2 + 4 + 8x^2 = 12x^2 + 4$

Application activity 8.3



Work in groups.

1. Find the first and second derivatives of the following functions:

a) $f(x) = x^4$

b) $f(x) = 3x^3 - 2x^2 + x$

c) $y = (x^2 + 2)^3$

2. Determine the second derivative of $y = x^3 + x^2$ and evaluate it at $x = 1$.

8.4 Applications of differentiation

8.4.1 Geometric interpretation of derivatives:

A. Tangent to a curve

Activity 8.4.1. A



Consider the function $f(x) = x^2$

- Find the points $(1, f(1))$ and $(2, f(2))$.
- Find the slope of the line joining these two points.
- Now consider a point close to $x = 1$, for example $x = 1.1$. Find the slope of the line joining $(1, f(1))$ and $(1.1, f(1.1))$.
- What happens to the slope as the second point approaches $x = 1$.
- Estimate the slope of the curve at $x = 1$.
- Using your result, write the equation of the tangent line to the curve at $x = 1$.



Content Summary

Geometric interpretation of the derivative

The derivative $f'(x)$ represents the slope of the tangent to the curve $y = f(x)$ at a given point.

Equation of the tangent

If P is the point with x -coordinate a , then the slope of the tangent at this point is $f'(a)$, that is,

$\frac{y - f(a)}{x - a} = f'(a)$. Hence, the equation of the tangent to the curve $y = f(x)$ at the point $(a, f(a))$ is

given by $y - f(a) = f'(a)(x - a)$.

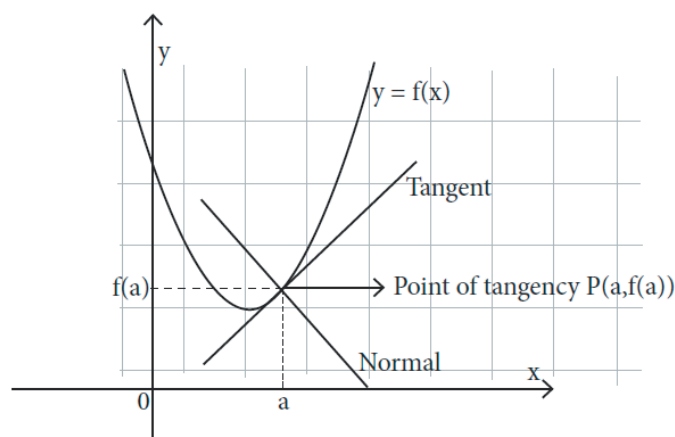


Fig from old book pg 184

Steps to find the equation of a tangent

- Find the derivative $f'(x)$.
- Substitute $x = a$ to get the slope $f'(a)$.
- Use the point $(a, f(a))$.
- Substitute into the tangent formula.

Example 1

Find the equation of the tangent to $f(x) = x^2 + 2$ at the point where $x = 1$.

Solution:

$f(x) = x^2 + 2$ and $f(1) = 1^2 + 2 = 3$. So, the point of contact is $(1, 3)$.

$f'(x) = 2x$ and $f'(1) = 2(1) = 2$. The tangent has equation

$$\frac{y-3}{x-1} = 2 \Rightarrow y-3 = 2(x-1) \Rightarrow y-3 = 2x-2, y = 2x+1.$$

Example 2

Find the equation of the tangent to the curve $y = x^3$ at $x = 2$.

Solution:

$f'(x) = 3x^2 \Rightarrow f'(2) = 12$. Point: $f(2) = 8$

So the point is $(2, 8)$. Equation: $y - 8 = 12(x - 2) \Rightarrow y = 12x - 24 + 8, y = 12x - 16$

Application activity 8.4.1. A



Find the equation of the tangent to each of the following functions at an indicated point:

- $y = x - 2x^2 + 3$ at $x = 2$
- $y = \sqrt{x+1}$ at $x = 4$
- $y = x^3 - 5x$ at $x = 1$
- $y = \frac{4}{x}$ at $(1, 4)$

3. Determine the slope of the tangent to the curve $y = x^2 + 1$ at $x = 3$.
4. A curve is given by $y = 2x^2 + x$
 - a) Find the equation of the tangent at $x = 1$
 - b) Find the slope at that point.

B. Normal to a curve

Activity 8.4.1. B



Consider the curve $y = x^2$

- Find the derivative of the function.
- Determine the slope of the tangent to the curve at $x = 1$.
- Recall that two lines are perpendicular if the product of their slopes is equal to -1.
- Find the slope of a line perpendicular to the tangent at $x = 1$
- What do you think this perpendicular line represents on the curve?



Content Summary

The normal to a curve at a given point is the line that is perpendicular to the tangent to the curve at that point.

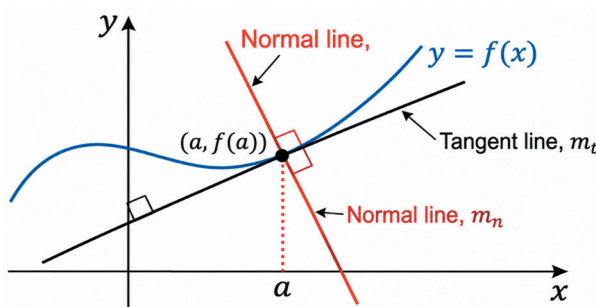
If the slope of the tangent at $x = a$ is:

$m_t = f'(a)$, then the slope of the normal

is: $m_n = -\frac{1}{f'(a)}$, provided $f'(a) \neq 0$.

The equation of the normal to the curve $y = f(x)$ at the point $(a, f(a))$ is:

$$y - f(a) = -\frac{1}{f'(a)}(x - a).$$



Remark: If $f'(a) = 0$, the tangent is horizontal and the normal is vertical: $x = a$.

Example 1

Find the equation of the normal to the curve $y = x^2 + 2x$, at $x = 0$.

Solution:

$f'(x) = 2x + 2 \Rightarrow f'(0) = 2$. Slope of normal: $m_n = -\frac{1}{2}$. Point: $f(0) = 0$, So the point is $(0, 0)$.

Equation: $y - 0 = -\frac{1}{2}(x - 0) \Rightarrow y = -\frac{1}{2}x$

Example 2

Find the equation of the normal to $y = \frac{8}{x}$ at the point where $x = 4$.

Solution:

When $x = 4$, $y = \frac{8}{4} = 2$, so the point of contact is $(4, 2)$. Now $y = 8x^{-1}$ and therefore $\frac{dy}{dx} = -8x^{-2}$.

When $x = 4$, $\frac{dy}{dx} = -8(4^{-2}) = -8\left(\frac{1}{16}\right) = -\frac{1}{2}$. Hence the slope of the normal is $-\frac{1}{-\frac{1}{2}} = 2$.

So, the equation of the normal is $\frac{y-2}{x-4} = 2 \Rightarrow y-2 = 2(x-4) \Rightarrow y-2 = 2x-8 \Leftrightarrow y = 2x-6$

Application activity 8.4.1. B



1. Find the equation of the normal to each of the following functions at an indicated point:

- $y = x^2$ at the point $(3, 9)$
- $y = x^3 - 5x + 2$ at $x = -2$
- $y = \frac{5}{\sqrt{x}} - \sqrt{x}$ at the point $(1, 4)$
- $y = \frac{8}{\sqrt{x}} - \frac{1}{x^2}$ at $x = 1$

2. Determine the slope of the normal to the curve $y = x^2 + 1$ at $x = 2$.

3. A function is given by $y = 2x^2 + x$

- Find the equation of the normal at $x = 1$
- Find the slope of the normal.

8.4.2 Mean value theorems for derivatives

Activity 8.4.2



Consider the function $f(x) = x^2$ on the interval $[1, 3]$.

- Compute $f(1)$ and $f(3)$.
- Find the slope of the line joining the points $(1, f(1))$ and $(3, f(3))$.
- Find the derivative $f'(x)$.
- Determine a value $c \in (1, 3)$ such that: $f'(c) = \text{slope of the secant line}$.
- What do you observe?



Content Summary

A. Lagrange's theorem

If a function $f(x)$ satisfies:

- f is continuous on $[a, b]$,
- f is differentiable on (a, b) , then there exists at least one $c \in (a, b)$ such that:

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

B. Rolle's theorem

If a function $f(x)$ satisfies:

If a function $f(x)$ satisfies:

- f is continuous on $[a, b]$,
- f is differentiable on (a, b) ,
- $f(a) = f(b)$, then there exists at least one $c \in (a, b)$ such that: $f'(c) = 0$.

Remark: Rolle's Theorem is a special case of Lagrange's Theorem when $f(a) = f(b)$.

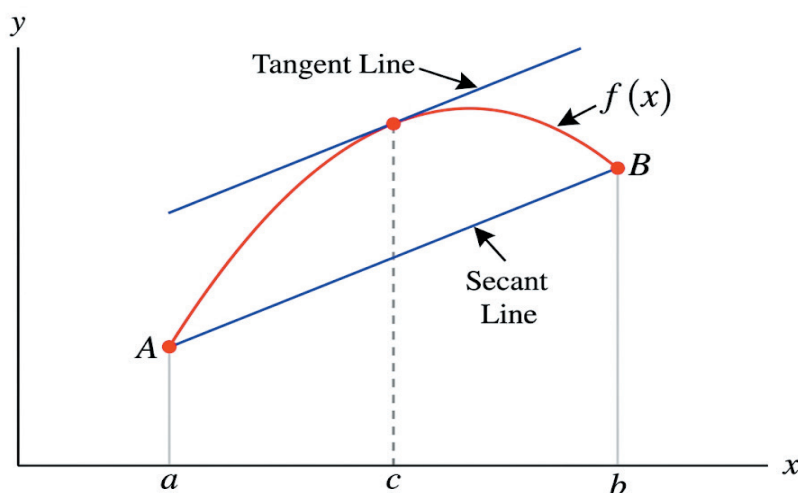
Geometrical interpretation of the Mean Value Theorem.

First define $A = (a, f(a))$ and $B = (b, f(b))$ and then we know from the Lagrange's Theorem that there is a c such that $a < c < b$ and that $f'(c) = \frac{f(b) - f(a)}{b - a}$.

Now, if we draw in the secant line connecting A and B then we can know that the slope of the secant line is, $\frac{f(b) - f(a)}{b - a}$.

Likewise, if we draw the secant line to $f(x)$ at $x = c$ we know that its slope is $f'(c)$. What the Lagrange's Theorem tells us is that these two slopes must be equal or in other words the secant line connecting A and B and the tangent line at $x = c$ must be parallel.

We can see this in the following sketch.



C. Cauchy's Mean Value Theorem

Also known as the Extended Mean Value Theorem, this version involves two functions, $f(x)$ and $g(x)$. If both are continuous on $[a, b]$ and differentiable on (a, b) , and $g'(x) \neq 0$ for all x in (a, b) , then there exists some c in (a, b) such that: $\frac{f'(c)}{g'(c)} = \frac{f(b) - f(a)}{g(b) - g(a)}$

Geometrically, this can be interpreted as the slope of a parametric curve $(g(t), f(t))$ at some point $t = c$ being equal to the slope of the secant line connecting the endpoints $(g(a), f(a))$ and $(g(b), f(b))$.

Example

Suppose we have the functions $f(x) = x^2$ and $g(x) = x^3$ on the interval $[1, 2]$. We want to find a value c that satisfies the theorem.

Calculate the components:

- $f(1) = 1, f(2) = 4$
- $g(1) = 1, g(2) = 8$
- $f'(x) = 2x, g'(x) = 3x^2$

Set up the Cauchy equation: $\frac{2c}{3c^2} = \frac{4-1}{8-1} \quad \frac{2}{3c} = \frac{3}{7}$

Solve for c : $14 = 9c$ where $c = \frac{14}{9} \approx 1.56$.

Since 1.56 lies within the interval $(1, 2)$, the theorem is satisfied.

In a professional educational setting, this theorem is often used to prove **L'Hôpital's Rule** that will be learnt here below.

Example 1 (Lagrange's Theorem)

Verify Lagrange's theorem for $f(x) = x^2$ on $[1, 3]$.

Solution:

$f(1) = 1, f(3) = 9$. Slope of secant: $\frac{f(3) - f(1)}{3 - 1} = \frac{9 - 1}{2} = 4$

Derivative: $f'(x) = 2x$. Set: $2c = 4 \Rightarrow c = 2$. Since $c \in (1, 3)$, theorem is verified, then $c = 2$

Example 2 (Rolle's Theorem)

Verify Rolle's Theorem for the function $f(x) = x^2 - 4x + 3$ on the interval $[1, 3]$.

Solution:

Check the conditions:

- f is a polynomial $\Rightarrow$ continuous and differentiable.
- $f(1) = 1 - 4 + 3 = 0$
- $f(3) = 9 - 12 + 3 = 0$

Thus $f(1) = f(3)$. Find derivative: $f'(x) = 2x - 4$. Set $f'(c) = 0$: $2c - 4 = 0 \Rightarrow c = 2$

Since $2 \in (1, 3)$, Rolle's theorem is verified.

Example 3

Find the value of c satisfying Lagrange's theorem for $f(x) = x^3$ on $[0, 2]$.

Solution:

$f(0) = 0$, $f(2) = 8$, Slope: $\frac{8-0}{2-0} = 4$. Derivative: $f'(x) = 3x^2$.

$$\text{Set: } 3c^2 = 4 \Rightarrow c^2 = \frac{4}{3} \Rightarrow c = \sqrt{\frac{4}{3}} = \frac{2}{\sqrt{3}}, c = \frac{2}{\sqrt{3}}$$

Application activity 8.4.2



1. Verify Lagrange's Mean Value theorem for the function $f(x) = x^3 - 2x^2 - x + 3$ defined in the interval $[0, 1]$.
2. Verify Rolle's theorem for the function $f(x) = x(x-3)^2$, $0 \leq x \leq 3$.
3. Verify Rolle's theorem for the function $f(x) = 2(x+1)(x-2)$ defined in the interval $[-1, 2]$.

8.4.3 L'Hôpital's theorem

Activity 8.4.3



Evaluate the following limits:

1) $\lim_{x \rightarrow 0} \frac{\sin x}{x}$

2) $\lim_{x \rightarrow 1} \frac{x^2 - 1}{x - 1}$

- Substitute directly in each expression.
- What form do you obtain?
- Can you easily determine the limit?
- Differentiate the numerator and denominator separately
- Apply the limit after differentiation.



Content Summary

Let f and g be differentiable functions such that:

$\lim_{x \rightarrow a} f(x) = 0$ and $\lim_{x \rightarrow a} g(x) = 0$, or both approach $\pm\infty$, $g'(x) \neq 0$ near a , then:

$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$ provided the limit on the right exists.

Remark:

- The theorem applies only to indeterminate forms $\frac{0}{0}$ or $\frac{\infty}{\infty}$.
- It may be applied more than once if necessary.

Example 1

Evaluate $\lim_{x \rightarrow \infty} \frac{2x^2 + 1}{x^2 + 3}$.

Solution:

$\lim_{x \rightarrow \infty} \frac{2x^2 + 1}{x^2 + 3} = \frac{\infty}{\infty}$. By applying the L'Hospital's Rule: $\lim_{x \rightarrow \infty} \frac{2x^2 + 1}{x^2 + 3} = \lim_{x \rightarrow \infty} \frac{(2x^2 + 1)'}{(x^2 + 3)'} = \lim_{x \rightarrow \infty} \frac{4x}{2x} = \frac{\infty}{\infty}$

Again, apply L'Hospital's Rule: $\lim_{x \rightarrow \infty} \frac{4x}{2x} = \lim_{x \rightarrow \infty} \frac{(4x)'}{(2x)'} = \frac{4}{2} = 2$

Example 2

By using Hospital rule, evaluate the following limit $\lim_{x \rightarrow 2} \frac{\sqrt{7+x}-3}{x-2}$

Solution:

$\lim_{x \rightarrow 2} \frac{\sqrt{7+x}-3}{x-2} = \frac{0}{0}$ (I.F); $\lim_{x \rightarrow 2} \frac{\sqrt{7+x}-3}{x-2} = \lim_{x \rightarrow 2} \frac{(\sqrt{7+x}-3)'}{(x-2)'} = \lim_{x \rightarrow 2} \frac{\frac{1}{2\sqrt{7+x}}}{1} = \frac{1}{2} \lim_{x \rightarrow 2} \frac{1}{\sqrt{7+x}} = \frac{1}{2} \cdot \frac{1}{3} = \frac{1}{6}$

Example 3

Determine $\lim_{x \rightarrow 2} \left(\frac{4}{x^2 - 4} - \frac{1}{x - 2} \right)$

Solution:

$\lim_{x \rightarrow 2} \left(\frac{4}{x^2 - 4} - \frac{1}{x - 2} \right) = \lim_{x \rightarrow 2} \left(\frac{4 - (x + 2)}{x^2 - 4} \right) = \lim_{x \rightarrow 2} \left(\frac{2 - x}{x^2 - 4} \right) = \frac{0}{0}$ I.F

$\lim_{x \rightarrow 2} \left(\frac{4}{x^2 - 4} - \frac{1}{x - 2} \right) = \lim_{x \rightarrow 2} \frac{(2 - x)'}{(x^2 - 4)'} = \lim_{x \rightarrow 2} -\frac{1}{2x} = -\frac{1}{4}$

Application activity 8.4.3

Use L'Hôpital's rule, to find the following limits:

1) $\lim_{x \rightarrow \infty} \frac{4x + 3}{3x^2 + 5}$ 2) $\lim_{x \rightarrow 3} \frac{9 - x^2}{x^3 - 3x^2 + 2x - 6}$ 3) $\lim_{x \rightarrow -1} \frac{x^3 - 2x - 1}{3x^2 - 2x - 5}$

4) $\lim_{x \rightarrow \infty} \frac{x^3 - 7x^4}{3x^4 - 5x^3}$ 5) $\lim_{x \rightarrow 2} \frac{4 - x^2}{x^2 - 3x + 2}$ 6) $\lim_{x \rightarrow 0} \frac{1 - \cos x}{x}$

8.4.4 Variations of a function (Increasing or decreasing function) and critical points: stationary and turning point (maximum and minimum values)

Activity 8.4.4



Consider the function $f(x) = x^2 - 4x + 3$;

- Calculate $f(x)$ for $x = 0, 1, 2, 3, 4$.
- Plot the corresponding points and sketch the graph.
- Identify the intervals where the function increases and where it decreases. Note that the function $f(x)$ increases if for $a > b$, $f(a) > f(b)$. It decreases when for $a > b$, $f(a) < f(b)$.
- Find the derivative $f'(x)$.
- Determine the value(s) of x where $f'(x) = 0$.
- What do you observe about the graph at that point?



Content Summary

A. Increasing and decreasing functions

Let $f(x)$ be a differentiable function:

- If $f'(x) > 0$, then $f(x)$ is increasing.
- If $f'(x) < 0$, then $f(x)$ is decreasing.

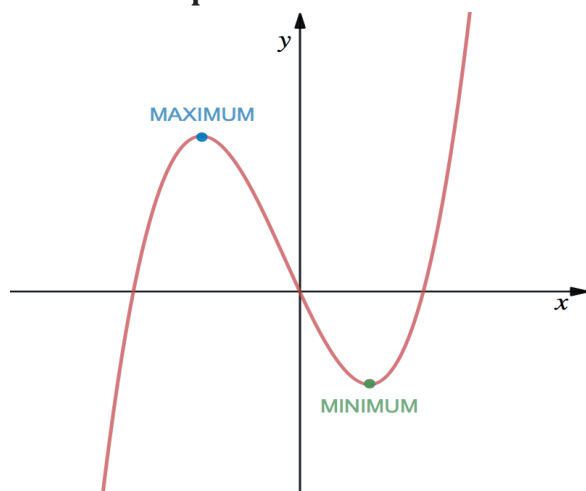
B. Critical (stationary) points

A point $x = c$ is called a **critical point** if: $f'(c) = 0$ or $f'(c)$ is undefined.

A **stationary point** occurs when: $f'(c) = 0$.

C. Maximum and Minimum points

- A **maximum point** occurs when the function changes from increasing to decreasing.
- A **minimum point** occurs when the function changes from decreasing to increasing.



D. First derivative test

- If $f(x)$ changes from positive to negative at $x = c$, then $f(c)$ is a maximum.
- If $f(x)$ changes from negative to positive at $x = c$, then $f(c)$ is a minimum.

Example 1

Given the function $f(x) = 2x^3 + 9x^2 + 12x + 20$, determine the interval where the graph of the function is increasing and where it is decreasing.

Solution:

$$f'(x) = 6x^2 + 18x + 12; f'(x) = 0 \Rightarrow 6x^2 + 18x + 12 = 0 \text{ and } 6(x^2 + 3x + 2) = 0 \Rightarrow 6(x+2)(x+1) = 0. \text{ Hence } x = -2 \text{ or } x = -1.$$

Test intervals:

- For $x < -2$, take $x = -3$: $f'(-3) = 12 > 0$ (Increasing)
- For $-2 < x < -1$, take $x = -1.5$: $f'(-1.5) = -1.5 < 0$ (decreasing)
- For $x > -1$, take $x = 0$: $f'(0) = 12 > 0$ (Increasing)

Thus, $f(x)$ is increasing on $(-\infty, -2) \cup (-1, +\infty)$, and $f(x)$ is decreasing on $(-2, -1)$.

Example 2

Find the stationary points of $f(x) = x^3 - 3x^2$

Solution:

$$f'(x) = 3x^2 - 6x = 3x(x - 2). \text{ Set } f'(x) = 0: x = 0 \text{ or } x = 2$$

$$\text{Second derivative: } f''(x) = 6x - 6, f''(0) = -6 < 0 \Rightarrow \text{maximum}$$

$$f''(2) = 6 > 0 \Rightarrow \text{minimum; Values: } f(0) = 0, f(2) = 8 - 12 = -4$$

Maximum at $(0, 0)$, Minimum at $(2, -4)$.

Application activity 8.4.4



1) Determine the intervals where the following functions are increasing or decreasing:

a) $f(x) = x^2 - 2x$

b) $f(x) = x^3 - 3x$

2) Find the stationary points and classify them:

a) $f(x) = x^2 + 4x + 1$

b) $f(x) = x^3 - 6x^2$

8.4.5 Concavity of a function and inflexion points

Activity 8.4.5



Consider the function $f(x) = x^3$

- Calculate the values of $f(x)$ for $x = -2, -1, 0, 1, 2$
- Determine the first derivative $f'(x)$
- Determine the second derivative $f''(x)$.
- Evaluate $f'''(x)$ at $x = -1, 0, 1$ and state whether it is positive, negative, or zero.
- Describe how the sign of $f''(x)$ changes as x passes through 0
- Based on the sign of $f''(x)$ describe how the function bends:
when $f''(x) > 0$ and $f''(x) < 0$
- Identify the value of x at which the change in curvature occurs.

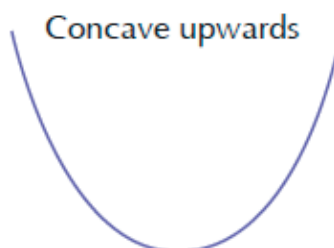
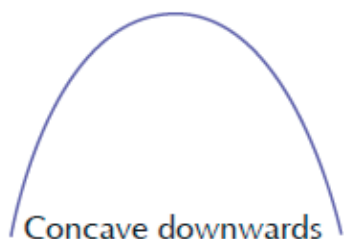


Content Summary

Concavity of a function

Let $f(x)$ be twice differentiable.

- If $f''(x) > 0 \quad \forall x \in]a, b[$, the graph is **concave upward** (concave) in interval $]a, b[$.
- If $f''(x) < 0 \quad \forall x \in]a, b[$, the graph is **concave downward** (convex) in interval $]a, b[$.



Inflection point

A point $x = c$ is called an **inflection point** if:

- $f''(c) = 0$ or is undefined, and the concavity changes at $x = c$.

Procedure to determine concavity

- Find the second derivative $f''(x)$.
- Solve $f''(x) = 0$ to find possible inflection points.
- Test intervals around these points.
- Determine where the function is concave up or down.

Example 1

Determine the concavity of $f(x) = x^2$.

Solution:

$$f'(x) = 2x, \quad f''(x) = 2 > 0$$

Thus the function is concave upward for all x . No inflection point

Example 2

Find the inflection point of $f(x) = x^3 - 3x^2$.

Solution:

$$f'(x) = 3x^2 - 6x, \quad f''(x) = 6x - 6$$

$$\text{Set: } f''(x) = 0 \Rightarrow 6x - 6 = 0 \Rightarrow x = 1$$

Test intervals:

- For $x < 1$, take $x = 0$: $f''(0) = -6 < 0$ (concave down)
- For $x > 1$, take $x = 2$: $f''(2) = 6 > 0$ (concave up)

Thus, concavity changes at $x = 1$. $f(1) = 1 - 3 = -2$. Inflection point $(1, -2)$

Application activity 8.4.5



Given $f(x) = x^4 - 4x^3$

- i. state the values of x for which f is increasing;
- ii. state the values of x for which f is decreasing;
- iii. find the critical points.
- iv. state the values of x for which the curve of f is concave upwards;
- v. state the values of x for which the curve of f is concave downwards;
- vi. find the inflection points.

8.4.6 Curve sketching

A. Polynomial function

Activity 8.4.6.A



Consider the function $f(x) = (x-1)^2(x+2)$

- Find $f'(x)$ and determine where $f'(x) = 0$.
- Identify intervals where the function increases or decreases.
- Find $f''(x)$ and determine where $f''(x) = 0$.
- Identify intervals of concavity.
- How can this information be represented in a table?



Content Summary

First Derivative Table (Variation)

This table shows:

- Sign of $f'(x)$
- Increasing/decreasing intervals
- Maximum and minimum points

Second Derivative Table (Concavity)

This table shows:

- Sign of $f''(x)$
- Concavity (upward or downward)
- Inflection points

Example

Given the function $y = f(x) = \frac{x^3}{3} + \frac{x^2}{2}$,

- state the values of x for which f is increasing;
- find the x -coordinate of each extreme point of f ;
- state the values of x for which the curve of f is concave upwards;
- find the x -coordinate of each point of inflection;
- sketch the general shape of the graph of f indicating the extreme points and points of inflection.

Solution:

$$f(x) = \frac{x^3}{3} + \frac{x^2}{2}, \quad f'(x) = x^2 + x = x(x+1).$$

$$\text{Set: } f'(x) = 0 \Rightarrow x(x+1) = 0 \Rightarrow x = 0 \quad \text{or} \quad x = -1$$

Sign of $f'(x)$

x	$-\infty$	-1	0	$+\infty$	
$f'(x)$	$+$	0	$-$	0	$+$
$f(x)$	$-\infty$	$\nearrow \frac{1}{6}$	$\searrow 0$	$\nearrow +\infty$	

Therefore f is increasing for $x < -1$ or $x > 0$ i.e. $]-\infty, -1[\cup]0, +\infty[$.

The extreme/critical points are $(x = -1, f(-1)) = (-1, \frac{1}{6})$ and $(x = 0, f(0)) = (0, 0)$.

Now $f''(x) = 2x + 1$.

Set: $f''(x) = 0 \Rightarrow 2x + 1 = 0 \Rightarrow x = -\frac{1}{2}$

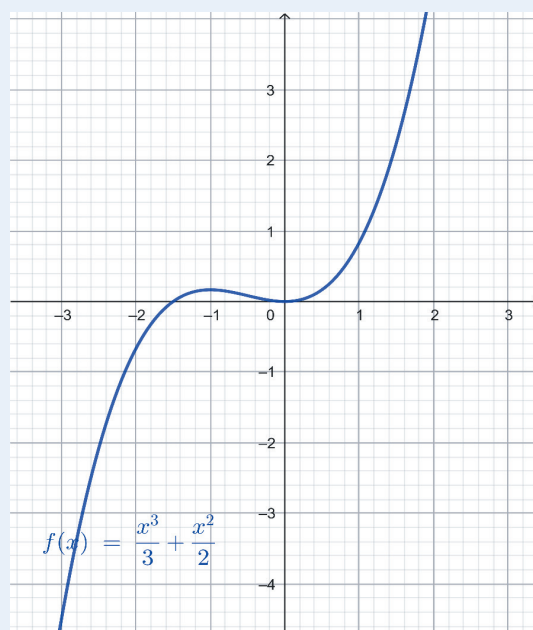
Sign of $f''(x)$

x	$-\infty$	$-\frac{1}{2}$	$+\infty$
$f''(x)$	$-$	0	$+$
$f(x)$			

Thus, the curve is concave upwards for $x > -\frac{1}{2}$ i.e. $]-\frac{1}{2}, +\infty[$

Also, $(-\frac{1}{2}, f(-\frac{1}{2})) = (-\frac{1}{2}, \frac{1}{12})$ is a point of inflection.

The graph of $f(x) = \frac{x^3}{3} + \frac{x^2}{2}$ is presented below.



Application activity 8.4.6. A



For the function $f(x) = x^3 - 6x^2$

1. Find:

- Intervals of increase and decrease
- Maximum and minimum points
- Intervals of concavity
- Inflection points

2. Construct

- the first derivative table.
- the second derivative table.

3. Sketch the graph using both tables.

B. Rational functions

Activity 8.4.7. B



Let f be the function $f(x) = \frac{1}{x}$

- Calculate the first and second derivative of $f(x)$
- In one table, present the sign table of $f'(x)$ and $f''(x)$ showing also the intervals where $f(x)$ is increasing or decreasing and the sense of concavity.
- Try to sketch the function according to the above table



Content Summary

Variation table or synthetic table summarize all information about the function: domain of definition, limits at the boundaries of domain, asymptotes to the curve, extrema points, variation of function (increasing or decreasing), inflection points and the sense of concavity (opens up or down).

Example:

Establish the variation table and sketch of the function $f(x) = \frac{x^2 + 1}{x}$

Condition of existence: $x \neq 0$

1) $Domf = \mathbb{R} \setminus \{0\} =]-\infty, 0[\cup]0, +\infty[$

2) Limits at the boundaries of $Domf$

a) $\lim_{x \rightarrow -\infty} \frac{x^2 + 1}{x} = \frac{\infty}{\infty} \quad (IF)$

Removing IF: $\lim_{x \rightarrow -\infty} \frac{x^2 + 1}{x} = \lim_{x \rightarrow -\infty} \frac{x^2(1 + \frac{1}{x^2})}{x} = \lim_{x \rightarrow -\infty} x = -\infty$

$\Rightarrow$ There is no Horizontal asymptote

b) $\lim_{x \rightarrow 0} \frac{x^2 + 1}{x} = \frac{1}{0} \Leftrightarrow \lim_{x \rightarrow 0^-} \frac{x^2 + 1}{x} = \frac{1}{0^-} = -\infty \Leftrightarrow \lim_{x \rightarrow 0^+} \frac{x^2 + 1}{x} = \frac{1}{0^+} = +\infty$

$\Rightarrow$ There is a vertical asymptote : $x = 0$

c) $\lim_{x \rightarrow +\infty} \frac{x^2 + 1}{x} = \frac{\infty}{\infty} \quad (IF) \Leftrightarrow \lim_{x \rightarrow +\infty} \frac{x^2 + 1}{x} = \lim_{x \rightarrow +\infty} \frac{x^2(1 + \frac{1}{x^2})}{x} = \lim_{x \rightarrow +\infty} x = +\infty$

$\Rightarrow$ There is no Horizontal asymptote

Let find the oblique asymptote (OA)

$OA \equiv y = ax + b$ with $a = \lim_{x \rightarrow \pm\infty} \frac{f(x)}{x}$ and $b = \lim_{x \rightarrow \pm\infty} [f(x) - ax]$

$$a = \lim_{x \rightarrow \pm\infty} \frac{x^2 + 1}{x^2} = \lim_{x \rightarrow \pm\infty} \frac{x^2}{x^2} = 1$$

$$b = \lim_{x \rightarrow \pm\infty} \left(\frac{x^2 + 1}{x} - x \right) = \lim_{x \rightarrow \pm\infty} \left(\frac{x^2 + 1 - x^2}{x} \right) = \lim_{x \rightarrow \pm\infty} \left(\frac{1}{x} \right) = 0$$

$a = 1, b = 0 \Rightarrow$ there is no OA

3) Parity of $f(x)$

a) $f(x)$ is even function if $f(x) = f(-x)$

$$f(-x) = \frac{(-x)^2 + 1}{-x} = -\frac{x^2 + 1}{x} \neq f(x), \text{ so } f(x) = \frac{x^2 + 1}{x} \text{ is not even}$$

b) $f(x)$ is odd function if $-f(x) = f(-x)$

$$f(-x) = \frac{(-x)^2 + 1}{-x} = -\frac{x^2 + 1}{x}$$

c) $-f(x) = -\frac{x^2 + 1}{x}$, so $-f(x) = f(-x)$ and $f(x)$ is odd function

4) Periodicity of $f(x)$. The given function is not periodic

5) First derivative of $f(x)$

$$f'(x) = \left(\frac{x^2 + 1}{x} \right)' = \frac{2x \cdot x - 1 \cdot (x^2 + 1)}{x^2} = \frac{2x^2 - x^2 - 1}{x^2} \Leftrightarrow f'(x) = \frac{x^2 - 1}{x^2}$$

Critical points:

d) Condition of existence of $f'(x)$, $x^2 \neq 0 \Leftrightarrow x \neq 0$

$$f'(x) = 0 \Leftrightarrow \frac{x^2 - 1}{x^2} = 0 \Leftrightarrow x^2 - 1 = 0 \Rightarrow x = -1 \text{ or } x = 1$$

Extrema points:

$x = -1 \Rightarrow y = -2$, function has a maximum point $(-1, -2)$

$x = 1 \Rightarrow y = 2$, function has a minimum point $(1, 2)$

Sign table of $f'(x)$

x	$-\infty$	-1	0	1	$+\infty$
$f'(x)$	+ 0 -			- 0 +	
$f(x)$	$-\infty \nearrow -2 \searrow -\infty$			$+\infty \searrow 2 \nearrow +\infty$	

6) Second derivative of $f(x)$

$$f''(x) = \left(\frac{x^2 - 1}{x^2} \right)' = \frac{2x \cdot x^2 - 2x \cdot (x^2 - 1)}{x^4} = \frac{2x^3 - 2x^3 + 2x}{x^4} = \frac{2x}{x^4} \Leftrightarrow f''(x) = \frac{2}{x^3}$$

Critical points:

- Existence condition of $f''(x)$: $x^3 \neq 0 \Rightarrow x \neq 0$
- $f''(x) = 0 \Leftrightarrow \frac{1}{x^3} = 0$ It doesn't exist in $\mathbb{R}$

Sign table of $f''(x)$

x	$-\infty$	0	$+\infty$
$f''(x)$	-		+
$f(x)$			

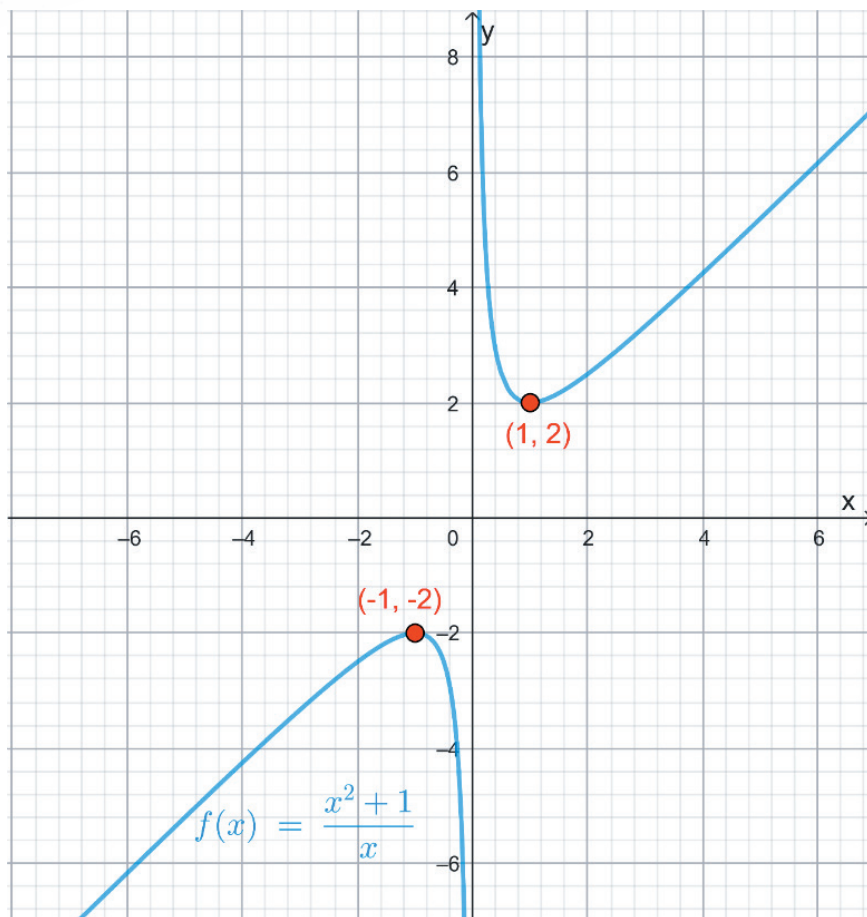
7) Variation table or synthetic table

x	$-\infty$	-1	0	1	$+\infty$	
$f'(x)$	$+$	0	$-$	$-$	0	$+$
$f''(x)$	$-$	$-$	$-$	$+$	$+$	$+$
$f(x)$	$-\infty$		$-\infty$		$+\infty$	$+\infty$

8) Supplementary points

x	2	-2	-3	3	-4	4
$f(x)$	2.5	-2.5	$-\frac{10}{3}$	$\frac{10}{3}$	$-\frac{17}{4}$	$\frac{17}{4}$

9) Graph of function $f(x) = \frac{x^2 + 1}{x}$



Application activity 8.4.6. B



Study and sketch the curve of $y = f(x) = x + \frac{1}{4x}$

C. Irrational function

Activity 8.4.6. C



Consider the function $f(x) = \sqrt{x}$

- Determine the domain of the function.
- Compute $f(x)$ for $x = 0, 1, 4, 9$, put the points on cartesian plan and join them to form a graph.
- Does the function increase or decrease?
- Can you describe its shape (concavity)?



Content Summary

An **irrational function** is a function that contains a variable inside a root, for example:

$$f(x) = \sqrt{x}, \quad f(x) = \sqrt{x+1}, \quad f(x) = \sqrt{x^2+1}.$$

Steps for Studying and Sketching the Curve

- Determine the domain (expression inside the root must be ≥ 0).
- Find the first derivative $f'(x)$ and determine intervals of increase or decrease.
- Find the second derivative $f''(x)$ to study concavity.
- Construct the table of variation.
- Sketch the curve.

Example

Study and sketch the curve of: $f(x) = \sqrt{x+1}$.

Solution

1. Domain: $x+1 \geq 0 \Rightarrow x \geq -1$. $\text{Dom}f = [-1, +\infty[$
2. First Derivative $f'(x) = \frac{1}{2\sqrt{x+1}}$. Since $f'(x) > 0$ for all $x > -1$, the function is increasing.

First Derivative Table

x	-1	$+\infty$
$f'(x)$	+	
$f(x)$	0	$+\infty$

3. Second Derivative

$$f''(x) = -\frac{1}{4(x+1)^{3/2}}, \text{ then } f''(x) < 0, \text{ the function is concave downward.}$$

Second Derivative Table

x	-1	$+\infty$
$f''(x)$	$-$	
$f(x)$		

4. Key Points

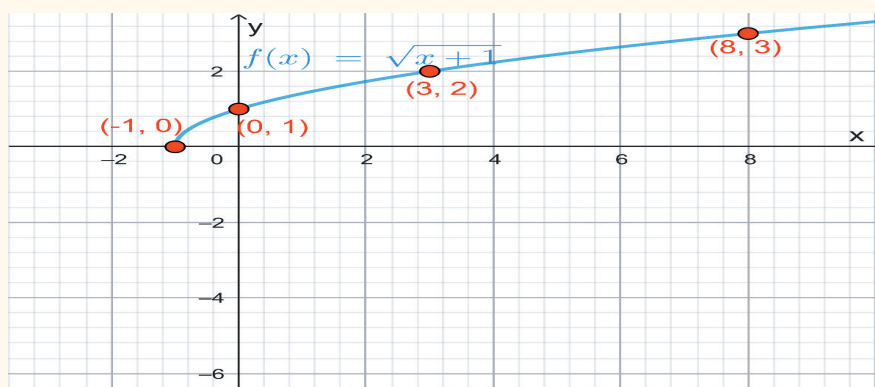
$$f(-1) = 0, \quad f(0) = 1, \quad f(3) = 2$$

5. Sketch of the Curve

The curve starts at $(-1, 0)$.

It increases for all $x > -1$.

It is concave downward.



Application activity 8.4.6. C



Work in groups.

Give the functions: 1) $f(x) = \frac{1}{\sqrt{x}}$ 2) $f(x) = \sqrt{x-2}$ 3) $f(x) = \sqrt{x^2+1}$

For each function:

- Determine the domain
- Construct the table of variation
- Sketch the graph

8.4.7 Gradient as a measure of rate of change

Activity 8.4.7



A car moves along a straight road and its position (in meters) after t seconds is given by:

$$x(t) = t^2$$

1. Find the average speed of the car between:
 - $t = 1$ and $t = 2$
 - $t = 2$ and $t = 3$
2. What do you notice about the change in speed?
3. Find the derivative $x'(t)$.
4. What does $x'(t)$ represent?



Content Summary

Gradient and Rate of Change

The **gradient** (slope) of a curve at a point is given by the derivative:

$$\text{Gradient} = f'(x)$$

- It represents the **rate of change** of y with respect to x .
- It shows how fast one quantity changes relative to another.

Interpretation

- In motion: $x'(t)$ **represents velocity**
- In economics: derivative represents **marginal cost or profit**
- In science: rate of change of temperature, population, etc.

Average vs Instantaneous Rate of Change

Average rate of change

$$\frac{f(b) - f(a)}{b - a}$$

Instantaneous rate of change $f'(x)$

Example

Example 1 (Motion)

A particle moves according to: $x(t) = t^2 + 2t$

Find the velocity at $t = 3$.

Solution:

$$x'(t) = 2t + 2 \Rightarrow x'(3) = 6 + 2 = 8. \text{ Velocity at } t = 3 \text{ is } 8 \text{ m/s}$$

Example 2 (Gradient)

Find the gradient of the curve. $y = x^2 + 3x$ at $x = 1$.

Solution:

$$y' = 2x + 3 \Rightarrow y'(1) = 5. \text{ Gradient at } x = 1 \text{ is } 5$$

Example 3 (Average vs Instantaneous)

$$\text{Given: } f(x) = x^2$$

Find:

- Average rate of change between $x = 1$ and $x = 3$
- Instantaneous rate at $x = 2$

Solution:

$$\frac{f(3) - f(1)}{3 - 1} = \frac{9 - 1}{2} = 4. f'(x) = 2x \Rightarrow f'(2) = 4$$

Both rates are equal at $x = 2$

Application activity 8.4.7



Work in groups.

1. A particle moves according to: $x(t) = t^2 + 4t$
 - a) Find the velocity function
 - b) Find the velocity at $t = 2$
2. Find the gradient of the curve: $y = x^2 - 2x$ at $x = 1$.
3. For the function $f(x) = x^3$:
 - a) Find the average rate of change between $x = 1$ and $x = 2$
 - b) Find the instantaneous rate at $x = 1$

8.4.8 Kinematic meaning of derivatives

Activity 8.4.8



A particle moves along a straight line and its position (in meters) after t seconds is given by:

$$x(t) = t^2.$$

- Find the derivative $x'(t)$ and $x''(t)$.
- Evaluate $x'(t)$ and $x''(t)$ at $t = 2$.
- What do these values represent?



Content Summary

Position, Velocity, and Acceleration

Let $x(t)$ be the position of a particle at time t .

- Velocity is the rate of change of position: $v(t) = x'(t)$
- Acceleration** is the rate of change of velocity: $a(t) = v'(t) = x''(t)$

The acceleration $a(t)$: change in velocity (speeding up or slowing down)

Key relationships: Speed $v(t) = \frac{dx}{dt}$, Acceleration $a(t) = \frac{d^2x}{dt^2}$.

Motion Analysis:

- If $v(t) > 0$: particle moves forward
- If $v(t) < 0$: particle moves backward
- If $v(t) = 0$: particle is at rest
- If $a(t) > 0$: velocity increases
- If $a(t) < 0$: velocity decreases.

Example 1

A particle moves according to: $x(t) = t^2 + 2t$

Find: Velocity and acceleration at $t = 2$.

Solution:

$$v(t) = x'(t) = 2t + 2; \quad a(t) = x''(t) = 2 \quad \text{and} \quad v(2) = 2(2) + 2 = 6$$

Example 2

Given: $x(t) = t^3 - 3t^2$. Find when the particle is at rest.

Solution:

$$v(t) = x'(t) = 3t^2 - 6t = 3t(t - 2)$$

$$\text{Set: } v(t) = 0 \Rightarrow t = 0 \quad \text{or} \quad t = 2$$

Example 3

$$v(t) = 0 \Rightarrow t = 0 \quad \text{or} \quad t = 2$$

Given: $x(t) = t^3$. Determine whether the particle is accelerating or decelerating at $t = 1$.

Solution:

$v(t) = 3t^2$, $a(t) = 6t$, $a(1) = 6 > 0$. The particle is accelerating at $t = 1$

Application activity 8.4.8



- 1) Given: $x(t) = t^3 - 6t^2$
 - a) Find when the particle is at rest
 - b) Determine whether it is accelerating or decelerating at $t = 2$
- 2) Explain the physical meaning of:
 - a) $x'(t)$
 - b) $x''(t)$
- 3) A particle moves in a straight line so that its position x cm relative to O at time t seconds is given by $x(t) = t^3 - 6t^2 + 5, t \geq 0$. Find
 - a) Its initial position, velocity and acceleration and hence describe its motion
 - b) The times when it is instantaneously at rest and its position and acceleration.

8.4.9 Optimization problems

Activity 8.4.9



A farmer has 40 meters of fencing and wants to enclose a rectangular field.

- Let the length be x and the width be y . Write an equation relating x and y .
- Express the area A in terms of one variable.
- Try different values of x and compute the area.
- Which value gives the largest area?



Content Summary

Optimization involves finding the **maximum** or **minimum** value of a function.

Key Steps in Solving Optimization Problems

- Identify the quantity to optimize (area, volume, cost, etc.).
- Express it as a function of one variable.
- Find the derivative of the function.

- Solve: $f'(x) = 0$
- Use:
 - First derivative test, or
 - Second derivative test
- State the final answer clearly.

Critical Points

A critical point occurs when: $f'(x) = 0$ or $f'(x)$ is undefined

Second Derivative Test

- If $f''(x) > 0$: minimum
- If $f''(x) < 0$: maximum

Example 1 (Maximum Area)

A rectangle has perimeter 40 m. Find dimensions that maximize the area.

Solution:

$$2x + 2y = 40 \Rightarrow y = 20 - x. \quad A = x(20 - x) = 20x - x^2$$

$$A'(x) = 20 - 2x, \text{ then } 20 - 2x = 0 \Rightarrow x = 10 \text{ or } y = 20 - 10 = 10. \text{ Thus}$$

$$A''(x) = -2 < 0 \Rightarrow \text{maximum}$$

Example 2 (Minimum Cost)

The cost function is: $C(x) = x^2 - 8x + 20$. Find the minimum cost.

Solution:

$$C'(x) = 2x - 8. \quad 2x - 8 = 0 \Rightarrow x = 4, \text{ then } C''(x) = 2 > 0 \Rightarrow \text{minimum or } C(4) = 16 - 32 + 20 = 4$$

Minimum cost is 4 at $x = 4$

Example 4

Determine two positive numbers whose sum is 15 and the sum of whose squares is minimum.

Solution:

Let one number be x . Then the other is $15 - x$. Since the numbers are positive, $0 < x < 15$.

Let S be the sum of the squares of the numbers: $S = x^2 + (15 - x)^2$; $S = x^2 + 225 - 30x + x^2$
or $S = 2x^2 - 30x + 225$.

Differentiate: $\frac{dS}{dx} = 4x - 30$.

For a maximum or minimum, $\frac{dS}{dx} = 0$. So, $4x - 30 = 0 \Rightarrow 4x = 30$, $x = \frac{15}{2}$. Now, $\frac{d^2S}{dx^2} = 4 > 0$.

Hence S is minimum when $x = \frac{15}{2}$. The other number is $15 - \frac{15}{2} = \frac{15}{2}$.

Hence the sum of squares is minimum when the numbers are $\frac{15}{2}$ and $\frac{15}{2}$.

Application activity 8.4.9



1. A rectangle has area 100m^2 . Find dimensions with minimum perimeter.
2. A farmer wants to fence three sides of a rectangular field (one side along a river). If 60 m of fencing is available, find dimensions that maximize the area.
3. Find the minimum value of $f(x) = x^2 + 6x + 10$.

End of unit assessment



A. Knowledge and Understanding

1. Multiple Choice Questions

Choose the correct answer.

- a) The derivative of $f(x) = x^2$ from first principle is:
A. x B. $2x$ C. x^2 D. 2
- b) The derivative of $f(x) = x^2 \sin x$ requires:
A. Chain rule C. Quotient rule
B. Product rule D. Sum rule
- c) If $f'(x) = 0$ at a point, that point is:
A. Always maximum C. Critical point
B. Always minimum D. Point of inflection
- d) Rolle's theorem applies when:
A. $f(a) = f(b)$ C. $f(x)$ is not continuous
B. $f'(x) > 0$ D. $f'(x) = 1$
- e) The second derivative is used to determine:
A. Slope only C. Intercept
B. Concavity D. Domain
- f) If $\lim_{x \rightarrow a} f(x) = \infty$, then
A. Horizontal asymptote C. Minimum
B. Vertical asymptote D. Inflection

B. Skills

1. Differentiate:

a) $f(x) = 3x^2 + 5x - 1$

b) $f(x) = x^2 \ln x$

c) $f(x) = \frac{x^2 + 1}{x}$

2. Find the equation of the tangent and normal to the curve $y = x^2$ at $x = 1$.

3. Given: $f(x) = x^3 + 3x^2 - 24x$

a) Find critical points

b) Determine intervals of increase/decrease

4. For: $f(x) = x^3 - 3x$

a) Find $f''(x)$

b) Determine concavity

c) Find inflection point

5. Find the maximum value of: $f(x) = x(10 - x)$

6. A particle moves according to: $x(t) = t^3 - 6t^2$

a) Find velocity

b) Find acceleration

c) Find when the particle is at rest

C. Attitudes and Values

1) A local cooperative wants to build a water storage tank using the minimum amount of material possible while maintaining a specific volume to withstand dry seasons. Why is using derivatives the most responsible and valuable approach for a student helping this cooperative?

a) It allows the student to practice complex algebraic formulas to impress the community leaders.

b) It provides an exact mathematical method to optimize resources, saving money and minimizing material waste for the community.

c) It proves that abstract classroom mathematics is completely different from practical, real-world construction work.

d) It guarantees that the tank will automatically fill up faster because derivatives deal with rates of change.

2) When finding the derivative of a complex function using the Chain Rule, a student writes down only the final answer without showing the intermediate steps, but the answer turns out to be incorrect. Why is cultivating the habit of showing systematic steps highly valued in mathematics?

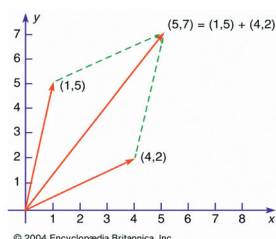
a) It is the only way to satisfy the teacher and avoid losing marks on the final exam layout.

b) It proves to the teacher that the student did not copy the final answer from a classmate's paper.

- c) It develops a mindset of logical precision and clarity, allowing the student and others to trace the reasoning process, spot specific misconceptions, and self-correct.
 - d) It makes the solution look longer and more complex, which gives the impression of hard work.
- 3) Your group is given a complex problem to maximize the profit of a small student-led business using derivatives. Group members hold different opinions on how to set up the revenue function. What is the most productive attitude to adopt during this collaborative task?
- a) Argue firmly until the group adopts your own function, as individual confidence is the key to leadership.
 - b) Actively listen to and test every member's proposed function against the problem constraints, valuing diverse perspectives to build a more accurate and robust model together.
 - c) Let the top-performing student in the class design the function alone while the rest of the group quietly copies the final layout.
 - d) Suggest changing the problem to an easier topic so that the group can avoid any disagreements or lengthy discussions.

UNIT 9

VECTOR SPACE OF REAL NUMBERS



Key Unit Competence:

Determine the magnitude of vectors, the angle between two vectors and the dot product of two vectors.

Learning objectives

By the end of this unit, you should be able to:

Knowledge and understanding

1

- Define a vector and explain the concept of the magnitude of a vector.
- State the formula for the dot product of two vectors.
- Describe how the angle between two vectors can be determined using the dot product.

Skills

2

- Calculate the magnitude of a vector from its components.
- Determine the dot product of two vectors in component form.
- Find the angle between two vectors using the dot product formula.

Attitudes and values

3

- Demonstrate accuracy and logical reasoning when solving vector problems.
- Show perseverance and confidence when performing vector calculations.
- Appreciate the importance of vectors in solving real-life problems in science and technology.

Introduction

This unit introduces the fundamental ideas of vectors in the vector space of real numbers, which are important in many areas of Mathematics and science. Vectors are mathematical objects that have both magnitude and direction. They are useful for representing quantities such as displacement, velocity, and force. The purpose of this unit is to help you understand how vectors can be represented, compared, and operated on using mathematical rules. These ideas will help you develop logical thinking, analytical skills, and problem-solving abilities that are essential for advanced mathematics and real-life applications.

In this unit, you will study how to determine the magnitude of vectors, compute the dot product of two vectors, and find the angle between two vectors. These concepts will help you understand relationships between vectors and how they interact with each other in the vector space of real numbers.

The key words to be explained in this unit are the following: vector, vector space, component of a vector, magnitude (length) of a vector, unit vector, dot product (scalar product), angle between vectors, orthogonal vectors, parallel vectors, direction, and vector notation.

Introductory activity



Given the point A with coordinates $(1,2)$ and the point B with coordinates $(-3,1)$.

- Represent the point A and B in the same plane XY
- Draw an arrow from the point A to point B
- Find the component of the vector $\overrightarrow{AB}$

9.1 Definitions and operations on vector in the vector spaces $\mathbb{R}^2$

Activity 9.1



On a Cartesian plane, let $O(0,0)$ be the origin. Consider two vectors defined by the points $A(3,2)$ and $B(2,4)$.

- Write the vectors $\vec{a}$ and $\vec{b}$ in component form, where $\vec{a} = \overrightarrow{OA}$ and $\vec{b} = \overrightarrow{OB}$.
- Determine the vector $\vec{a} + \vec{b}$ and present the result in component form.
- Find the vector $\vec{a} - \vec{b}$.
- If vector $\vec{a}$ is multiplied by the scalar 2, determine $2\vec{a}$.



Content Summary

A **vector** is a quantity that has both **magnitude (length)** and **direction**.

For example, vector $\mathbf{a} = (3, 2)$ represents a movement of 3 units in the horizontal direction and 2 units in the vertical direction.

- one which has a defined magnitude but for which direction has no meaning (example: length of a piece of wire). In physics, length, mass, and speed have magnitude but no direction. Such quantities are defined as **scalars**.
- The other direction is of fundamental significance. Quantities such as force and wind velocity depend very much on the direction in which they act. These are **vector** quantities.

Example 1:

Vector $\mathbf{a} = (2, 3)$ represents a vector that moves 2 units in the x-direction and 3 units in the y-direction.

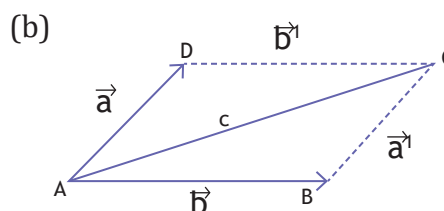
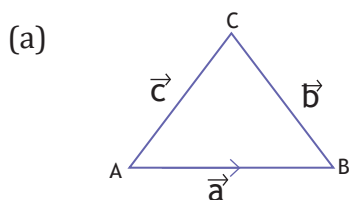
Example 2:

Vector $\mathbf{b} = (-1, 4)$ represents a vector that moves 1 unit to the left and 4 units upward.

Vectors can undergo different operations, such as addition, subtraction, and scalar multiplication.

Vector addition:

Two vectors can be added by adding their corresponding components. Vectors are added by the triangle law or end-on rule or the parallelogram law of addition.



In Figure (a), Vector $\vec{a} = \overrightarrow{AB}$ followed by a vector $\vec{b} = \overrightarrow{BC}$ giving a vector $\vec{c} = \overrightarrow{AC}$

$$\overrightarrow{AB} + \overrightarrow{BC} = \overrightarrow{AC} \text{ It means } \vec{c} = \vec{a} + \vec{b}$$

In Figure (b), ABCD is a parallelogram and $\overrightarrow{BC} = \overrightarrow{AD}$

$$\overrightarrow{AB} + \overrightarrow{AD} = \overrightarrow{AD} + \overrightarrow{AC} \text{ It means } \vec{c} = \vec{a} + \vec{b}$$

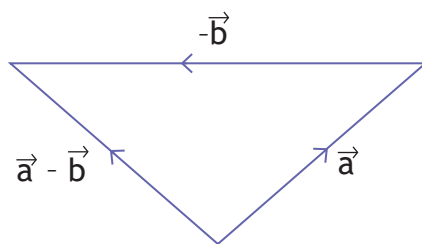
Example 3:

If $\mathbf{a} = (2, 3)$ and $\mathbf{b} = (1, 4)$ then, $\mathbf{a} + \mathbf{b} = (2 + 1, 3 + 4) = (3, 7)$

Vector subtraction:

A vector can be subtracted from another by subtracting their corresponding components.

The vector $\vec{a} - \vec{b}$ is equal to $\vec{a} + (-\vec{b})$. Thus $\vec{a} - \vec{b}$ can be found by adding vector $-\vec{b}$ to vector $\vec{a}$



Example 4:

If $a = (2, 3)$ and $b = (1, 4)$ then, $a - b = (2 - 1, 3 - 4) = (1, -1)$.

Scalar multiplication:

A vector can be multiplied by a real number (scalar). This changes the magnitude of the vector and may change its direction.

Example 5:

If $a = (2, 3)$ and the scalar is 2, then $2a = 2(2, 3) = (4, 6)$.

Components of a 2D vector

The Components:

A vector $\mathbf{v}$ in $\mathbb{R}^2$ is written as an ordered pair: $\mathbf{v} = (x, y)$

- **The x-component (x):** This is the horizontal displacement from the origin along the x-axis.
- **The y-component (y):** This is the vertical displacement from the origin along the y-axis.

In linear algebra, these are often represented as a **column vector**: $\mathbf{v} = \begin{pmatrix} x \\ y \end{pmatrix}$

Example:

Let's look at the vector $\mathbf{W} = (3, 4)$.

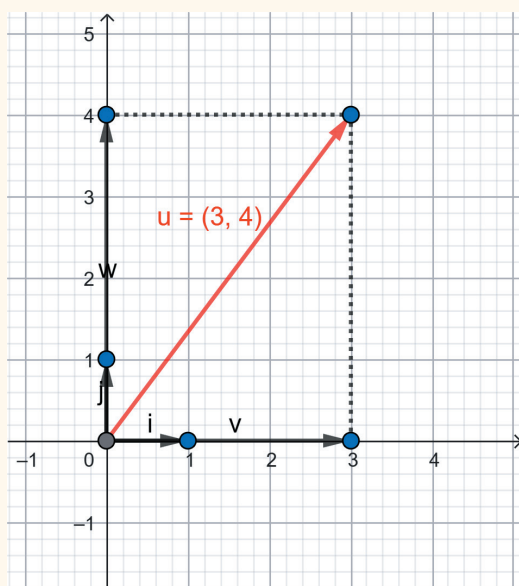
- **Components:** The x-component is 3 and the y-component is 4.
- **Geometric Meaning:** To draw this vector, you start at the origin (0,0), move 3 units to the right, and then move 4 units up.

Relation to the standard basis

Any 2D vector can also be written as a combination of the standard basis vectors, $\vec{i} = (1, 0)$ and $\vec{j} = (0, 1)$.

Example:

$$\vec{u} = 3\vec{i} + 4\vec{j} = \vec{v} + \vec{w}$$



$\vec{v} = 3\vec{i}$ is the x-component and $\vec{w} = 4\vec{j}$ is the y-component.

Application activity 9.1



1. On a Cartesian plane, consider the vectors $a = (3, 2)$ and $b = (-1, 4)$.
 - a) Find the vector $a + b$ and draw it.
 - b) Find the vector $a - b$ and draw it.
 - c) Find $2a + (-1)b$
 - d) Represent the resulting vector graphically (in c).

9.2 Properties of operations on vectors in the Vector spaces $\mathbb{R}^2$

Activity 9.2



Consider the following vectors in the vector space of real numbers:

$$a = (2, 3), b = (4, -1), c = (-1, 2)$$

- a) Calculate $a + b$ and $b + a$. Verify that they are equal.
- b) Calculate $(a + b) + c$ and $a + (b + c)$. Verify that the results are equal.
- c) Calculate $k(a + b)$ and compare it with $ka + kb$.



Content Summary

A vector space is a set V whose elements are called vectors, together with a set of scalars $\mathbb{R}$ (but in general, this set of scalars can be any field F). With two operations: vector addition and scalar multiplication, the vector space V satisfies the following 10 axioms, for all $\vec{X}, \vec{Y}, \vec{Z}$ and $r, s \in \mathbb{R}$.

1. $\vec{X} + \vec{Y} \in V$ (Closure under vector addition)
2. $\vec{X} + \vec{Y} = \vec{Y} + \vec{X}$ (Commutativity)
3. $(\vec{X} + \vec{Y}) + \vec{Z} = \vec{X} + (\vec{Y} + \vec{Z})$ (Associativity of vector addition)
4. There exists a $\vec{0} \in V$ such that $\vec{0} + \vec{X} = \vec{X} + \vec{0}$ (Existence of additive identity)
5. There exists $-\vec{X} \in V$ such that $\vec{X} + (-\vec{X}) = \vec{0}$ (Existence of additive inverse)
6. $r\vec{X} \in V$ (Closure under scalar multiplication)
7. $r(\vec{X} + \vec{Y}) = r\vec{X} + r\vec{Y}$ (Distributivity of vector sum)
8. $(r + s)\vec{X} = r\vec{X} + s\vec{X}$ (Distributivity of scalar sum)
9. $r(s\vec{X}) = (rs)\vec{X}$ (Associativity of scalar multiplication)
10. $1\vec{X} = \vec{X}$ (Scalar multiplication identity)

Example:

Show that $\mathbb{R}^2$ is a vector space over $\mathbb{R}$, $\mathbb{R}^2 = \mathbb{R} \times \mathbb{R} = \{(x, y) : x \in \mathbb{R} \text{ and } y \in \mathbb{R}\}$

The vector addition in $\mathbb{R}^2 = \mathbb{R}^2 \times \mathbb{R}^2 \rightarrow \mathbb{R}^2$:

$$(x_1, y_1) \times (x_2, y_2) \rightarrow (x_1, y_1) + (x_2, y_2) = (x_1 + x_2, y_1 + y_2).$$

The scalar multiplication $\mathbb{R}^2 \times \mathbb{R}^2 \rightarrow \mathbb{R}^2$: $a(x_1, y_1) \rightarrow a(x_1, y_1) = (ax_1, ay_1)$

Solution:

Let $\vec{u} = (x_1, y_1), \vec{v} = (x_2, y_2), \vec{w} = (x_3, y_3) \in \mathbb{R}^2$ and $\alpha, \beta \in \mathbb{R}$

- i. We verify if $\mathbb{R}^2, +$ is an abelian group. That is

$$\vec{u} + \vec{v} = (x_1, y_1) + (x_2, y_2) = (x_1 + x_2, y_1 + y_2) \in \mathbb{R}^2 \text{ (closure)}$$

$$\begin{aligned} 1. (\vec{u} + \vec{v}) + \vec{w} &= [(x_1, y_1) + (x_2, y_2)] + (x_3, y_3) = (x_1 + x_2, y_1 + y_2) + (x_3, y_3) \\ &= (x_1 + x_2 + x_3, y_1 + y_2 + y_3) = (x_1, y_1) + [(x_2, y_2) + (x_3, y_3)] \\ &= \vec{u} + [\vec{v} + \vec{w}] \text{ (associativity for addition)} \end{aligned}$$

$$\begin{aligned} 2. \vec{u} + \vec{v} &= (x_1, y_1) + (x_2, y_2) = (x_1 + x_2, y_1 + y_2) = (x_2 + x_1, y_2 + y_1) = (x_2, y_2) + (x_1, y_1) \\ &\text{(commutativity)} \end{aligned}$$

$$3. \vec{u} + \vec{0} = (x_1, y_1) + (0, 0) = (x_1, y_1) = \vec{u} \text{ (Additive identity is } (0, 0))$$

$$4. \vec{u} + (-\vec{u}) = (x_1, y_1) + (-x_1, -y_1) = (x_1 - x_1, y_1 - y_1) = (0, 0) = \vec{0}$$

(x_1, y_1) has an additive inverse $(-x_1, -y_1)$

ii. We verify if the scalar multiplication satisfies the following:

$$1. \alpha(\beta\vec{u}) = \alpha\beta(x_1, y_1) = \alpha(\beta x_1, \beta y_1) = (\alpha\beta x_1, \alpha\beta y_1) = \alpha\beta(x_1, y_1) = \alpha\beta\vec{u} \text{ (Associativity for multiplication)}$$

$$2. 1\vec{u} = 1(x_1, y_1) = (1x_1, 1y_1) = (x_1, y_1) = \vec{u} \text{ (Multiplication identity is 1)}$$

$$3. a(\vec{u} + \vec{v}) = a[(x_1, y_1) + (x_2, y_2)] = a[(x_1, y_1) + (x_2, y_2)]$$

$$= (ax_1 + ax_2, ay_1 + ay_2) = (ax_1, ay_1) + (ax_2, ay_2)$$

$$= a(x_1, y_1) + a(x_2, y_2) = a\vec{u} + a\vec{v} \text{ (right distributivity)}$$

$$4. (\alpha + \beta)\vec{u} = (\alpha + \beta)(x_1, y_1) = [(\alpha + \beta)x_1, (\alpha + \beta)y_1]$$

$$= (\alpha x_1 + \beta x_1, \alpha y_1 + \beta y_1) = (\alpha x_1, \alpha y_1) + (\beta x_1, \beta y_1)$$

$$= \alpha(x_1, y_1) + \beta(x_1, y_1) = \alpha\vec{u} + \beta\vec{u} \text{ (Left distribution)}$$

Hence, $\mathbb{R}^2$ is a vector space over $\mathbb{R}$.

Properties

- The sum of an infinite list of vectors $V_1, V_2, \dots, V_k$ can be computed in any order, and fully parenthesized in any way, and the sum will be the same.
- $\vec{V} + \vec{W} = \vec{0}$ if and only if $\vec{W} = -\vec{V}$
- The negation of $\vec{0}$ is $\vec{0}$: $-\vec{0} = \vec{0}$
- The negation of the negation of a vector is the vector itself: $-(-\vec{V}) = \vec{V}$
- If $\vec{V} + \vec{Z} = \vec{V}$, then $\vec{Z} = \vec{0}$
- Zero times any vector is the zero vector: $0\vec{V} = \vec{0}$ for every vector $\vec{V}$
- Any scalar times the zero vector: $c\vec{0} = \vec{0}$ for every real number c .
- The only ways that the product of a scalar and a vector can equal the zero vector are when either the scalar is 0 or the vector is $\vec{0}$. That is $c\vec{v} = \vec{0}$, then either $c = 0$ or $\vec{v} = \vec{0}$
- The scalar -1 times the negation of the vector: $(-1)\vec{v} = -\vec{v}$
- We define subtraction in terms of addition by defining $\vec{V} - \vec{W} = \vec{V} + (-\vec{W})$
- All the usual properties of subtraction follow, such as $\vec{u} + \vec{v} = \vec{w}$ if and only if $\vec{u} = \vec{w} - \vec{v}$
- $c(\vec{v} - \vec{w}) = c\vec{v} - c\vec{w}$
- $(c - d)\vec{v} = c\vec{v} - d\vec{v}$

Application activity 9.2



1. Let $f(\mathbb{R})$ be the set of functions: $\mathbb{R} \rightarrow \mathbb{R}$. Define the operations $(f + g)(x) = f(x) + g(x)$ and $(\alpha f)(x) = \alpha f(x)$. Show that $f(\mathbb{R})$ is a vector space under these operations.

9.3 Sub-vector spaces of the Vector spaces $\mathbb{R}^2$

Activity 9.3



Consider the vector space $\mathbb{R}^2$; Let $W = \{(x, y) \in \mathbb{R}^2 \mid y = 2x\}$.

- a) Show that the vector $(0, 0)$ belongs to W
- b) Let $u = (x_1, 2x_1)$ and $v = (x_2, 2x_2)$ be two vectors in W .
Show that $u + v$ also belongs to W .
- c) Let k be a real number and $u = (x, 2x) \in W$. Show that $ku \in W$.



Content Summary

The set of all ordered pairs of real numbers $\mathbb{R}^2$ is a vector space of dimension two. A vector in $\mathbb{R}^2$ is written as (x, y) where x and y are real numbers. In this vector space, vectors can be added together and multiplied by real numbers (scalars).

A sub-vector space (subspace) of $\mathbb{R}^2$ is a subset of vectors that is itself a vector space under the same operations of vector addition and scalar multiplication.

For a subset W of $\mathbb{R}^2$ to be a sub-vector space, the following conditions must be satisfied:

1. The Zero Vector Requirement

The additive identity (the origin) must be an element of W . $\mathbf{0} = (0, 0) \in W$

If the set does not pass through the origin, it cannot be a sub-vector space.

2. Closure Under Addition

If you take any two vectors in W , their sum must also stay within W .

For any $\mathbf{u}, \mathbf{v} \in W$: $\mathbf{u} + \mathbf{v} \in W$

3. Closure Under Scalar Multiplication

If you take any vector in W and multiply it by any real number (scalar), the resulting vector must stay within W .

For any $\mathbf{u} \in W$ and any $c \in \mathbb{R}$: $c\mathbf{u} \in W$

If any of these conditions is not satisfied, then the set is not a sub-vector space.

Example 1:

Show that the following is a subset $W = \{(x, y) \in \mathbb{R}^2 \mid y = 0\}$

Solution

Step 1: Check the zero vector

If $x = 0$, then $y = 0$. Thus, $(0, 0) \in W$

Step 2: Closure under addition

Let $u = (x_1, 0)$, $v = (x_2, 0)$. Then, $u + v = (x_1 + x_2, 0)$

Since the second component is still 0, the result belongs to W

Step 3: Closure under scalar multiplication. Let k be a real number $ku = k(x_1, 0) = (kx_1, 0)$

This vector still belongs to W . Therefore, W is a sub-vector space of $\mathbb{R}^2$

Example 2:

Check whether the following is a sub-vector space, $W = \{(x, y) \in \mathbb{R}^2 \mid x = y\}$

Solution

Step 1: Check the zero vector $0 = 0$. Thus, $0 = 0 \in W$

Step 2: Closure under addition

Let $u = (x_1, x_1)$, $v = (x_2, x_2)$. Then, $u + v = (x_1 + x_2, x_1 + x_2)$. The result still satisfies $x = y$ it belongs to W

Step 3: Closure under scalar multiplication

Let k be a real number then $ku = (kx_1, kx_1)$. This also satisfies $x = y$. Therefore, W is a sub-vector space.

Example 3:

Check whether W is a sub-vector space or not $W = \{(x, y) \in \mathbb{R}^2 \mid x + y = 1\}$

Solution

Step 1: Check the zero vector

For the zero vector $(0, 0)$: $0 + 0 = 0 \neq 1$

Thus, $(0, 0) \notin W$. Since the set does not contain the zero vector, it cannot be a sub-vector space.

Therefore, W is not a sub-vector space.

Example 4:**Solution**

Check whether W is a sub-vector space. $W = \{(x, y) \in \mathbb{R}^2 \mid xy = 0\}$

Step 1: Check the zero vector

For the zero vector $(0, 0)$. We have $0 \times 0 = 0$ and thus, $(0, 0) \in W$

Step 2: Check closure under vector addition

Take two vectors $(2, 0)$ and $(0, 3)$. Both belong to $2 \times 0 = 0$, $0 \times 3 = 0$. Now, add them

$u + v = (2 + 0, 0 + 3) = (2, 3)$. Check if this result belongs to W : $2 \times 3 = 6$

Since the set is not closed under vector addition, the subset fails one of the required conditions.

Therefore, W is not a sub-vector space of because it is not closed under vector addition.

Application activity 9.3



Let $\mathbb{R}^2$ be the vector space of all ordered pairs of real numbers. Consider the subset

$$W = \{(x, y) \in \mathbb{R}^2 \mid x - 2y = 0\}$$

Determine whether W is also a sub-vector space of $\mathbb{R}^2$

9.4 Linear combination of vectors in the Vector spaces $\mathbb{R}^2$

Activity 9.4



A linear combination of vectors $\vec{V}_1, \vec{V}_2, \dots, \vec{V}_n$ of a vector space $(\mathbb{R}, V, +)$ is a sum of the form $a_1V_1 + a_2V_2 + \dots + a_nV_n$ where $a_1, a_2, \dots, a_n$ are scalars. In the following case, decide by observation whether the vector w can be written as a linear combination of the two given vectors u and v . Explain your reasoning.

1. $u = (1, 0), \quad v = (0, 1), \quad w = (3, 2).$
2. $u = (2, 2), \quad v = (4, 4), \quad w = (6, 6).$



Content Summary

A linear combination of vectors $\vec{V}_1, \vec{V}_2, \dots, \vec{V}_n$ of a vector space $(\mathbb{R}, V, +)$ is each sum of the form $\alpha V_1 + \beta V_2 + \dots + \gamma V_n$ where $\alpha, \beta, \dots, \gamma$ are scalars which are called **coefficients of a linear combination**.

Example:

Show that the vector $\vec{w} = (14, 3)$ is a linear combination of vectors $\vec{u} = (4, 3)$ and $\vec{v} = (2, -1)$, illustrate all three vectors.

Solution:

Let $\vec{w} = \alpha\vec{u} + \beta\vec{v}$ then $(14, 3) = \alpha(4, 3) + \beta(2, -1)$, then we form the system
$$\begin{cases} 14 = 4\alpha + 2\beta \\ 3 = 2\alpha - \beta \end{cases}$$

Solve for α and β , then
$$\begin{cases} 4\alpha + 2\beta = 14 \\ 3\alpha - \beta = 3 \end{cases} \Leftrightarrow \begin{cases} 4\alpha + 2\beta = 14 \\ 6\alpha - 2\beta = 6 \end{cases} \Leftrightarrow \begin{cases} 10\alpha = 20 \\ \beta = 3\alpha - 3 \end{cases} \Leftrightarrow \begin{cases} \alpha = 2 \\ \beta = 3 \end{cases}$$

As $\alpha = 2$ and $\beta = 3$ are real numbers, $\vec{w} = \alpha\vec{u} + \beta\vec{v} = (14, 3)$ is a linear combination of $(4, 3)$ and $(2, -1)$.

Application activity 9.4



Let $u = (2,1)$ and $v = (1,3)$ and let $w = (7,8)$;

Find scalars α and β such that $\vec{w} = \alpha\vec{u} + \beta\vec{v}$

1. Write the equation using coordinates $(7,8) = \alpha(2,1) + \beta(1,3)$
2. Form the system of equations and solve for α and β
3. Verify your answer by substituting the values of α and β back into $\alpha\vec{u} + \beta\vec{v}$

9.5 Linear dependent and linear independent vectors

Activity 9.5



Let $u = (1,1)$ and $v = (3,3)$

1. Can you find a number k such that $v = ku$?
2. If you set $c_1 = 3$ and $c_2 = -1$, calculate $c_1u + c_2v$. What is the result?



Content Summary

(i) Linearly dependent vectors

We say that the vectors $\vec{V}_1, \vec{V}_2, \dots, \vec{V}_n \in V$ are linearly dependent if we can find the scalars $\alpha, \beta, \dots, \gamma$ which are not all equal to zero, such that $\alpha\vec{V}_1 + \beta\vec{V}_2 + \dots + \gamma\vec{V}_n = \vec{0}$

Example 1:

Show that the vectors $\vec{u} = (1, -1)$ and $\vec{v} = (-2, 2)$ are linearly dependent.

Solution:

$\forall, \alpha, \beta \in \mathbb{R} : \alpha\vec{u} + \beta\vec{v} = \vec{0}$. Let us find α and β we have $\alpha(1, -1) + \beta(-2, 2) = (0, 0)$, which gives us the system $\begin{cases} 1\alpha - 1\beta = 0 \\ -1\alpha + 2\beta = 0 \end{cases} \Leftrightarrow \begin{cases} \alpha - 2\beta = 0 \\ -1\alpha + 2\beta = 0 \end{cases}$. Thus $\alpha = 2\beta$

For $\alpha \in \mathbb{R}$, $\beta = \frac{\alpha}{2}$. This means that there exist many values of α and β different from zero ($\alpha = 2\beta$), hence $\vec{u}$ and $\vec{v}$ are linearly dependent.

(ii) Linearly independent vectors

We say that the vectors $\vec{V}_1, \vec{V}_2, \dots, \vec{V}_n \in V$ are linearly independent if $\alpha\vec{V}_1 + \beta\vec{V}_2 + \dots + \gamma\vec{V}_n = \vec{0}$ holds only if $\alpha = \beta = \dots = \gamma = 0$. Otherwise, the set is linearly dependent.

Example: Show that the vectors $\vec{u} = (1, -1)$ and $\vec{v} = (-2, 3)$ are linearly independent.

Solution:

$\forall, \alpha, \beta \in \mathbb{R} : \alpha\vec{u} + \beta\vec{v} = \vec{0}$. We find α and β .

$\alpha(1, -1) + \beta(-2, 3) = (0, 0)$, which gives the system:
$$\begin{cases} 1\alpha + 2\beta = 0 \\ -1\alpha + 3\beta = 0 \end{cases}$$

$0\alpha + 5\beta = 0 \Rightarrow \alpha = 0$ and $\beta = 0$. Hence, the vectors $\vec{u}$ and $\vec{v}$ are linearly independent.

(iii) Spanning vectors

Let $S = \{\vec{V}_1, \vec{V}_2, \dots, \vec{V}_n\}$ be a set of vectors in a vector space V . If every vector in V can be expressed in at least one way as a linear combination of the vectors of S , we say that the set S spans V , or that V is spanned by S .

Example 3:

Let V be the vector space $\mathbb{R}^2$ and let $S = \{\vec{v}, \vec{w}\}$, where $\vec{v} = (2, 1)$ and $\vec{w} = (1, 3)$

a) Determine whether S spans $\mathbb{R}^2$.

b) For $\vec{\gamma} = (-2, 4)$, determine c_1 and c_2 such that $\vec{\gamma} = c_1\vec{v} + c_2\vec{w}$

Solution:

a) We let $\vec{x} = (a, b)$ represents any vector in $\mathbb{R}^2$ (a, b are any arbitrary real numbers).

We must determine if there are constant c_1 and c_2 such that $c_1\vec{v} + c_2\vec{w} = \vec{x}$. This leads to $c_1(2, 1) + c_2(1, 3) = (a, b)$. From which, we obtain the following system in unknown c_1 and c_2

$$\begin{cases} 2c_1 + c_2 = a \\ c_1 + 3c_2 = b \end{cases}$$
 We find the unique solution for c_1 and c_2 to be $c_1 = \frac{3a-b}{5}$ and

$$c_2 = \frac{2b-a}{5}$$

Thus, $S = \{\vec{v}, \vec{w}\}$, spans $\mathbb{R}^2$.

b) Since $a = -2$ and $b = 4$, then $c_1 = \frac{3a-b}{5} = \frac{-6-4}{5} = -\frac{10}{5} = -2$ or

$$c_2 = \frac{2b-a}{5} = \frac{8-(-2)}{5} = \frac{10}{5} = 2$$

Thus, $\vec{\gamma} = -2\vec{v} + 2\vec{w}$

Application activity 9.5



Consider the vectors in the vector space $\mathbb{R}^2$, $u_1 = (1, 2)$, $u_2 = (2, 4)$, and $v = (3, 1)$

1. Show that the vectors u_1 and u_2 are linearly dependent by finding scalars α and β , both non-zero, such that $\alpha u_1 + \beta u_2 = (0, 0)$
2. Determine whether vectors u_1 and v are linearly independent by using $cu_1 + dv = (0, 0)$ for scalars c and d .

9.6 Basis and Dimension

Activity 9.6



Consider the vectors $u_1 = (1, 2)$ and $u_2 = (3, 1)$

- a) Check linear independence. Determine whether u_1 and u_2 are linearly independent.
- b) Check spanning: Can any vector $w = (x, y)$ in $\mathbb{R}^2$ be written as a linear combination of u_1 and u_2 ? Write the system of linear equations and explain.
- c) When u_1 and u_2 are linearly independent such that any vector $w = (x, y)$ in $\mathbb{R}^2$ can be written as a linear combination of u_1 and u_2 we say that $\{u_1, u_2\}$ form a basis of $\mathbb{R}^2$. Based on your answer above, state whether $\{u_1, u_2\}$ is a basis of $\mathbb{R}^2$.



Content Summary

A set of vectors $S = \{\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n\}$ in a vector space V is called the basis for V if

- a) S spans V and
- b) S is made by linearly independent vectors $\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n$.

Example: Let $\vec{v}_1 = (3, 2)$ and $\vec{v}_2 = (1, 4)$. Show that the set $S = \{\vec{v}_1, \vec{v}_2\}$ is a basis for $\mathbb{R}^2$.

Solution:

To show that S spans $\mathbb{R}^2$, we must show that any vector $\vec{x} = (x_1, x_2)$ in $\mathbb{R}^2$ can be expressed as a linear combination of $\vec{v}_1$ and $\vec{v}_2$. That is $\vec{x} = c_1 \vec{v}_1 + c_2 \vec{v}_2$. This leads to the system of linear equations $\begin{cases} 3c_1 + c_2 = x_1 \\ 2c_1 + 4c_2 = x_2 \end{cases}$. For S to span $\mathbb{R}^2$, we must show that the

system has a solution for c_1 and c_2 for any possible choice of $\vec{x}$. Solving for c_1 and c_2 gives us:

$$\begin{aligned} \begin{cases} 12c_1 + 4c_2 = 4x_1 \\ -2c_1 - 4c_2 = -x_2 \end{cases} &\Leftrightarrow \begin{cases} 12c_1 + 4c_2 = 4x_1 \\ -2c_1 - 4c_2 = -x_2 \end{cases} \Leftrightarrow \begin{cases} 10c_1 = 4x_1 - x_2 \\ c_2 = x_1 - 3c_1 \end{cases} \Leftrightarrow \begin{cases} c_1 = \frac{4x_1 - x_2}{10} \\ c_2 = x_1 - 3\left(\frac{4x_1 - x_2}{10}\right) \end{cases} \\ &\Leftrightarrow \begin{cases} c_1 = \frac{4x_1 - x_2}{10} \\ c_2 = \frac{10x_1 - 12x_1 + 3x_2}{10} \end{cases} \Leftrightarrow \begin{cases} c_1 = \frac{4x_1 - x_2}{10} \\ c_2 = \frac{-2x_1 + 3x_2}{10} \end{cases} \text{ which are the value of } c_1 \text{ and } c_2. \end{aligned}$$

Thus, S spans $\mathbb{R}^2$. To prove that S is linearly independent, we must show that the only solution of $c_1\vec{v}_1 + c_2\vec{v}_2 = \vec{0}$ is $c_1 = c_2 = 0$. So $\begin{cases} 3c_1 + c_2 = 0 \\ 2c_1 + 4c_2 = 0 \end{cases}$ and solving the system we obtain,

$$\begin{cases} 12c_1 + 4c_2 = 0 \\ -2c_1 - 4c_2 = 0 \end{cases} \Leftrightarrow \begin{cases} 10c_1 = 0 \\ 4c_2 = -2c_1 \end{cases} \Leftrightarrow \begin{cases} c_1 = 0 \\ c_2 = \frac{1}{2}c_1 \end{cases} \Leftrightarrow \begin{cases} c_1 = 0 \\ c_2 = 0 \end{cases} \Leftrightarrow c_1 = c_2 = 0.$$

Hence, the set $S = \{\vec{v}_1, \vec{v}_2\}$ is a basis for $\mathbb{R}^2$.

The dimension of a vector space is the number of vectors in its basis. Since the basis contains 2 vectors, $\dim(\mathbb{R}^2) = 2$

Thus, the dimension of the vector space $(\mathbb{R}, \mathbb{R}^2, +, \times)$ is 2.

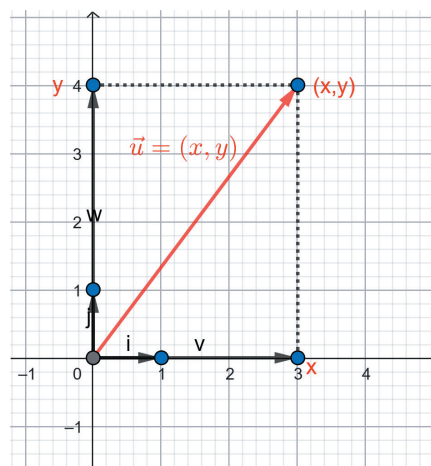
The standards basis of $\mathbb{R}^2$

The standard basis of the vector space $\mathbb{R}^2$ is the pair of unit vectors $\vec{i} = (1, 0)$ on x-axis and the vector $\vec{j} = (0, 1)$ on y-axis.

Indeed, every vector $\vec{u} = (x, y) \in \mathbb{R}^2$ can be written as a linear combination of $\vec{i} = (1, 0)$ and $\vec{j} = (0, 1)$.

$$(x, y) = (x, 0) + (0, y) = x(1, 0) + y(0, 1)$$

$$\text{i.e: } \vec{u} = x\vec{i} + y\vec{j}$$



..

Application activity 9.6



Consider the vectors

$$u_1 = (2, 1), \quad \text{and } u_2 = (1, -1)$$

- Let $w = (5, 0)$. Find scalars a and b such that $w = au_1 + bu_2$
- Write the equation using coordinates and form the corresponding system.
- Solve the system to determine the values of a and b .
- Determine whether the set $S = \{u_1, u_2\}$ form a basis of $\mathbb{R}^2$
- Hence determine the dimension of the vector space spanned by the vectors in S .

9.7 Introduction to Euclidian Vector space $\mathbb{R}^2$

Activity 9.7



You are helping someone navigate a rectangular park. The park is represented as a 2D plane. There are three key points:

- Entrance at $O = (0, 0)$
 - Birdwatching Zone at $A = (4, 3)$
 - Playground at $B = (1, 5)$
- Represent the path from the entrance to the Birdwatching Zone as a vector.
 - Represent the path from the entrance to the playground as a vector.
 - Is the path from the Birdwatching Zone to the playground in the same direction as from the entrance to the Birdwatching Zone? (Think of the angle between vectors.)
 - Can you get from the entrance to the playground by first walking to the Birdwatching Zone and then along another straight path? (Hint: Think in terms of vector addition.)
 - Without calculating exact numbers, which path is longer: Entrance $\rightarrow$ Birdwatching Zone $\rightarrow$ Playground or Entrance $\rightarrow$ Playground directly?



Content Summary

A 2D vector space is strictly Euclidean if it satisfies these four inner product axioms for all vectors $\mathbf{u}, \mathbf{v}, \mathbf{w}$ and scalar c :

- Positivity:** $\mathbf{v} \cdot \mathbf{v} \geq 0$, and $\mathbf{v} \cdot \mathbf{v} = 0$ only if $\mathbf{v} = \mathbf{0}$.
- Symmetry:** $\mathbf{u} \cdot \mathbf{v} = \mathbf{v} \cdot \mathbf{u}$.
- Additivity:** $(\mathbf{u} + \mathbf{v}) \cdot \mathbf{w} = \mathbf{u} \cdot \mathbf{w} + \mathbf{v} \cdot \mathbf{w}$.
- Homogeneity:** $(c\mathbf{u}) \cdot \mathbf{v} = c(\mathbf{u} \cdot \mathbf{v})$.

These rules can also be expressed as: For any vectors $\vec{u}, \vec{v}$, and $\vec{w}$ and any scalar α , there corresponds a real number $(\vec{u}, \vec{v})$ such that the following conditions are satisfied:

- 1) $(\vec{u} + \vec{v}, \vec{w}) = (\vec{u}, \vec{w}) + (\vec{v}, \vec{w})$
- 2) $(\alpha \vec{u}, \vec{v}) = \alpha(\vec{u}, \vec{v})$
- 3) $(\vec{u}, \vec{v}) = (\vec{v}, \vec{u})$
- 4) $(\vec{u}, \vec{u}) > 0$ for $\vec{u} \neq 0$

The expression $(\vec{u}, \vec{u})$ is frequently denoted by $(\vec{u})$; it is called the scalar square of the vector $\vec{u}$. Thus, property (4) implies that the quadratic form corresponding to the bilinear form $(\vec{u}, \vec{u})$ is positive definite; it is called the scalar square of the vector $\vec{u}$.

In two dimensions, the **Euclidean vector space** is usually written as $\mathbb{R}^2 = \{(x, y) \mid x, y \in \mathbb{R}\}$ which is the set of all ordered pairs of real numbers.

Example:

Let $u = (2, 1)$ and $v = (3, 4)$

- i. Vector addition: $u + v = (2, 1) + (3, 4) = (5, 5)$
- ii. Scalar multiplication:
 $2u = 2(2, 1) = (4, 2)$.

Application activity 9.7



Give two vectors $\vec{u} = (2, 1)$ and $\vec{v} = (1, 3)$

1. On a Cartesian plane, draw the vectors $\vec{u}$ and $\vec{v}$ starting from the origin $(0, 0)$.
2. Observe the vectors and describe the direction of each vector.
3. Identify which vector appears longer.
4. Look and measure the angle formed between $\vec{u}$ and $\vec{v}$.
5. Decide whether the vectors point in similar directions or different directions.

9.8 Dot product and properties

Activity 9.8



The dot product of vectors (a, b) and (c, d) is the scalar $(a, b) \cdot (c, d) = ac + bd$. Let $\vec{u} = (2, 3)$ and $\vec{v} = (4, -1)$ be two vectors, compute the dot product of vectors $\vec{u}$ and $\vec{v}$.



Content Summary

The **dot product** (also called the **scalar product**) is an operation between two vectors that results in a **single number (a scalar)**. If two vectors are given as $a = (a_1, a_2)$ and $b = (b_1, b_2)$ their dot product is calculated by multiplying corresponding components and then adding the results $a \cdot b = a_1 b_1 + a_2 b_2$

Example:

Let $a = (2, 3)$, $b = (4, 1)$, then $a \cdot b = (2 \times 4) + (3 \times 1) = 8 + 3 = 11$

Properties

Let's define our primary vectors as: $\mathbf{u} = (3, 2)$, $\mathbf{v} = (-1, 4)$, $\mathbf{w} = (5, 0)$; Here are the key **properties of the dot product** (also called the scalar product) in a Euclidean vector space:

1. Commutative property

$$u \cdot v = v \cdot u$$

$$\mathbf{u} \cdot \mathbf{v} = (3)(-1) + (2)(4) = -3 + 8 = 5$$

$$\mathbf{v} \cdot \mathbf{u} = (-1)(3) + (4)(2) = -3 + 8 = 5$$

The results are identical. The order of vectors does not affect the result.

2. Distributive over Vector Addition

$$u \cdot (v + w) = u \cdot v + u \cdot w$$

- First, find $\mathbf{v} + \mathbf{w} = (-1 + 5, 4 + 0) = (4, 4)$.
- $\mathbf{u} \cdot (\mathbf{v} + \mathbf{w}) = (3)(4) + (2)(4) = 12 + 8 = 20$
- Now, calculate the parts: $\mathbf{u} \cdot \mathbf{v} = 5$ (from Example 1) and $\mathbf{u} \cdot \mathbf{w} = (3)(5) + (2)(0) = 15$.
- $\mathbf{u} \cdot \mathbf{v} + \mathbf{u} \cdot \mathbf{w} = 5 + 15 = 20$

Therefore, the dot product distributes across addition.

3. Scalar Multiplication

$$(ku) \cdot v = k(v \cdot u) = u \cdot (kv)$$

$$c\mathbf{u} = 2(3, 2) = (6, 4) \text{ and } (c\mathbf{u}) \cdot \mathbf{v} = (6)(-1) + (4)(4) = -6 + 16 = 10. \text{ Now check : } c(\mathbf{u} \cdot \mathbf{v}) = 2(5) = 10.$$

You can factor out a scalar from either vector.

4. Positive-Definiteness

$u \cdot u \geq 0$ i.e. $u \cdot u = (3)(3) + (2)(2) = 9 + 4 = 13$.

The number $\vec{u} \cdot \vec{u} = \|\vec{u}\|^2$: The dot product of a vector with itself is always non-negative, and it equals zero only if the vector is the zero vector.

5. Orthogonality Property

$u \cdot v = 0$ if and only if $u \perp v$: Two vectors are perpendicular if their dot product is zero.

Let us take $\mathbf{a} = (2, 3)$ and $\mathbf{b} = (-3, 2)$, $\mathbf{a} \cdot \mathbf{b} = (2)(-3) + (3)(2) = -6 + 6 = 0$

Because the dot product is exactly zero, vectors $\mathbf{a}$ and $\mathbf{b}$ are orthogonal (perpendicular).

Application activity 9.8



1. Let $\mathbf{a} = (3, -2)$ and $\mathbf{b} = (1, 4)$.

a) Compute the dot product $\mathbf{a} \cdot \mathbf{b}$

b) Determine the angle θ between the two vectors using $\cos \theta = \frac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{a}| |\mathbf{b}|}$

9.9 Modulus or Magnitude of Vectors

Activity 9.9



Consider the following vectors: $\mathbf{a} = (3, 4)$, $\mathbf{b} = (-2, 5)$, $\mathbf{c} = (-3, -4)$, $\mathbf{d} = (5, -12)$

a) Calculate the magnitude (modulus) of each vector using the formula: For $\mathbf{v} = (x, y)$,

$$|\mathbf{v}| = \sqrt{x^2 + y^2}$$

b) Compare the magnitudes of the vectors and arrange them from smallest to largest.



Content Summary

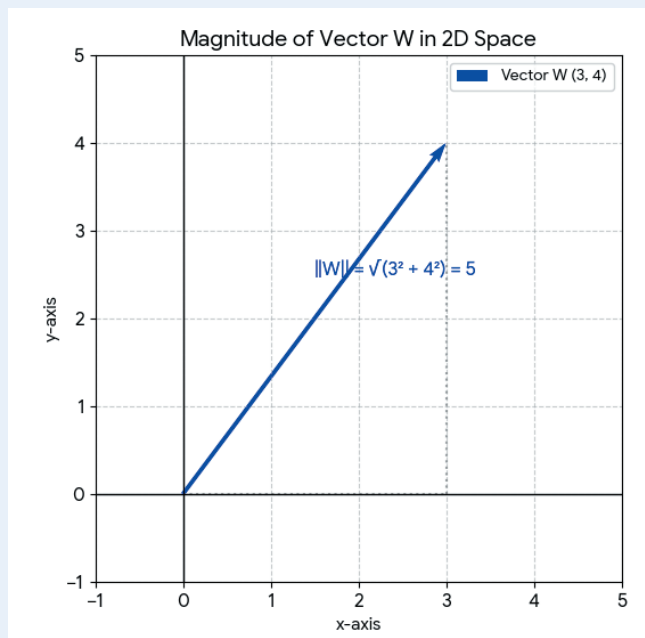
If $A(x_1, y_1)$ and $B(x_2, y_2)$, then we have the vector $\overrightarrow{AB} = (x_2 - x_1, y_2 - y_1)$. The magnitude of the vector $\overrightarrow{AB}$ is given by $|\overrightarrow{AB}| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$. Particularly, for $A(0, 0)$ and $B(x, y)$, $|\overrightarrow{AB}| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} = \sqrt{x^2 + y^2}$

Example 1:

let the magnitude of $\vec{w} = (3, 4)$

Solution:

$$\|\vec{w}\| = \sqrt{3^2 + 4^2} = \sqrt{9 + 16} = 5$$



Example 2:

If $A = (3, 1)$ and $B = (6, 3)$. Find (a) $\overrightarrow{AB}$ (b) $|\overrightarrow{AB}|$

Solution:

$$\text{a) } \overrightarrow{AB} = \begin{pmatrix} 6-3 \\ -3-1 \end{pmatrix} = \begin{pmatrix} 3 \\ -4 \end{pmatrix}$$

$$\text{b) } |\overrightarrow{AB}| = \sqrt{(3)^2 + (-4)^2} = \sqrt{9 + 16} = 5$$

Unit vector

A unit vector has a length of one unit.

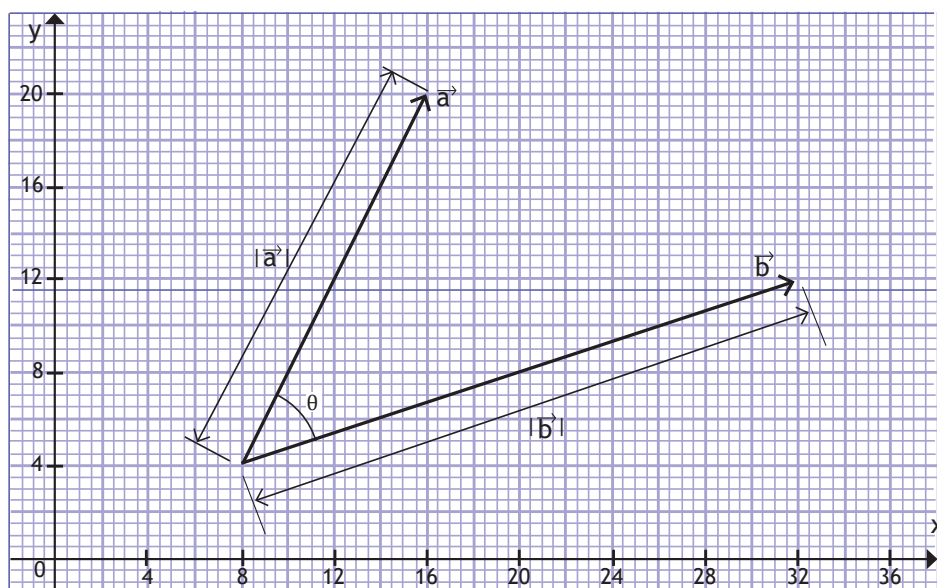
The unit vector in the direction of the vector a is the vector $\vec{a} = \frac{1}{|\vec{a}|} \vec{a}$

The unit vector in the direction of x and y are denoted by $\vec{i}$ and $\vec{j}$ respectively.

Thus, $\vec{i} \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $\vec{j} \begin{pmatrix} 0 \\ 1 \end{pmatrix}$.

The dot product of two vectors (given an angle formed between)

Let us use Figure 9.3 to show how we can calculate the dot product of two vectors:



$$\vec{a} \cdot \vec{b} = |\vec{a}| \times |\vec{b}| \times \cos \theta$$

$|\vec{a}|$ is the magnitude (length) of vector $\vec{a}$.

$|\vec{b}|$ is the magnitude (length) of vector $\vec{b}$

θ is the angle between $\vec{a}$ and $\vec{b}$

Relation to angle between vectors

Given that for $\vec{u} = (a, b)$; $\|\vec{u}\| = \sqrt{\vec{u} \cdot \vec{u}} = \sqrt{a^2 + b^2}$, we find $u \cdot v = \|u\| \|v\| \cos \theta$.

If $\mathbf{u} = (3, 2)$, $\mathbf{v} = (-1, 4)$ let us find the angle between them:

$$\|\mathbf{u}\| = \sqrt{3^2 + 2^2} = \sqrt{13}; \quad \|\mathbf{v}\| = \sqrt{(-1)^2 + 4^2} = \sqrt{17}$$

We know that $\mathbf{u} \cdot \mathbf{v} = 5$. Therefore, $5 = \sqrt{13} \cdot \sqrt{17} \cos \theta$

$$\cos(\theta) = \frac{5}{\sqrt{13} \cdot \sqrt{17}} = \frac{5}{\sqrt{221}} \approx 0.336 \quad \theta = \cos^{-1}(0.336) \approx 70.35^\circ$$

Where θ is the angle between u and v . This is often used to find angles or check if vectors are perpendicular $u \cdot v = 0$.

Application activity 9.9



Let $a = (2, 3)$ and $b = (4, -1)$ be two vectors.

- Compute the dot product of vectors a and b .
- Draw and determine the angle θ between the two vectors using the formula:

$$\cos \theta = \frac{a \cdot b}{|a| |b|}$$

- Verify the commutative property of the dot product by computing $b \cdot a$



Assessing knowledge

1. Write two examples of vectors in the vector space $\mathbb{R}^2$.
2. Identify the components of the vector $(4, -3)$.

Skills

3. If $|a| = 5$, $|b| = 10$ and the angle between them is 60°

Find $a \cdot b$

4. Let $u = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$, $v = \begin{pmatrix} 2 \\ 4 \end{pmatrix}$, $w = \begin{pmatrix} 3 \\ 1 \end{pmatrix}$ be vectors in $\mathbb{R}^2$

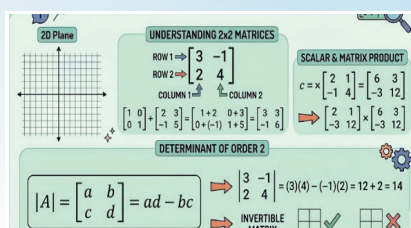
- a) Determine whether the set $\{u, v, w\}$ is linearly independent or linearly dependent.
- b) Find the set of vectors from $\{u, v, w\}$ that span $\mathbb{R}^2$.

Attitudes and values

5. When structural engineers design a suspension bridge, they use the magnitude of vectors, the angles between cables, and the dot product to balance forces. Why should a student value mastering these vector concepts beyond just passing an exam?
 - a) Because it proves that mathematical concepts are completely abstract and should only be handled by computers rather than human design.
 - b) Because these concepts provide the exact geometric and algebraic tools needed to analyze forces, optimize materials, and ensure the safety of real-world structures.
 - c) Because knowing these definitions allows a student to solve textbook equations faster than their classmates during timed competitions.
 - d) Because vector operations are designed to make simple directional problems look more complicated to external observers.
- 5) A student is modeling the work done by a constant force vector moving an object along a displacement vector. They calculate a dot product of zero and immediately conclude, "*No work is being done because the force vector has a magnitude of zero.*" However, looking at the system, the force is quite large. What attitude or reflective habit is missing from this student's approach?
 - a) The patience to accept that mathematical models often contradict reality, so the physical observation should be ignored.
 - b) The desire to change the initial data so that the dot product yields a non-zero number that matches their expectation.
 - c) The habit of critical precision, which would lead them to realize that a zero dot product means the force is acting perfectly perpendicular 90° to the direction of motion, not that the force itself is zero.
 - d) The compliance to simply copy a classmate's work without questioning why the calculations and physical observations do not seem to align.

UNIT 10

MATRICES AND DETERMINANTS OF ORDER 2



Key Unit Competence:

Use matrices and determinants of order 2 in solving simultaneous linear equations.

Learning objectives

By the end of this unit, you should be able to:

Knowledge and understanding

1

- Describe 2×2 matrices and their role in linear and geometric transformations.
- Explain matrix operations: equality, addition, scalar multiplication, multiplication, transpose, and inverse.
- Describe 2×2 determinants, their properties, and applications in solving linear systems and transformation problems.

Skills

2

- Compute and represent linear and geometric transformations using matrices.
- Perform matrix operations accurately and efficiently.
- Calculate determinants and use them to solve problems
- Apply matrix operations and determinants to real-life and mathematical problems.

Attitudes and values

3

- Appreciate the role of matrices in modelling and problem-solving.
- Apply curiosity and analytical thinking when exploring matrix applications.
- Build confidence and responsibility in performing matrix-related tasks.

Introduction

This unit introduces you to the fundamental concepts of matrices and determinants of order 2. The concepts acquired in this unit are essential tools in mathematics, physics, engineering, and computer science. A matrix of order 2 is a tabular array of numbers with 2 rows and 2 columns, used to organize data, represent linear equations, and perform transformations in two-dimensional space. The determinant of a 2×2 matrix is a special number associated with the matrix that provides important information about the system it represents.

In this unit, you will learn how to perform operations on 2×2 matrices, including addition, scalar multiplication, and multiplication, as well as how to calculate determinants. You will also explore how matrices and determinants can be used to solve systems of linear equations, which is a practical skill for modelling and analysing real-world problems. Understanding these concepts will help you see the connection between algebraic computations and geometric interpretations in 2D.

You will acquire key words such as matrix of order 2, determinant, system of linear equations, scalar multiplication, matrix addition, matrix multiplication, and inverse matrix.

Introductory activity



Search in the reference books or on internet and find the meaning of a matrix. Then, consider the system of linear equations:
$$\begin{cases} 2x + 3y = 8 \\ x - y = 2 \end{cases}$$

1. Write the matrix of coefficients: Look at the numbers in front of x and y . Place them in a 2×2 matrix A .
2. Write the constant as a column matrix: Take the numbers on the right side of the equations and write them as a 2×1 matrix B .
3. Write the variables as a column matrix: Take x and y and write them as a 2×1 matrix X .
4. Now, write the system as a matrix equation $AX = B$.

10.1 Linear transformation in two dimensions

Activity 10.1



What is transformation as it is used in mathematics? What is a linear transformation? Research on the different types of 2D linear transformations.

Discuss your findings with the rest of the class.



Content Summary

A linear transformation T defined from a vector space V to a vector space W is a function $T : V \rightarrow W$ that satisfies the following conditions:

- 1) $\forall \vec{u}, \vec{v} \in V : T(\vec{u} + \vec{v}) = T(\vec{u}) + T(\vec{v})$
- 2) $\forall \vec{u} \in V$ and scalar $\alpha : T(\alpha \vec{u}) = \alpha T(\vec{u})$

Example 1:

Show that $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined by $T(x, y) = (x, x + y)$ is a linear transformation.

Solution:

We verify the two conditions of the definition. For the first condition, we let (x_1, y_1) and (x_2, y_2) in $\mathbb{R}^2$ and compute.

$$T((x_1, y_1) + (x_2, y_2)) = T(x_1 + x_2, y_1 + y_2) = (x_1 + x_2, x_1 + x_2 + y_1 + y_2)$$

$$= (x_1, x_1 + y_1) + (x_2, x_2 + y_2) = T(x_1, y_1) + T(x_2, y_2). \text{ This proves the first condition.}$$

$T(\alpha(x, y)) = T(\alpha x, \alpha y) = (\alpha x, \alpha x + \alpha y) = \alpha(x, x + y) = \alpha T(x, y)$. Hence T is a linear transformation.

Example 2:

Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be given by $T(x, y) = (x, 1)$. Show that T is not linear.

Solution:

We show that the first condition of the definition is violated. Indeed, for any two vectors (x_1, y_1) and (x_2, y_2) , we have $T((x_1, y_1) + (x_2, y_2)) = T(x_1 + x_2, y_1 + y_2) = (x_1 + x_2, 1) \neq (x_1, 1) + (x_2, 1)$

$$= T(x_1, y_1) + T(x_2, y_2)$$

Hence, the given transformation is not linear.

Application activity 10.1



1. In pairs, determine whether the following are linear transformations or not.

a) $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2 : T(x, y) = (y, 0)$

b) $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2 : T(x, y) = (x^2, 0)$

10.2 Matrix of a linear transformation

Activity 10.2



A transformation f : defined in 2D is such that $f(x, y) = (30x + 25y, 20x + 15y)$.

This information was written in the following way:

$$f\begin{pmatrix} x \\ y \end{pmatrix} = \begin{bmatrix} 30 & 20 \\ 25 & 15 \end{bmatrix} \begin{pmatrix} x \\ y \end{pmatrix}. \text{ If this expression is equivalent to } f(\vec{v}) = A\vec{v}$$

- Write the vector $\vec{v}$
- What does A represent?
- Give the example of real-life situation where you found the expressions written like A .



Content Summary

Image of a vector under a transformation:

Consider a transformation $f: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ that associates (x, y) to (x', y') such that

$$f: \mathbb{R}^2 \rightarrow \mathbb{R}^2 : f(x, y) = (ax + by, cx + dy). \text{ We can write it as } \begin{pmatrix} x' \\ y' \end{pmatrix} = f\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} ax + by \\ cx + dy \end{pmatrix} \text{ or}$$

writing in the form $\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}.$

$$\text{This expression can be written as } \begin{pmatrix} x' \\ y' \end{pmatrix} = A \begin{pmatrix} x \\ y \end{pmatrix} \text{ where } A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}.$$

Table A is called a matrix.

Example:

Consider the reflection (symmetry) about x -axis noted by R_x defined in the vector space

$$\mathbb{R}^2. \text{ It is defined by: } f(\vec{i}) = f\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \text{ and } f(\vec{j}) = f\begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ -1 \end{pmatrix}.$$

The images can be written using the table called the standard transformation matrix

$$M = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

The transformation matrix is the matrix obtained by using the images of the standard basis of the vector space.

(i) Definition

A matrix is a rectangular arrangement of numbers, expressions, and algebraic symbols that are arranged in rows and columns. A matrix is denoted with a capital letter: $A, B, C, \dots$, and the elements are enclosed by parentheses $()$ or square brackets $[]$.

Example:

$$A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} \quad B = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} \quad C = \begin{pmatrix} 1 & 3 & 4 \\ 5 & 12 & 13 \\ 7 & 6 & 0 \end{pmatrix} \quad M = \begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} & \cdots & a_{mn} \end{pmatrix}$$

To identify an element of a matrix, we consider which row and column the element is positioned.

Example:

matrix B, a_{32} is an element of the third row and second column.

From matrix C, 4 is an element of the first row and the third column.

(ii) The dimension, size, or order of the matrix

From the previous example, Matrix A is of order 2×2 ("read three by three") because there are 2 rows and 2 columns

Matrix B is of order 3×3 ("read three by three") because there are 3 rows and 3 columns

Matrix M is of order $m \times n$, ("read m by n") because there are m rows and n , columns

A square matrix is a matrix formed by the same number of rows and columns.

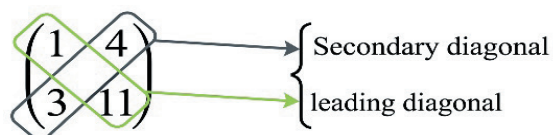
Square matrix of **order 2** has the form $\begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}$

Square matrix of order three has the form $\begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}$

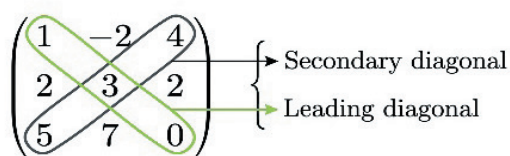
The elements of the form (a_{ij}) , where the two subscripts i and j are equal, constitute the **principal diagonal** (or **leading diagonal** or **main diagonal** or **major diagonal** or **primary diagonal**).

The **secondary diagonal** (or **minor diagonal** or **antidiagonal** or **counter diagonal**) is formed by the elements with $i + j = n + 1$.

Example of Matrix of order two:



Example of Matrix of order three:



(iii) Types of matrices

There are several types of matrices, but the most used are:

1) Row matrix: A matrix formed by one row

Example: $(2 \quad 4 \quad 7)$

2) Column matrix or Vector matrix: A matrix formed by one column

Example: $\begin{pmatrix} 1 \\ 3 \\ 6 \end{pmatrix}$

3) Zero matrix or null matrix: all elements are zero

Example: $\begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}$

4) Triangular matrix: A matrix whose elements located below or above the leading diagonal are zeros

a) Upper triangular matrix

In an upper triangular matrix, the elements located below the leading diagonal are zeros.

Example: 1) $\begin{pmatrix} 1 & 2 \\ 0 & -5 \end{pmatrix}$ 2) $\begin{pmatrix} 1 & -2 & 4 \\ 0 & 3 & 2 \\ 0 & 0 & 4 \end{pmatrix}$

b) Lower triangular matrix

In a lower triangular matrix, the elements above the leading diagonal are zeros.

Example: 1) $\begin{pmatrix} 1 & 0 \\ 18 & -5 \end{pmatrix}$ 2) $\begin{pmatrix} 1 & 0 & 0 \\ 2 & 3 & 0 \\ 5 & 7 & 9 \end{pmatrix}$

5) Diagonal matrix

In a diagonal matrix, all the elements above and below the leading diagonal are zeros.

Example: 1) $\begin{pmatrix} 10 & 0 \\ 0 & -5 \end{pmatrix}$ 2) $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 4 \end{pmatrix}$

6) Scalar matrix

A scalar matrix is a diagonal matrix in which the leading diagonal elements are equal.

Example: 1) $\begin{pmatrix} -5 & 0 \\ 0 & -5 \end{pmatrix}$ 2) $\begin{pmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{pmatrix}$

7) Identity matrix or Unity matrix

An identity matrix by multiplication of matrices (noted by I) is a diagonal matrix in which the leading diagonal elements are equal to 1.

(i) Identity matrix of order two $I_2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$

(ii) Identity matrix of order three $I = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$

8) Rectangular matrix: A matrix is said to be rectangular if the number of rows is not equal to the number of columns

Example: A 5×2 matrix

$$\begin{bmatrix} 1 & 2 & -3 & 4 & 5 \\ 2 & 0 & 6 & 7 & -\frac{1}{3} \end{bmatrix}$$

Application activity 10.2



Answer with “True” or “False” for each of the following statements.

1. A row matrix is a matrix that has only one row.
2. A column matrix is a matrix that has only one column.
3. A zero matrix is a matrix in which all elements are equal to 1.
4. In an upper triangular matrix, all elements below the main (leading) diagonal are zero.
5. In a lower triangular matrix, all elements below the main diagonal are zero.
6. In a diagonal matrix, all elements above and below the main diagonal are zero.
7. A scalar matrix is a diagonal matrix in which all diagonal elements are equal.
8. An identity matrix is a diagonal matrix in which all diagonal elements are equal to 1.
9. A rectangular matrix has the same number of rows and columns.
10. The transpose of a matrix is obtained by interchanging rows and columns.

10.3 Operations on matrices and their properties: Equality and Addition of matrices.

Activity 10.3



Given $A = \begin{pmatrix} 2 & 4 \\ 1 & 3 \end{pmatrix}$ and $B = \begin{pmatrix} 5 & 2 \\ 3 & 6 \end{pmatrix}$,

- Find $A + B$ by adding the values of the same position $a_{ij} + b_{ij}$ where for example we add $a_{21} + b_{21}$.
- State the order of the resulting matrix.



Content Summary

(i) Equality of matrices

Two matrices are equal if the elements of the two matrices that occupy the same position are equal i.e

a) If $\begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} = \begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{pmatrix}$, then $\begin{cases} a_{11} = b_{11} \\ a_{21} = b_{21} \\ a_{12} = b_{12} \\ a_{22} = b_{22} \end{cases}$

Example:

Given the matrices $A = \begin{pmatrix} 2 & 5 \\ x & 7 \end{pmatrix}$ and $B = \begin{pmatrix} 2 & y \\ 3 & 7 \end{pmatrix}$. If $A = B$, find the values of x and y

Solution:

If two matrices are equal, the elements in the same positions must be equal.

$$x = 3 \text{ and } y = 5$$

(ii) Addition of matrices

Given two matrices of the same dimension, $A = (a_{ij})$ and $B = (b_{ij})$, the matrix sum is defined as: $A + B = (a_{ij} + b_{ij})$. That is, the resultant matrix's elements are obtained by adding the elements of the two matrices that occupy the same position.

(i) If $A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}$ and $B = \begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{pmatrix}$, then

$$A + B = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} + \begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{pmatrix} = \begin{pmatrix} a_{11} + b_{11} & a_{12} + b_{12} \\ a_{21} + b_{21} & a_{22} + b_{22} \end{pmatrix}$$

$$\text{And } A - B = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} - \begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{pmatrix} = \begin{pmatrix} a_{11} - b_{11} & a_{12} - b_{12} \\ a_{21} - b_{21} & a_{22} - b_{22} \end{pmatrix}$$

Example:

1) If $A = \begin{pmatrix} 3 & 2 \\ -1 & 1 \end{pmatrix}$ and $B = \begin{pmatrix} 1 & 2 \\ 4 & 6 \end{pmatrix}$, find the sum $A + B$ and the difference $A - B$

Solution:

$$A + B = \begin{pmatrix} 3 & 2 \\ -1 & 1 \end{pmatrix} + \begin{pmatrix} 1 & 2 \\ 4 & 6 \end{pmatrix} = \begin{pmatrix} 4 & 4 \\ 3 & 7 \end{pmatrix} \text{ and } A - B = \begin{pmatrix} 3 & 2 \\ -1 & 1 \end{pmatrix} - \begin{pmatrix} 1 & 2 \\ 4 & 6 \end{pmatrix} = \begin{pmatrix} 2 & 0 \\ -5 & -5 \end{pmatrix}$$

Properties of addition of matrices

- 1) **Closure:** The sum of two matrices of order two is another matrix of order two.
- 2) **Associative:** $A + (B + C) = (A + B) + C$
- 3) **Additive identity:** $A + 0 = A$, where 0 is the zero-matrix of the same dimension.
- 4) **Additive inverse:** $A + (-A) = 0$

The opposite matrix of A written as $-A$ is a matrix obtained by changing the sign for every value of A .

- 5) **Commutative:** $A + B = B + A$

Example:

1) Given the matrices: $A = \begin{pmatrix} 5 & -2 \\ 3 & 1 \end{pmatrix}$ and $B = \begin{pmatrix} 4 & 7 \\ -3 & 0 \end{pmatrix}$.

Calculate $A + B$ and verify the Commutative Property ($A + B = B + A$).

Solution:

First, calculate **A + B**: $A + B = \begin{pmatrix} 5+4 & -2+7 \\ 3+(-3) & 1+0 \end{pmatrix} = \begin{pmatrix} 9 & 5 \\ 0 & 1 \end{pmatrix}$

Next, calculate **B + A**: $B + A = \begin{pmatrix} 4+5 & 7+(-2) \\ -3+3 & 0+1 \end{pmatrix} = \begin{pmatrix} 9 & 5 \\ 0 & 1 \end{pmatrix}$

Conclusion: Since both results are $\begin{pmatrix} 9 & 5 \\ 0 & 1 \end{pmatrix}$, the commutative property is verified.

2) Given the matrix: $M = \begin{pmatrix} -6 & 2 \\ 0 & -4 \end{pmatrix}$

- i. Find the **additive inverse** of M (written as $-M$).
- ii. Show that $M + (-M) = 0$, where 0 is the zero-matrix.

Solution:

i. To find $-M$, change the sign of every element in M : $-M = \begin{pmatrix} 6 & -2 \\ 0 & 4 \end{pmatrix}$

ii. **Finding the sum $M + (-M)$:** $M + (-M) = \begin{pmatrix} -6+6 & 2+(-2) \\ 0+0 & -4+4 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$

Conclusion: The sum results in the zero-matrix, confirming the property of the additive inverse.

Application activity 10.3



Given $A = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 2 & 0 \\ 1 & 3 \end{bmatrix}$

- Compute $A + B$
- Compute $A - B$
- Compute $2B = B + B$ and find $3B$.

10.4 Operations on matrices and their properties: Multiplication by a scalar on matrices

Activity 10.4



A small school canteen sells notebooks and pens for two days. The daily sales (in number of items) are recorded in a matrix (in rows) as follows: $S = \begin{pmatrix} 10 & 8 \\ 12 & 7 \end{pmatrix}$

If the number of items triples due to high demand, find the number of items sold.



Content Summary

Given a matrix, $A = (a_{ij})$, and a real number, $k \in \mathbb{R}$, the product of a real number by a matrix is a matrix of the same dimension as A , and each element is multiplied by k .

$$k \cdot A = (k a_{ij})$$

$$\text{If } A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} \text{ then } kA = \begin{pmatrix} ka_{11} & ka_{12} \\ ka_{21} & ka_{22} \end{pmatrix}$$

Example:

If $A = \begin{pmatrix} -3 & 6 \\ 5 & 2 \end{pmatrix}$, find $2A$.

Solution:

$$2A = 2 \begin{pmatrix} -3 & 6 \\ 5 & 2 \end{pmatrix} = \begin{pmatrix} -6 & 12 \\ 10 & 4 \end{pmatrix}$$

Example:

Consider the matrix $A = \begin{pmatrix} 2 & 0 & 1 \\ 3 & 0 & 0 \\ 5 & 1 & 1 \end{pmatrix}$, find $7A$.

Solution:

$$7A = 7 \begin{pmatrix} 2 & 0 & 1 \\ 3 & 0 & 0 \\ 5 & 1 & 1 \end{pmatrix} = \begin{pmatrix} 14 & 0 & 7 \\ 21 & 0 & 0 \\ 25 & 7 & 7 \end{pmatrix}$$

Properties:

- 1) $\alpha(\beta A) = (\alpha\beta)A$, $A \in M_{m \times n}$, $\alpha, \beta \in \mathbb{R}$
- 2) $\alpha(A + B) = \alpha A + \alpha B$, $A, B \in M_{m \times n}$, $\alpha \in \mathbb{R}$
- 3) $(\alpha + \beta)A = \alpha A + \beta A$, $A \in M_{m \times n}$, $\alpha, \beta \in \mathbb{R}$
- $1A = A$, $A \in M_{m \times n}$

Application activity 10.4

In a school, the number of students participating in math and science clubs on two different days is recorded as: $C = \begin{pmatrix} x & 6 \\ 8 & y \end{pmatrix}$

If the number of students triples on a special event day, the resulting matrix becomes

$3.C = \begin{pmatrix} 9 & 18 \\ 24 & 21 \end{pmatrix}$. Find the value of x and y .

10.5 Operations on matrices and their properties: Multiplication of matrices

Activity 10.5

Consider the following matrices: $A = \begin{pmatrix} 2 & 3 \end{pmatrix}$ and $B = \begin{pmatrix} 4 \\ 5 \end{pmatrix}$.

- a) Multiply the row matrix A by the column matrix B .
- b) State the result.

**Content Summary**

Matrix multiplication is an operation where **rows of the first matrix are multiplied by columns of the second matrix**, and the products are added. Two matrices can be multiplied only if the number of columns of the first matrix equals the number of rows of the second matrix.

Let $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ and $B = \begin{pmatrix} e & f \\ g & h \end{pmatrix}$. Then the product $AB = \begin{pmatrix} ae + bg & af + bh \\ ce + dg & cf + dh \end{pmatrix}$

This means:

- Multiply row 1 of A by column 1 of B to get the first element.
- Multiply row 1 of A by column 2 of B to get the second element.
- Multiply row 2 of A by column 1 of B to get the third element.
- Multiply row 2 of A by column 2 of B to get the fourth element.

Example 1:

If $A = \begin{pmatrix} 1 & 3 \\ 2 & 5 \end{pmatrix}$ and $B = \begin{pmatrix} 2 & 0 \\ 1 & 1 \end{pmatrix}$, find the product $A \cdot B$

Solution:

$$A \cdot B = \begin{pmatrix} 1 & 3 \\ 2 & 5 \end{pmatrix} \cdot \begin{pmatrix} 2 & 0 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 1 \cdot 2 + 3 \cdot 1 & 1 \cdot 0 + 3 \cdot 1 \\ 2 \cdot 2 + 5 \cdot 1 & 2 \cdot 0 + 5 \cdot 1 \end{pmatrix} = \begin{pmatrix} 5 & 3 \\ 9 & 5 \end{pmatrix}$$

Example 2:

If $A = \begin{pmatrix} 2 & 3 \\ 1 & 4 \end{pmatrix}$ and $B = \begin{pmatrix} 5 & 2 \\ 0 & 1 \end{pmatrix}$, Find the product AB

Solution:

Step 1: Multiply the first row of A by the first column of B . Then we have

$$(2 \times 5) + (3 \times 0) = 10 + 0 = 10$$

Step 2: Multiply the first row of A by the second column of B . Then we have

$$(2 \times 2) + (3 \times 1) = 4 + 3 = 7$$

Step 3: Multiply the second row of A by the first column of B .

$$(1 \times 5) + (4 \times 0) = 5 + 0 = 5$$

Step 4: Multiply the second row of A by the second column of B .

$$(1 \times 2) + (4 \times 1) = 2 + 6 = 6$$

Final answer $AB = \begin{pmatrix} 10 & 7 \\ 5 & 6 \end{pmatrix}$

Therefore, the product of A and B is $\begin{pmatrix} 10 & 7 \\ 5 & 6 \end{pmatrix}$.

Note: Idempotent matrix or projection matrix: A matrix that is equal to its square i.e

$$A^2 = A \times A = A$$

Example: $P = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$. We have $P^2 = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} = P$.

Application activity 10.5



Two schools organize buses to transport students to a sport competition. The number of buses and the number of students each bus can carry are represented using matrices.

The number of buses used by School A and School B is given by $A = \begin{pmatrix} 2 & 1 \\ 3 & 2 \end{pmatrix}$

Where: Row 1 represents School A

Row 2 represents School B

Column 1 represents small buses

Column 2 represents large buses

The number of students each type of bus can carry is given by $A = \begin{pmatrix} 20 & 0 \\ 0 & 40 \end{pmatrix}$

Where: A small bus carries 20 students and a large bus carries 40 students.

1. Multiply the matrices A and B
2. Interpret the result of the multiplication in terms of the number of students transported by each school.

10.6 Determinant of a matrix of order 2: Definition and properties

Activity 10.6



Given that for $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, the determinant of A is $\det(A) = ad - bc$.

- 1) Find $\det B$ if $B = \begin{pmatrix} 1 & 4 \\ -1 & 2 \end{pmatrix}$
- 2) Search and tell where the determinant is used.



Content Summary

(i) Meaning of determinant

The **determinant** of a 2×2 matrix is a scalar value (a single number) that is calculated from the elements of the square matrix. It provides important information about the matrix, such as whether it has an inverse. For a 2×2 matrix, $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, the determinant is defined as $\det(A) = ad - bc$

Example 1: Non-Zero determinant

$A = \begin{pmatrix} 2 & 3 \\ 1 & 4 \end{pmatrix}$, $\det(A) = (2 \times 4) - (3 \times 1) = 8 - 3 = 5$. Since $\det(A) \neq 0$, matrix A is **invertible**, it has an inverse matrix.

Example 2: Zero determinant

$$B = \begin{pmatrix} 2 & 4 \\ 1 & 2 \end{pmatrix}, \det(B) = (2 \times 2) - (4 \times 1) = 4 - 4 = 0$$

Here, the second row is a multiple of the first row. The determinant is 0, so the matrix B is **not invertible, it has not an inverse matrix**.

(ii) Properties

- 1. Invertibility:** A 2×2 matrix is **invertible** (has an inverse) if and only if its determinant is not zero ($\det(A) \neq 0$).

A Singular Matrix (No Inverse)

A matrix is **singular** if its determinant is **exactly zero** ($\det(A) = 0$). In this case, the matrix cannot be inverted.

Problem: Determine if matrix $B = \begin{pmatrix} 6 & 3 \\ 4 & 2 \end{pmatrix}$ is singular. $\det(B) = (6 \times 2) - (3 \times 4) = 0$

- 2. Row multiples:** If one row (or column) is a multiple of another row (or column), the determinant is 0.

Example: Let $A = \begin{pmatrix} 2 & 3 \\ 4 & 6 \end{pmatrix}$. Notice that Row 2 is exactly $2 \times$ Row 1).

Calculation: $\det(A) = (2 \times 6) - (3 \times 4) = 12 - 12 = 0$

- 3. Row swap:** Swapping the two rows (or columns) of a matrix changes the sign of the determinant.

Example: Let $B = \begin{pmatrix} 5 & 1 \\ 2 & 3 \end{pmatrix}$.

First, find the original determinant: $\det(B) = (5 \times 3) - (1 \times 2) = 15 - 2 = 13$

Now, swap Row 1 and Row 2 to get $B' = \begin{pmatrix} 2 & 3 \\ 5 & 1 \end{pmatrix}$: $\det(B') = (2 \times 1) - (3 \times 5) = 2 - 15 = -13$

Result: The determinant changed from 13 to -13.

- 4. Row scaling:** Multiplying a row (or column) by a number k multiplies the determinant by k .

Example: Let $C = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$.

Original determinant: $\det(C) = (1 \times 4) - (2 \times 3) = 4 - 6 = -2$. Now, multiply **Row 1** by

$k = 5$ to get $C' = \begin{pmatrix} 5 & 10 \\ 3 & 4 \end{pmatrix}$: $\det(C') = (5 \times 4) - (10 \times 3) = 20 - 30 = -10$

Verification: -10 is exactly 5 times (-2).

5. Zero row: If one row (or a column) is zero, the determinant is 0.

Example: Let $D = \begin{pmatrix} 0 & 0 \\ 7 & 8 \end{pmatrix}$, $\det(D) = (0 \times 8) - (0 \times 7) = 0 - 0 = 0$.

Application activity 10.6



1. Decide whether each statement is True or False

- a) The determinant of a 2×2 matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ is given by $ad - bc$
- b) If the determinant of a 2×2 matrix is 0, the matrix is invertible.
- c) Swapping the two rows of a 2×2 matrix changes the sign of a determinant
- d) Multiplying a row of a 2×2 matrix by a number multiplies the determinant by the same number.
- e) If one row of a 2×2 matrix is a multiple of the other row, the determinant is non-zero.

2. For each of the following matrices, calculate the determinant and determine whether the matrix is invertible.

(a) $A = \begin{pmatrix} 3 & 5 \\ 2 & 4 \end{pmatrix}$ (b) $B = \begin{pmatrix} 3 & 6 \\ 5 & 10 \end{pmatrix}$ (c) $C = \begin{pmatrix} 0 & 1 \\ -2 & 3 \end{pmatrix}$ (d) $D = \begin{pmatrix} 0 & 0 \\ 6 & 3 \end{pmatrix}$

10.7 Operations on matrices and their properties: Transpose and inverse of a square matrix

Activity 10.7



1. Consider the following row matrix showing the number of books read by two students in a week: $A = (3 \ 5)$,

- a) i) Change the row matrix A into a column matrix.
ii) Do the same for $M = \begin{pmatrix} 3 & 5 \\ 1 & 2 \end{pmatrix}$ by interchanging its rows and columns.
- b) In each case, compare the two matrices. What do you notice about the position of the numbers?

2. Consider the mathematical expression:

$$f(x) = y = 2x + 3$$

- a) Use the inverse operations to express x in terms of y get $f^{-1}(x)$. What happens when you find $f \circ f^{-1}(x)$?
- b) Given the Matrix $M = \begin{pmatrix} 3 & 5 \\ 1 & 2 \end{pmatrix}$ discuss how you can find the matrix P such that

$$\begin{pmatrix} 3 & 5 \\ 1 & 2 \end{pmatrix} \times P = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I$$

(i) Transpose of a matrix

The transpose of a matrix is obtained by interchanging its rows and columns. If a matrix is denoted by A , its transpose is written as A' . In other words, if the matrix A has order $m \times n$ then, its transpose A' has order $n \times m$.

Example 1: Transpose the following matrix $A = \begin{pmatrix} 2 & 5 \\ 3 & 7 \end{pmatrix}$

Solution:

Row (3 7) becomes column $\begin{pmatrix} 3 \\ 7 \end{pmatrix}$

Row (2 5) becomes column $\begin{pmatrix} 2 \\ 5 \end{pmatrix}$

Example 2: Transpose the following matrix $B = \begin{pmatrix} 1 & 4 & 6 \\ 2 & 5 & 8 \\ 3 & 8 & 9 \end{pmatrix}$

Solution:

Row (1 4 6) becomes column $\begin{pmatrix} 1 \\ 4 \\ 6 \end{pmatrix}$

Row (2 5 8) becomes column $\begin{pmatrix} 2 \\ 5 \\ 8 \end{pmatrix}$

Row (3 8 9) becomes column $\begin{pmatrix} 3 \\ 8 \\ 9 \end{pmatrix}$

Therefore, $A' = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 8 \\ 6 & 7 & 9 \end{pmatrix}$

(ii) Inverse of a matrix

The **inverse of a matrix** is another matrix that reverses the effect of the original matrix in multiplication.

If a matrix A has an inverse, it is written as A^{-1} . A matrix A has an inverse if: $AA^{-1} = A^{-1}A = I$

Where I is the identity matrix. For a 2×2 matrix $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$

The inverse exists only if **the determinant is not equal to zero**.

The determinant of A is $\det(A) = ad - bc$

If $ad - bc \neq 0$, the inverse of A is given by: $A^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$

To get the new matrix from the original one:

- **Swap** the positions of a and d (the main diagonal).
- **Change the signs** of b and c (the opposite diagonal).
- **Multiply** the entire matrix by the reciprocal of the determinant $\left(\frac{1}{\det(A)}\right)$.

Example 3: Find the inverse of: $A = \begin{pmatrix} 2 & 1 \\ 3 & 4 \end{pmatrix}$

Solution:

Step 1: Find the determinant: $\det(A) = (2 \times 4) - (1 \times 3) = 8 - 3 = 5$

Since $5 \neq 0$, the matrix has an inverse.

Step 2: The inverse is $A^{-1} = \frac{1}{5} \begin{pmatrix} 4 & -1 \\ -3 & 2 \end{pmatrix}$

Example 4: Find the inverse of $M = \begin{pmatrix} 4 & 7 \\ 2 & 6 \end{pmatrix}$.

Solution:

Determinant: $(4 \times 6) - (7 \times 2) = 24 - 14 = 10$.

Swap and negate: $\begin{pmatrix} 6 & -7 \\ -2 & 4 \end{pmatrix}$. Result: $M^{-1} = \frac{1}{10} \begin{pmatrix} 6 & -7 \\ -2 & 4 \end{pmatrix} = \begin{pmatrix} 0.6 & -0.7 \\ -0.2 & 0.4 \end{pmatrix}$

Note:

a) Invertible matrix: A square matrix having a multiplicative inverse, that is, a matrix B such that $AB = BA = I$

b) Involutory matrix: A square matrix which is its own inverse, i.e $AA = I$

Example: $A = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$. Verification: $A^2 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I$

c) Orthogonal matrix: a matrix whose inverse is equal to its transpose $A^{-1} = A^t$

Example: $A = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$. The transpose is $A^t = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$, then

$A^t A = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I$. Therefore, A is an orthogonal matrix.

Application activity 10.7



Consider the matrix $A = \begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix}$

1. Find the determinant of matrix A
2. Find A^{-1}
3. Multiply A by A^{-1} and state what type of matrix you obtain.

10.8 Applications of determinants

Activity 10.8



- 1) Solve the following system of equations. You may use any method you are comfortable with: substitution or elimination if given $\begin{cases} 2x + 3y = 11 \\ x - 2y = -3 \end{cases}$
- 2) Use the reference book or internet and discuss how to use matrices to solve simultaneous linear equations such as $\begin{cases} 2x + 3y = 11 \\ x - 2y = -3 \end{cases}$.



Content Summary

Solving simultaneous linear equations

As discussed in S3, Cramer's Rule is a method for solving a system of linear equations using determinants. It works for two or more equations with the same number of unknowns.

For a system of two equations with two unknowns $\begin{cases} a_1x + b_1y = c_1 \\ a_2x + b_2y = c_2 \end{cases}$

Step 1: Write the coefficient matrix $A = \begin{pmatrix} a_1 & b_1 \\ a_2 & b_2 \end{pmatrix}$

The determinant of A is $\det(A) = a_1b_2 - a_2b_1$

Important: The system has a unique solution only if $\det(A) \neq 0$

Step 2: Create matrices for x and y

Replace the x -column with the constants $(c_1 \ c_2)$ to get A_x . We have $A_x = \begin{pmatrix} c_1 & b_1 \\ c_2 & b_2 \end{pmatrix}$

Replace the y -column with the constants $(c_1 \ c_2)$ to get A_y . We have $A_y = \begin{pmatrix} a_1 & c_1 \\ a_2 & c_2 \end{pmatrix}$

Step 3: Apply Cramer's Rule we obtain $x = \frac{\det(A_x)}{\det(A)}$, $y = \frac{\det(A_y)}{\det(A)}$

Example: Solve the system $\begin{cases} 2x + 3y = 11 \\ x - 2y = -3 \end{cases}$

Solution:

Step 1: Coefficient matrix $A = \begin{pmatrix} 2 & 3 \\ 1 & -2 \end{pmatrix}$, $\det(A) = (2 \times -2) - (1 \times 3) = -4 - 3 = -7$

Step 2: Matrix for x and y : $A_x = \begin{pmatrix} 11 & 3 \\ -3 & -2 \end{pmatrix}$, $\det(A_x) = (11 \times -2) - (-3 \times 3) = -22 + 9 = -13$

and $A_y = \begin{pmatrix} 2 & 11 \\ 1 & -3 \end{pmatrix}$, $\det(A_y) = (2 \times -3) - (1 \times 11) = -6 - 11 = -17$

$$x = \frac{\det(A_x)}{\det(A)} = \frac{-13}{-7} = \frac{13}{7}, y = \frac{\det(A_y)}{\det(A)} = \frac{-17}{-7} = \frac{17}{7}$$

Step 1: Coefficient matrix $A = \begin{pmatrix} 2 & 3 \\ 1 & -2 \end{pmatrix}$, $\det(A) = (2 \times -2) - (1 \times 3) = -4 - 3 = -7$

Step 2: Matrix for x and y : $A_x = \begin{pmatrix} 11 & 3 \\ -3 & -2 \end{pmatrix}$, $\det(A_x) = (11 \times -2) - (-3 \times 3) = -22 + 9 = -13$

and $A_y = \begin{pmatrix} 2 & 11 \\ 1 & -3 \end{pmatrix}$, $\det(A_y) = (2 \times -3) - (1 \times 11) = -6 - 11 = -17$

$$x = \frac{\det(A_x)}{\det(A)} = \frac{-13}{-7} = \frac{13}{7}, y = \frac{\det(A_y)}{\det(A)} = \frac{-17}{-7} = \frac{17}{7}$$

Step 3: Solving for x and y we get

Answer $(x, y) = \left(\frac{13}{7}, \frac{17}{7}\right)$.

Area calculation in surveying and construction

In land surveying or architecture, the determinant is used to find the area of a parallelogram (or triangle) formed by two vectors. If you have two sides of a plot of land represented by vectors $\vec{v} = (a, b)$ and $\vec{w} = (c, d)$, the area is the absolute value of the determinant.

Example: Imagine a small triangular garden plot in Kigali defined by two paths starting from a corner:

- Path 1: 4 meters East, 1 meter North (4, 1).
- Path 2: 2 meters East, 5 meters North (2, 5).

The area of the parallelogram formed by these paths is: $\det \begin{pmatrix} 4 & 2 \\ 1 & 5 \end{pmatrix} = (4 \times 5) - (2 \times 1) = 18 \text{ m}^2$

Since a triangle is half of a parallelogram, the garden area is exactly 9 m^2 .

Application activity 10.8



Solve the following system of linear equations using Cramer's Rule $\begin{cases} 2x + y = 7 \\ x + 3y = 11 \end{cases}$



Knowledge and understanding

1. Define a 2×2 matrix and give an example.
2. Given, $A = \begin{bmatrix} 2 & 5 \\ 1 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 5 \\ 0 & 4 \end{bmatrix}$
 - (a) Determine whether $A = B$ and explain.
3. Classify the following matrices and justify your answer:

(a) $A = \begin{pmatrix} 3 & 0 \\ 0 & 3 \end{pmatrix}$ (b) $B = [1 \quad 2]$

(c) $C = \begin{bmatrix} 2 \\ 5 \end{bmatrix}$ (d) $D = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$

Skills

4. A point $P(2,3)$ in the plane is rotated 90° counterclockwise about the origin.
 - a) Write the transformation matrix for a 90° counterclockwise rotation
 - b) Compute the image in the point P after the rotation
5. Given matrices $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$, $B = \begin{bmatrix} 2 & 0 \\ 1 & 3 \end{bmatrix}$
 - a) Compute $A + B$
 - b) Compute AB
6. Solve the system of equations using determinants $\begin{cases} 2x + 3y = 5 \\ 4x - y = 1 \end{cases}$
 - a) Find the determinant of the coefficient matrix
 - b) Use Cramer's Rule to find x and y
7. A school has two buses transporting students:
 - School A: 3 small buses, 2 large buses
 - School B: 2 small buses, 3 large buses

Each small bus carries 20 students, and each large bus carries 40 students. Represent the situation using matrices and compute the total number of students transported by each school.

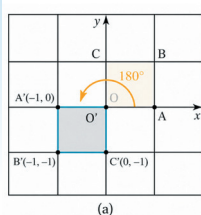
Attitudes and values

8. A student claims that "matrices of order 2×2 are just abstract numbers in brackets that only exist in mathematics notebooks." Which of the following pairs of real-life scenarios provides the best evidence to challenge this mindset and show the practical value of matrices?
 - a) Calculating the total weight of a single brick and finding the height of a classroom door.

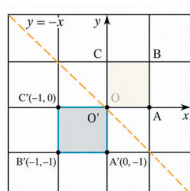
- b) Comparing the monthly sales records of two different school canteens and transforming two-dimensional coordinates in mobile phone video games.
 - c) Estimating the time it takes for rain to fall and measuring the temperature of a single cup of tea.
 - d) Writing down a random list of student names in alphabetical order without any numerical data.
9. A cooperative manager is overwhelmed by trying to manually calculate and compare the changing costs of seeds and fertilizers across two different agricultural seasons for two distinct sectors. How does arranging this data into a system of 2×2 matrices change and improve the decision-making process?
- a) It automatically predicts the exact weather patterns for the upcoming agricultural year, removing all future risks.
 - b) It makes the reports look more complicated, which allows the cooperative to charge higher prices to external buyers.
 - c) It condenses multiple separate arithmetic operations into a single structured system, allowing for rapid, error-free comparison and clearer strategic planning.
 - d) It proves that manual counting is always superior to mathematical modeling when dealing with community finances.

UNIT 11

OPERATIONS ON LINEAR TRANSFORMATION IN 2D



(a)



(b)



Key Unit Competence:

Apply linear and geometric transformations in two dimensions in solving mathematical problems.

Learning objectives

By the end of this unit, you should be able to:

Knowledge and understanding

1

- Explain the concepts of linear and geometric transformations in two dimensions.
- Describe the properties of orthogonal and orthonormal transformations.
- Identify the types of symmetry in geometric figures.
- Identify the identical (identity) transformation and its effect on figures.

Skills

2

- Apply linear and geometric transformations to solve mathematical problems
- Construct and interpret reflections, rotations, and symmetries on coordinate planes
- Determine the kernel and range of given linear transformations.
- Represent transformations algebraically and geometrically.

Attitudes and values

3

- Appreciate the importance of transformations in solving geometric and real-life problems
- Show accuracy, patience, and critical thinking when performing transformations
- Demonstrate logical reasoning and consistency in problem-solving.

Introduction

This unit introduces you to the fundamental concepts of linear transformations in two-dimensional space, which play a central role in mathematics, physics, engineering, and computer graphics.

A linear transformation is a mathematical operation that maps vectors from one position in a plane to another while preserving vector addition and scalar multiplication. These transformations are useful for representing rotations, reflections, scaling, and shapes of objects in 2D.

In this unit, you will learn how to add and scale linear transformations, compute the composition of two transformations, and apply these transformations to vectors using matrix representations. These concepts will help you understand how transformations interact and how they can be used to manipulate geometric objects in two-dimensional space.

The key words to be explained in this unit are: Linear transformations, 2D vector, matrix representation, vector image, vector pre-image, rotation, reflection, scaling, and shape.

Introductory activity



Search in the reference books or on internet and find the meaning of a matrix. Then, Draw an 'L' shape on a Cartesian plane using the coordinates $O(0,0)$, $B(2,0)$, and $H(0,3)$.

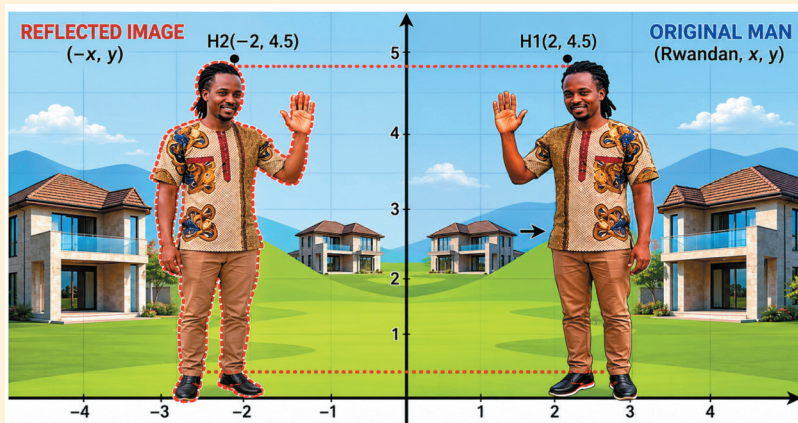
1. Reflect the 'L' shape across the y-axis and write down the new coordinates for each vertex.
2. Rotate the original shape 90 degrees clockwise around the origin and identify where the point $H(0,3)$ lands.

11.1 Geometric transformation in 2D: Reflection

Activity 11.1



1. What does it mean to reflect a shape across a line? Research about reflection.



2. In real life, where can you observe examples of reflection? Can you give an example using mirrors, water, or objects in the environment?



Content Summary

A geometric transformation is an operation that modifies the position, size, shape, and orientation of the geometric object with respect to its current state and position.

Generally, **reflection** is a transformation representing a flip of a figure. Figures may be reflected in a point, a line, or a plane. When reflecting a figure in a line or in a point, the image is congruent to the preimage.

We say that the point $P = \begin{pmatrix} x \\ y \end{pmatrix}$ has an image $P' = \begin{pmatrix} x' \\ y' \end{pmatrix}$ under the reflection.

Every linear **transformation** in 2D can be expressed using a 2x2 matrix, and it is called the transformation matrix.

Image of a vector under a transformation:

The transformation f can change (x, y) into (x', y') such that

$f: \mathbb{R}^2 \rightarrow \mathbb{R}^2 : f(x, y) = (ax + by, cx + dy)$ we can write it as

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = f \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} ax + by \\ cx + dy \end{pmatrix} \text{ or writing in matrix form } \begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$$

Reflection (symmetry) about x-axis: R_x is defined by:

$$f(\vec{i}) = f \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \text{ and } f(\vec{j}) = f \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ -1 \end{pmatrix}. \text{ The standard transformation matrix is } M = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

The transformation matrix is the matrix obtained by using the image of the standard basis of the vector space.

Example 1:

Reflection Under the x-axis: When reflecting an object across the **x-axis**, the horizontal position (x) stays exactly the same, but the vertical position (y) is flipped to its opposite value.

The geometric rule for this reflection the image of (x, y) is $(x, -y)$.

To achieve this result, we use the identity matrix but negate the value responsible for the y -coordinate (the bottom-right entry): $R_x = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$.

Suppose we want to reflect a point on his hand, located at **(2, 4)**, across the **x-axis** instead

$$\text{of the } y\text{-axis. } R_x(2, 4) = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 2 \\ 4 \end{bmatrix} = \begin{bmatrix} (1 \times 2) + (0 \times 4) \\ (0 \times 2) + (-1 \times 4) \end{bmatrix} = \begin{bmatrix} 2 \\ -4 \end{bmatrix}$$

Example 2:

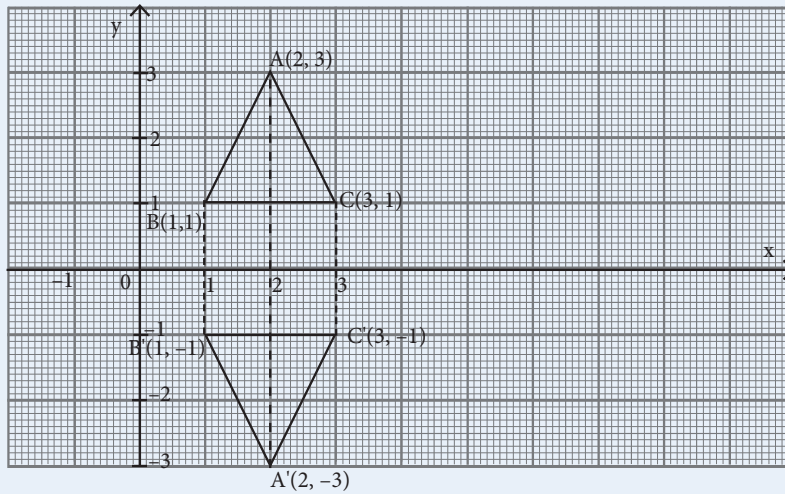
Find the image of the point $(6, -3)$ reflect in the x -axis.

Solution:

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 6 \\ -3 \end{pmatrix} = \begin{pmatrix} 6 - 0 \\ 0 + 3 \end{pmatrix} = \begin{pmatrix} 6 \\ 3 \end{pmatrix}$$

Illustration

Consider the triangular shape with vertices $A(2,3)$, $B(1,1)$, and $C(3,1)$

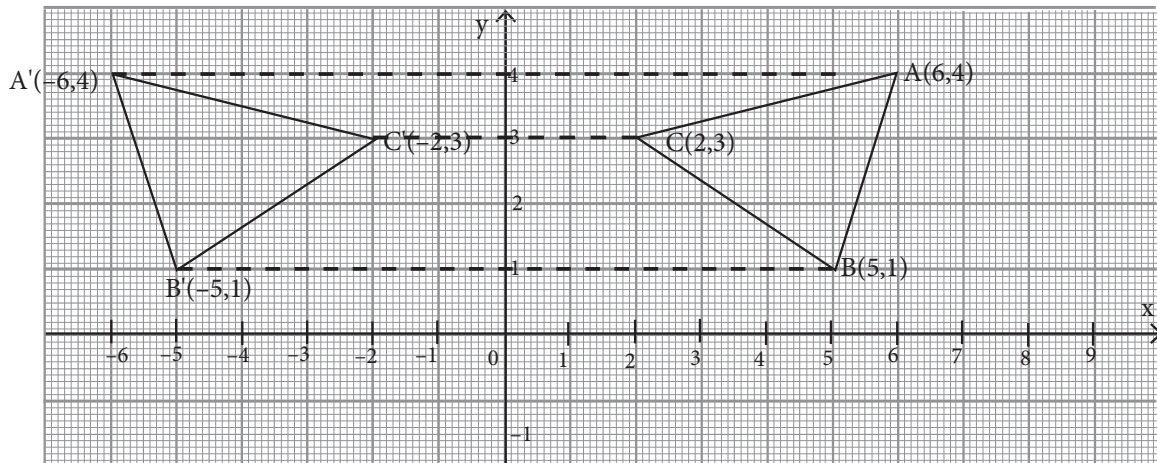


(Reflection (symmetry) about y-axis: $f(\vec{i}) = f\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} -1 \\ 0 \end{pmatrix}$ and $f(\vec{j}) = f\begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$

The standard transformation matrix is $M = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$.

Illustration:

Consider the triangular shape with vertices $A(6,4)$, $B(5,1)$, and $C(2,3)$



Example 3:

Find the image of the point $(1,2)$ reflected in the y-axis.

Solution:

$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 2 \end{pmatrix} = \begin{pmatrix} -1+0 \\ 0+2 \end{pmatrix} = \begin{pmatrix} -1 \\ 2 \end{pmatrix}$. Thus, the image of $(1,2)$ reflected in the y-axis is $(-1,2)$

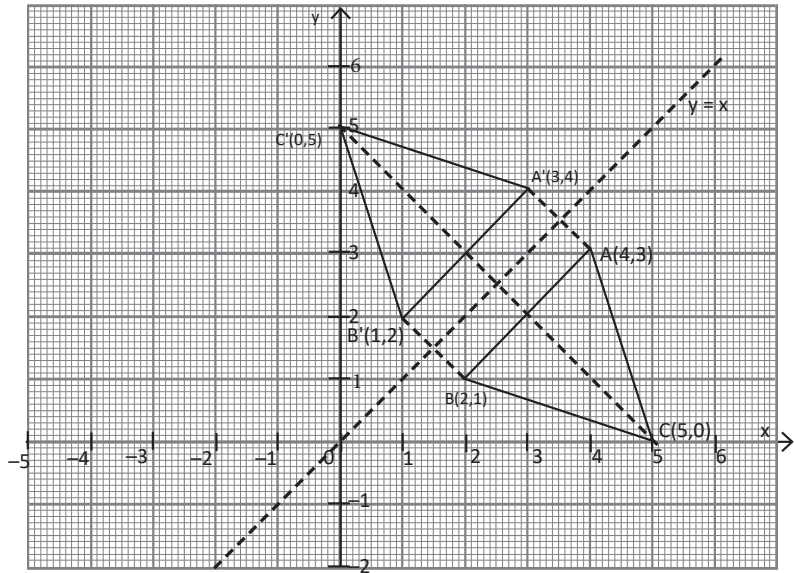
Generally, if f is the reflection in the y-axis in $\mathbb{R}^2$ then, $f(x,y) = (-x,y)$

Reflection (symmetry) about the line $x = y$

The standard transformation matrix is $M = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$

Illustration

Consider the triangular shape with vertices $A(4,3)$, $B(2,1)$, and $C(0,5)$.



Example 4:

Use the standard transformation matrix in the line $x = y$ to compute the image of the point $(-3,4)$

Solution:

$$f(-3,4) = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} -3 \\ 4 \end{pmatrix} = \begin{pmatrix} 0+4 \\ -3+0 \end{pmatrix} = \begin{pmatrix} 4 \\ -3 \end{pmatrix}$$

Generally, if f is the reflection in the line $x = y$ in $\mathbb{R}^2$ then, $f(x,y) = (y,x)$

Summary: Use the following rule to find the reflected image across x-axis, y-axis, or across a line of symmetry using a reflection matrix.

For a reflection over the:	x-axis	y-axis	line $x = y$
Multiply the vertex on the left by the transformation matrix	$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$	$\begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$	$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$
Generally, if f is the reflection	$f(x,y) = (x,-y)$	$f(x,y) = (-x,y)$	$f(x,y) = (y,x)$

Application activity 11.1



Consider a triangle ABC with vertices $A(1,2)$, $B(3,1)$, and $C(2,4)$

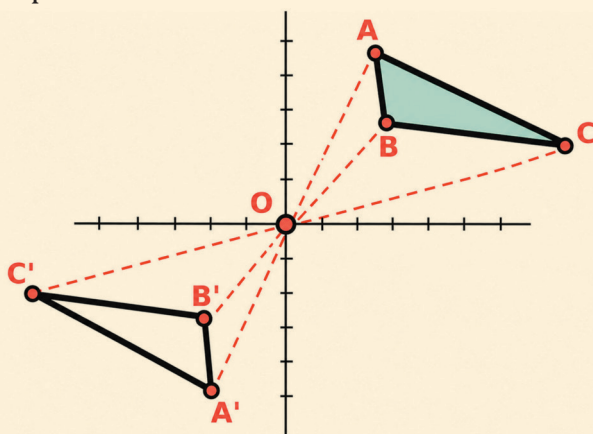
1. Plot the triangle ABC on the coordinate plane
2. Reflect the triangle across the y-axis.
3. Write the coordinates of the new vertices A' , B' , and C' .

11.2 Central symmetry

Activity 11.2



If we turn the shape half a turn (180°) around point O , what do you think will happen to the shape?



If a shape remains the same after turning it half a turn around a point, what do you think that point is called? Discuss with your peers.



Content Summary

We call the central symmetry of point A about the origin O if the distance AO is equal to AO , and we say that the image of point A about the origin O is A' . $f(\vec{i}) = f\left(\begin{pmatrix} 1 \\ 0 \end{pmatrix}\right) = \begin{pmatrix} -1 \\ 0 \end{pmatrix}$ and

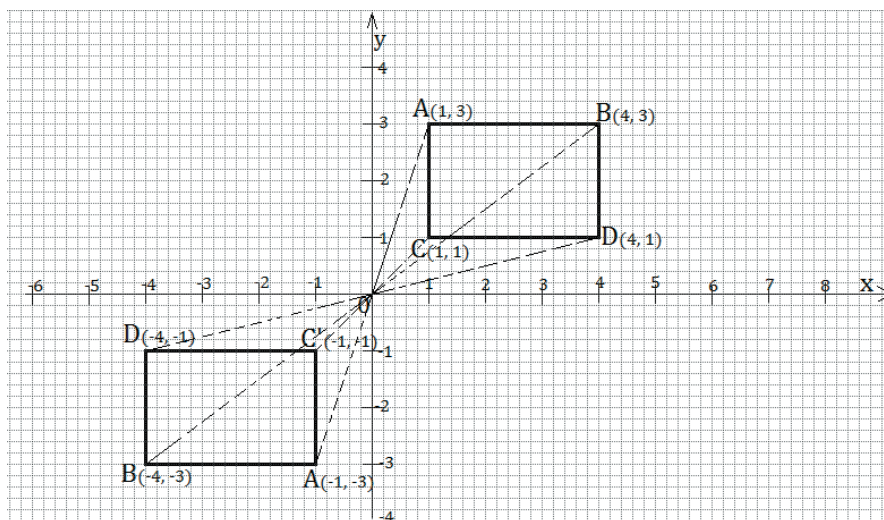
$f(\vec{j}) = f\left(\begin{pmatrix} 0 \\ 1 \end{pmatrix}\right) = \begin{pmatrix} 0 \\ -1 \end{pmatrix}$. The standard transformation matrix is $\begin{bmatrix} f(\vec{i}) & f(\vec{j}) \end{bmatrix} = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$

Generally, if f is a central symmetry of center origine $(0,0)$ in $\mathbb{R}^2$ then, $f(x, y) = (-x, -y)$.

This means $f\left(\begin{pmatrix} x \\ y \end{pmatrix}\right) = \begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} \cdot \begin{pmatrix} x \\ y \end{pmatrix}$

Illustration

Consider the rectangular shape with vertices $A(1, 3)$, $B(4, 3)$, $C(1, 1)$, and $D(4, 1)$.



In this figure, point $O(0,0)$ is called the **center of symmetry** because every part of the rectangle moves to the opposite side through that point.

Example:

Given the segment AB formed by the points $A = (1, 0)$ and $B = (2, 3)$, we are going to calculate its symmetry regarding the centre of coordinates.

Solution:

To do it, we will calculate the symmetry of the points A and B . Precisely, the symmetrical of A is A' :

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 0 \end{pmatrix} \Rightarrow \begin{cases} x' = -1 \\ y' = 0 \end{cases}. \text{ Therefore } A' = (-1, 0). \text{ The symmetrical of the point } B \text{ is } B':$$

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} \cdot \begin{pmatrix} 2 \\ 3 \end{pmatrix} \Rightarrow \begin{cases} x' = -2 \\ y' = -3 \end{cases}$$

Therefore, the symmetry of the segment AB is the segment $A'B'$ that passes through the points $A' = (-1, 0)$ and $B' = (-2, -3)$.

Application activity 11.2



Triangle ABC has vertices $A(2, 3)$, $B(4, 1)$, and $C(1, 5)$. Reflect triangle ABC through the point $O(0, 0)$ (the origin).

1. Draw triangle ABC and its image $A'B'C'$ on the coordinate plane.
2. What do you notice about the relationship between the coordinates of each point and its image after reflection through $O(0, 0)$?

11.3 Rotation around the origin (0,0) of axes

Activity 11.3



Answer the following with “Yes” or “No” and explain to your peers about your choice

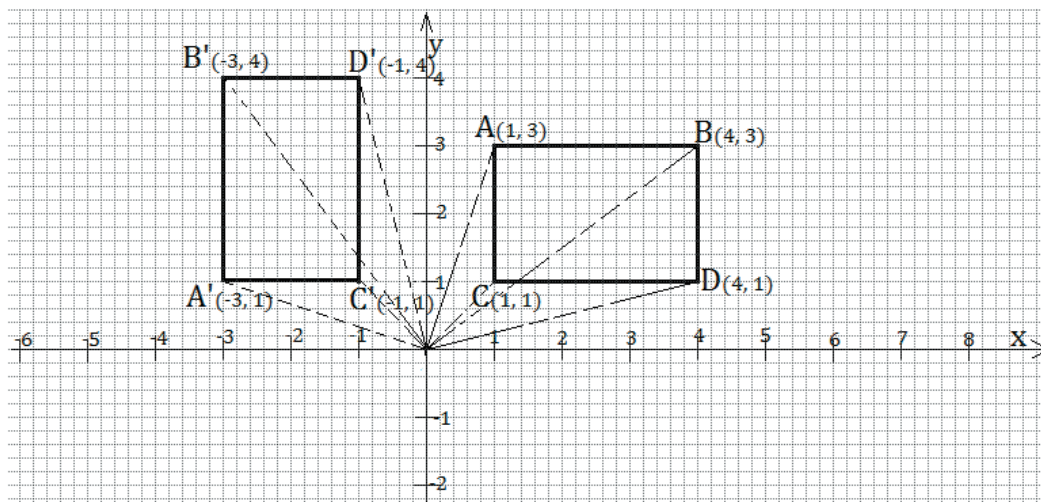
1. If a triangle is rotated 360° around its center:
 - a) Its position remains the same.
 - b) Its size reduces
 - c) Its vertices remain the same.
2. If a square is rotated 90° clockwise around its center:
 - a) Its position remains the same
 - b) Its size changes
 - c) Its vertices occupy new positions



Content Summary

A **rotation** is a transformation in a plane that turns every point of a pre-image through a specified angle and direction about a **fixed point**. The fixed point is called **the center of rotation**. The amount of rotation is called **the angle of rotation**, and it is measured in degrees.

Consider the following illustration,



In this figure, the angle of rotation for each vertex is 90° .

A rotation maps every point of a preimage to an image rotated about a centre point, usually the origin, using a rotation matrix for 90° , 180° , and 270° .

The Transformation matrix Perspective

We can also prove this using the standard rotation matrix. The matrix for a rotation around the origin by angle θ is: $R_\theta = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$. This means that to rotate counterclockwise about the origin, multiply the vertex matrix by the given matrix.

Angle of rotation	90°	180°	270°
Multiply on the left the transformation matrix	$\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$	$\begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$	$\begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$

Example:

Given, the point $P(2,3)$. Rotate it 90° , 180° , and 270° clockwise around the origin $O(0,0)$.

Solution:

Considering the rule for rotation

- a) Rotation 90° clockwise around the origin: $(x, y) \rightarrow (y, -x)$ i.e $P'(3, -2)$
- b) Rotation 180° around the origin: $(x, y) \rightarrow (-x, -y)$ i.e $P'(-2, -3)$
- c) Rotation 270° clockwise around the origin: $(x, y) \rightarrow (-y, x)$ i.e $P'(-3, 2)$

Application activity 11.3



A triangle ABC with vertices $A(2,1)$, $B(4,1)$, and $C(3,3)$ was rotated 90° anti-clockwise around the origin $O(0,0)$. Find the coordinates of the rotated triangle $A'B'C'$.

11.4 Orthogonal transformations

Activity 11.4



An **orthogonal transformation** is a linear transformation that preserves the lengths of vectors and the angles between them. In simpler terms, it is a transformation that moves or flips an object without stretching, shrinking, or warping its shape. In 2D or 3D space, these transformations correspond to **rigid motions** (rotations and reflections) that keep the origin fixed. Now, answer with “Yes” or “No” to the following and explain your answer.

- Does an orthogonal transformation change the length of a vector?
- Does an orthogonal transformation change the angle between two vectors?
- Is the rotation of a triangle about the origin an example of an orthogonal transformation?
- Does an orthonormal transformation require that the axes stay at right angles and have length 1?
- Can a reflection of a square across a line be considered an orthogonal transformation?
- Does an orthonormal transformation change the size or shape of a figure?



Content Summary

Orthogonal transformations are linear transformations that preserve the lengths of vectors and the angles between them. In terms of matrices, a square matrix A is orthogonal if its transpose is also its inverse: $A^T A = I$ or $A^T = A^{-1}$.

This property ensures that the transformation does not “stretch” or “squash” the space; it only moves it. In 2D space, there are two primary types of orthogonal transformations.

- 1. Rotation:** A rotation moves every point in the plane by a fixed angle θ around the origin. This transformation preserves the “orientation” of the space (i.e., it doesn’t flip the object).

The Transformation Matrix: $R_\theta = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$

Properties: The determinant $\det(R_\theta) = 1$. This is often called a “proper” orthogonal transformation.

Example: For a 90° ($\pi / 2$ radians) counter-clockwise rotation, the matrix becomes: $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$

- 2. Reflection:** A reflection “flips” the plane across a line that passes through the origin.

This transformation reverses the orientation of the space (a right-handed coordinate system becomes left-handed).

The Transformation Matrix: If the line of reflection makes an angle θ with the positive x-axis, the matrix is: $Ref_\theta = \begin{pmatrix} \cos(2\theta) & \sin(2\theta) \\ \sin(2\theta) & -\cos(2\theta) \end{pmatrix}$

Properties: The determinant $\det(Ref_\theta) = -1$. This is an “improper” orthogonal transformation.

Common Cases:

- Reflection across the x-axis ($\theta = 0^\circ$): $\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$
- Reflection across the y-axis ($\theta = 90^\circ$): $\begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$
- Reflection across the line $y = x$ ($\theta = 45^\circ$): $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$

Based on the **determinant** of the matrix:

Type	Determinant	Description
Rotation	$\det(A) = 1$	The figure is turned around the origin. It preserves “handedness” (orientation).
Reflection	$\det(A) = -1$	The figure is flipped over a line. It reverses the orientation (like the Rwandan man reflection we discussed).

Why “Orthogonal”?

The name comes from the fact that the columns (and rows) of these matrices are **orthonormal vectors**. If you look at the columns of R_θ :

- **Length:** $\sqrt{\cos^2 \theta + \sin^2 \theta} = 1$ (Unit length).
- **Dot Product:** $(\cos \theta)(-\sin \theta) + (\sin \theta)(\cos \theta) = 0$ (They are perpendicular/orthogonal).

This is why orthogonal transformations never distort the actual shape of an object they only change its position or facing.

Key characteristics for an orthogonal transformation

- **Distance preservation (Isometry):** The distance between any two points remains exactly the same after the transformation.
- **Angle Preservation:** The dot product of two vectors remains unchanged, meaning the angle between them is preserved.
- **Origin fixed:** Unlike a translation (sliding), an orthogonal transformation always maps the origin (0, 0) to itself.

Example 1:

Consider the reflection across the y-axis that we used earlier: $A = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$

Testing for Orthogonality: Transpose (A^T): $\begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$

Multiply $A^T A$: $\begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I$

Since $A^T A$ results in the **Identity Matrix**, the reflection is a confirmed orthogonal transformation.

Example 2:

Use matrix transformation to reflect the point P(3, 1) across the line $y = \sqrt{3}x$

Solution:

Find the Angle θ : The line $y = \sqrt{3}x$ is in the form $y = (\tan \theta)x$.

Since $\tan \theta = \sqrt{3}$, the angle the line makes with the x-axis is $\theta = 60^\circ$.

For the reflection matrix, we need $2\theta = 120^\circ$.

- $\cos(120^\circ) = -0.5$
- $\sin(120^\circ) = \frac{\sqrt{3}}{2} \approx 0.866$

Reflection Matrix: $Ref_{60^\circ} = \begin{pmatrix} \cos 120^\circ & \sin 120^\circ \\ \sin 120^\circ & -\cos 120^\circ \end{pmatrix} = \begin{pmatrix} -0.5 & 0.866 \\ 0.866 & 0.5 \end{pmatrix}$

Matrix Multiplication: $\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} -0.5 & 0.866 \\ 0.866 & 0.5 \end{pmatrix} \begin{pmatrix} 3 \\ 1 \end{pmatrix}$

$$x' = (-0.5 \times 3) + (0.866 \times 1) = -1.5 + 0.866 = -0.634$$

$$y' = (0.866 \times 3) + (0.5 \times 1) = 2.598 + 0.5 = 3.098$$

Result: The reflected point is approximately $P'(-0.634, 3.098)$.

Application activity 11.4



A vector $\mathbf{v} = \begin{pmatrix} 4 \\ 2 \end{pmatrix}$ is rotated counter-clockwise by 60° about the origin. Use the transformation matrix to find the coordinates of the transformed vector $\mathbf{v}'$.

11.5 Operations on transformations

Activity 11.5



Consider the linear transformation $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined by:

$$T_1(x, y) = (x, 0), \quad T_2(x, y) = (0, y)$$

Answer with “True” or “False” for the following:

1. $T_1 + T_2$ is a linear transformation
2. $T_1 \circ T_2$ is a linear transformation



Content Summary

As we discussed for functions, the main operations possible for linear transformations are the addition and the composition.

a) Addition

$$T : V \rightarrow W \text{ and } f : V \rightarrow W \text{ and } g : V \rightarrow W$$

$f + g : V \rightarrow W : (f + g)(a) = f(a) + g(a)$. The sum of two linear transformations is a linear transformation.

Example:

Given two linear transformations $T_1, T_2 : \mathbb{R}^2 \rightarrow \mathbb{R}^2 : T_1(x, y) = (2x, y), \quad T_2(x, y) = (x, -y)$

Compute the sum.

$$(T_1 + T_2)(x, y) = T_1(x, y) + T_2(x, y) \Leftrightarrow (T_1 + T_2)(x, y) = (2x, y) + (x, -y) = (3x, 0)$$

Observation:

The sum $T_1 + T_2$ maps $(x, y) \rightarrow (3x, 0)$. This is still a linear transformation.

Check linearity

- **Zero vector:** $(T_1 + T_2)(0, 0) = (3 \cdot 0, 0) = (0, 0)$
- **Additivity:** Let (x_1, y_1) and (x_2, y_2) be vectors:
 $(T_1 + T_2)(x_1, y_1) + (x_2, y_2) = (3(x_1 + x_2), 0) = (3x_1, 0) + (3x_2, 0) = (T_1 + T_2)(x_1, y_1) + (T_1 + T_2)(x_2, y_2)$
- **Homogeneity for scalar c:** $(T_1 + T_2)(c(x, y)) = (3(cx), 0) = c(3x, 0) = c(T_1 + T_2)(x, y)$
- **Conclusion:** The sum $T_1 + T_2$ is linear.

b) Composition of two linear transformations

Let $f : V \rightarrow W$ and $g : V \rightarrow W$ then the composite of f and g , $f \circ g$ is a linear transformation.

Proof

1. $(f \circ g)(a + b) = g(f(a + b)) = g(f(a) + f(b)) = g(f(a)) + g(f(b)) = g \circ f(a) + g \circ f(b)$
2. $((g \circ f)(\beta a)) = g(\beta f(a)) = \beta(g \circ f)(a), \beta \in \mathbb{R}$

Therefore, the composition of two linear transformations is also a linear transformation.

Example 1:

Let the linear transformations T_1 and T_2 defined by $T_1(x, y) = (x, -y)$, $T_2(x, y) = (2x, y)$

- a) Compute the composition $(T_1 \circ T_2)(x, y)$
- b) Find the image of the vector $(1, 2)$ under the composite transformation

Solution:

The composition of functions means we apply the second function first and then apply the first function to that result. Formally:

$$(T_1 \circ T_2)(x, y) = T_1(T_2(x, y))$$

Step 1: Find the output of $T_2(x, y)$

From the problem $T_2(x, y) = (2x, y)$

Step 2: Apply T_1 in the new vector $(2x, y)$

Since T_1 takes any input (a, b) and returns $(a, -b)$, we substitute $a = 2x$ and $b = y$

The result for $(T_1 \circ T_2)(x, y) = (2x, -y)$

(b) Now we simply plug the specific coordinates $x = 1$ and $y = 2$

Substitute the values into $(2x, -y)$

$(T_1 \circ T_2)(1, 2) = (2(1), -(2))$, then $(T_1 \circ T_2)(1, 2) = (2, -2)$. The image of the vector $(1, 2)$ is $(2, -2)$

One-to-one linear transformation

A linear transformation is said to be one-to-one if for each element in the range, there is a unique element in the domain that maps to it.

Onto linear transformation

A linear transformation f is said to be onto if for every element in the range space, there exists an element in the domain that maps to it.

The word isomorphism comes from the Greek word, which means 'equal shape'. An isomorphism is a linear transformation that is both one-to-one and onto. If two vector spaces V and W have the same basic structures, then there is an isomorphism between them.

$$\dim V = \dim W.$$

Note: A linear transformation has an inverse if and only if it is an isomorphism.

Example 1:

State whether the following transformation is an isomorphism or not.

$$f : \mathbb{R}^2 \rightarrow \mathbb{R}^2 : f(x, y) = (x + 2y, 3x + 4y)$$

Solution:

We must verify if f is one-to-one and onto.

Let $u = (x_1, x_2)$ and $v = (y_1, y_2)$ then, if $f(u) = f(v)$ we have,

$$(x_1 + 2x_2, 3x_1 + 4x_2) = (y_1 + 2y_2, 3y_1 + 4y_2) \text{ and the system}$$

$$\begin{cases} x_1 + 2x_2 = y_1 + 2y_2 \\ 3x_1 + 4x_2 = 3y_1 + 4y_2 \end{cases} \text{ and solving, we have } \begin{cases} 2x_1 + 4x_2 = 2y_1 + 4y_2 \\ -3x_1 - 4x_2 = -3y_1 - 4y_2 \end{cases} \Rightarrow -x_1 = -y_1$$

$x_1 = y_1$ and $x_2 = y_2$. Thus, $u = v$, and f is one-to-one.

Let (a, b) be an arbitrary element in the range space, then we have to verify if the following system has a solution for all values of a and b .

$$\begin{cases} x + 2y = a \\ 3x + 4y = b \end{cases} \Leftrightarrow \begin{cases} 2x + 4y = 2a \\ -3x - 4y = -b \end{cases} \Leftrightarrow \begin{cases} -x = 2a - b \\ 2y = a - x \end{cases} \Leftrightarrow \begin{cases} x = b - 2a \\ y = \frac{a - x}{2} \end{cases}$$

Thus, the transformation is onto.

Hence, the transformation is an isomorphism in 2D

Example 2:

State whether the following transformation is an isomorphism or not.

$$f : \mathbb{R}^2 \rightarrow \mathbb{R}^2 : f(x, y) = (x + 2y, 2x + 4y)$$

Solution:

We must verify if f is one-to-one or onto.

Let $u = (x_1, x_2)$ and $v = (y_1, y_2)$ and if $f(u) = f(v)$, we have

$(x_1 + 2x_2, 2x_1 + 4x_2) = (y_1 + 2y_2, 2y_1 + 4y_2)$ and the system

$$\begin{cases} x_1 + 2x_2 = y_1 + 2y_2 \\ 2x_1 + 4x_2 = 2y_1 + 4y_2 \end{cases} \text{ and solving, we have } \begin{cases} 2x_1 + 4x_2 = 2y_1 + 4y_2 \\ -2x_1 - 4x_2 = -2y_1 - 4y_2 \end{cases} \text{ (adding) we get}$$

$$0x_1 + 0x_2 = 0y_1 + 0y_2$$

Some different vectors of the domain have the same image (zero vectors). Thus, f is not one-to-one. Hence, f is not an isomorphism. There is no need to verify if the transformation is onto.

Remark

For the linear application: $f: V \rightarrow W$ when $V = W$ it means $V \rightarrow V$, then f is called an **endomorphism**.

f is a one-to-one linear transformation (bijection), then it is called an **isomorphism**.

$V = W$ (endomorphism) is a one-to-one transformation, then f is called an **automorphism**.

$V = W$ and f is an injection, then f is called **monomorphism**.

$V = W$ and f is a surjection, then f is called **epimorphism**.

Note: Injection (one-to-one): If $f(\vec{a}) = f(\vec{b})$, then $\vec{a} = \vec{b}$

Surjection (onto): $V \rightarrow W$ for every $f(\vec{a}) \in W$ there exists $\vec{a} \in V$

A bijection verifies both injection and surjection.

Application activity 11.5



Let $T_1, T_2: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be two linear transformations defined by

$$T_1(x, y) = (x + y, y), \quad T_2(x, y) = (x, 2y)$$

1. Compute the sum of the two linear transformations $(T_1 + T_2)(x, y)$
2. Find $(T_1 + T_2)(1, 2)$
3. State whether $(T_1 + T_2)(x, y)$ is a linear transformation.

11.6 Matrix of a composite linear transformation in 2D

Activity 11.6



Suppose we want to transform a geometric figure by first reflecting it across the x-axis and then rotating it 90° counterclockwise. Given that the standard transformation

matrices for reflection M_1 and rotation M_2 are Reflection (M_1): $\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ and $M_2 = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$,

- Replace find M_2 .
- Calculate the **composite Matrix** ($M_2 \times M_1$).
- Discuss the resulting matrix and verify which type of reflection it is. Across X-axis, y-axis or across the line $y=x$?



Content Summary

Mathematically, if T_1 and T_2 are linear transformations represented by matrices M_1 and M_2 , the composition $T_2 \circ T_1$ (applying T_1 first, then T_2) is equivalent to a single transformation represented by the product of their matrices $M_{combined} = M_2 \times M_1$.

In other words, a **composite transformation** is the result of applying two or more linear transformations in succession. For any vector $\vec{v}$, if we apply transformation A and then transformation B, the result is $B(A(\vec{v}))$.

In matrix terms, if M_A and M_B are the matrices representing these transformations, the composite matrix M_C is: $M_C = M_B \times M_A$.

Crucial Rule: The matrix of the transformation performed **first** is always placed on the **right**.

To build a composite linear transformation, we use the standard matrices for 2×2 systems:

Rotation (counter-clockwise by angle θ): $\begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$

Scaling (s_x and s_y factors): $\begin{pmatrix} s_x & 0 \\ 0 & s_y \end{pmatrix}$

Reflection (across the x-axis): $\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$.

Example 1:

A geometric figure is first scaled by a factor of 2 horizontally and 3 vertically, then reflected across the y-axis. Find the composite matrix and the image of the point P(2, 4).

Solution:

Scaling (M_1): $\begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}$

Reflection across y-axis (M_2): $\begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$

The composite matrix ($M_2 \times M_1$) = $M_c = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix} = \begin{pmatrix} -2 & 0 \\ 0 & 3 \end{pmatrix}$

Applying it to point P: $\begin{pmatrix} -2 & 0 \\ 0 & 3 \end{pmatrix} \begin{pmatrix} 2 \\ 4 \end{pmatrix} = \begin{pmatrix} (-2)(2) + (0)(4) \\ (0)(2) + (3)(4) \end{pmatrix} = \begin{pmatrix} -4 \\ 12 \end{pmatrix}$

Therefore, image of P (2,4) is **P'(-4, 12)**.

Example 2: A vector is rotated by 30° and then further rotated by 60° counter-clockwise. Show that the composite matrix is equivalent to a single 90° rotation.

Solution:

$M_1(30^\circ): \begin{pmatrix} \cos 30^\circ & -\sin 30^\circ \\ \sin 30^\circ & \cos 30^\circ \end{pmatrix} = \begin{pmatrix} \frac{\sqrt{3}}{2} & -\frac{1}{2} \\ \frac{1}{2} & \frac{\sqrt{3}}{2} \end{pmatrix}$

$M_2(60^\circ): \begin{pmatrix} \cos 60^\circ & -\sin 60^\circ \\ \sin 60^\circ & \cos 60^\circ \end{pmatrix} = \begin{pmatrix} \frac{1}{2} & -\frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \frac{1}{2} \end{pmatrix}$

The composite matrix ($M_2 \times M_1$) = $M_c = \begin{pmatrix} \frac{1}{2} & -\frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \frac{1}{2} \end{pmatrix} \begin{pmatrix} \frac{\sqrt{3}}{2} & -\frac{1}{2} \\ \frac{1}{2} & \frac{\sqrt{3}}{2} \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$

This is exactly the matrix for a 90° **rotation**, confirming that $M(\alpha) \times M(\beta) = M(\alpha + \beta)$.

Note:

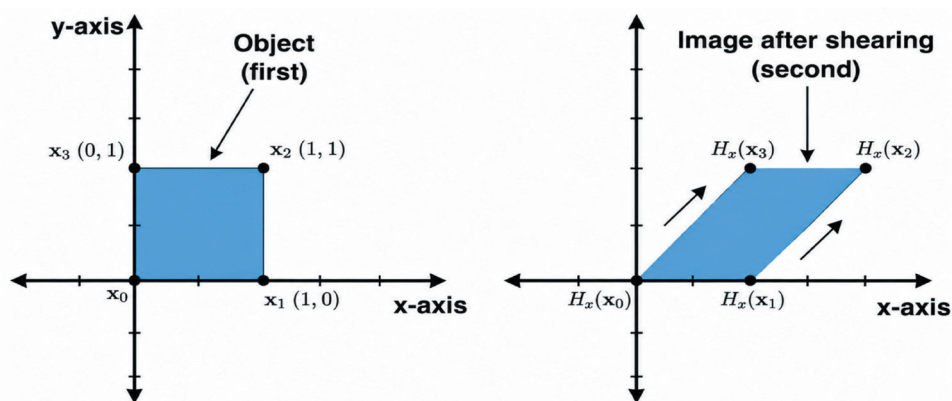
- **Order Matters:** $BA \neq AB$, changing the order of transformations usually changes the final result.
- The area scale factor of the composite transformation is the product of individual determinants. For example, in Example 1, $\det(M_c) = -6$, meaning the area increases 6-fold (the negative sign indicates a change in orientation).

Practically, calculate the composite matrix first before transforming a set of coordinates; it is much more efficient than transforming points step-by-step.

Note: A horizontal shear

A horizontal shear (also known as a shear parallel to the x-axis) shifts each point in the plane horizontally by an amount proportional to its y-coordinate. Points on the x-axis remain fixed, while points further away from the axis shift further.

The X-Shear preserves the Y coordinate and changes are made to X coordinates, which causes the vertical lines to tilt right or left as shown in below figure.



For a horizontal shear, every point moves left or right by k times its y -coordinate, while its y -coordinate stays fixed. Points on the x -axis don't move at all because their y -value is zero. The farther a point is from the invariant line, the more it slides. Rectangles become parallelograms, but the base and height remain the same, so the area is preserved.

The Transformation Matrix

For a horizontal shear with a shear factor k , the transformation matrix is: $H_k = \begin{pmatrix} 1 & k \\ 0 & 1 \end{pmatrix}$.

To find the new coordinates (x', y') of a point (x, y) , you perform the following multiplication:

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} 1 & k \\ 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x + ky \\ y \end{pmatrix}$$

Example: Let's apply a horizontal shear with a factor of $k = 2$ to a square with vertices at $(0,0)$, $(1,0)$, $(1,1)$, and $(0,1)$.

Solution:

$$\text{Origin } (0, 0): \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

(Points on the x -axis do not move)

$$\text{Point } (1, 0): \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$\text{Point } (1, 1): \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1+2(1) \\ 1 \end{pmatrix} = \begin{pmatrix} 3 \\ 1 \end{pmatrix}$$

$$\text{Point } (0, 1): \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0+2(1) \\ 1 \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$$

Summary of Results

The original unit square is transformed into a **parallelogram** with vertices at $(0,0)$, $(1,0)$, $(3,1)$, and $(2,1)$. Notice that while the shape has “leaned” to the right, the height (y -coordinates) and the area of the shape remain unchanged.

Application:

It is applied in computer graphics and image processing: If you have ever used the “Italic” button in a text editor or the “Skew” tool in Photoshop, you have applied a horizontal shear matrix.

- **Font styling:** Many “oblique” or “faux italic” fonts are created by applying a horizontal shear to a standard upright font. The base of the letter stays on the line ($y=0$), while the top shifts.
- **Perspective correction:** When a camera takes a photo at an angle, the image can look distorted. Graphic software uses shear transformations (often combined with scaling and rotation) to “straighten” the image and simulate a front-on view.

Application activity 11.6



Let $T_1, T_2 : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be two linear transformations defined by

$$T_1(x, y) = (x + y, y), \quad T_2(x, y) = (x, 2y)$$

1. Compute the sum of the two linear transformations $(T_1 + T_2)(x, y)$
2. Find $(T_1 + T_2)(1, 2)$
3. State whether $(T_1 + T_2)(x, y)$ is a linear transformation.

A shape is rotated 90° counter-clockwise about the origin, followed by a horizontal shear with factor $k = 2$.

Find the single matrix that represents this composition if the matrix transformation of the horizontal shear with factor k is $\begin{pmatrix} 1 & k \\ 0 & 1 \end{pmatrix}$.

11.7 Kernel of a linear transformation

Activity 11.7



1. In your words, explain what the kernel of a linear transformation represents.
2. If T represents the transformation and $v = (x, y)$. Interpret the following:
 $T(x, y) = (0, 0)$.
3. Discuss some examples of linear transformations T for which $T(x, y) = (0, 0)$.



Content Summary

Let $T : V \rightarrow W$ be a linear transformation of a vector space V into the vector space W , the kernel of T denoted by $\ker T$ is the set of all vectors of V which have zero vectors of W as their images.

$$\ker T = \{\vec{u} \in V : f(\vec{v}) = \vec{0} \in W\}$$

Example:

Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be an endomorphism of $T : \mathbb{R}^2$ such that $f(x, y) = (2x - y, -2x + y)$. Find:

1. $\ker T$
2. The basis of $\ker T$ and $\dim \ker T$

Solution:

$$\begin{aligned} \ker T &= \{\vec{v} \in \mathbb{R}^2 : T(\vec{v}) = \vec{0} \in \mathbb{R}^2\} = \{(x, y) \in \mathbb{R}^2 : (2x - y, -2x + y) = (0, 0)\} \\ &= \left\{ (x, y) \in \mathbb{R}^2 : \begin{cases} 2x - y = 0 \\ -2x + y = 0 \end{cases} \right\} = \{(x, y) \in \mathbb{R}^2 : 2x - y = 0\} = \{(x, y) \in \mathbb{R}^2 : y = 2x\} \\ &= \{(x, 2x) \in \mathbb{R}^2 : x \in \mathbb{R}\} = \{(x(1, 2) \in \mathbb{R}^2 : x \in \mathbb{R}\}. \text{ Thus } \ker T = \{(x(1, 2) \in \mathbb{R}^2 : x \in \mathbb{R}\}. \end{aligned}$$

We can express the solution in vector form by letting $y = t$ (where t is any real number):

$$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} t \\ 2t \end{pmatrix} = t \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

Geometrically, this represents a **straight line** passing through the origin in the 2D plane with the equation $y = 2x$. The basis of $\ker T$ is $\{(1, 2)\}$. Thus, $\dim \ker T = 1$.

Application activity 11.7



- 1) If a linear transformation is $T(x, y) = (x - y, x - y)$. Find the kernel.
- 2) Consider a linear transformation $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined by the following matrix:

$$A = \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}$$

- a) Define what is meant by the **Kernel** of this transformation.
- b) Find the Kernel of T and describe its geometric representation in the Cartesian plane.

11.8 Range of a linear transformation

Activity 11.8



1. In your words, explain what the range of a linear transformation represents.

2. Consider the linear transformation $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined by

$$T(x, y) = (x + y, x + y)$$

Find the following:

a) $T(1, 0)$

b) $T(0, 1)$

c) $T(2, 3)$

d) Based on your observations, describe the set of vectors $T(x, y)$ for all $(x, y) \in \mathbb{R}^2$.



Content Summary

Let $V \rightarrow W$ be a linear transformation of a vector space V into the vector space W , the range of T denoted by $\text{im}T$ is the set of all vector images in T , thus $\text{im}T = \{T(\vec{v}) \in W : \vec{v} \in V\}$

Example: Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be an endomorphism of $\mathbb{R}^2$ such that $T(x, y) = (x - y, y)$; find:

a) $\text{im}T$

b) $\text{Dim im}T$

Solution:

a) Now, let $(a, b) \in \text{im}T$ be given. There exists a vector $(x, y) \in \mathbb{R}^2$ such

that $T(x, y) = (x - y, y) = (a, b)$. This yields the following system

$$\begin{cases} x - y = a \\ y = b \end{cases} \rightarrow \begin{cases} x = a + y \\ y = b \end{cases} \text{ and } (a, b) \text{ leads to the solution given by } T(x, y)$$

$$= (a + b, b) = (a, 0) + (b, b) = a(1, 0) + b(1, 1); a, b \in \mathbb{R}. \text{ Hence, } \{(1, 0), (1, 1)\} \text{ spans } \text{im}T.$$

b) $\text{Dim im}T = 2$

Nullity and rank

The dimension of the kernel is called the **nullity** of T (denoted $\text{nullity } T$) and the dimension of the range of T is called the **rank** of T (denoted $\text{rank } T$).

Example:

Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be an endomorphism of $\mathbb{R}^2$ such that $T(x, y) = (x, x + y)$; Find:

- a) nullity T
- b) rank T

Solution:

$$\text{Ker } T = \{v \in \mathbb{R}^2 : T(\vec{v}) = 0 \in \mathbb{R}^2\} = \{(x, y) \in \mathbb{R}^2 : (x, x + y) = (0, 0)\}$$

$$= \left\{ (x, y) \in \mathbb{R}^2 : \begin{cases} x = 0 \\ x + y = 0 \end{cases} \right\} = \left\{ (x, y) \in \mathbb{R}^2 : \begin{cases} x = 0 \\ y = 0 \end{cases} \right\} \text{ or } (x, y) = (0, 0)$$

$\text{Ker } T = \{(0, 0) \in \mathbb{R}^2\}$. Thus, $\dim \text{Ker } T = 0$ and Nullity $T = \dim \text{Ker } T = 0$

b) Now, let $(a, b) \in \text{Im}$ be given. There exists a vector $(x, y) \in \mathbb{R}^2$ such that

$T(x, y) = (x, x + y) = (a, b)$. This yields the following system:

$$\begin{cases} x = a \\ x + y = b \end{cases} \Leftrightarrow \begin{cases} x = a \\ y = -x + b \end{cases} \Leftrightarrow \begin{cases} x = a \\ y = -a + b \end{cases} \text{ and } (a, b) \text{ leads to the solution given by}$$

$$(a, b) = (a, -a + b) = (a, -a) + (0, b) = a(1, -1) + b(0, 1); a, b \in \mathbb{R}. \text{ Hence,}$$

$$\text{Im } T = \text{span}\{(1, -1), (0, 1)\}.$$

$$\dim \text{Im } T = 2. \text{ Rank } T = \dim \text{Im } T = 2$$

Application activity 11.8



Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be a linear transformation defined by $T(x, y) = (x + y, x + y)$

Find:

- a) $\text{Im } T$
- b) $\dim \text{Im } T$.



Knowledge and understanding

Write “True” or “False” for each statement.

1. Linear transformation maps vectors from one vector space to another while preserving vector addition and scalar multiplication.
2. The sum of two linear transformations is always a linear transformation.
3. The order of composition of two linear transformations always gives the same result.
4. A reflection across the x-axis in the plane is a linear transformation.
5. Rotating a vector in $\mathbb{R}^2$ changes its direction but preserves its magnitude.
6. A reflection across the y-axis changes the sign of the x-coordinate of a vector.

Skills

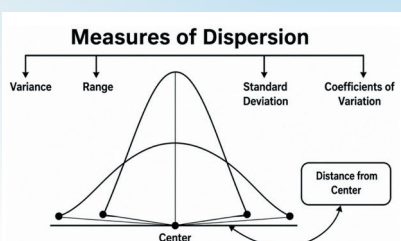
7. Let $T_1, T_2 : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be defined by $T_1(x, y) = (x + y, y)$, $T_2(x, y) = (2x, -y)$
 - a) Compute the sum $(T_1 + T_2)$
 - b) Compute the scalar multiplication $2(T_1(x, y))$
8. Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be a linear transformation defined by $f(x, y) = (x + y, x + y)$
 - a) Find the kernel $\ker(f)$
 - b) Determine the nullity or f of $\dim \ker(f)$
 - c) Find the range of f and determine the rank of f .

Attitudes and values

9. Why should a student value understanding how geometric transformations like scaling and rotation affect 2D vectors beyond just passing a mathematics test?
 - a) Because these operations form the essential mathematical foundation used to build computer graphics, video game animations, and digital image filters.
 - b) Because manual vector mapping is the only way to prove that a computer-generated image is completely accurate.
 - c) Because learning these rules makes simple coordinate geometry look more complicated and advanced to outside observers.
 - d) Because it guarantees that any 2D drawing can automatically be converted into a physical 3D object without errors.
10. When applying a linear transformation to a set of vectors representing a architectural blueprint, a student states, “It doesn’t matter if the transformation alters the origin $(0,0)$, as long as the shape looks mostly correct.” Why does this mindset lack mathematical value?
 - a) It ignores the rule that a true linear transformation must fix the origin, which ensures the entire spatial grid scales and rotates with absolute precision.
 - b) It spends too much time worrying about the origin instead of focusing on making the visual design look creative.
 - c) It prevents the student from using different colors to represent different transformation steps on the graph.
 - d) It makes the final drawing too simple for the teacher to grade effectively.

UNIT 12

MEASURES OF DISPERSION



Key Unit Competence:

Evaluate measures of dispersion for real-life data to formulate logical conclusions.

Learning objectives

By the end of this unit, you should be able to:

Knowledge and understanding

1

- Recall and explain measures of central tendency.
- Describe and compute measures of dispersion (variance, standard deviation, coefficient of variation).
- Interpret measures of central tendency and dispersion to analyse data variability.

Skills

2

- Construct and interpret box plots using given data
- Calculate variance, standard deviation, and coefficient of variation accurately.
- Apply measures of dispersion to analyze and interpret data sets meaningfully.
- Compare different data sets using appropriate statistical measures.

Attitudes and values

3

- Appreciate the importance of summarizing data for analysis and decision-making.
- Demonstrate accuracy, patience, and attention to detail in statistical calculations.
- Show critical thinking and curiosity in exploring patterns and trends in data.

Introduction

This unit introduces you to the fundamental concepts of measures of dispersion, which describe how data values are spread or scattered around a central value. In statistics, understanding the variability of data is important for interpreting information accurately and making meaningful comparisons between different data sets. Measures of dispersion such as range, variance, standard deviation, and coefficient of variation help quantify the extent to which data values differ from one another and from the mean.

In this unit, you will learn how to calculate and interpret the measures of dispersion and understand their relationship with measures of central tendency. You will also explore graphical data representation to display the distribution and spread of data. The knowledge and skills acquired in this unit will help you analyse and interpret data from real-life situations to draw informed conclusions. By the end of this unit, you will be able to use measures of dispersion to support data analysis and make reasoned decisions based on statistical evidence.

You will acquire key words such as range, variance, standard deviation, coefficient of variation, box plot, data distribution, and variability.

Introductory activity



During 6 consecutive days, a fruit-seller has recorded the number of fruits sold per type.



The table below shows the types and the number of fruits sold in one week.

Type of fruits	A (banana)	B (orange)	C (pineapple)	D (avocado)	E (mango)	F (apple)
Number of fruits sold	1100	962	1080	1200	884	900

- Which type of fruit has the highest number of sales?
- Which type of fruit had the least number of fruits sold?
- What was the total number of fruits sold that week?
- Find out the average number of fruits sold per day.
- Observe the distance between each fruit's sales and the average (1,021). If a different seller had the same average but their sales ranged from 884 to 1,200, which seller's business is more "predictable" or "consistent"? Why might a business owner want a low "spread" in their daily sales?

12.1 Recall on measures of central tendency and presentation of data using box plot

Activity 12.1



The table below shows the marks (out of 20) obtained by some students in a mathematics test: 8, 10, 12, 9, 11, 10, 13

Answer the following questions:

- Arrange the data in ascending order.
- Find the mean (average) of the marks.
- Find the median of the marks.
- Identify the mode of the data.
- State the minimum and maximum values.



Content Summary

A measure of central tendency, also called average, is a value about which the distribution of data is approximately balanced. There are three types of measures of central tendency namely the mean, the median, and the mode.

Mean

The mean value of a set of data is the sum of all the items in the set of data divided by the number of items.

For discrete raw data mean = $\frac{\text{sum of items}}{\text{number of items}}$ i.e. $\bar{x} = \frac{\sum_{i=1}^n x_i}{n}$ where n is the number of items.

Example 1:

The mean of the numbers 8, 10, 11, 13, 15, 16, 19, and 22 is given by

$$\bar{x} = \frac{8+10+11+13+15+16+19+22}{8} = \frac{144}{8} = 14.25$$

For data in an ungrouped frequency distribution $\bar{x} = \frac{\sum(f \cdot x)}{\sum f}$

Where f = the frequency of each value or class; x = the value (or the midpoint of a class interval); $\sum f$ = the total frequency (N).

The median

The median of the data is the middle value when all values are arranged in order of size. When the number of items is odd, then the median is the item in the middle. When the number of items is even, the median is the mean of the two numbers in the middle.

First, arrange the data in ascending or descending order. Let n be the total number of data points.

If n is odd: The median is the value at the middle position: $\text{Median} = \left(\frac{n+1}{2}\right)^{\text{th}}$ term

If n is even: The median is the average of the two middle values:

$$\text{Median} = \frac{\left(\frac{n}{2}\right)^{\text{th}} \text{ term} + \left(\frac{n}{2} + 1\right)^{\text{th}} \text{ term}}{2}$$

Example 2: Find the median of:

a) 6 2 8 12 3 5 20 15 3

b) 9 4 5 6 8 10 2 7

Solution:

a) Arranging the numbers in ascending order, we have 2 3 3 5 6 8 12 15 20. These are 9 numbers, and the fifth is the middle of them, thus the median is that number. Alternative formula:

$\text{Median} = x_{\frac{n+1}{2}}$ where n = number of items.

$\text{Median} = x_{\frac{9+1}{2}} = x_5$ number (or fifth number)

So, for our case, we have a median = 6

b) Arranging the numbers in ascending order, we have 2 4 5 6 7 8 9 10. There are 8 numbers. The median is the mean of the two middle numbers, or simply between the 4th and the 5th number.

We have $\frac{6+7}{2} = 6.5$

Generally, we have the formula $\text{Median} = \frac{x_{\frac{n}{2}} + x_{\frac{n+1}{2}}}{2}$ or simply the $\left(\frac{n+1}{2}\right)^{\text{th}}$ value.

Mode

The mode of a set of data is the value of the highest frequency in the distribution of marks.

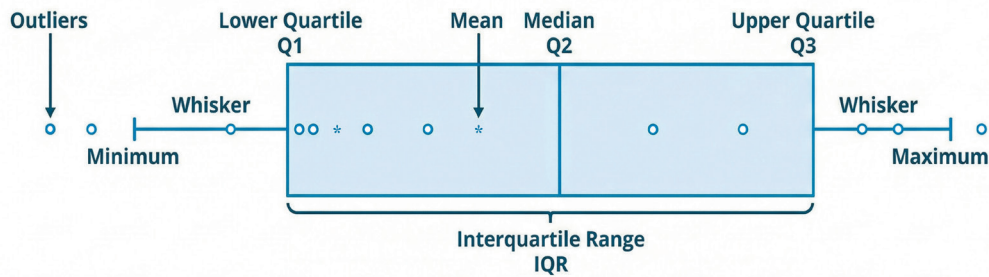
In Example 14.3, the mode is 45 because it has the highest frequency, 12.

Box plot

A box plot is an excellent tool for understanding data distribution, concisely summarizing key metrics. It provides a clear visual representation of the data's central tendency, variability, and potential outliers using five key statistics: **The minimum, first quartile (Q1), median (Q2), third quartile (Q3), and maximum.**

The “box” represents the **interquartile range** ($\text{IQR} = Q3 - Q1$), which contains the middle 50% of the data, while the “**whiskers**” extend to the smallest and largest values (excluding outliers). Box plots are especially useful for comparing distributions across different groups and identifying skewness or unusual values.

Box plot



Example

Consider the following dataset representing students' test scores:
22, 25, 28, 30, 35, 12, 20, 18, 15

Solution:

Step 1: Arrange the data (already in ascending order)

Step 2: Find the median (Q2)

Median = 22

Step 3: Find Q1 (median of lower half)

Lower half: 12, 15, 18, 20, $Q_1 = \frac{(15 + 18)}{2} = 16.5$

Step 4: Find Q3 (median of upper half)

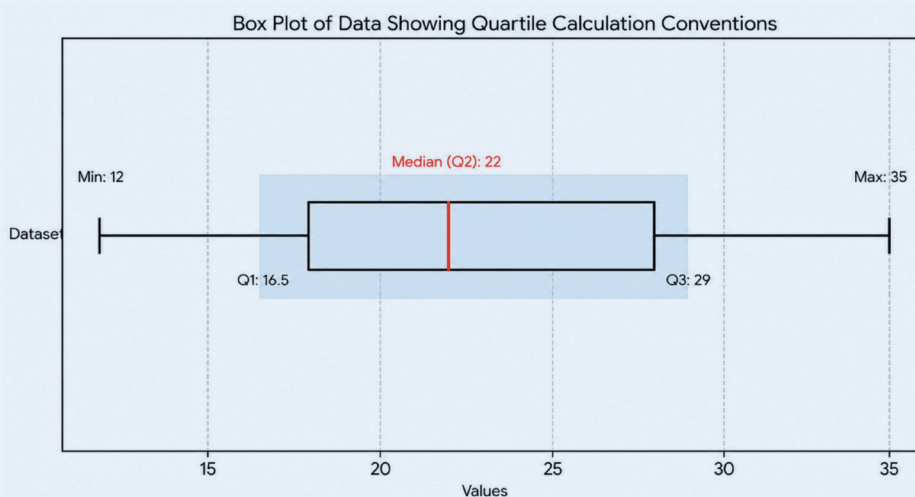
Upper half: 25, 28, 30, 35, $Q_3 = \frac{(28 + 30)}{2} = 29$

Step 5: Determine minimum and maximum

Minimum = 12, Maximum = 35

Step 6: Calculate IQR

$$IQR = Q_3 - Q_1 = 29 - 16.5 = 12.5$$



Interpretation:

The box plot would show a box from 16.5 to 29 with a median line at 22. The whiskers extend to 12 and 35. Since the upper spread is slightly larger than the lower spread, the data shows a slight positive skew. No outliers are observed.

Example 2: Observe the data set. Determine the Q_1 , Q_2 , Median and mean and represent them on a box plot.

8.2	8.8	9.2	10.6	12.7
8.4	9	9.7	11.6	14
8.5	9.2	10.4	11.8	15.9
8.8	9.2	10.5	12.6	16.1

Solution:

1. Ordered data set

With $n = 20$ values, we first arrange them in ascending order: 8.2, 8.4, 8.5, 8.8, 8.8, 9.0, 9.2, 9.2, 9.2, 9.7, 10.4, 10.5, 10.6, 11.6, 11.8, 12.6, 12.7, 14.0, 15.9, 16.1.

2. Statistical Calculations

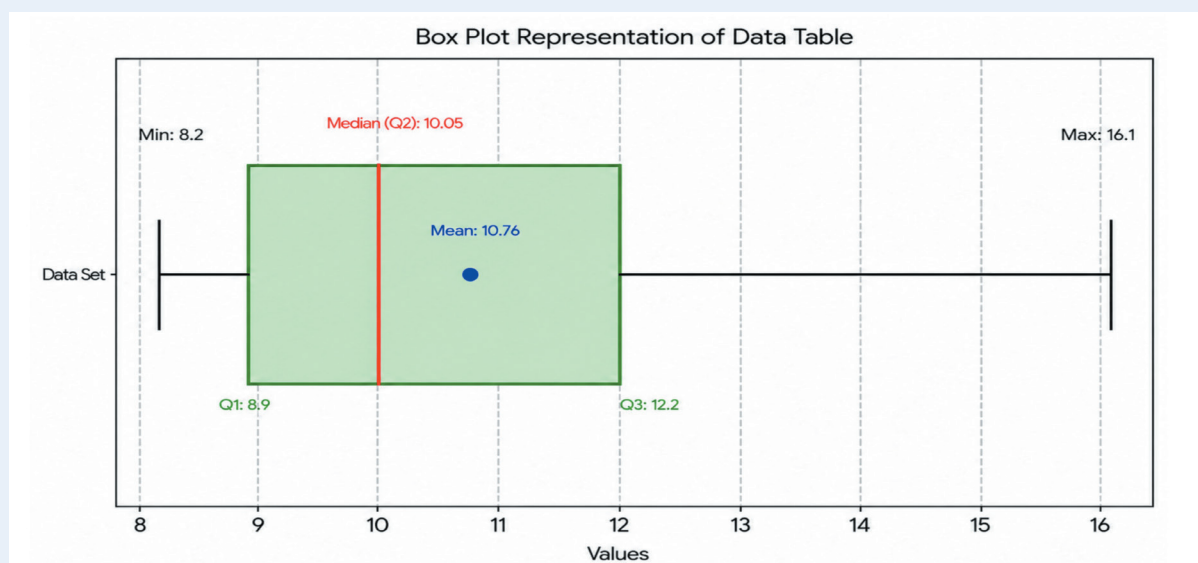
- **Mean:** The sum of all values divided by 20. $\mu = \frac{215.2}{20} = \mathbf{10.76}$
- **Median (Q_2):** Since n is even, it is the average of the 10^{th} and 11^{th} values (9.7 and 10.4).
 $Q_2 = \frac{9.7 + 10.4}{2} = \mathbf{10.05}$
- **Lower Quartile (Q_1):** The median of the first 10 values (average of 5^{th} and 6^{th} values, 8.8 and 9.0). $Q_1 = \frac{8.8 + 9.0}{2} = \mathbf{8.9}$
- **Upper Quartile (Q_3):** The median of the last 10 values (average of 15^{th} and 16^{th} values, 11.8 and 12.6). $Q_3 = \frac{11.8 + 12.6}{2} = \mathbf{12.2}$

3. Box Plot Visualization

The box plot below shows the distribution of the data. The red line represents the **median**, the blue text indicates the **mean**, and the green box shows the **Interquartile Range (Q_1 to Q_3)**.

Key summary: Minimum: 8.2; $Q_1=8.9$; Median (Q_2)=10.5; Mean=10.76; $Q_3=12.2$; Maximum=16.1.

Box plot:



Application activity 12.1



For the data below 5, 3, 7, 6, 3, 4, 2, 7, 5, 6, 5, 4, 9, 8, 3, 6, 5

Find the

- first (lower) quartile
- second quartile (the median)
- third (upper) quartile
- inter-quartile range (IQR)
- quartile deviation.

12.2 Variance

Activity 12.2



- Before starting the third term, a class teacher calculated the mean mark of five students got in second term in Mathematics and he/she obtained that the mean mark is $\bar{x} = 16.875$. Use this mean to complete the table below:

x	f	$(x - \bar{x})$	$(x - \bar{x})^2$	$f(x - \bar{x})^2$
12	4			
13	2			
15	1			
19	4			
21	5			
	$\sum f =$			$\sum f(x - \bar{x})^2 =$

Explain the expression $f(x - \bar{x})^2$ in your own words.

- 2) Discuss the meaning of a variance and answer the following statements with True or False
 - a) Variance measures how spread out the data values are around the mean.
 - b) Variance can be negative if some data values are smaller than the mean.
 - c) A higher variance indicates that the data points are more spread out.
 - d) Variance is calculated by averaging the squared deviations from the mean.



Content Summary

Variance is a statistical measure that describes how much the values in a dataset deviate from the mean. In other words, it tells us how the data points are spread out. It represents the average of the squared differences between each data point and the mean of the dataset.

A small variance means the values are close to the mean, while **a large variance** indicates that the values are more spread out.

Formula for variance $\sigma^2 = \frac{\sum(x_i - \mu)^2}{N}$

Where:

x_i = each data point

μ = mean of the data set

N = total number of data points

To calculate the variance, σ^2

- (i) For each reading of x calculate $x - \bar{x}$, its deviation from the mean.
- (ii) Square this deviation to give $(x - \bar{x})^2$, and note that, irrespective of whether the deviation was positive or negative, this is now positive.
- (iii) Find $\sum(x - \bar{x})^2$, the sum of all those values.
- (iv) Find the average by dividing the sum by n , the number of readings; this gives

$$\sigma^2 = \frac{\sum(x_i - \mu)^2}{N}, \text{ known as variance.}$$

Note: When data points are listed with their corresponding frequencies (f), you must account for how many times each value occurs.

Formula: $\sigma^2 = \frac{\sum f(x - \bar{x})^2}{\sum f}$ where $N = \sum f$ and $\bar{x} = \frac{\sum fx}{\sum f}$.

This formula is not usually used, however, since in most cases the purpose of calculating the statistic is to estimate the corresponding parameter. For example, the sample mean $\bar{X}$ is

used to estimate the population mean μ . The expression $\frac{\sum (X - \bar{X})^2}{n}$ does not give the best estimate of the population variance because when the population is large and the sample is small (usually less than 30), the variance computed by this formula usually underestimates the population variance. Therefore, instead of dividing by n , find the variance of the sample by dividing by $(n - 1)$, giving a slightly larger value and an *unbiased* estimate of the population variance.

The formula for the sample variance, denoted by s^2 , is $s^2 = \frac{\sum (X - \bar{X})^2}{n - 1}$, where $\bar{X}$ = sample mean, n = sample size.

Developing this formula $\sigma^2 = \frac{\sum (x_i - \mu)^2}{N}$ we have $\sigma^2 = \frac{\sum_{i=1}^n (x_i^2 - 2x_i\bar{x} + (\bar{x})^2)}{n}$, thus

$$\sigma^2 = \frac{1}{n} \sum_{i=1}^n x_i^2 - (\bar{x})^2 \text{ where } n = N \text{ and } \bar{x} = \mu$$

Example 1: Consider Dataset of number of books read by 5 students in a week: 2, 4, 4, 6, 8

Solution:

Step 1: Find the mean

$$\text{Mean} = \frac{2 + 4 + 4 + 6 + 8}{5} = \frac{24}{5} = 4.8$$

Step 2: Find the deviation from the mean:

$$x_i - \text{Mean} = [2 - 4.8, 4 - 4.8, 4 - 4.8, 6 - 4.8, 8 - 4.8] = [-2.8, -0.8, -0.8, 1.2, 3.2]$$

Step 3: Square each deviation

$$(-2.8)^2 = 7.84, (-0.8)^2 = 0.64, (-0.8)^2 = 0.64, (1.2)^2 = 1.44, (3.2)^2 = 10.24$$

Step 4: Find the average of squared deviations (Variance)

$$\text{Variance} = \frac{7.84 + 0.64 + 0.64 + 1.44 + 10.24}{5} = \frac{20.8}{5} = 4.16$$

Step 5: Interpretation

The variance of the number of books read is 4.16, which measures how much the students' reading counts differ from the average (mean) of 4.8 books.

Interpreting the values of variance and standard deviation allows us to understand the **spread** and **consistency** of a dataset relative to its mean.

a) Interpreting the Spread of Data

- **Small Value:** A low variance or standard deviation indicates that the data points are clustered closely around the mean. This suggests that the data is **consistent** and has little variability.
- **Large Value:** A high value indicates that the data points are spread out far from the mean. This suggests that the data is **diverse** or has high variability.

b) Practical Interpretation of Standard Deviation

While variance is useful for mathematical calculations, the **standard deviation** is often easier to interpret in real-world contexts because it is expressed in the **same units** as the original data.

- **Distance from the Mean:** You can think of the standard deviation as the “typical” or “average” distance of a data point from the mean.
- **Comparing Datasets:** If two classes take the same exam and Class A has a standard deviation of 5 while Class B has a standard deviation of 15, you can conclude that Class A’s performance was more uniform, whereas Class B had a much wider range of high and low scores.

The number of customers served lunch in a restaurant over a period of 60 days is as follows:

Number of customers served lunch	20 – 29	30 – 39	40 – 49	50 – 59	60 – 69	70 – 79
Number of days in the 60-day period	6	12	16	14	8	4

Find the mean and variance of the number of customers served lunch using this grouped data.

Solution:

To find the mean and variance for the number of customers served lunch, we use the methods for grouped data with frequencies.

1. Organizing the Data Table

First, we identify the class midpoints ($\bar{x}$) for each group and multiply them by their respective frequencies (f) to find the mean.

Number of Customers (Class)	Frequency (f)	Midpoint (x)	fx	$(x - \bar{x})^2$	$f(x - \bar{x})^2$
10 – 14	6	12	72	64	384
15 – 19	12	17	204	9	108
20 – 24	16	22	352	4	64
25 – 29	14	27	378	49	686
30 – 34	8	32	256	144	1,152
35 – 39	4	37	148	289	1,156
Totals	$\sum f = 60$		$\sum fx = 1,410$		$\sum f(x - \bar{x})^2 = 3,550$

2. Calculating the Mean ($\bar{x}$)

The mean for grouped data is calculated by dividing the sum of fx by the total frequency.

$$\bar{x} = \frac{\sum fx}{\sum f} \quad \bar{x} = \frac{1,410}{60} = 23.5$$

The average number of customers served per day is **23.5**.

3. Calculating the Variance (s^2)

Since this data represents a 60-day period (a sample of the restaurant's total operation), we use the formula for **sample variance** to provide an unbiased estimate.

Formula: $s^2 = \frac{\sum f(x - \bar{x})^2}{n - 1}$

Calculation: Sum of squared deviations: 3,550; **Sample size (n - 1):** 60 - 1 = 59.

$$s^2 = \frac{3,550}{59} \approx 60.17.$$

Application activity 12.2



- 1) Given the number of pencils sold in a week by 5 students: 2, 4, 4, 6, 8

Answer with True or False for the following:

- a) The mean of the dataset is 4.8.
 - b) The variance of the dataset is 5.2
 - c) The variance of the dataset is approximately 4.16
 - d) Variance can be negative if some values are below the mean.
- 2) Find the variance and the standard deviation for the frequency distribution of the data. The data represent the number of miles that 20 runners ran during one week.

Class	Frequency	Midpoint
5.5–10.5	1	8
10.5–15.5	2	13
15.5–20.5	3	18
20.5–25.5	5	23
25.5–30.5	4	28
30.5–35.5	3	33
35.5–40.5	2	38

12.3 Standard deviation (including a combined set of data)

Activity 12.3



A class of students took a math test. The mean score was 70, and the variance of the scores is 16.

1. Calculate the standard deviation by finding the square root of the variance.
2. Interpret what the standard deviation tells you about the test scores.



Content Summary

The **standard deviation (SD)** is a measure of how spread out the values in a dataset are around the mean. It is the **square root of the variance**, which brings the measure back to the same units as the original data.

(i) Standard deviation (single dataset)

Formula (Population SD) for a single data set is given by:

$$\sigma = \sqrt{\frac{\sum(x_i - \mu)^2}{N}}. \text{ Remember that } \sigma^2 = \frac{1}{n} \sum_{i=1}^n x_i^2 - (\bar{x})^2 \text{ where } n = N \text{ and } \bar{x} = \mu.$$

Example:

A group of 6 students took a science test. The mean score was 50, and the variance of the scores is 25.

- a) Find the standard deviation.
- b) Interpret what the standard deviation tells you about the students' scores in relation to the mean

Solution:

Step 1: Find the standard deviation

$$\text{Standard deviation} = \sqrt{\text{variance}} = \sqrt{25} = 5$$

Step 2: Interpretation

The standard deviation is 5, which means that, on average, the students' scores deviate 5 points from the mean score of 50. Most students' scores are likely to fall between 45 and 55 (mean $\pm$ standard deviation). This indicates that the students' performance is relatively close to the average, with a small spread around the mean.

(ii) Standard deviation for the combined data set

Sometimes, you have two or more separate datasets, and you want the SD of the combined dataset.

Steps:

1. Find the mean and variance of each dataset.
2. Use the formula for combined variance
3. Take the square root of σ_c^2 to get the combined SD.

The formula of the combined variance is given by:

$$\sigma_c^2 = \frac{N_1\sigma_1^2 + N_2\sigma_2^2 + N_1(\mu_1 - \mu_c)^2 + N_2(\mu_2 - \mu_c)^2}{N_1 + N_2}$$

Where: N_1, N_2 = size of the two data sets

σ_1^2, σ_2^2 = variance of two data sets

μ_1, μ_2 = means of two data sets

μ_c = mean of the combined data sets

Example:

Consider the combined standard deviation for the following two data set:

Data set 1: 2, 4, 6

Data set 2: 3, 5, 7

Solution:

Step 1: Mean

$$\mu_1 = \frac{2+4+6}{3} = 4$$

$$\mu_2 = \frac{3+5+7}{3} = 5$$

Step 2: Variances

$$\sigma_1^2 = \frac{[(2-4)^2 + (4-4)^2 + (6-4)^2]}{3} = \frac{4+0+4}{3} = \frac{8}{3} \approx 2.67$$

$$\sigma_2^2 = \frac{[(3-5)^2 + (5-5)^2 + (7-5)^2]}{3} = \frac{4+0+4}{3} = \frac{8}{3} \approx 2.67$$

Step 3: Combined mean:

$$\mu_c = \frac{2+4+6+3+5+7}{6} = \frac{27}{6} = 4.5$$

$$\sigma_c^2 = \frac{3 \times 2.67 + 3 \times 2.67 + 3 \times (4-4.5)^2 + 3 \times (5-4.5)^2}{6} = \frac{8.01+8.01+3 \times 0.25+3 \times 0.25}{6} = 2.92$$

Step 4: Combined SD

$$\sigma_c = \sqrt{2.92} \approx 1.71$$

Interpretation:

The combined dataset has a standard deviation of about 1.71, which reflects the spread of all six values around the combined mean.

Application activity 12.3



A dataset has a mean of 40 and a variance of 9. Answer with True or False for the following

- The standard deviation of the dataset is 81.
- Most data values lie approximately between 37 and 43.
- The standard deviation is larger than the variance.
- The standard deviation shows how far values are from the mean of 40.

12.4 Coefficient of variation

Activity 12.4



A class has a mean score of 50 and a standard deviation of 5.

- Divide the standard deviation by the means
- Find the percentage of the result obtained in 1.



Content Summary

The coefficient of variation shows the extent of variability in relation to the mean of the population. The coefficient of variation should be computed only for data measured on a ratio scale, as these are the measurements that can only take non-negative values. The coefficient of variation (C.V), unlike the previous measures we have studied, is a relative measure of dispersion. It is expressed as a percentage rather than in terms of the unit of the data. It is useful when comparing the variable of two or more batches of data. Those are expressed in different units of measurement.

The coefficient of variation is given by:

$$CV = \frac{\sigma}{\bar{x}} \times 100 \quad \text{where } \sigma \text{ is the standard deviation and } \bar{x} \text{ is the mean.}$$

Example 1:

Given that $\sigma = 6.26$ and $\bar{x} = 20$ Then $CV = \frac{6.26}{20} \times 100 = 31.3\%$

That is, for this sample, the relative size of the average spread around the mean is 31.3%.

Example 2:

A data set of $[100, 100, 100]$ has constant values. Its standard deviation is 0, and the average is 100:

$$\text{The CV is } 100\% \times \frac{0}{100} = 0\%$$

Example 3:

The data set of [1, 5, 6, 8, 10, 40, 65, 88] has more variability again. Its standard deviation is 30.78, and its average is 27.875:

The CV is $100\% \times \frac{30.78}{27.875} = 110.4\%$. The coefficient of variation (CV) greater than 100%, means that the standard deviation is larger than the mean.

Interpretation:

The data shows very high variability relative to the mean.

- The values are widely spread out, not clustered around the average.
- The dataset is highly inconsistent or unstable.
- The mean is not a reliable representation of the data because the variation is too large.

Application activity 12.4



A student measures the number of pages they read per day over a week. The mean number of pages read is 20, and the standard deviation is 4.

- Calculate the coefficient of variation (CV).
- Interpret what the CV tells you about the student's reading habits in relation to the mean.

12.5 Application of measures of dispersion and data interpretation

Activity 12.5



Carry out research to identify real-world applications of measures of dispersion.
Discuss your findings with the rest of the class.



Content Summary

Measures of dispersion are widely used in real-life and professional contexts to assess variability and consistency in data. In education, they help evaluate variability in students' test scores, allowing teachers to identify consistency or gaps in learning. For example, a teacher can compare the standard deviation of two classes' scores to determine which class is more consistent in performance.

Example 1:

A teacher in a Rwandan secondary school recorded the test scores of 7 students in a mathematics quiz: Scores: 55, 60, 62, 58, 60, 65, 70

- Find the mean, median, and mode of the scores.
- Find the interquartile range (IQR)
- Find the standard deviation.
- Interpret the standard deviation in relation to the mean.

Solution:

$$\text{Mean} = \frac{55 + 60 + 62 + 58 + 60 + 65 + 70}{7} = \frac{430}{7} = 61.43$$

Median = 60

Mode = 60

Interquartile range (IQR): $Q_1 = 58, Q_2 = 65 \rightarrow IQR = 65 - 58 = 7$

Standard deviation = 4.62

Interpretation: Students' scores deviate on average by about 4.6 points from the mean (≈ 61). This indicates that the scores are moderately spread, with most students performing close to the average. The teacher can conclude that most students understand the material reasonably well.

Example 2:

A small shop in Kigali recorded its weekly sales (in Rwandan Francs, FRW):

Sales: 150,000, 160,000, 170,000, 165,000, 155,000, 175,000, 180,000 FRW

- Find the mean and standard deviation of the weekly sales.
- Calculate the coefficient of variation (CV).
- Interpret the CV in relation to the mean.

Solution:

$$1. \text{Mean} = \frac{150,000 + 160,000 + 170,000 + 165,000 + 155,000 + 175,000 + 180,000}{7} = \frac{1,155,000}{7} = 165,000$$

2. Standard deviation:

Deviation from the mean: $-15,000; -5,000; 5,000; 5,000; 0; -10,000; 10,000; 15,000$

Squared deviation:

$225,000,000; 25,000,000; 25,000,000; 0; 100,000,000; 100,000,000; 225,000,000$

$$\text{Variance} = \frac{700,000,000}{7} = 100,000,000, \text{SD} = \sqrt{100,000,000} = 10,000 \text{RWF}$$

$$3. \text{Coefficient of variation } CV = \frac{10,000}{165,000} \times 100\% \approx 6.06\%$$

4. Interpretation of CV:

A CV of 6.06% indicates that weekly sales vary by about 6% of the average sales (165,000 RWF). This shows that the shop's sales are relatively consistent and stable throughout the week. The shopkeeper can be confident that sales performance is predictable, making it easier to plan inventory, staff, and cash flow.

Application activity 12.5



Read each statement carefully and indicate whether it is True (T) or False (F).

- A high standard deviation indicates that the data points are closely clustered around the mean.
- The coefficient of variation is useful for comparing variability between datasets with different means or units.
- In a classroom, a small range of test scores indicates that students' performance is consistent.
- In business, a low CV in weekly sales suggests that sales fluctuate widely.
- Standard deviation can be used in health studies to determine how much patient measurements vary from the average.

End of unit assessment



Assessing knowledge and understanding

Read each statement carefully and indicate whether it is True (T) or False (F).

- A small standard deviation in students' test scores indicates that most students performed close to the mean.
- The coefficient of variation (CV) is only useful for data measured in the same units.
- A high range in sales data indicates high variability and may suggest inconsistent business performance.
- Standard deviation cannot be used to make conclusions about consistency in patient measurements.
- Measures of dispersion help in interpreting data and making informed decisions in real-life situations.
- The mean alone is sufficient to understand the spread and consistency of data.

Assessing skills

- A Rwandan secondary school teacher recorded the marks of 6 students in a mathematics quiz: Scores: 58, 62, 60, 55, 61, 64
 - Calculate the mean of the scores.
 - Calculate the standard deviation (SD).
 - Interpret the SD in relation to the mean and explain what it tells you about students' performance.
 - Suggest one action the teacher could take based on the interpretation.
- A small shop in Kigali recorded its weekly sales (in FRW):
Sales: 150,000; 160,000; 155,000; 165,000; 170,000
 - Calculate the mean of the weekly sales.
 - Calculate the standard deviation (SD).

- c) Calculate the coefficient of variation (CV).
- d) Interpret the CV in relation to the mean and explain what it tells the shopkeeper about sales consistency.
- e) Suggest one business decision the shopkeeper can make based on the CV.

Assessing attitudes and values

9. A shopkeeper wants to restock shoes but faces highly variable daily sales. Why should a student value calculating accurate measures of dispersion (like standard deviation) instead of just relying on the average (mean)?
 - a) To prove that advanced mathematical calculations are always more trustworthy than a shopkeeper's practical experience.
 - b) To make the inventory spreadsheet look more complex so that the shop can secure higher loans from local banks.
 - c) Because dispersion measures reveal sales consistency and variability; small calculation errors can lead to bad planning, causing stock shortages or wasted capital.
 - d) Because calculating standard deviation automatically increases daily customer demand and business profits.
10. Two schools have the exact same mean score of 65% on a national assessment, but School A has a high standard deviation while School B has a standard deviation near zero. What does a student with a habit of critical thinking and curiosity conclude?
 - a) Both schools perform identically, so there is no need to investigate further or look at the data again.
 - b) School A has a wide gap between high and low performers, while School B has highly uniform student progress, prompting a curious mind to investigate the teaching methods used in each.
 - c) Standard deviation is just an abstract exam formula that has no actual meaning when comparing school performance.
 - d) School A is automatically better because a higher standard deviation means the school has more money and better resources.

PROJECT: Investigating academic diversity



1. Project Objective

You will collect, organize, and analyze end-of-term assessment data to determine the consistency of academic performance across different subjects or classes using Box Plots, Variance, and Standard Deviation.

2. Required Tools

- Comprehensive assessment score sheets (anonymized for privacy).
- Graph paper, rulers, and calculators.
- Spreadsheet software (optional, e.g., Excel or Google Sheets).

3. Implementation Phases

Phase I: Data Collection & Organization

- Target Population: Select two distinct groups for comparison (e.g., Senior 4 Science vs. Senior 4 Arts, or Mathematics scores vs. English scores).
- Sample Size: Randomly select 30 scores from each group to ensure the sample is manageable yet statistically significant.
- Data Cleaning: Sort the data in ascending order to prepare for quartile calculations.

Phase II: Statistical Analysis

You must calculate the following for both selected groups:

1. Five-Number Summary: Find the Min, Medi Q_1 , Q_2 , Q_3 , and Max.
2. Measures of Dispersion:
 - Calculate the Range and Interquartile Range (IQR).
 - Calculate the Mean ($\bar{X}$).
 - Calculate the Sample Variance (s^2) and Standard Deviation (s) using the unbiased formula:
$$s^2 = \frac{\sum (X - \bar{X})^2}{n - 1}$$

Phase III: Visualization

Construct a Parallel Box Plot on a single axis. This allows students to visually compare the “spread” and “skewness” of the two groups simultaneously.

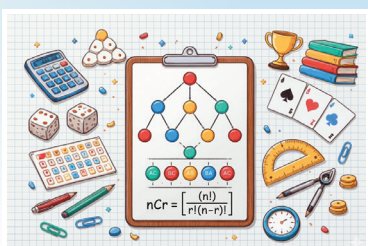
3. Critical Thinking and discussion Questions: You should include a written report answering the following:
 - Consistency: Which group has a smaller standard deviation? What does this tell you about the consistency of those students’ performances compared to the other group?
 - The “Average” Trap: Find two groups with similar means but very different variances. Why is the mean alone a poor representative of the students’ performance in this case?
 - Outliers: Identify any scores that fall far outside the whiskers of your box plot. What might explain these extreme results?

4. Assessment Rubric

Criteria	Excellent (5 pts)	Satisfactory (3 pts)	Needs Improvement (1 pt)
Accuracy	Calculations for s^2 , S , and quartiles are error-free.	Minor calculation errors present.	Significant errors in statistical formulas.
Visualization	Box plot is neatly drawn to scale with all 5 points labeled.	Box plot is present but lacks clear labels or scale.	Diagram is messy or mathematically incorrect.
Interpretation	Deep insight into what the variance says about student performance.	Basic understanding of spread is shown.	No connection made between numbers and school performance.

UNIT 13

COMBINATORICS



Key Unit Competence:

Use arrangements and combinations in determining the number of ways a random experiment occurs.

Learning objectives

By the end of this unit, you should:

Knowledge and understanding

1

- Explain the basic counting principles (addition and multiplication).
- Distinguish between arrangements and combinations (with or without repetitions or conditions).
- Describe Pascal's triangle and its relationship with binomial expansion.

Skills

2

- Construct Venn diagrams, tree diagrams, and contingency tables to solve counting problems.
- Apply multiplication principles to determine the number of possible outcomes in different situations.
- Calculate permutations and combinations with or without repetitions or conditions.
- Solve real-life problems involving binomial expansion and counting.

Attitudes and values

3

- Appreciate the importance of systematic and logical thinking in problem solving.
- Develop accuracy and persistence when solving counting and combinatorial problems.
- Show interest and curiosity in exploring different counting strategies.

Introduction

This unit introduces you to the fundamental concepts of combinatorics, which focus on counting and arranging objects in a systematic way. Combinatorics is a branch of mathematics that helps determine the number of possible outcomes in a random experiment, making it essential for solving problems in probability, statistics, and real-life decision-making. In this unit, you will learn how to use permutations and combinations to calculate the number of ways events can occur, considering different arrangements and selections. You will explore scenarios where order matters (permutations) and where order does not matter (combinations) and learn to apply these concepts to solve practical problems.

The knowledge and skills acquired in this unit will help you approach problems that involve counting possibilities, arranging objects, and selecting groups in an organized and logical manner. By the end of this unit, you will be able to apply combinatorial methods to analyze random experiments and make informed predictions based on the number of possible outcomes.

You will also acquire key terms such as permutation, combination, factorial, arrangement, selection, and counting principles, which are essential for understanding and solving combinatorial problems.

Introductory activity



1. A market in Kigali sells 3 popular fruits: Banana (B), Pineapple (P), and Mango (M).
 - a) If you want 1 fruit for your snack, how many choices do you have?
 - b) If you want 2 different fruits, list all possible combinations. How many are there?
2. You have 2 Rwandan storybooks: Inka y'Ishuri (I) and Umwana n'Igiti (U). You want to place them on a shelf. How many ways can you arrange them? List them.

13.1 Introduction to counting techniques

Activity 13.1



Two roads join A and B, and 3 roads join B and C. Write down different roads from A to C via B. How many are they? Discuss the different methods you can apply to find the correct answer.

A

B

C

A

B1

C1

B2

C2

C3



Content Summary

Counting in mathematics is the process of determining how many ways something can happen. It is the first step in solving problems involving arrangements, selections, and organization of objects. Everyday examples include choosing fruits at a market, arranging books on a shelf, or selecting students for a school committee.

Fundamental principle of counting

The fundamental principle of counting states that if there are n ways to do one thing and m ways to do another, then there are $n \times m$ ways to do both. For example, the number of ways of choosing a shirt (2 colours) and a pair of shoes (3 types) gives $2 \times 3 = 6$ possible outfits.

Techniques of counting

There are two main techniques of counting.

- The first is **listing, or enumeration**, which involves writing all possible outcomes to see the total. For instance, choosing 2 fruits from Banana (B), Mango (M), and Pineapple (P) gives the combinations B & M, B & P, and M & P, totalling 3 ways.
- The second technique is **multiplication, or the tree diagram method**, which is useful when making sequential choices.

Example

A student wants to have 1 fruit and 1 drink for breakfast. The available options are:

- (i) First option for fruits: Banana, Mango, Pineapple
- (ii) Second option for drinks: Juice, Milk

How many different breakfast combinations can the student make? List all possible combinations.

Solution:

Step 1: Identify the number of choices:

- Number of fruits = 3 (Banana, Mango, Pineapple)
- Number of drinks = 2 (Juice, Milk)

Step 2: Use the counting principle

To find the total number of breakfast combinations, multiply the number of choices for fruits by the number of choices for drinks:

The total combination $3 \times 2 = 6$

Step 3: List all possible combinations

1. Banana and Juice $\rightarrow$ (B,J)
2. Banana and Milk $\rightarrow$ (B,M)
3. Mango and Juice $\rightarrow$ (M,J)
4. Mango and Milk $\rightarrow$ (M,M)
5. Pineapple and Juice $\rightarrow$ (P,J)
6. Pineapple and Milk $\rightarrow$ (P,M).

Application activity 13.1



A student has 2 pencils (Red, Blue) and 2 notebooks (Math, English).

1. How many ways can the student choose 1 pencil and 1 notebook?
2. List all possible combinations.

13.2 Venn diagram

Activity 13.2



In a class survey about favourite subjects, 10 students like Mathematics, 8 students like Science, and 4 students like both Mathematics and Science.

1. Draw a Venn diagram with two overlapping circles labelled Mathematics and Science.
2. Fill in the diagram with the correct numbers.
3. Use your diagram to answer the following questions:
 - a) How many students like only Mathematics?
 - b) How many students like only Science?
 - c) How many students like at least one of the two subjects?



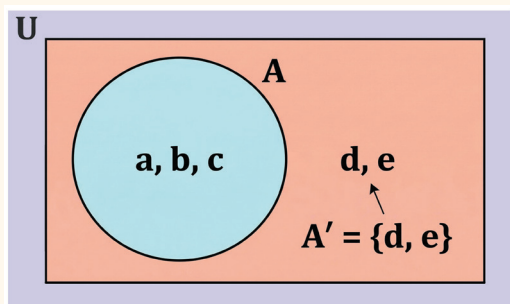
Content Summary

Venn diagram is a way of representing sets in a closed curve. They are usually circular or oval shaped. The universal set, written with U is represented by a rectangle and is a set containing all the subsets considered in the problem.

The complement of a set say A is the set written A' or $\overline{A}$ of an element in the universal set but not the set A i.e. $A' = U - A$ and we can write $A \cup A' = U$.

Example 1: If $U = \{a, b, c, d, e\}$ and $A = \{a, b, c\}$ then $A' = \{d, e\}$

The Venn diagram solution is presented as:



$n(A)$ or $|A|$ reads as the number of the element in set A . This is also called the cardinal number of A .

The intersection of sets A and B is denoted by $A \cap B$. This represents the set of all elements common to both A and B.

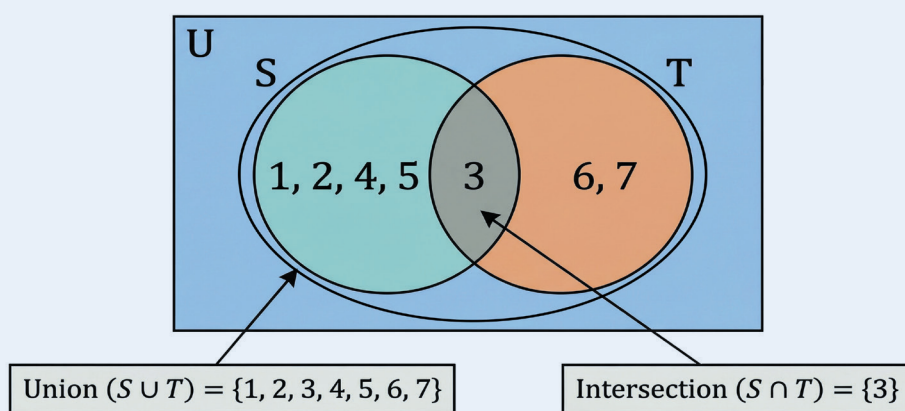
The union of sets A and B is denoted by $A \cup B$. This represents the set of all elements belonging to A or B or both A and B.

Example 2:

Let set $S = \{1, 2, 3, 4, 5\}$ and set $T = \{3, 6, 7\}$. Based on these two sets, identify their union ($S \cup T$) and their intersection ($S \cap T$).

Solution:

The Venn diagram solution is presented as



Disjoint sets are sets with no elements in common. If A and B are disjoint sets, $A \cap B = \phi$. Also, A and B are said to be **mutually exclusive**.

Application activity 13.2



Consider the Universal set $U = \{1, 2, 3, 4, 5, 6, 7, 8\}$

We have two subsets: Set $S = \{1, 2, 3, 4, 5\}$ and $T = \{6, 7, 8\}$

- Find the intersection $S \cap T$
- Find the union $S \cup T$
- Are sets S and T disjoint? Explain why.

13.3 Tree diagram and number of outcomes

Activity 13.3



- 1) There are 2 roads joining A and B and 3 roads joining B and C. Write down different roads from A to C via B. How many are they?

A

B

C

A

B1
B2

C1
C2
C3

- 2) A family has now three children and the wife was giving either one boy or one girl. If F=female or Girl and M= male or Boy, use a tree diagram to illustrate the eventual gender for 3 children in that family.
- 3) You are going to a restaurant for dinner. You can either get a crunchy or a soup. For meat, you can choose either beef, chicken, or fish.
- a) Create a tree diagram and list all possible request;
- b) Find the total number of all possible selections.



Content Summary

A **tree diagram** is simply a way of representing a sequence of events. Tree diagrams are particularly useful in probability since they record all possible outcomes in a clear and uncomplicated manner.

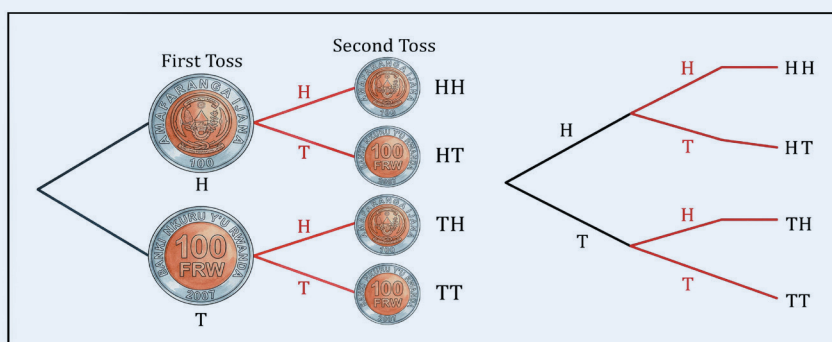
It has branches and sub-branches which help us to see the sequence of events and all the possible outcomes at each stage.

Example:

1. Using a tree diagram, determine all the possible outcomes that can be obtained when a coin is tossed twice.

Solution:

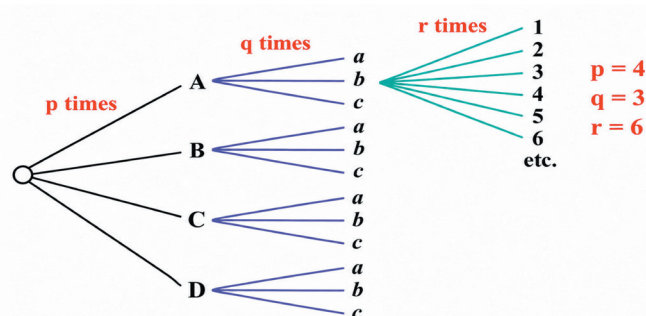
In the first toss, we get either a head (H) or a tail (T). On getting a H in the first toss, we can get a H or T in the second toss. Likewise, after getting a T in the first toss, we can get a H or T in the second toss.



$$S = \{HH, HT, TH, TT\}$$

We see that the number of possible outcomes is 4. There 2 selections of the first choice and 2 selections of the second. This is $2 \times 2 = 4$.

Multiplication principle: If a task consists of a sequence of choices in which there are p selections for the first choice, q selections for the second choice, r selections for the third choice, and so on, then the task of making these selections can be done in $n = p \times q \times r$.

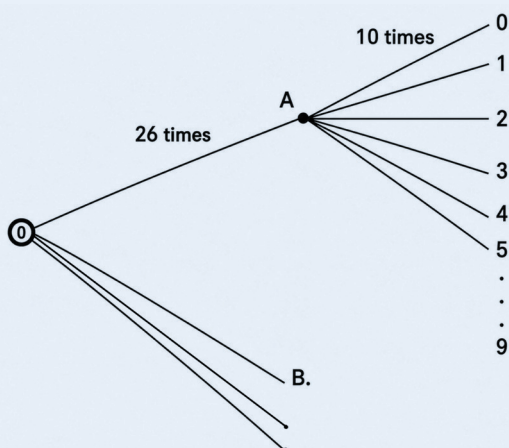


Example 1:

How many two-symbol code words can be formed if the first symbol is a letter (uppercase) and the second symbol is a digit?

Solution:

It sometimes helps to begin by listing some of the possibilities. The code consists of a letter (uppercase) followed by a digit, so some possibilities are A0, A1, A2, ..., B0, B1, B2, B3, ..., X9.



The task consists of making two selections.

The first selection requires choosing an uppercase letter (26 choices) and the second task requires choosing a digit (10 choices).

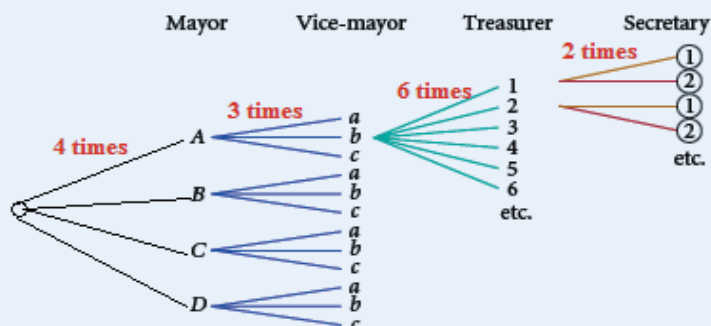
By the Multiplication Principle, there are:

$n = 26 \times 10 = 260$ different code words of the type described.

Example 2: In a city election there are four candidates for mayor, three candidates for vice-mayor, six candidates for treasurer, and two for secretary. In how many ways can these four offices be filled?

Solution:

The task of filling an office can be divided into four choices:

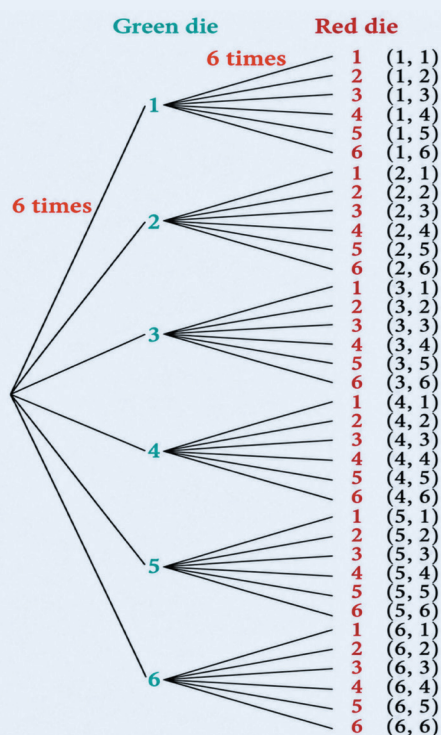


Corresponding to each of the four possible mayors, there are three vice-mayors. These two offices can be filled in $4 \times 3 = 12$ different ways. Also, corresponding to each of these 12 possibilities, we have six different choices for treasurer, giving $12 \times 6 = 72$ different possibilities. Finally, to each of these 72 possibilities there correspond two choices for secretary. In all, these offices can be filled in $4 \times 3 \times 6 \times 2 = 144$ different ways.

Example 3: Consider an experiment in which two dice are rolled, one green and the other red. The set of outcomes consists of all the different ways that the dice may come to rest. Find the number of all possible outcomes. It is called the cardinal of the sample space for this experiment.

Solution:

Since there are 6 possible ways for the green die to come up and 6 ways for the red die to come up, the number of outcomes of rolling the two dice is $6 \times 6 = 36$.



We use a tree diagram to obtain a list of the 36 outcomes.

Example 4: Find the cardinal of a sample space for the experiment of selecting one family from the set of all possible three-child families.

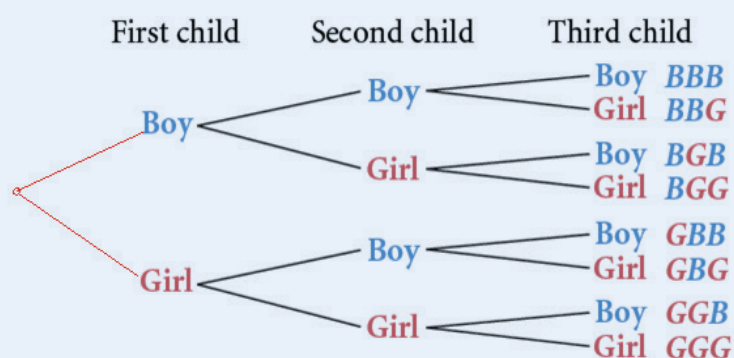
Solution:

One way of describing a sample space for this experiment is to list the number of girls in the family. The only possibilities are members of the set $\{0, 1, 2, 3\}$.

That is, a three-child family can have 0, 1, 2, or 3 girls. This sample space has four outcomes.

Another way of describing a sample space for this experiment is to denote B as “boy” and G as “girl.” Then the sample space can be given as $\{BBB, BBG, BGB, BGG, GBB, GBG, GGB, GGG\}$ where BBB means first child is a boy, second child is a boy, third child is a boy, and so on.

This sample space has $2 \times 2 \times 2 = 8$ outcomes.



Application activity 13.3



- 1) Angelus has asked his girlfriend to make all the decisions for their date on her birthday. She will pick a restaurant and an activity for the date. Angelus will choose a gift for her. The local restaurants include Mexican, Chinese, Seafood, and Italian. The activities she can choose from are swimming, bowling, and movies. Angelus will buy her either candy or flowers.
 - a) Draw a tree diagram to illustrate the choices if Angelus must start by choosing a gift, they choose a food at the restaurant and then they choose an activity to be done there.
 - b) How many outcomes are there for these three decisions?
- 2) A designer has 3 fabric colours he/she may use for a dress: red, green, and blue. Four different patterns are available for the dress. If each dress design requires one colour and one pattern, how many different dress designs are possible?

13.4 Contingency table

Activity 13.4



Below is the contingency table based on the school club survey. This table shows the relationship between two categories: Grade Level and Favourite Activity.

Grade Level	Favourite: Sports	Favourite: Music	Totals
Juniors	12	8	20
Seniors	18	12	30
Column totals	30	20	50

Use the table above to answer these three direct questions:

1. Reading a specific cell: Exactly how many seniors chose music as their favorite activity?
2. Calculating totals: If you look at the “Column Totals,” what is the total number of students who prefer Sports across both grade levels?
3. Data analysis: What is the total number of all students who participated in this survey?



Content Summary

This is a method for presenting the frequencies of outcomes from an experiment in which sample observations are categorised according to two criteria. Each cell of the table gives the number of occurrences for a particular combination of categories. An example of a contingency table in which individuals are categorised by gender and performance is shown below.

	Male	Female	Total
Pass	32	43	75
Fail	8	15	23
	40	58	98

A final column giving the row sums and a final row giving the column sums may be added. Then the sum of the final column and the sum of the final row both equal the number in the sample.

If the sample is categorized according to three or more criteria, the information can be presented similarly in several such tables.

Application activity 13.4



In the Northern Province of Rwanda, a cooperative manager surveyed 200 farmers to see if the type of crop they planted depended on the size of their land. The collected data are presented in the following table.

Farm size	Irish potatoes	Pyrethrum	Total
Small scale	80	40	120
Large scale	30	50	80
Total	110	90	200

- How many large-scale farmers chose to plant pyrethrum as their primary crop?
- What is the total number of farmers in this survey who planted irish potatoes?

13.5 Multiplication principles

Activity 13.5



How many ways can 3 students line up in a row?

- List all the possible arrangements.
- How many arrangements are there in total?



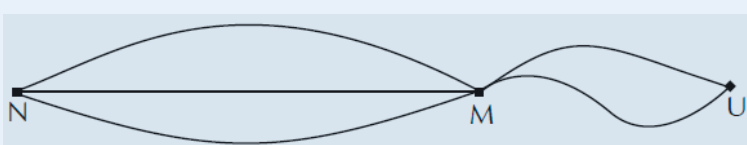
Content Summary

The multiplication principle is also known as the product rule or product principle. Suppose that an experiment (operation) is to be performed in two successive ways. If there are n_1 different ways of performing the first operation, and for each of these there are n_2 different ways of performing a second independent operation, then the two operations in succession can be performed in $n_1 \times n_2$ different ways. More generally, if the operation is composed of k successive steps, which may be performed in $n_1, n_2, \dots, n_k$ distinct ways, respectively, then the operation may be performed in $n_1 \times n_2 \times \dots \times n_k$ distinct ways.

Example 1:

There are 3 different paths from Nizeyimana's to Mukaneza's and 2 different paths from Mukaneza's to Uwimana's. How many different pathways could be taken from Nizeyimana's to Uwimana's via Mukaneza's?

Solution:



The number of different pathways is $3 \times 2 = 6$

Example 2:

Habimana has 4 shirts, 3 pairs of trousers, and 2 pairs of shoes. He chooses a shirt, a pair of trousers, and a pair of shoes to wear every day. Find the maximum number of days he does not need to repeat his clothing.

Solution:

The number of all possible different clothing items is $4 \times 3 \times 2 = 24$. Therefore, the maximum number of days Habimana does not need to repeat his clothing is 24.

Example 3:

Find how many ways the first three places (with no ties) can be filled in a race with 5 contestants.

Solution:

The first place can be filled in 5 ways, since any contestant can come first. When the first place has been filled, there are 4 more contestants to choose from for the second place. Hence the second place can be filled in 5×4 ways. Finally, for each of these ways, the third place can be filled by any of the remaining 3 contestants, and the total number of ways is $5 \times 4 \times 3 = 60$

Note 1: Factorial notation

The factorial notation of n integers denoted by $n!$ is the product of the first consecutive integers and is read as 'factorial n '. Thus, $n! = n \times (n-1) \times (n-2) \times \dots \times 3 \times 2 \times 1$

Thus, for example $6! = 6 \times 5 \times 4 \times 3 \times 2 \times 1 = 720$

Note 2: $0! = 1$

Application activity 13.5



A school wants to select a president, a vice-president, and a secretary from 4 students. Each student can hold only one position. How many ways can these positions be filled? Choose the corrected answer.

- A. 12
- B. 24
- C. 6
- D. 64

13.6 Arrangements with repetitions

Activity 13.6



You have 3 different letters: A, B, and C.

How many ways can you arrange them in a row if each letter is used only once?



Content Summary

An **arrangement with repetition** is an ordered selection of objects in which:

1. **Order matters** (e.g., $ABC \neq BAC$)
2. **Objects can be repeated** in different positions

This type of arrangement is also called a **permutation with repetition**.

If you have n distinct objects and want to arrange them into r positions, the formula is simply: n^r

Where: n = number of objects to arrange, r = number of positions.

Example:

A locker code has 3 digits. Each digit can be 0, 1, or 2. How many different 3-digit codes can be formed if digits can be repeated?

Solution:

Number of digits available (n) = 3(0,1,2)

Number of positions (r) = 3

Total arrangement = $3^3 = 3 \times 3 \times 3 = 27$

There are 27 different 3-digit codes.

Note: If you had n positions and you could reuse the objects infinitely (meaning you don't run out of choices), you would have n choices for all n positions. Following the n^r rule where $r = n$, the total arrangements with repetition would be: n^n .

(For example, making a 4-letter password using only the letters A, B, C, and D, where you can repeat letters like "AAAA", would give you $4^4 = 256$ possibilities). As you see, this case is not practical.

Application activity 13.6



A game company wants to create 5-symbol passwords for a new board game using the symbols ★, ▲, ●, ■, ◆. Each symbol can be used more than once. How many different 5-symbol passwords can be created?

13.7 Permutations

In standard combinatorics, “**arrangement**” and “**permutation**” mean the exact same thing. They are synonyms used to describe a grouping of items where the **order matters**.

13.7 1 Permutations with or without repetitions

Activity 13.7.1



You have 3 letters: A, B, C.

You want to make a 2-letter code.

- Letters cannot be repeated. Write all possible codes that can be made.
- Letters can be repeated. Write all possible codes that can be made.



Content Summary

Suppose that two of the three different pictures, A, B and C, are to be hung, in line, on a wall of three places. The pictures can be hung in different orders:

AB	AC
BA	BC
CA	CB

Each of these orders is a particular arrangement of the pictures and is called a permutation. Thus, a permutation is an ordered arrangement of the items in a set. Here, there are 6 permutations, or $\frac{3!}{(3-2)!} = \frac{3!}{1!} = 6$

The number of arrangements of r objects, taken from n unlike objects, can be considered as the number of ways of filling r places in order by the n given objects or

The number of ways to arrange r items from a set of n distinct items (**without repetition**) is given by: $P_r^n = \frac{n!}{(n-r)!}$

Note: A different notation can be used of permutation with no repetition: $P_r^n = P(n, r) = {}_nP_r$
Condition: $n \geq r$

Example 1:

The first object can be put in any of 6 places; following this, the second object can be put in any of 5 remaining places; following this, the last object can be put in any of 4 remaining places. Thus, by the product principle, there are $6 \times 5 \times 4 = 120$ possible permutations of 6 objects taken 3 at time. i.e $P_3^6 = \frac{6!}{(6-3)!} = 120$

Note 1: If $n = r$, requires us to arrange n **different objects** in n **different places** (without repetition) and the number of different ways is: $P_n^n = \frac{n!}{(n-n)!} = \frac{n!}{0!} = n!$

Example 2:

How many different ways can 10 students be arranged in a class of 10 places?

Solution:

The number of different ways is $10! = 3,628,800$ ways.

Note 2: The number of permutations of r objects, taken from n **unlike objects** where each of which may be **repeated** any number of times, is n^r . This is the case where You have n distinct types of objects available, and you want to fill r positions. Because you can reuse the same object as many times as you want, your choices never shrink. The number of permutation or arrangement of n objects in r positions is : n^r .

Example 3: Find how many three-letter codes can be formed from an alphabet of 26 letters (assuming that letters can be repeated).

Solution:

The first letter of the code may be any of the 26 in the alphabet. The same is true for the second letter and for the third letter. Therefore, the total number of codes is $26^3 = 26 \times 26 \times 26 = 1,757,626$ codes

Real-life Applications

a) Seating Arrangements

Example: In how many ways can 5 people sit in a row of 5 chairs?

Solution:

$5! = 120$ ways.

Application: Planning seating for events, weddings, or conferences where each person is distinct.

b) Rankings in Competitions

Example: 8 runners in a race. How many possible outcomes for gold, silver, and bronze (order matters)?

Solution:

$P(8,3) = 8!/5! = 8 \times 7 \times 6 = 336$ possible rankings.

Application: Sports analytics, predicting medal winners.

c) Password Creation (No repetition)

Example: Create a 4-digit PIN using digits 0–9 without repeating any digit.

Solution:

$P(10,4) = 10 \times 9 \times 8 \times 7 = 5040$ possible PINs.

Application: Security systems, generating unique codes.

d) Scheduling:

Example: A teacher wants to schedule 6 different subjects in 6 periods, one per period.

Solution:

$6! = 720$ possible schedules.

Application: School timetables, work shift assignments.

e) Permutations in Genetics

Example: Arranging 4 different genes on a chromosome (assuming linear order matters).

Solution:

$4! = 24$ possible gene orders.

Application: Understanding genetic sequences and mutations.

Example:

1) How many permutations of 3 letters chosen from eight letters of the word **RELATION** are there?

Solution:

We see that all those eight letters are distinguishable (unlike). So the required arrangements are given by ${}^8P_3 = \frac{8!}{(8-3)!} = \frac{8!}{5!} = 336$.

2) How many permutations are there of 2 letters chosen from letters A, B, C, D, E?

Solution:

There are 5 letters which are distinguishable (unlike). So the required arrangements are given by ${}^5P_2 = \frac{5!}{(5-2)!} = \frac{5!}{3!} = 20$.

3) How many different ways can a chairperson and an assistant chairperson be selected for a research project if there are seven scientists available?

Solution:

It is ${}^7P_2 = \frac{7!}{(7-2)!} = 42$

4) All we know about Shannon, Patrick, and Ryan is that they have different birthdays.

If we listed all the possible ways this could occur, how many would there be when we assume that there are 365 days in a year?

Solution:

This is an example of a permutation in which 3 birthdays are selected from a possible 365 days, and no birthday may repeat itself.

The number of ways that this can occur is $\frac{365!}{(365-3)!} = 365 \times 364 \times 363 = 48,228,180$

Application activity 13.7.1



Find how many ways the first three places (with no ties) can be filled in a race with 15 contestants.

13.7.2 Permutations of n unlike objects on a row and notation

Activity 13.7.2



I have a suitcase with a *number lock*, labelled with 3 first digits from 0 to 2. The lock can be opened if 3 specific digits are set in a particular sequence. Unfortunately, I forgot that specific sequence. In how many ways can I try to unlock my suitcase?



Content Summary

If I forgot the specific sequence to unlock this suitcase, I will immediately start listing all possible sequences of 3 digits taken from digits $\{0, 1, 2\}$ and reversed order of these digits is also included such as $\{012, 210, 021, 120, \dots\}$.

To get the number of ways in which some or all digits (in this case), at a time are arranged, we note that the first digit to be written down can be chosen in 3 ways; the second digit can be chosen in 2 ways because there are 2 remaining digits to be written down and then the third digit can be chosen in 1 way because it is only one digit remain to be written down. Thus, the three operations can be performed in $3 \times 2 \times 1 = 6$ ways.

Example:

Give all different ways three students: Cauchy, Emmanuel and Alexis can be sit on the same bench. Two ways were given in the table, complete others.

Cauchy	Emmanuel	Alexis
Cauchy	Alexis	Emmanuel
..	..	..

This suggests the following fact:

The number of different permutations of n different objects (unlike objects) in a row is $n \times (n-1) \times (n-2) \times (n-3) \times \dots \times 2 \times 1 = n!$

This corresponds to the number of ways of arranging n unlike objects in a line. It is a **permutation** or ordered arrangement of n *different* objects chosen from n objects.

A useful short hand of writing this operation is $n!$ (read **n factorial**). Then,

$$n! = n \times (n-1) \times (n-2) \times (n-3) \times \dots \times 2 \times 1$$

Thus, $1! = 1$, $2! = 2 \times 1 = 2$, $3! = 3 \times 2 \times 1 = 6$, $4! = 4 \times 3 \times 2 \times 1 = 24$, $5! = 5 \times 4 \times 3 \times 2 \times 1 = 120$ and so on.

Note that $0! = 1$

Example:

1) $\frac{6!}{2!4!} = \frac{6 \times 5 \times 4 \times 3 \times 2 \times 1}{2 \times 1 \times 4 \times 3 \times 2 \times 1} = 15$

2) $\frac{7!}{4!2!} = \frac{7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1}{4 \times 3 \times 2 \times 1 \times 2 \times 1} = 105$

3) Five children have to be seated on a bench.

a) In how many ways they can be seated?

b) How many arrangements are they, if the youngest child has to sit at the left end of the bench?

Solution:

a) Since there are five children, the first child can be chosen in 5 ways, the next child in 4 ways, the next in 3 ways, the next in 2 ways and the last in 1 way. Then, the number of ways is $5! = 5 \times 4 \times 3 \times 2 \times 1 = 120$.

b) Now, if the youngest child has to sit at the left end of the bench, this place can be filled in only 1 way. The next child can then be chosen in 4 ways, the next in 3 ways and so on. Thus, the number of total arrangement is $1 \times 4! = 1 \times 4 \times 3 \times 2 \times 1 = 24$.

4) What is the number of ways of sitting n_1 men and n_2 women on a bench if men sit together and women sit together?

Solution:

Note that the number of ways of arranging n objects in a line is $n! = 1 \times 2 \times 3 \times 4 \times 5 \times \dots \times n$.

Given that the total number $n = n_1 + n_2$ of people made by n_1 men and n_2 women, the total number of ways of sitting (arranging) n_1 men and n_2 women on a bench such that men sit together and women sit together considering that one group can sit at the left side or the right side of the other is $2 \times n_1! \times n_2!$.

5) Three different mathematics books and five other books are to be arranged on a bookshelf.

Find :

a) the number of possible arrangements of the book.

b) the number of possible arrangements if the three mathematics books must be kept together?

Solution:

We have 8 books altogether.

a) Since we have 8 books altogether, the first book can be chosen in 8 ways, the next in 7 ways, the next in 6 ways and so on. Thus, the total arrangement is $8! = 40320$

b) Since the 3 mathematics books have to be together, consider these bound together as one book. There are now 6 books to be arranged and these can be performed in $6! = 720$.

Note that we have taken the three mathematics book as one book; these three books can be arranged in $3! = 6$ ways. Thus, the total number of arrangements is $720 \times 6 = 4320$.

Application activity 13.7.2



1. Simplify

a. $\frac{5!}{2!}$ b. $\frac{10!}{6!7!}$

2. Four different English books, five different Biology books and ten other books are to be arranged on a bookshelf. Find

- The number of possible arrangements of the books.
- The number of possible arrangements if the three Biology books must be kept together?

13.7.3 Permutations of indistinguishable objects: Permutations with repetitions of n objects chosen from n objects

Activity 13.7.3



- Make 4 number cards numbered 1, 2, 3, 4. Arrange them on a desk in different ways by permuting the positions of each letter. How many arrangements are they possible?
- Consider the letter cards with letters that form the word BOOM. Change their arrangement in different ways.
 - Write down all possible *different* arrangements.
 - How many arrangements are they possible of four letters in the word “BOOM”?



Content Summary

A **permutation** is an ordered arrangement of r objects chosen from n objects.

Consider the arrangements of six letters in the word “**AVATAR**” (a title used for the movie).

We see that there are three **A**’s (or 3 alike letters).

- Let the three **A**’s in the word be distinguished as A_1A_1, A_2A_2 and A_3A_3 respectively. Then all the six letters are different, so the number of permutations of them (called labeled permutations) is $n! = 6!$.
- However, consider each of the real permutations without distinguishing the three **A**’s, for example **W=RATAVA**.
- The following are all of the 6 ($=3!$) labeled permutations among the $6!$ ones, which come from permuting the three labeled **A**’s in **W=RATAVA**:
 $RA_1TA_2VA_3, RA_1TA_3VA_2, RA_2TA_1VA_3, RA_2TA_3VA_1, RA_3TA_1VA_2, RA_3TA_2VA_1$
- All these six labeled permutations should be considered as an identical real permutation, which is **W=RATAVA**.

- Since each real permutation has six of such labeled permutations coming from the three A's, we conclude that the desired number of real permutations is just $\frac{6!}{3!} = \frac{6 \times 5 \times 4 \times 3!}{3!} = 6 \times 5 \times 4 = 120$

This suggests the following fact:

The number of different permutations of n **indistinguishable** objects with n_1 alike, n_2 alike, ..., is $\frac{n!}{n_1! \times n_2! \times \dots \times n_k!}$, where $n = n_1 + n_2 + \dots + n_k$ and n_i is the number of objects of type i .

In other words, if you have a total of n objects, where object 1 repeats n_1 times, object 2 repeats n_2 times, and so on, the total number of unique permutations is $\frac{n!}{n_1! \times n_2! \times \dots \times n_k!}$.

This corresponds to the number of arranging all n objects in line of which n_1 of one type are alike, n_2 of the second type are alike and so on.

Note: Alike means that the objects in a group are indistinguishable from one another. Note that if all $n_i = 1$ (no object is repeated), The number of different permutations becomes for n **distinguishable** objects which is $\frac{n!}{n_1! \times n_2! \times \dots \times n_k!} = \frac{n!}{1! \times 1! \times \dots \times 1!} = n!$.

Example:

1. How many distinguishable six-digit numbers can be formed from the digits 5, 4, 8, 5, 5, 4?

Solution:

There are 6 letters in total with three 5's and two 4's. Then the required numbers are

$$\frac{6!}{3!2!} = \frac{720}{12} = 60$$

2. How many arrangements can be made from the letters of the word TERRITORY?

Solution:

There are 9 letters in total with three R's and two T's.

$$\text{Thus, we have } \frac{9!}{3! \times 2!} = \frac{9 \times 8 \times 7 \times 6 \times 5 \times 4}{2 \times 1} = \frac{60,480}{2} = 30,240 \text{ arrangements.}$$

3. In how many different ways can 4 identical red balls, 3 identical green balls and a yellow ball be arranged in a row?

Solution:

Applications:

- Distinct colour sequences: Suppose you have 3 red balls and 2 blue balls, all in a row. You can need to know the number of colour sequences to be done on a row.
- In manufacturing, counting possible orders of products on a shelf if some are identical.
- Sitting on a bench: Different ways people can sit on a bench.

Application activity 13.7.3



1. How many different arrangements can be made from the letters of the word
a) ENGLISH b) MATHEMATICS
2. How many arrangements can be made from the letters of English alphabet?
3. In how many ways can 4 women, 3 children and 2 men be seated in a row, if women remain together, children together and men sit together?

13.7.4 Circular permutation

Activity 13.7.4



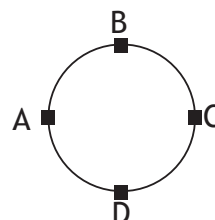
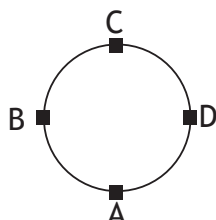
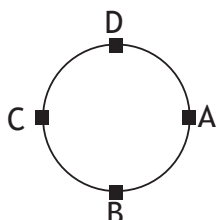
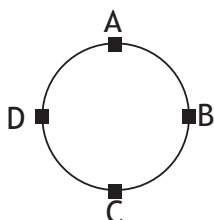
Imagine 4 friends A, B, C, and D are going to sit around a round table.

- a) If the table were straight (a row), how many different ways can they sit?
- b) If the table is round, do you think the number of arrangements will be the same? Why or why not?



Content Summary

Observe the arrangement of four letters A, B, C and D on a circle as shown below.



We can say that for all the arrangements:

- A is on the opposite side of C and its neighbours are B and D, and
- B is on the opposite side of D and its neighbours are A and C.

Thus all of the four arrangements are the same.

In arranging 4 different objects on a circle, 4 are indistinguishable. Therefore, the number of different ways of arranging 4 objects on a circle is $\frac{4!}{4} = 3!$

The number of arrangements of n unlike things in a circle will therefore be $(n-1)!$ In those cases where clockwise and anticlockwise arrangements are not considered to be different, this reduces to $\frac{1}{2}(n-1)!$

Example 1: In how many different ways can 5 people be arranged in a circle?

Solution:

The number of different ways of arranging 5 people is $(5 - 1)! = 4! = 24$

Example 2: Five friends A, B, C, D, and E are sitting around a round table. Clockwise and anticlockwise arrangements are considered identical.

How many distinct seating arrangements are possible?

Solution:

The number of friends $(n) = 5$

$$\text{Number of arrangements} = \frac{1}{2}(5 - 1)! = \frac{4!}{2} = \frac{24}{2} = 12$$

There are 12 distinct ways the 5 friends can sit around the table when clockwise and anticlockwise arrangements are considered the same.

Application activity 13.7.4



The team members Anna, Ben, Clara, David, Eva, Felix, and Grace are going to sit around a round table. Consider the following two cases:

Case 1: Clockwise and anticlockwise arrangements are considered different.

Case 2: Clockwise and anticlockwise arrangements are considered the same.

Find the number of distinct seating arrangements for each case.

13.8 Combinations:

13.8.1 Meaning and properties of combinations

Activity 13.8.1



Answer with True or False. Explain your answer to your colleagues

- Choosing 2 fruits from an apple and a banana is the same as choosing a banana and an apple.
- In some counting situations, the order of selection does not matter.
- Choosing A then B is always different from choosing B then A.
- When forming a group, the order of people in the group matters.



Content Summary

A combination is a selection of objects where the order is not taken into account. It means the order in which the objects are arranged is not important. When considering the number of combinations of r objects, the order in which they are placed is not important. Note that ABC, ACB, BAC, BCA, CAB, CBA are different arrangements, but they represent the same combination of letters. Denoting the number of combinations of three letters from the seven

letters A, B, C, D, E, F, G, by ${}^7P_3 = \frac{n!}{r!(n-r)!} = 35$

In combinatorics, a **combination** is a selection of items from a larger pool where the **order of selection does not matter**.

Imagine you are choosing 3 fruits from a basket containing an apple, a banana, a cherry, and a date.

- If you choose the apple, the banana, and the cherry, that is **one single combination**.
- It does not matter if you picked the apple first, second, or third; the resulting group is exactly the same.

The Mathematical Formula

To find the number of ways to choose r unique items from a total pool of n items, we use the combination formula (often read as “ n choose r ”): $\binom{n}{r} = C(n, r) = \frac{n!}{r!(n-r)!}$

In general, the number of combinations of r objects from n unlike objects is $\binom{n}{r} = \frac{n!}{r!(n-r)!}$

Condition: $n \geq r$

Example 1: How many different committees of 3 people can be chosen from a group of 12 people?

Solution:

The number of committees is $\binom{12}{3} = \frac{12!}{3!(12-3)!} = \frac{12 \times 11 \times 10 \times 9!}{3! \times 9!} = \frac{12 \times 11 \times 10}{3 \times 2 \times 1} = 220$

200 committees can be formed.

Note: Other notations that can be used are nC_r , C_r^n , ${}_nC_r$ or $C(n, r)$

Example 2: Suppose you want to select a committee of 3 students from a group of 5 students ($n = 5$, $r = 3$).

Solution:

Using the formula: $\binom{5}{3} = \frac{5!}{3!(5-3)!} = \frac{5 \times 4 \times 3 \times 2 \times 1}{(3 \times 2 \times 1) \times (2 \times 1)}$

By canceling out the common terms: $\binom{5}{3} = \frac{5 \times 4}{2 \times 1} = \frac{20}{2} = 10$

There are exactly **10 different combinations** (or distinct committees) you can form

Example 3: A bag contains 6 red, 4 white and 8 blue balls. If 3 balls are drawn at random, find the number of ways of selecting one red, one white and one blue.

Solution:

To find the number of ways to select exactly one red, one white, and one blue ball from the bag, we use the **Fundamental Counting Principle (Multiplication Rule)**. This rule states that if one event can occur in m ways and a second independent event can occur in n ways, then the two events can occur together in $m \times n$ ways.

Step 1: Find the number of ways to choose each color

- Selecting 1 red ball from the 6 available red balls: $\binom{6}{1} = 6$ ways
- Selecting 1 white ball from the 4 available white balls: $\binom{4}{1} = 4$ ways
- Selecting 1 blue ball from the 8 available blue balls: $\binom{8}{1} = 8$ ways

Step 2: Multiply the possibilities together

Since we are selecting one ball of each color simultaneously, we multiply the number of choices for each color: Total ways = $6 \times 4 \times 8 = 192$

There are exactly **192 ways** to select one red, one white, and one blue ball.

Properties

$$1) \binom{n}{r} = \binom{n}{n-r}$$

$$2) \binom{n}{n} = \binom{n}{0} = 1$$

$$3) \binom{n}{1} = n$$

There is a relationship between combinations and arrangements:

$$\binom{n}{r} = \frac{P_r^n}{r!} = \frac{n!}{(n-r)!} \times \frac{1}{r!} = \frac{n!}{r!(n-r)!}$$

Application activity 13.8.1



Answer with True or False for the following:

- a) A combination is a selection of objects where the order of arrangement matters
- b) In combinations, the arrangements ABC and CBA are considered the same.
- c) The formula for combinations is $nCr = \frac{n!}{r!(n-r)!}$
- d) The value of $nC0$ is always equal to 1 for any positive integer n .
- e) The number of combinations is always greater than the number of permutations for the same values of n and r

13.8.2 Combinations with repetitions or conditions

Activity 13.8.2



1. What is a combination? How is it different from a permutation?

Using the combination formula, find the number of ways to choose 2 fruits from {Apple, Banana, Orange}.

1) Condition in combination

Sometimes we are given a condition that must be considered in combination.

Example 1: A committee of 5 is to be chosen from 12 men and 8 women. In how many ways can this be done if there are to be 3 men and 2 women on the committee?

Solution:

The number of ways of choosing the men is

$$\binom{12}{3} = \frac{12!}{3!(12-3)!} = \frac{12 \times 11 \times 10 \times 9!}{3! \times 9!} = \frac{12 \times 11 \times 10}{3 \times 2 \times 1} = 220 \text{ ways}$$

The number of ways of choosing the women is $\binom{8}{2} = \frac{8!}{2!(8-2)!} = \frac{8!}{2! \times 6!} = \frac{8 \times 7}{2 \times 1} = 28 \text{ ways}$

Therefore, the total number of ways of choosing the committee (given by the product rule) is $220 \times 28 = 6160 \text{ ways}$.

Example 2: A group consists of 4 boys and 7 girls. In how many ways can a team of three be selected if it is to contain:

- a) 1 boys and 2 girls
- b) Girls only
- c) Boys only
- d) At least 2 boys
- e) At least one member of each gender?

Solution:

$$\text{a) } \binom{4}{1} \times \binom{7}{2} = \frac{4!}{1! \times (4-1)!} \times \frac{7!}{2! \times (7-2)!} = \frac{4 \times 3!}{1! \times 3!} \times \frac{7 \times 6 \times 5!}{2! \times 5!} = 4 \times 21 = 84 \text{ ways}$$

$$\text{b) } \binom{7}{3} = \frac{7!}{3!(7-3)!} = \frac{7 \times 6 \times 5 \times 4!}{3! \times 4!} = 35 \text{ ways}$$

$$\text{c) } \binom{4}{3} = \frac{4!}{3!(4-3)!} \times \frac{4 \times 3!}{3! \times 1!} = 4 \text{ ways}$$

$$\begin{aligned} \text{d) } \binom{4}{2} \times \binom{7}{1} + \binom{4}{3} \times \binom{7}{0} &= \frac{4!}{2! \times (4-2)!} \times \frac{7!}{1! \times (7-1)!} + 4 \times 1 \\ &= \frac{4 \times 3 \times 2!}{2! \times 2!} \times \frac{7 \times 6!}{1 \times 6!} + 4 = 6 \times 7 + 4 = 42 + 4 = 46 \text{ ways} \end{aligned}$$

$$\begin{aligned} \text{e) } \binom{4}{1} \times \binom{7}{2} + \binom{4}{2} \times \binom{7}{1} &= \frac{4!}{1! \times (4-1)!} \times \frac{7!}{2! \times (7-2)!} + \frac{4!}{2! \times (4-2)!} \times \frac{7!}{1! \times (7-1)!} \\ &= \frac{4 \times 3!}{3!} \times \frac{7 \times 6 \times 5!}{2! \times 5!} + \frac{4 \times 3 \times 2!}{2! \times 2!} \times \frac{7 \times 6!}{6!} = 4 \times 21 + 6 \times 7 = 84 + 42 = 126 \text{ ways} \end{aligned}$$

Example 3:

A football team of 11 is to be chosen from 15 players. How many different teams can be selected if:

- there is no condition (no restriction)?
- the captain was chosen before and must be in the team?
- the captain was chosen before must be in the team and one of the other players is injured and cannot play?

Solution:

a) The number of different teams possible is:

$$\binom{15}{11} = \frac{15!}{11! \times (15-11)!} = \frac{15 \times 14 \times 13 \times 12 \times 11!}{11! \times 4!} = \frac{15 \times 14 \times 13 \times 12}{4!} = \frac{15 \times 14 \times 13 \times 12}{4 \times 3 \times 2 \times 1} = 1365$$

b) Since the captain must be in the team, the selection is to have the other 10 players from the 14 available to join the captain. The number of teams possible is

$$\binom{14}{10} = \frac{14!}{10! \times (14-10)!} = \frac{14 \times 13 \times 12 \times 11 \times 10!}{10! \times 4!} = \frac{14 \times 13 \times 12 \times 11}{4!} = \frac{14 \times 13 \times 12 \times 11}{4 \times 3 \times 2 \times 1} = 1001$$

c) Since the captain must be in the team and one player cannot play because of injury, the selection is to now to choose the other ten players from the 13 available to join the captain. The number of teams possible is

$$\binom{13}{10} = \frac{13!}{10! \times (13-10)!} = \frac{13 \times 12 \times 11 \times 10!}{10! \times 3!} = \frac{13 \times 12 \times 11}{3!} = \frac{13 \times 12 \times 11}{3 \times 2 \times 1} = 286$$

2) Combination with repetition

In combinatorics, **combinations with repetition** (also known as **multi-choose** or **stars and bars**) occur when we are selecting r items from a pool of n distinct types of items, and we are allowed to choose the same type of item more than once.

Just like standard combinations, the **order of selection still does not matter**.

The Intuition: The Ice Cream Shop Example

Imagine an ice cream shop that offers 3 flavors ($n = 3$): **Vanilla (V)**, **Chocolate (C)**, and **Strawberry (S)**.

You want to buy a 4-scoop bowl ($r = 4$).

Because you can repeat flavors, you aren't limited to choosing just one of each. Possible combinations include:

- $\{V, V, V, V\}$ (all Vanilla)
- $\{V, V, C, S\}$ (two Vanilla, one Chocolate, one Strawberry)

- {C, C, S, S} (two Chocolate, two Strawberry)

Notice that {V, V, C, S} is exactly the same as {V, C, V, S} because the scoops sit together in the same bowl—the order you scooped them doesn't matter.

Formula

To find the number of ways to choose r items from a pool of n types with repetition allowed,

we use the following formula: $\binom{n+r-1}{r} = \frac{(n+r-1)!}{r!(n-1)!}$

Where:

- n is the number of **types** of items available.
- r is the number of items you are **choosing**.

Example

A bakery sells 5 different types of donuts. In how many ways can you select a box of 12 donuts?

Solution:

- **Identify the variables:** * $n = 5$ (the types of donuts available)
 $r = 12$ (the total number of donuts we are choosing)
- **Apply the formula:** $\binom{5+12-1}{12} = \binom{16}{12} = \binom{16}{4} = \frac{16 \times 15 \times 14 \times 13}{4 \times 3 \times 2 \times 1} = \frac{43,680}{24} = 1,820.$

There are exactly **1,820 different ways** to choose your box of donuts.

Comparison: Standard and repetition

Context	Formula	Condition
Without Repetition	$\binom{n}{r} = C(n, r) = \frac{n!}{r!(n-r)!}$	$r \leq n$ (You cannot choose more items than exist in the pool).
With Repetition	$\binom{n+r-1}{r} = \frac{(n+r-1)!}{r!(n-1)!}$	r can be larger than n (Like choosing 12 donuts from 5 types).

Application activity 13.8.2



A basketball team of 6 is to be chosen from 11 available players. In how many ways can this be done if

- there are no restrictions?
- 3 of the players are automatic selections?
- 3 of the players are automatic selections and 2 other players are injured and cannot play?

13.9 Pascal's triangles

Activity 13.9



Expand the following identities

a) $(a + b)^2 =$

b) $(a + b)^3 = (a + b)^2 (a + b) =$

c) $(a + b)^4 = (a + b)^3 (a + b) =$



Content Summary

It is sometimes necessary to expand expressions such as $(x + y)^4$ but the multiplication is tedious when the power is three or more. We now describe a far quicker way of obtaining such expansions. Consider the following expansions:

$$(x + y)^0 = 1$$

$$(x + y)^1 = x + y$$

$$(x + y)^2 = x^2 + 2xy + y^2$$

$$(x + y)^3 = x^3 + 3x^2y + 3xy^2 + y^3$$

$$(x + y)^4 = x^4 + 4x^3y + 6x^2y^2 + 4xy^3 + y^4$$

The first thing to notice is that the powers of x and y in the terms of each expansion, form a pattern. Looking at the expansion of $(x + y)^4$, we see that the first term is x^4 and then the power of x and then the power of x and y is 4 and the expansion ends with y^4 . Notice also that the number of terms is $4 + 1 = 5$. In general, for the expansion of $(x + y)^n$ where $n = 1, 2, 3, \dots$

As we move from left to right across the expansion, the powers of x decrease by 1 while the powers of y increase by 1.

The sum of the power of x and y in each term is n .

The number of terms in the expansion is $n + 1$.

To construct Pascal's triangle, start with 1 at the top, then continue placing numbers below it in a triangular pattern. Each number is the sum of the two numbers above it added together except for the edges, which are all ones. The following is the table showing Pascal's triangle. It helps find the coefficients of the terms.

Power	Coefficients						
0	→	1					
1	→	1	1				
2	→	1	2	1			
3	→	1	3	3	1		
4	→	1	4	6	4	1	
5	→	1	5	10	10	5	1
6	→	1	6	15	20	15	6
7	→	1	7	21	35	21	7
				...			

Example 1:

Expand $(x + y)^5$

Solution:

$$\begin{aligned}(x + y)^5 &= 1x^5y^0 + 5x^4y^1 + 10x^3y^2 + 10x^2y^3 + 5x^1y^4 + 1x^0y^5 \\ &= x^5 + 5x^4y + 10x^3y^2 + 10x^2y^3 + 5xy^4 + y^5\end{aligned}$$

Example 2:

Expand $(2x - 3)^4$

Solution:

Use Pascal's triangle:

$$\begin{aligned}(x + y)^4 &= 1x^4y^0 + 4x^3y^1 + 6x^2y^2 + 4x^1y^3 + 1x^0y^4 \\ &= x^4 + 4x^3y + 6x^2y^2 + 4xy^3 + y^4\end{aligned}$$

Replacing x by $2x$ and y by -3 gives

$$\begin{aligned}(2x - 3)^4 &= (2x)^4 + 4(2x)^3(-3) + 6(2x)^2(-3)^2 + 4(2x)(-3)^3 + (-3)^4 \\ &= 16x^4 - 96x^3 + 216x^2 - 216x + 81\end{aligned}$$

Application activity 13.9



1) $(2x + 3)^3$

2) $(3x - 1)^4$

13.10 Binomial expansion

Activity 13.10



Given the binomial $(a + b)^2 = a^2 + 2ab + b^2$

Show that:

a) The coefficients of $a^2 = \binom{2}{0} = 1$

b) The coefficient of $2ab = \binom{2}{1} = 2$

c) The coefficient of $b^2 = \binom{2}{2} = 1$



Content Summary

Expanding expressions such as $(1 + x)^{20}$, the coefficients of the terms can be obtained from Pascal's triangle, but the construction of the triangular array would be tedious. However, the **combination notation** could give a more general method to expand powers of any binomial bracket without constructing that triangular array. Thus, if $(x + y)^6$ is to be expanded, we have:

$$(x + y)^6 = \binom{6}{0}x^6y^0 + \binom{6}{1}x^5y^1 + \binom{6}{2}x^4y^2 + \binom{6}{3}x^3y^3 + \binom{6}{4}x^2y^4 + \binom{6}{5}x^1y^5 + \binom{6}{6}x^0y^6$$
$$= x^6 + 6x^5y + 15x^4y^2 + 20x^3y^3 + 15x^2y^4 + 6xy^5 + y^6$$

This argument can be generalized as follows: if n is any positive integer, $(x + y)^n$ can be expanded to give a series of terms in descending power of x and ascending power of y where the term containing x^{n-r} has the coefficient $\binom{n}{r}$. Thus, we have:

$$(x + y)^n = \binom{n}{0}x^n y^0 + \binom{n}{1}x^{n-1}y^1 + \binom{n}{2}x^{n-2}y^2 + \dots + \binom{n}{r}x^{n-r}y^r + \dots + \binom{n}{n}x^{n-n}y^n$$

In sigma (summation) notation, this gives us the theorem: For every integer

$$n \geq 1, (x + y)^n = \sum_{r=0}^n \binom{n}{r} x^{n-r} y^r$$

Example 1:

Expand $(2x + 3y)^3$

Solution:

Step 1: Set up the Expansion We write out each term using the binomial coefficients

$$(2x + 3y)^3 = \binom{3}{0}(2x)^3(3y)^0 + \binom{3}{1}(2x)^2(3y)^1 + \binom{3}{2}(2x)^1(3y)^2 + \binom{3}{3}(2x)^0(3y)^3$$

Step 2: Simplify the Terms

$$\text{Term 1: } 1 \times 8 \times 1 = 8x^3$$

$$\text{Term 2: } 3 \times 4 \times 3 = 36x^2y$$

$$\text{Term 3: } 3 \times 2 \times 9 = 54xy^2$$

$$\text{Term 4: } 1 \times 1 \times 27 = 27y^3$$

Step 3: Write the final result

$$(2x + 3y)^3 = 8x^3 + 36x^2y + 54xy^2 + 27y^3$$

Application activity 13.10

Write down the first three terms of the expansion in ascending powers of x : $\left(1 - \frac{x}{2}\right)^{10}$

13.11 Guided practice on binomial expansion**Activity 13.11**

1. $(a + b)^2 = a^2 + 2ab + b^2$
2. In the expression $(a + b)^3$, the coefficients are 1,3,4,1
3. $(a - b)^2 = a^2 + 2ab + b^2$
4. The coefficients in $(a + b)^n$ can be found using combinations

**Content Summary**

The concept of combinations is a method of selecting terms, leading to the general use of:

$\binom{n}{r}$ to determine the coefficients in the expansion of $(a \pm b)^n$.

Take the following steps:

- 1) Identify patterns in coefficients and powers
- 2) Use combinations to calculate coefficients
- 3) Write complete expansions of expressions such as $(a \pm b)^n$

Example 1:

Write down the expansions of each of the following:

(a) $(a + b)^4$ (b) $(a - b)^3$ (c) $(x + 2y)^5$

Solution:

$$\begin{aligned}(a) \quad (a + b)^4 &= \binom{4}{0}a^4b^0 + \binom{4}{1}a^3b^1 + \binom{4}{2}a^2b^2 + \binom{4}{3}a^1b^3 + \binom{4}{4}a^0b^4 \\ &= a^4 + 4a^3b + 6a^2b^2 + 4ab^3 + b^4\end{aligned}$$

$$(b) \quad (a - b)^3 = \binom{3}{0}a^3b^0 - \binom{3}{1}a^2b^1 + \binom{3}{2}a^1b^2 - \binom{3}{3}a^0b^3 = a^3 - 3a^2b + 3ab^2 - b^3$$

$$\begin{aligned}(c) \quad (x + 2y)^5 &= \binom{5}{0}x^5(2y)^0 + \binom{5}{1}x^4(2y)^1 + \binom{5}{2}x^3(2y)^2 + \binom{5}{3}x^2(2y)^3 + \binom{5}{4}x^1(2y)^4 + \binom{5}{5}x^0(2y)^5 \\ (x + 2y)^5 &= \binom{5}{0}x^5(2y)^0 + \binom{5}{1}x^4(2y)^1 + \binom{5}{2}x^3(2y)^2 + \binom{5}{3}x^2(2y)^3 + \binom{5}{4}x^1(2y)^4 + \binom{5}{5}x^0(2y)^5 \\ &= x^5 + 5x^4(2y) + 10x^3(4y^2) + 10x^2(8y^3) + 5x(16y^4) + 32y^5 \\ &= x^5 + 10x^4y + 40x^3y^2 + 80x^2y^3 + 80xy^4 + 32y^5\end{aligned}$$

Application activity 13.11



Expand $(x - 3y)^5$



Assessing knowledge and understanding

Section A: Answer by true or false

- 1) According to the Fundamental Principle of Counting, if there are m ways to do one thing and n ways to do another, the total number of ways to do both in succession is $m + n$.
- 2) In a permutation, the order of the items in an arrangement matters (e.g., $ABC \neq BAC$).
- 3) If a set of n objects contains some indistinguishable (alike) items, the number of permutations is found by dividing $n!$ by the factorials of the number of alike objects.

Section B: Chose the correct answer

- 1) A student wants to have 1 fruit and 1 drink for breakfast. If there are 3 types of fruits (Banana, Mango, Pineapple) and 2 types of drinks (Juice, Milk), how many different breakfast combinations are possible?
A) 5 B) 6 C) 9 D) 3
- 2) How many different 3-digit locker codes can be formed using the digits $\{0,1,2\}$ if digits can be repeated?
A) 6 B) 9 C) 27 D) 3
- 3) What is the value of $5!$ (5 factorial)?
A) 15 B) 25 C) 120 D) 60

Section C: Match the Counting Tool/Concept in Column A with its correct description or example in Column B.

Column A (Concept)	Column B (Description/Example)
1. Venn Diagram	a) A diagram with branches to show a sequence of events.
2. Tree Diagram	b) A way of representing sets and their intersections using closed curves.
3. Contingency Table	c) $n!$ represents the product of the first n consecutive integers.
4. Factorial Notation	d) A table that categorizes observations according to two criteria.

Assessing skills

- 1) A class has 6 students. A teacher wants to choose 2 students to represent the class in a meeting.
How many different pairs of students can be chosen?
- 2) A group of 7 students includes 4 boys and 3 girls. A committee of 3 students is to be selected. The committee must include **at least one girl**.
How many different committees can be formed?

Assessing attitudes and values

- 3) When creating a password for an online student registry or a mobile money account, why should a student value the principles of combinatorics (permutations and combinations)?
 - a) It helps calculate the exact number of possible arrangements to understand how password length and character variety prevent unauthorized access and hacking.
 - b) It provides a shortcut to bypass security protocols so that users never lock themselves out of their accounts.
 - c) It proves that digital security is entirely random and that calculating possibilities has no effect on real-life safety.
 - d) It allows users to create passwords that are long and complex purely to impress network administrators.
- 4) Your class needs to choose a student representative (classmate post) from a pool of eligible candidates. Why should the student election committee value using systematic counting techniques to find the exact number of ways this choice can be made?
 - a) It allows the committee to calculate the total number of valid possibilities beforehand, ensuring the ballot or voting slip is structured accurately, transparently, and free from administrative confusion.
 - b) It helps the committee secretly limit the options so that only the top-performing mathematics students can be elected.
 - c) It proves that calculating possibilities is a theoretical exercise designed to delay the election process.
 - d) It automatically tells the committee which candidate will win the election before any votes are actually cast.

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